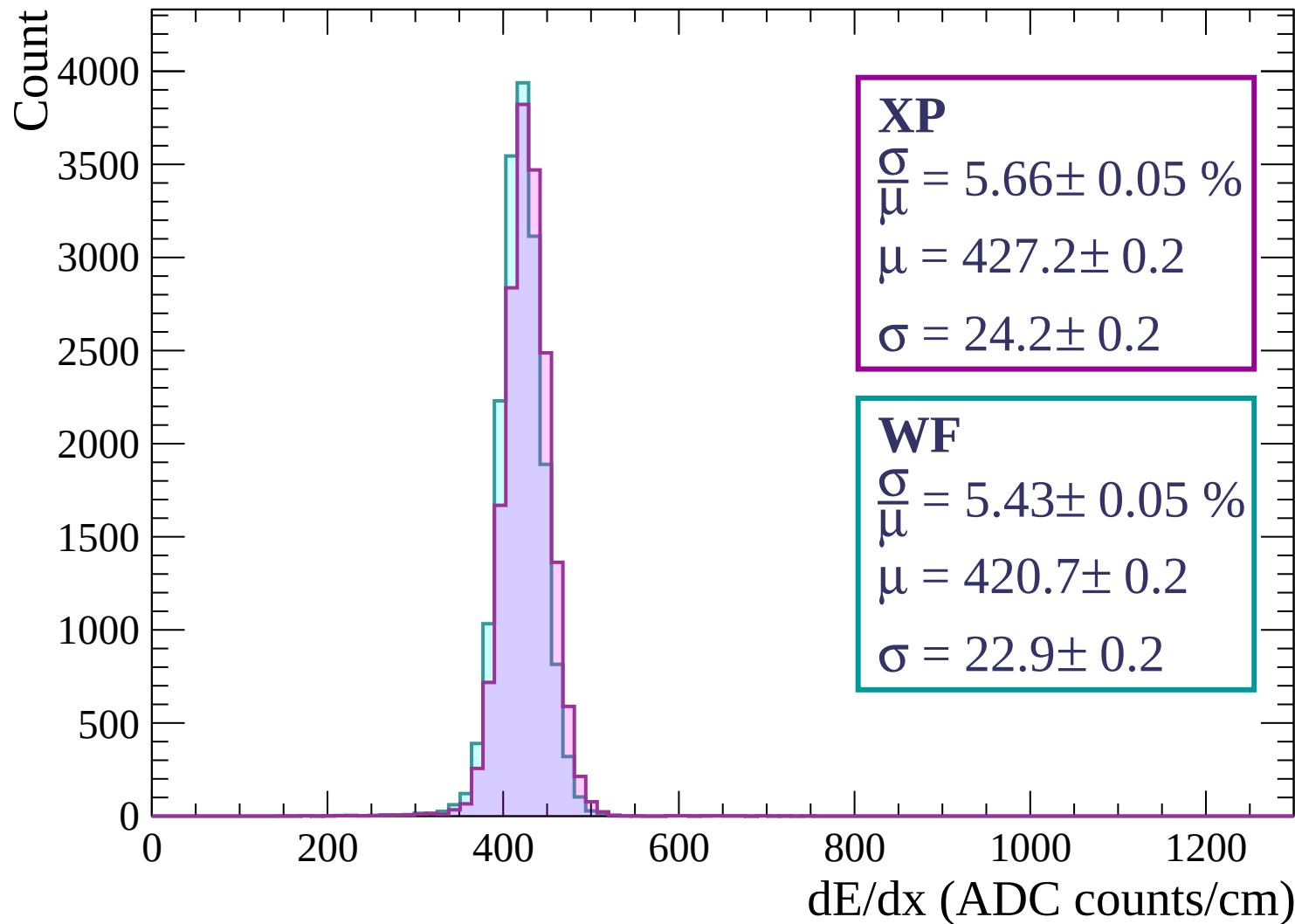
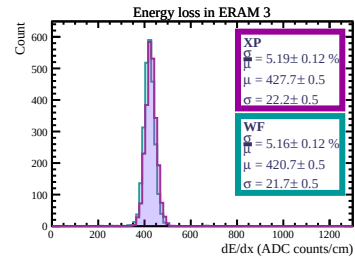
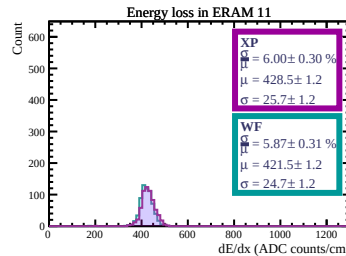
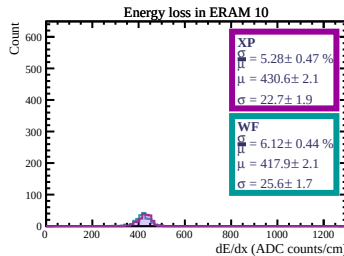
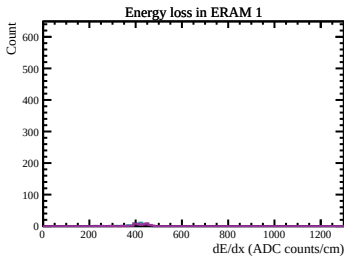
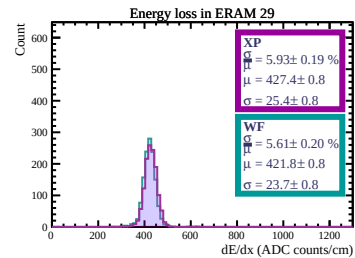
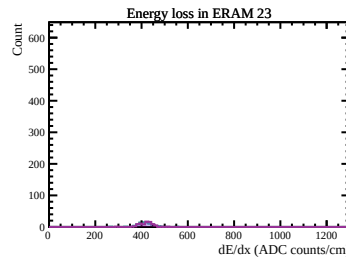
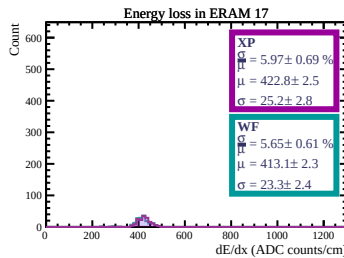
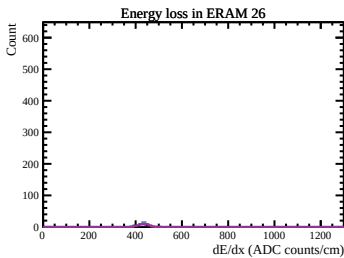
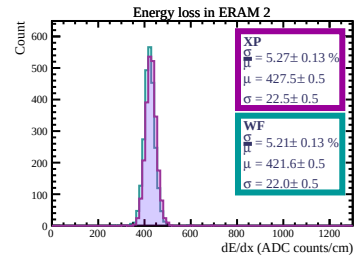
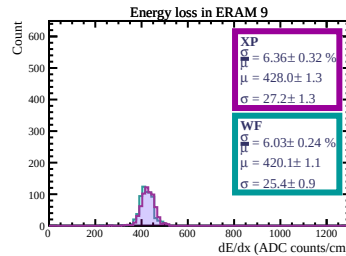
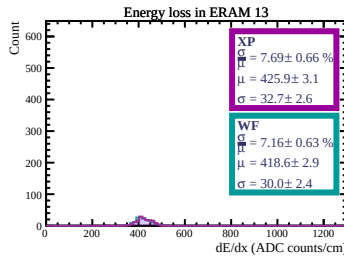
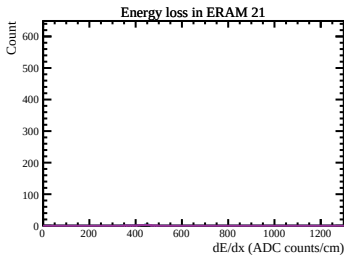
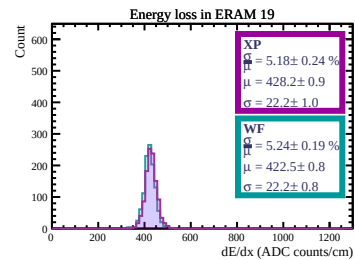
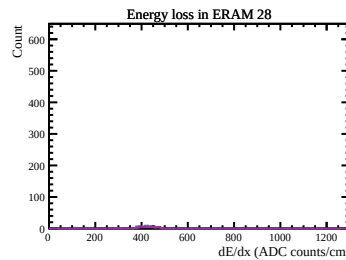
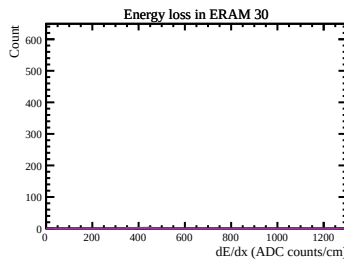
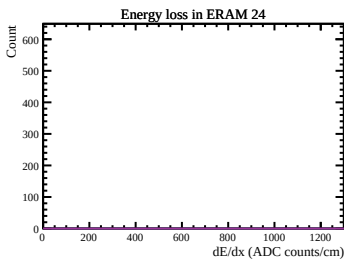
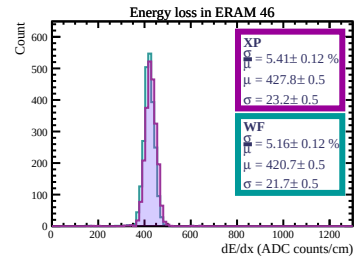
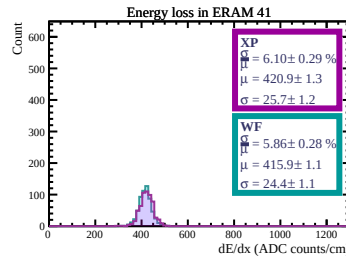
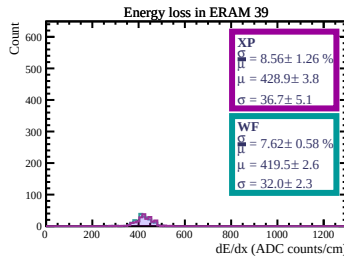
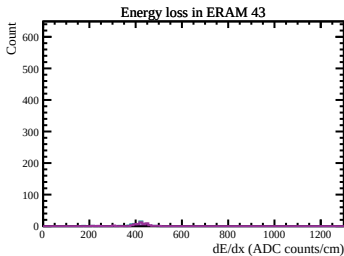
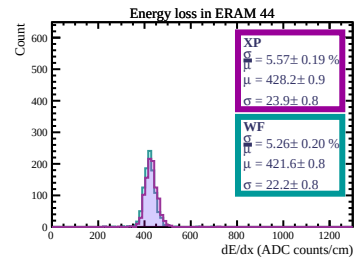
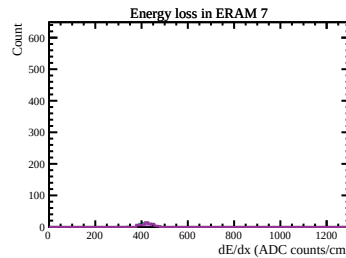
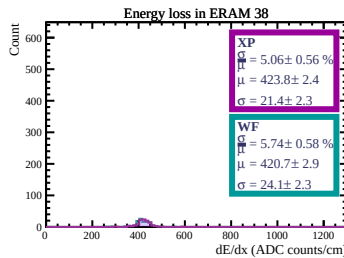
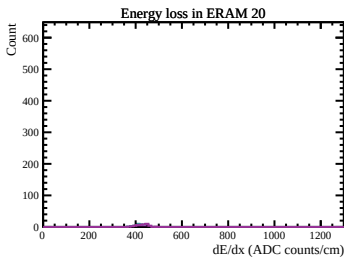
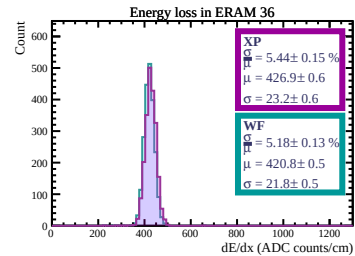
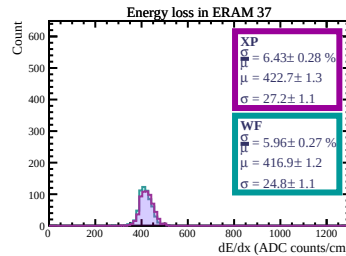
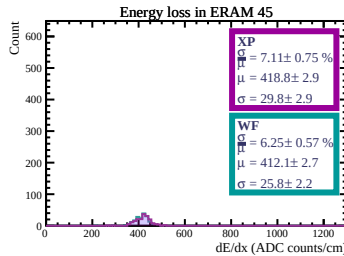
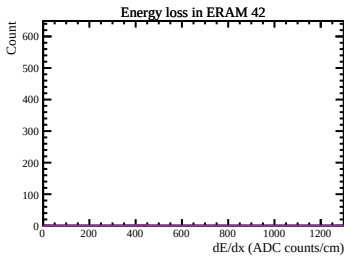
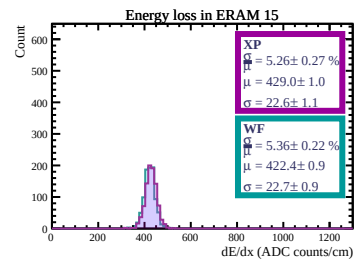
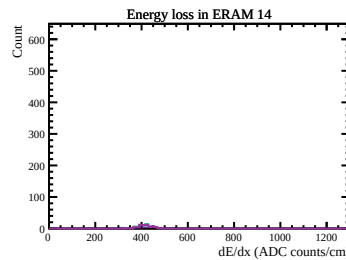
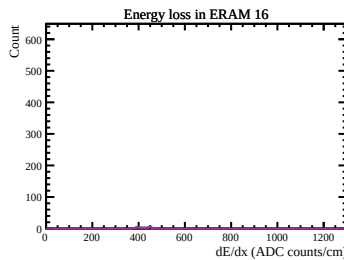
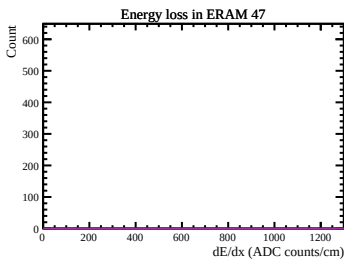
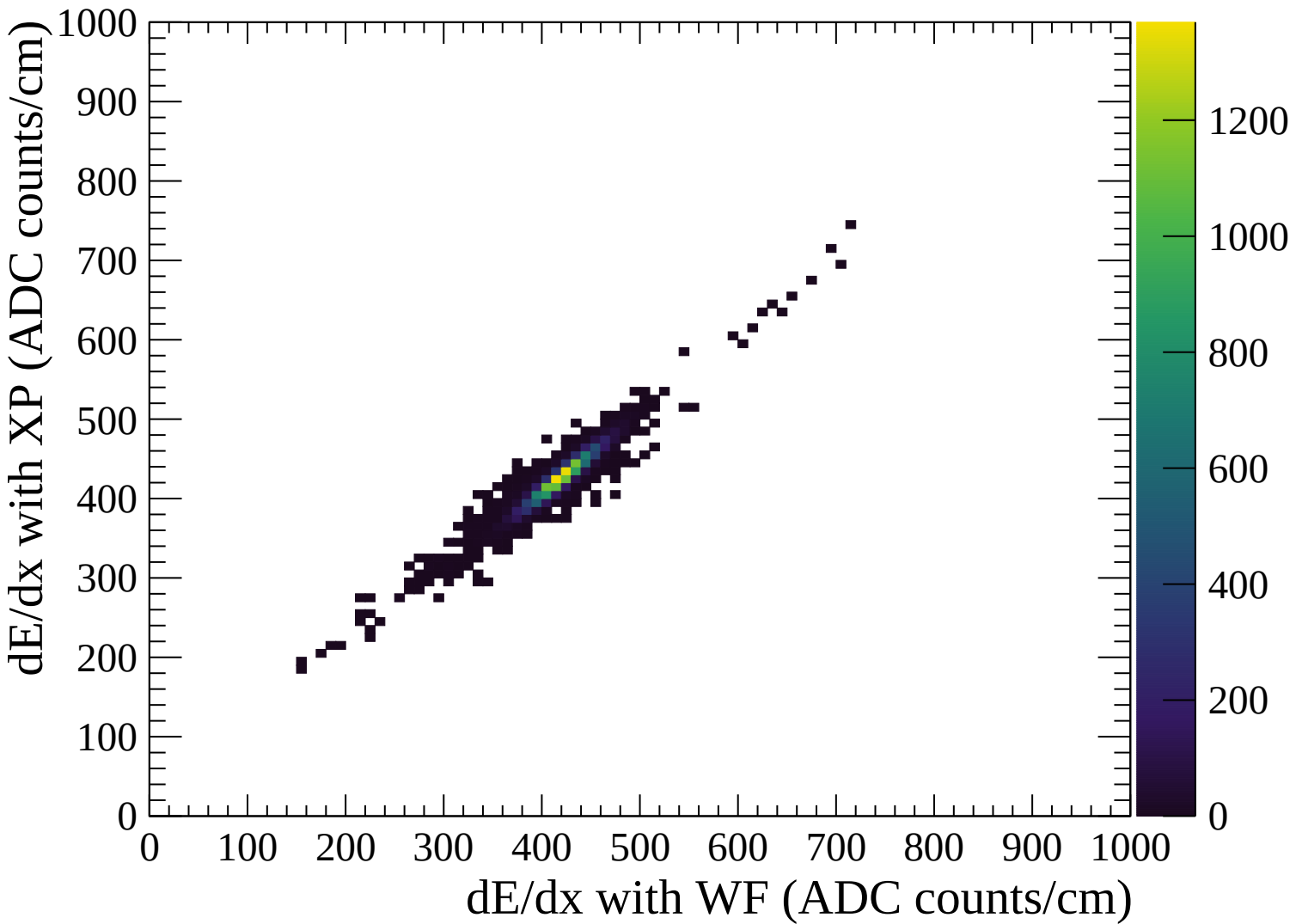
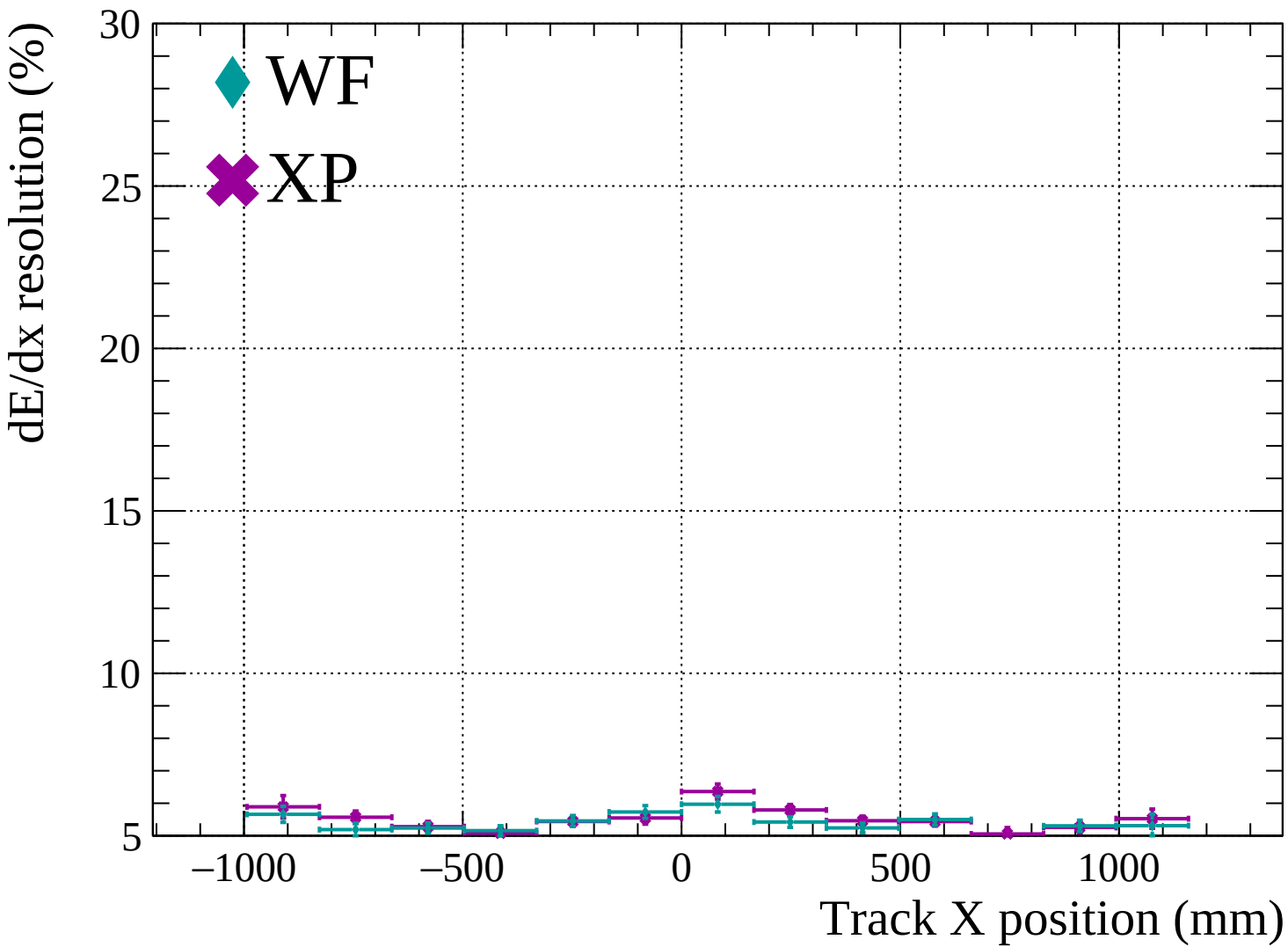


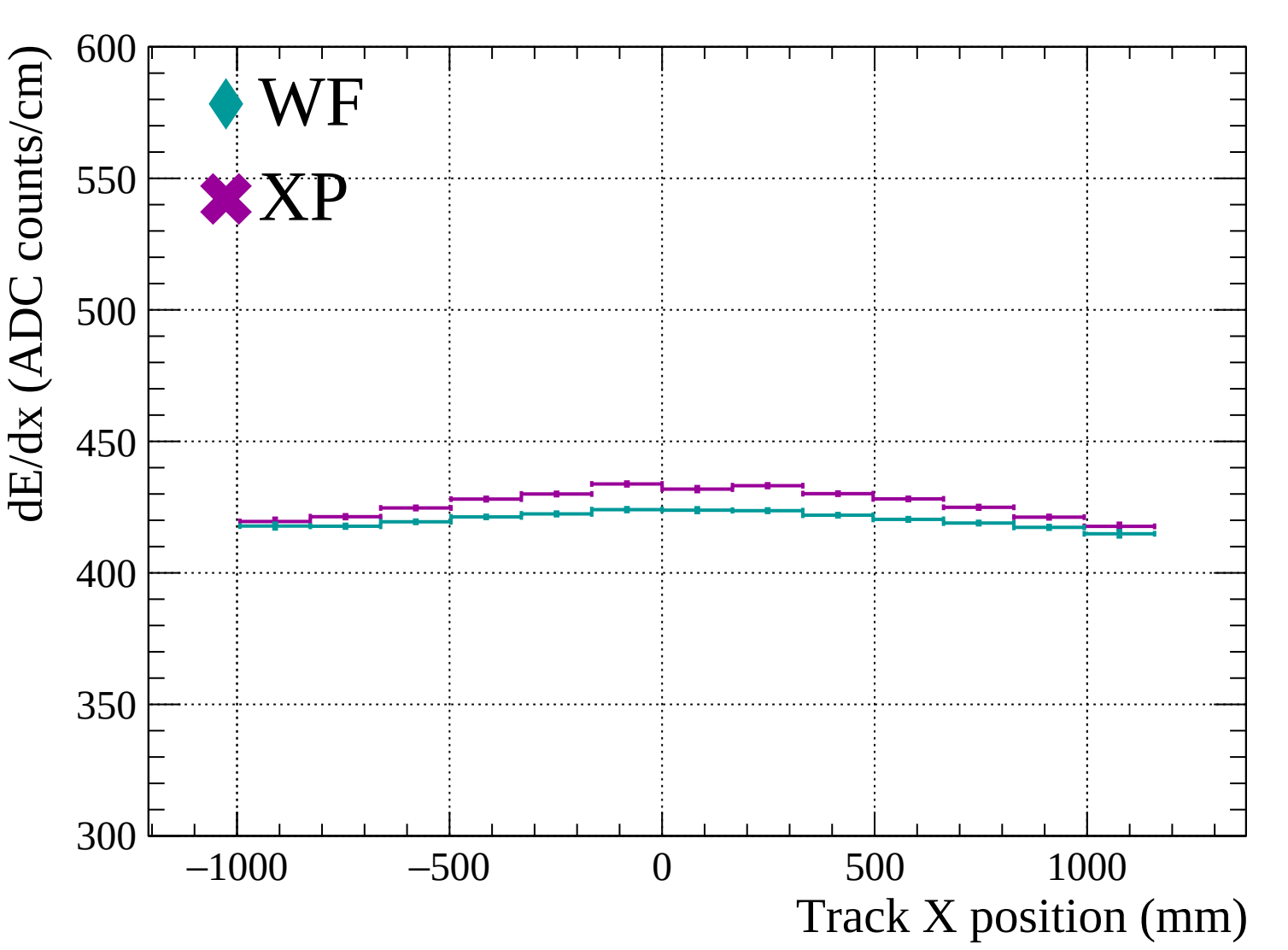


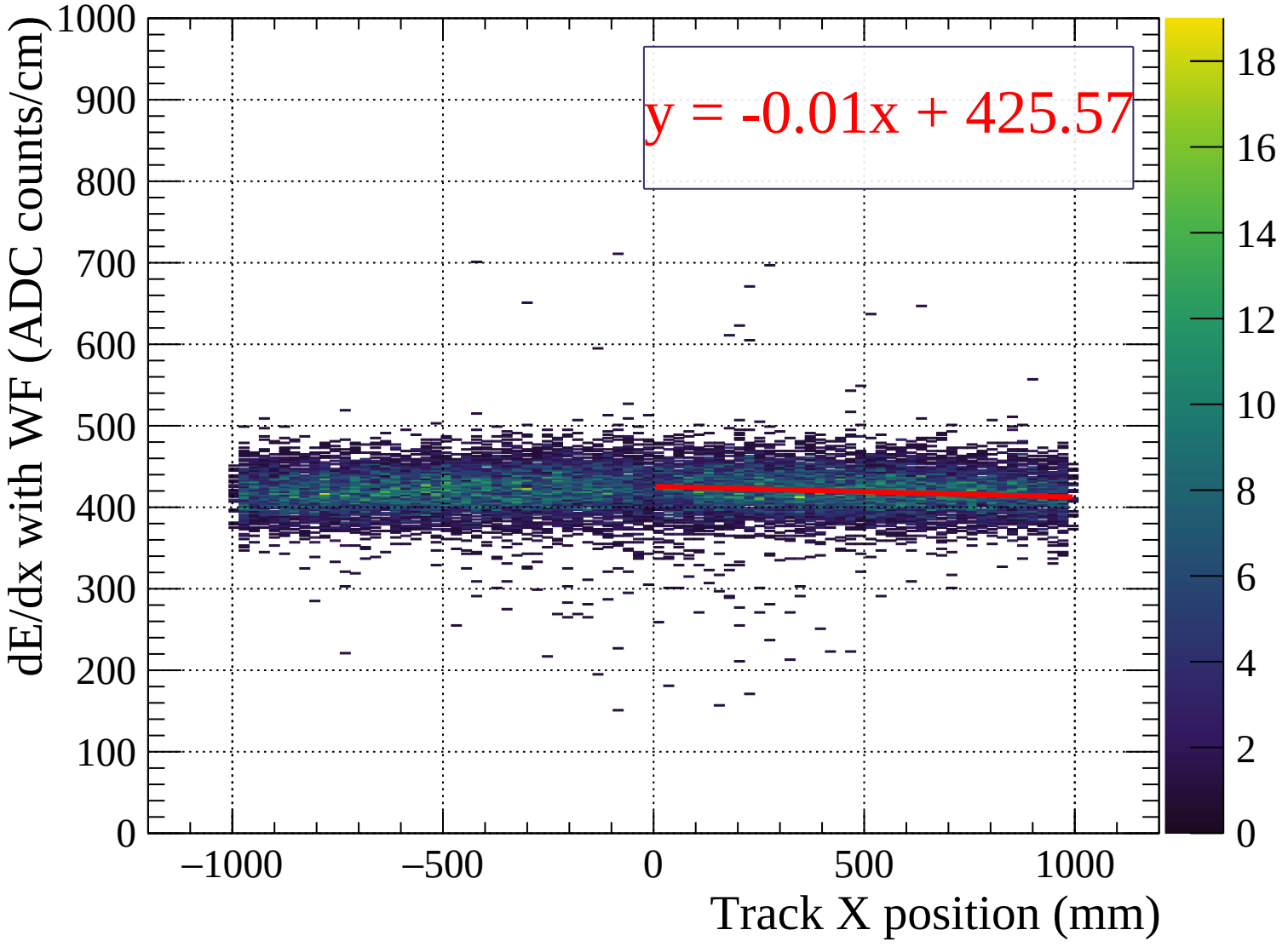


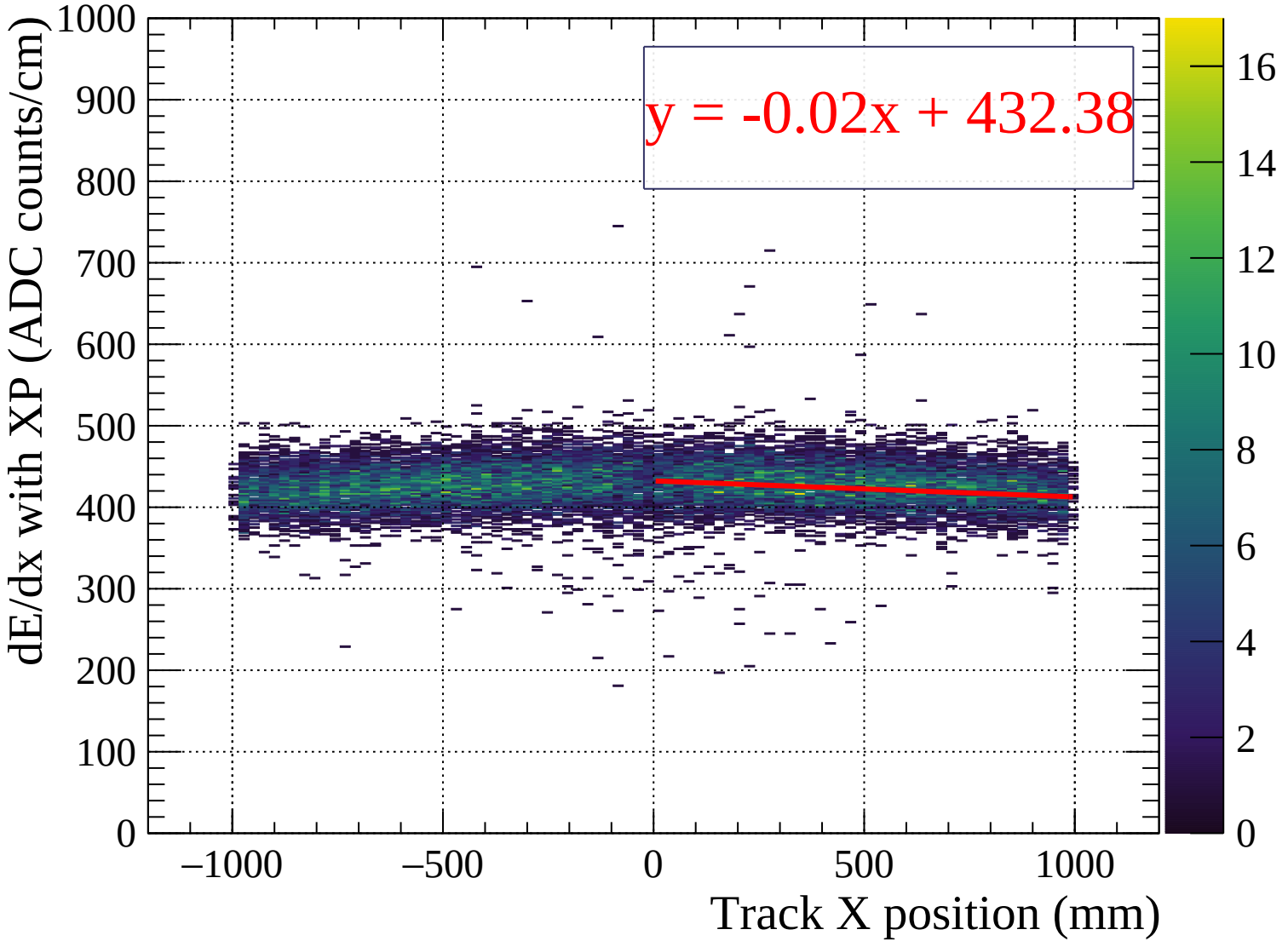


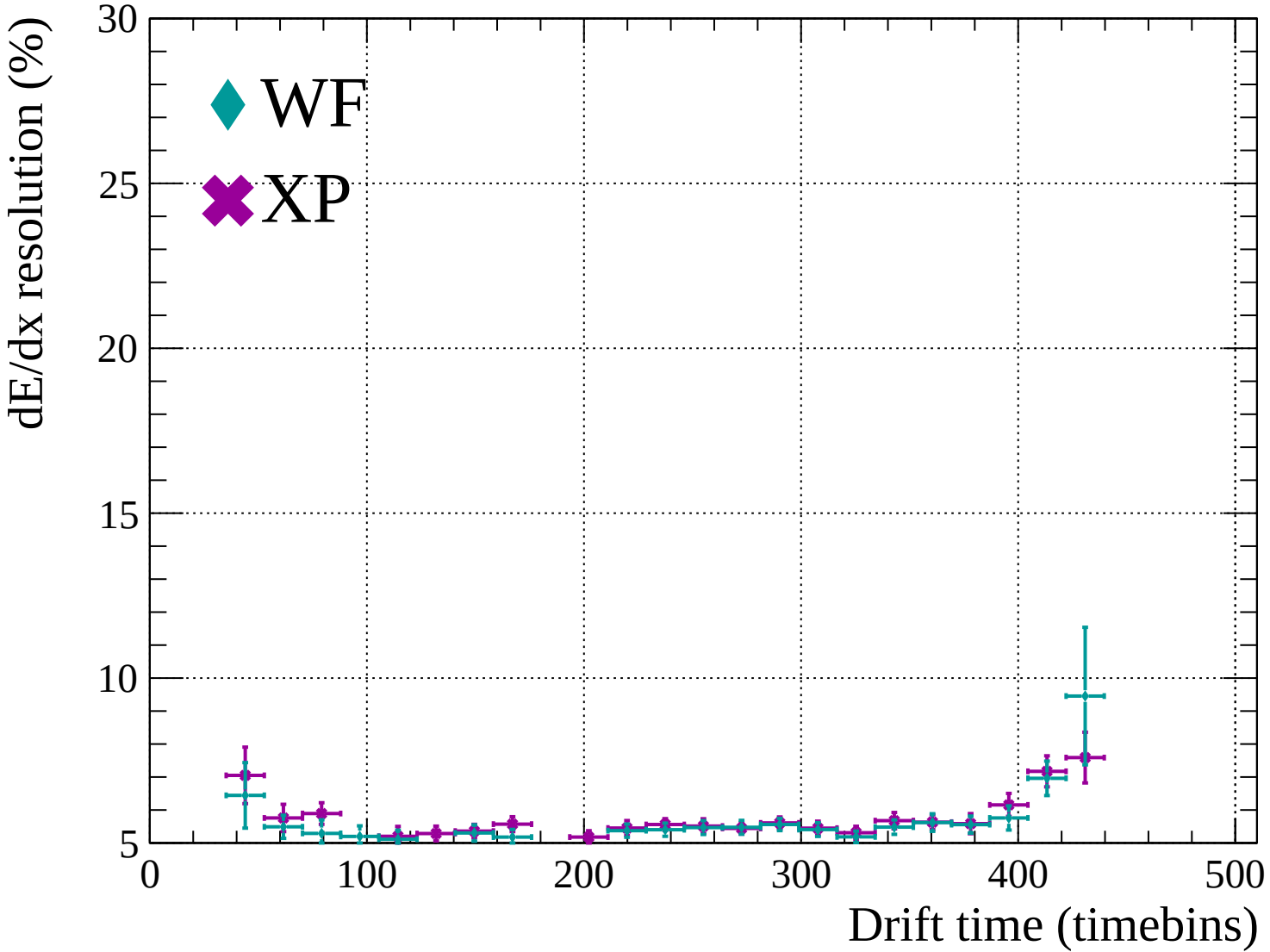


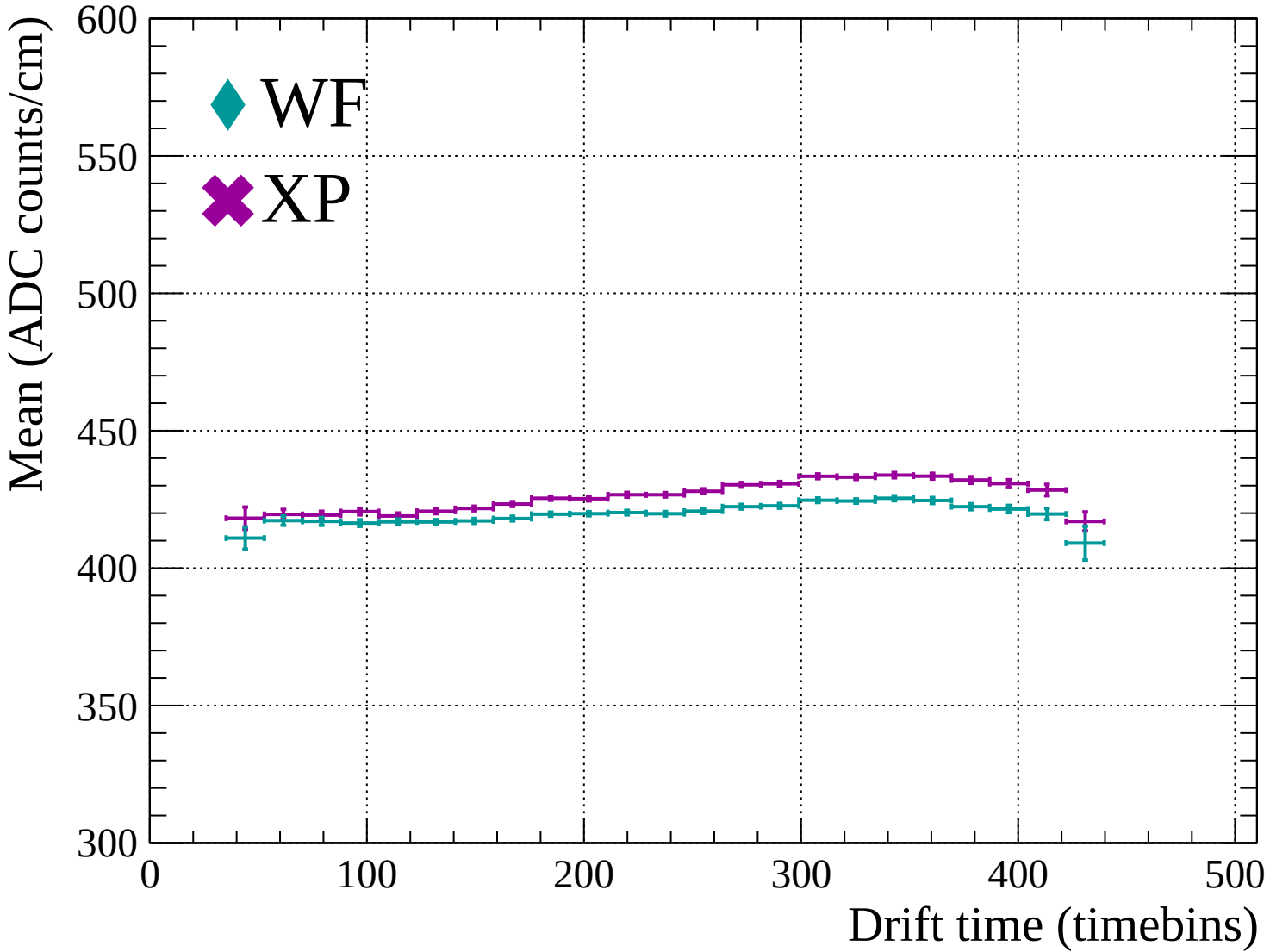


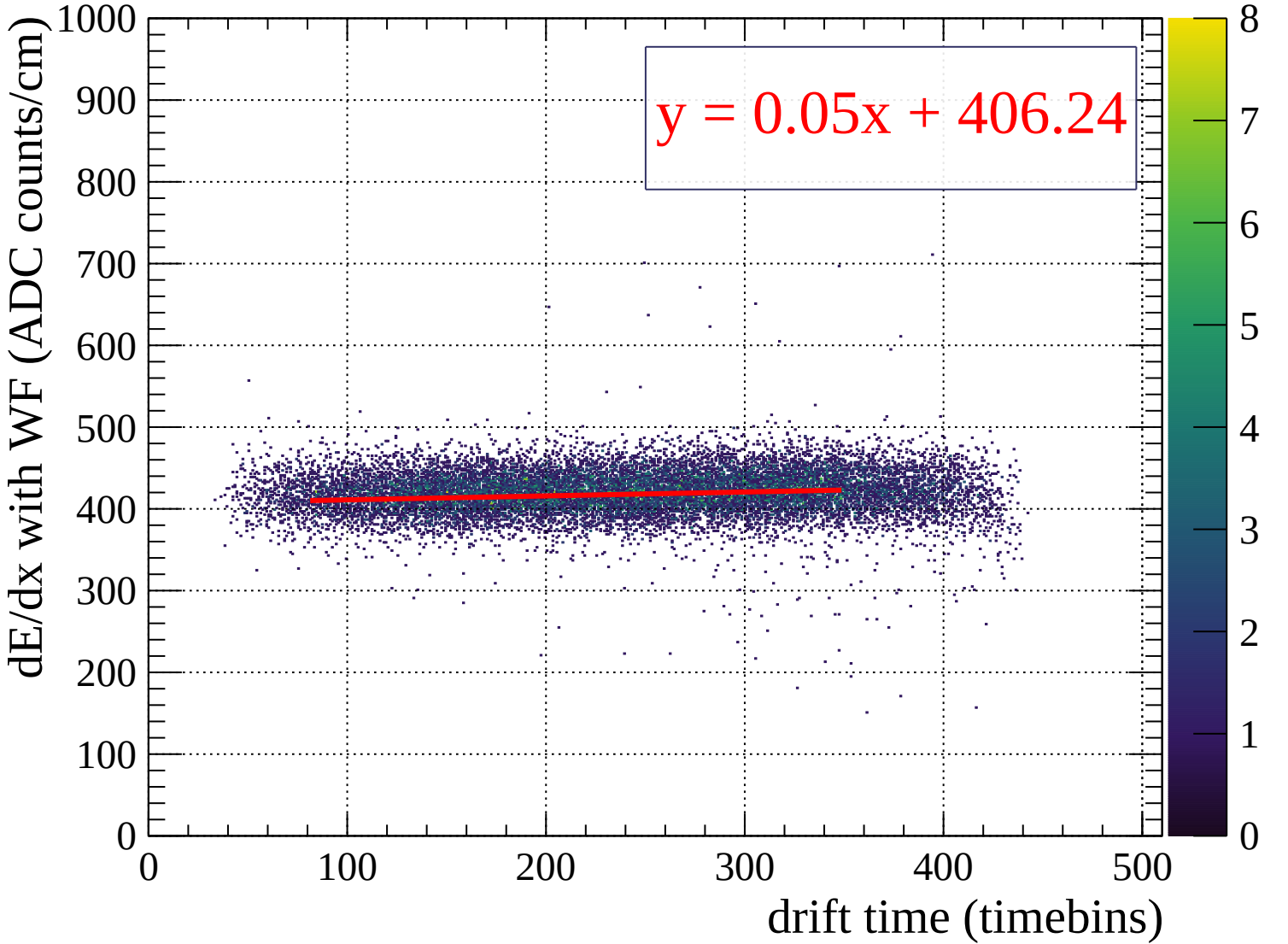


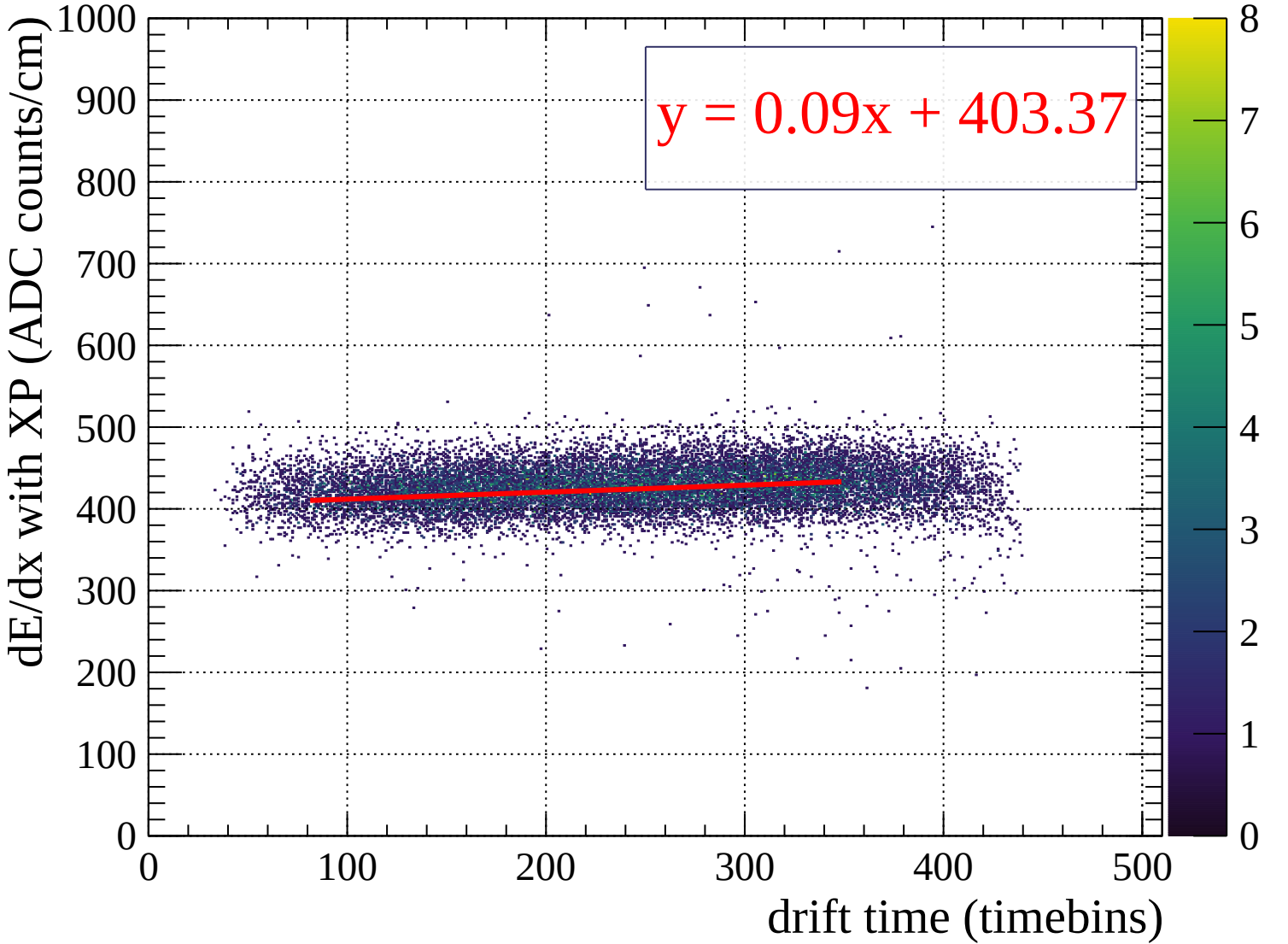


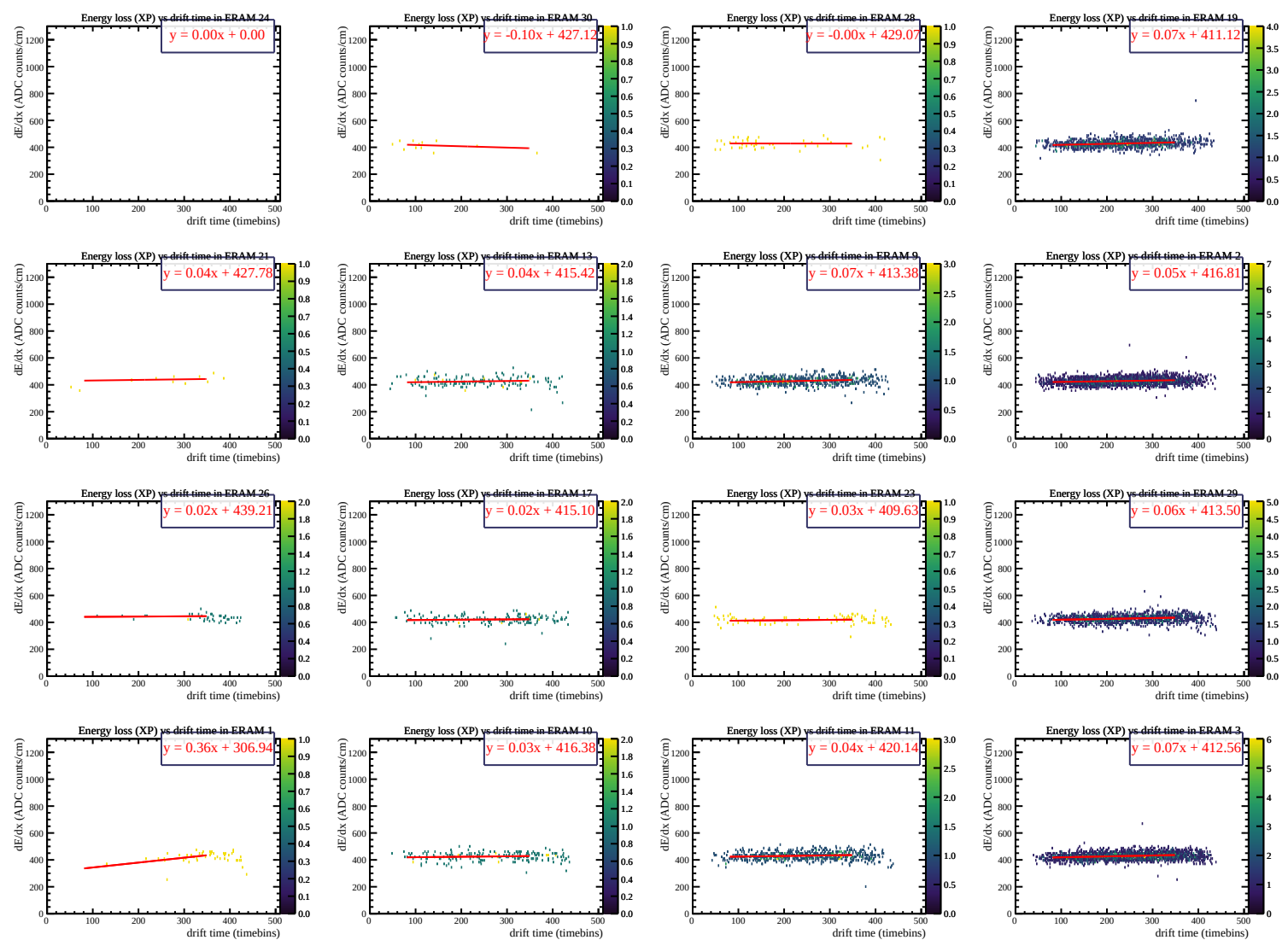


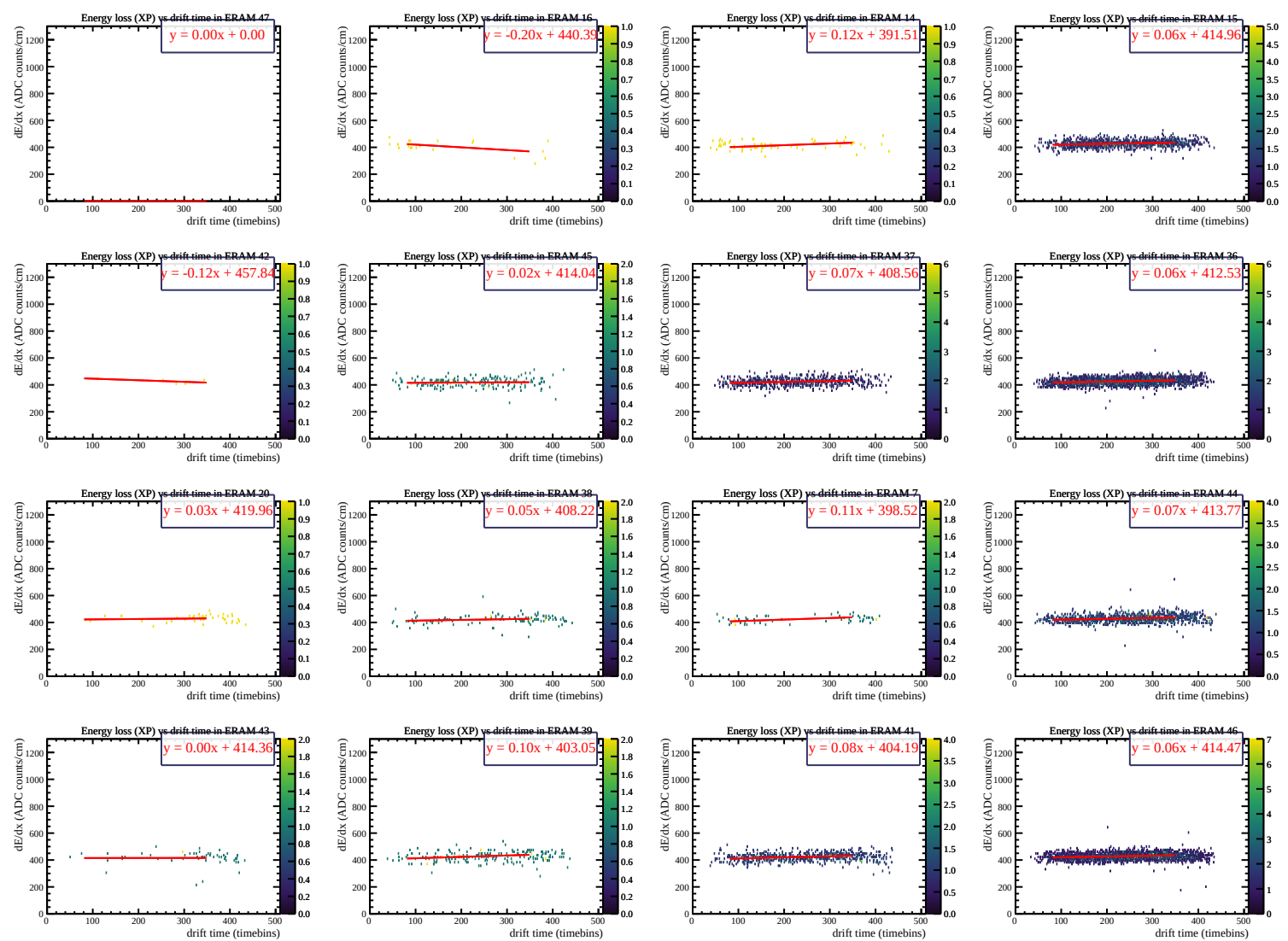


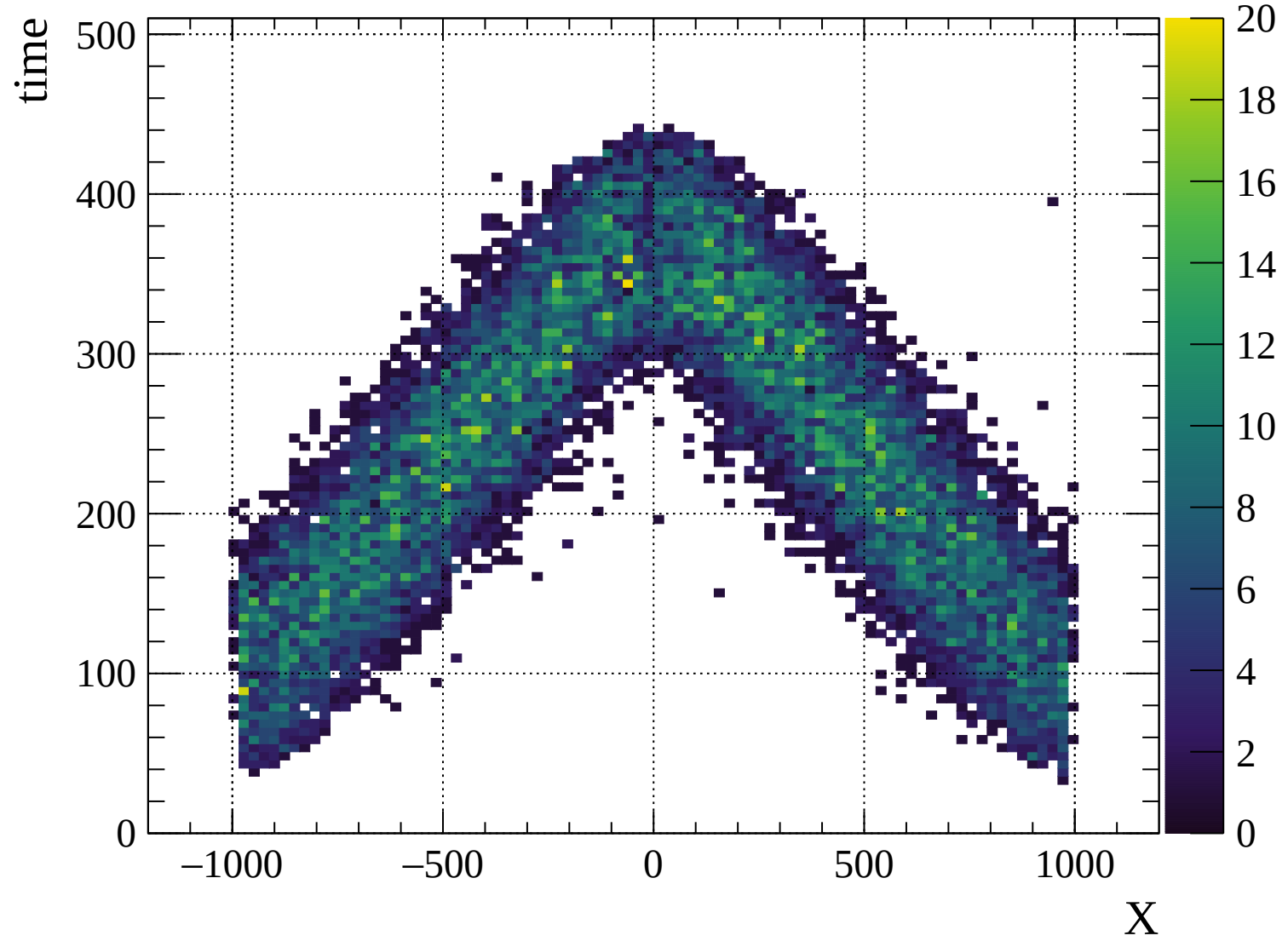


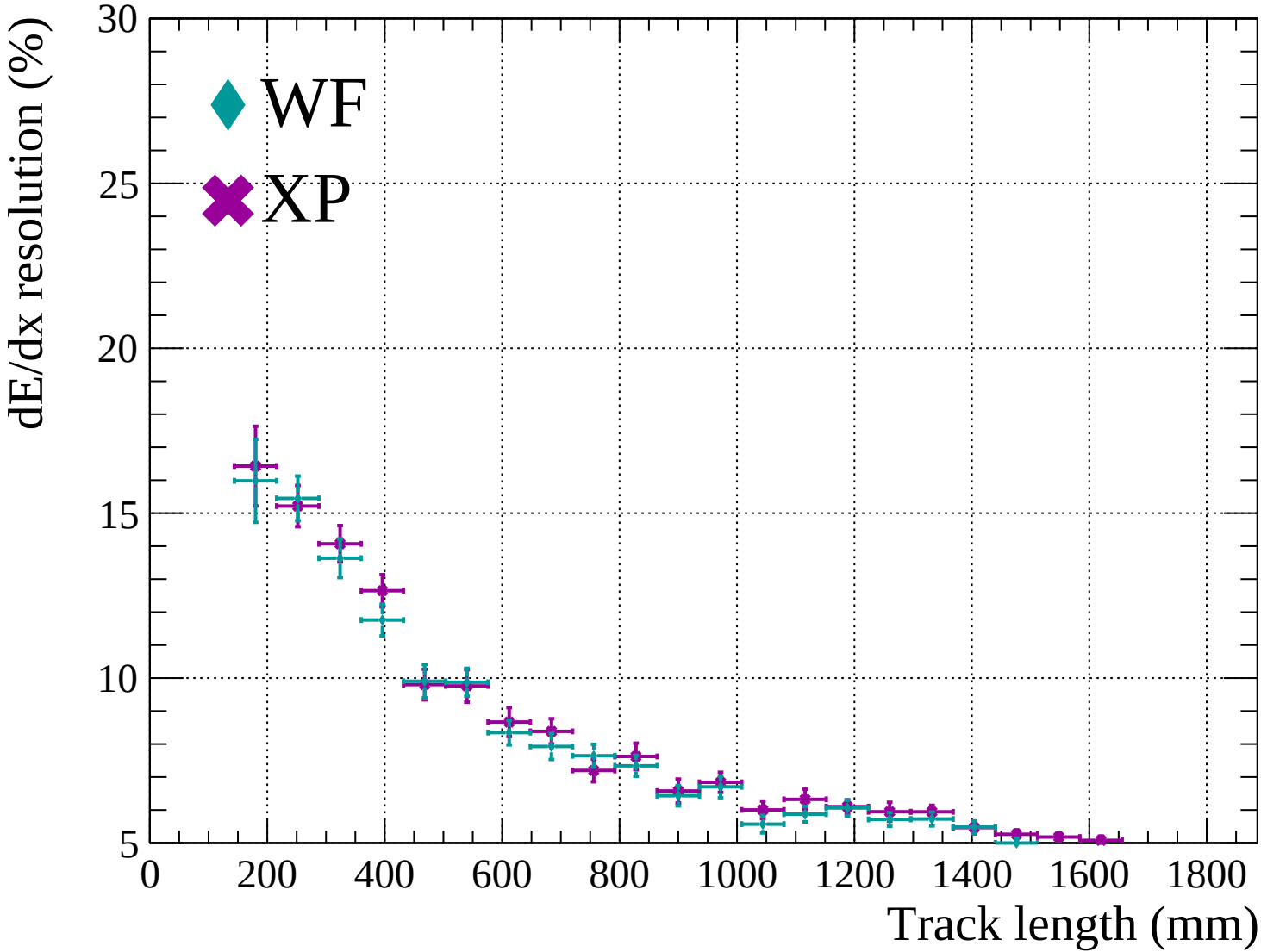


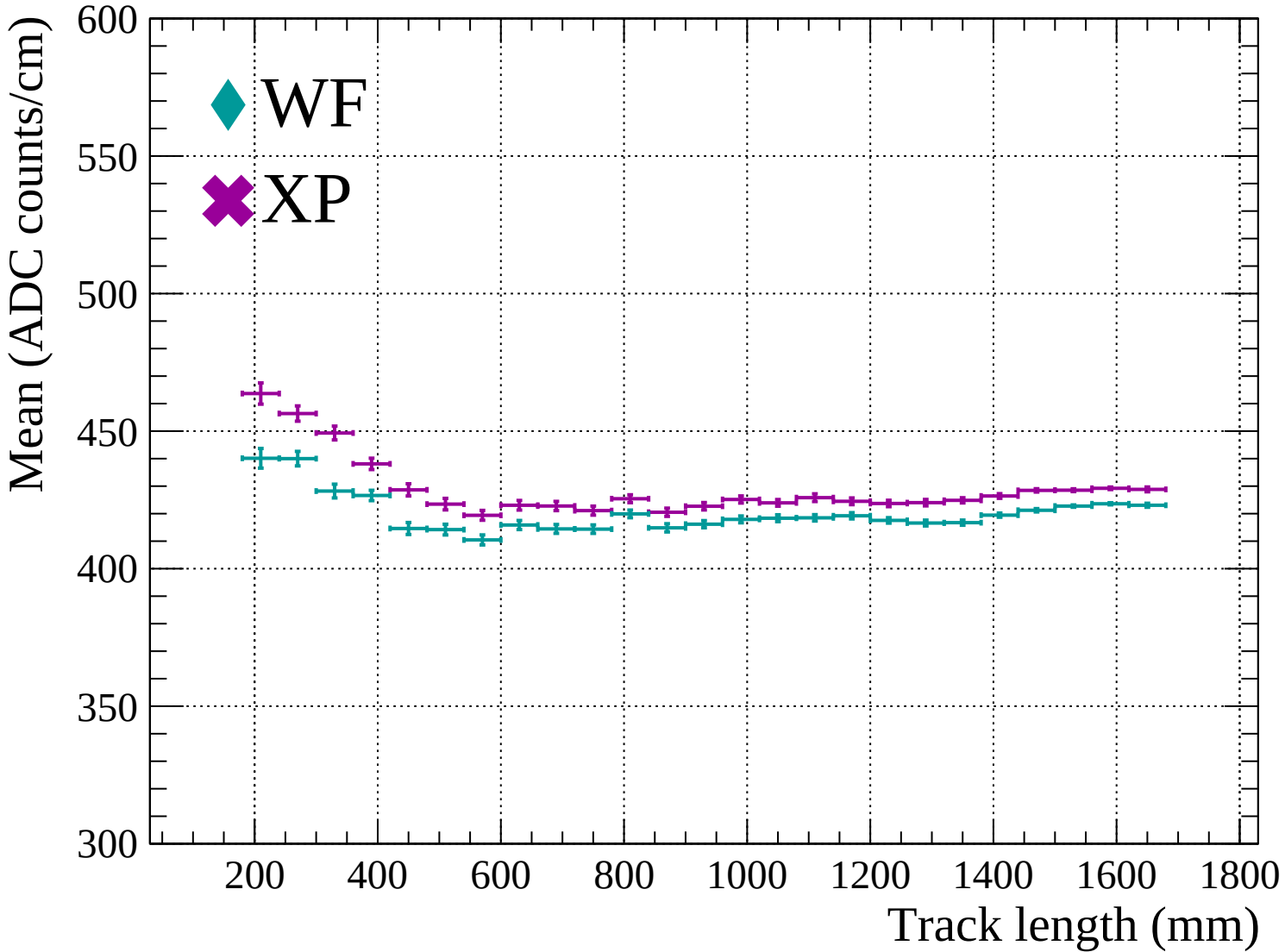


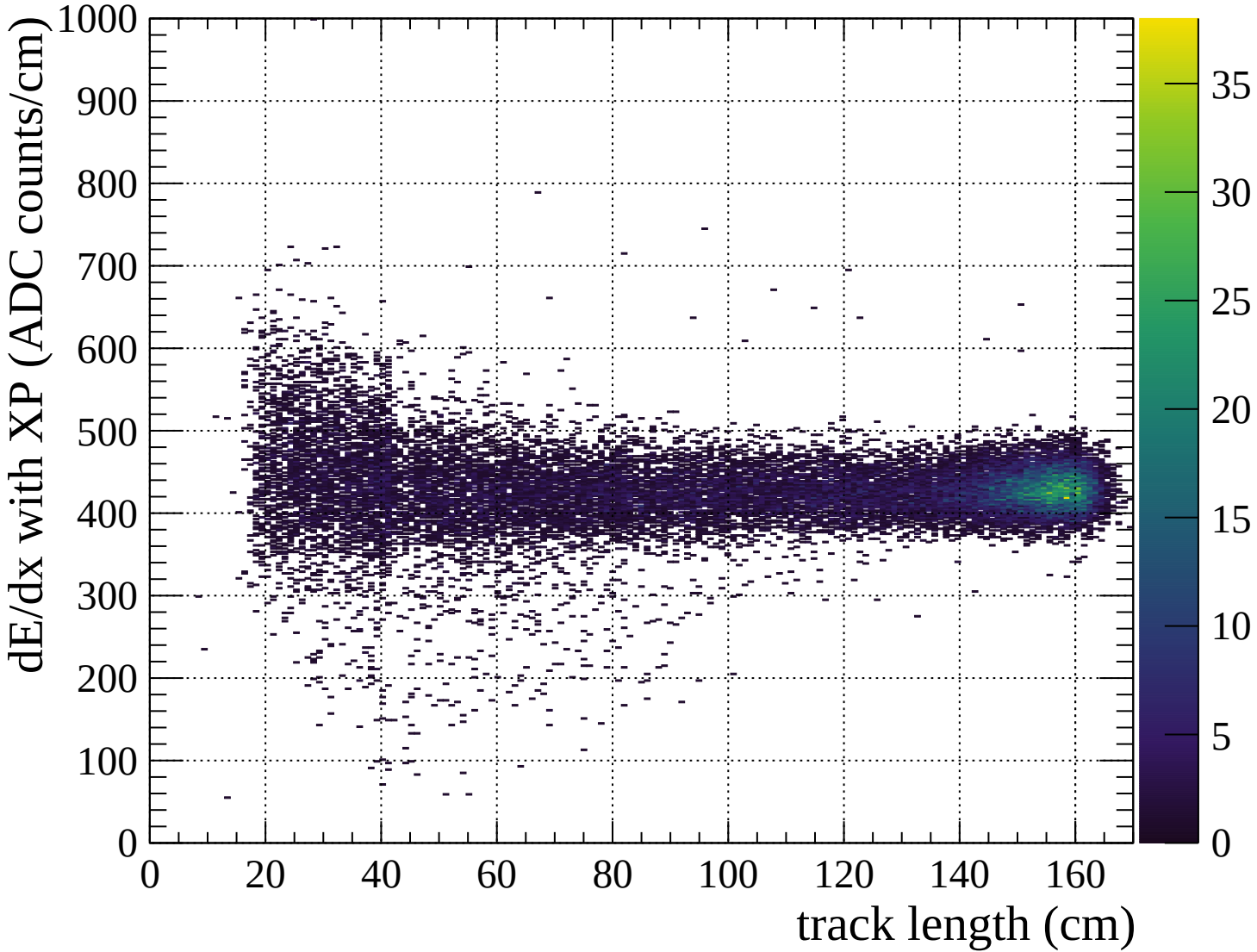


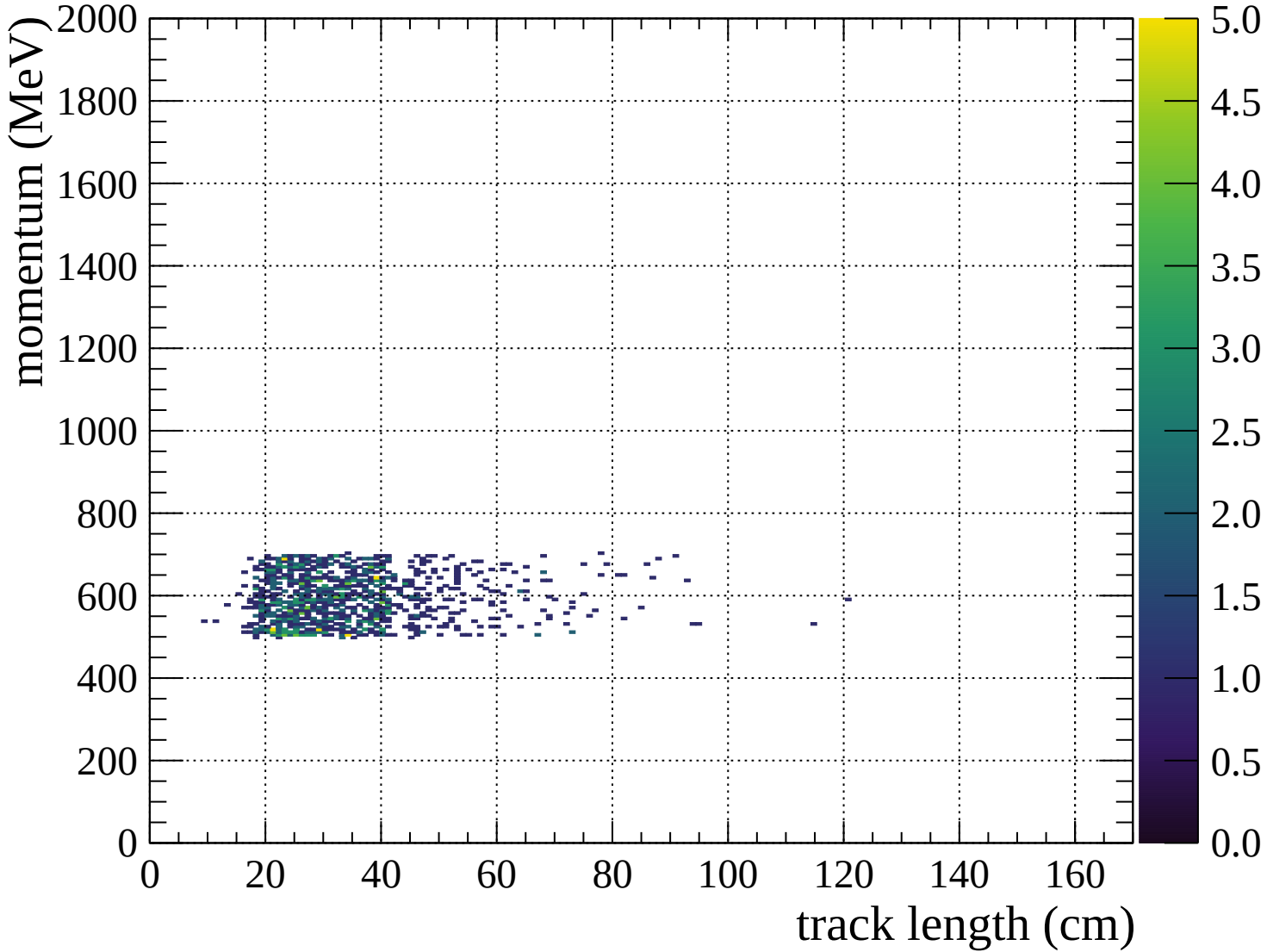


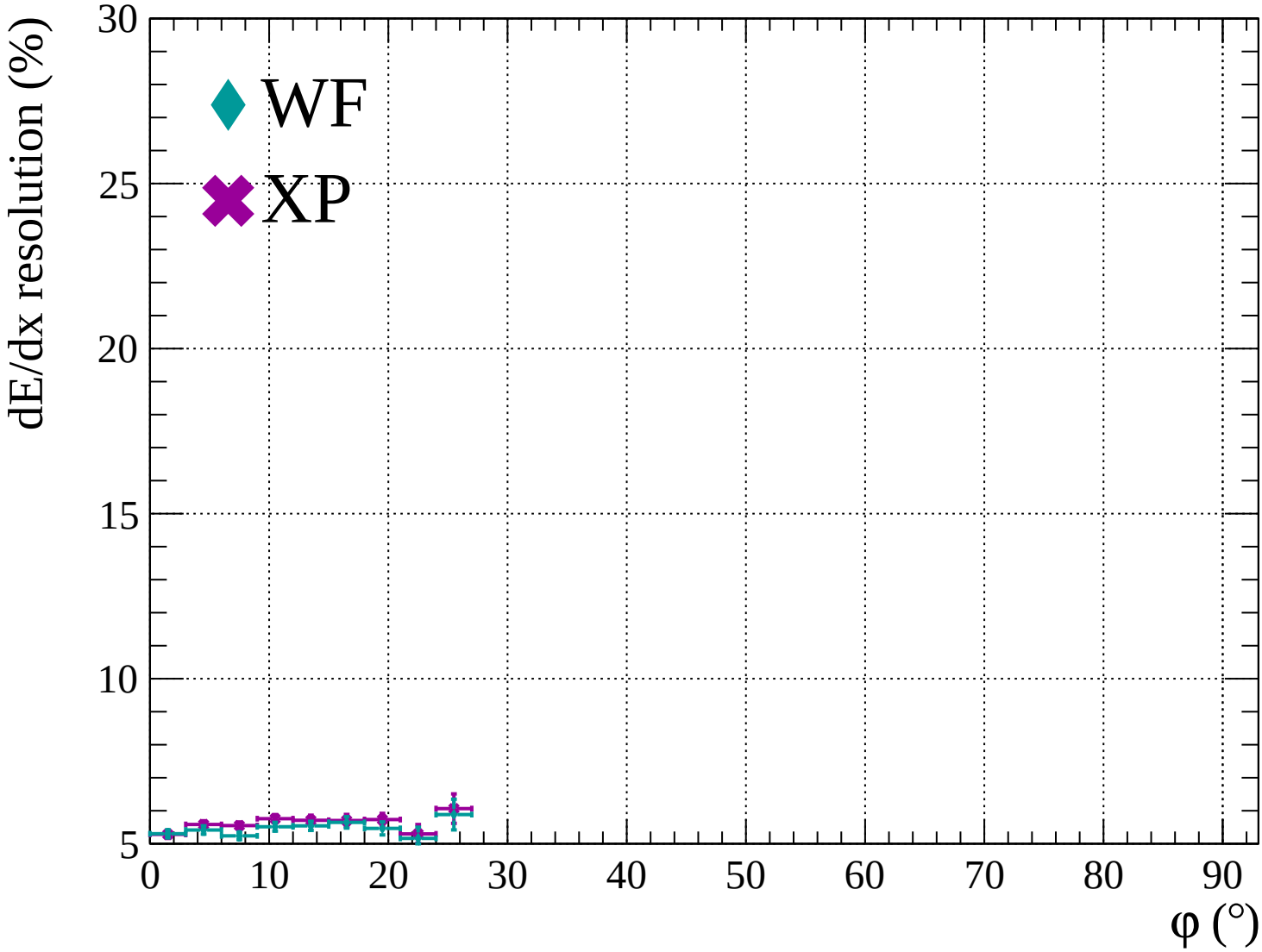


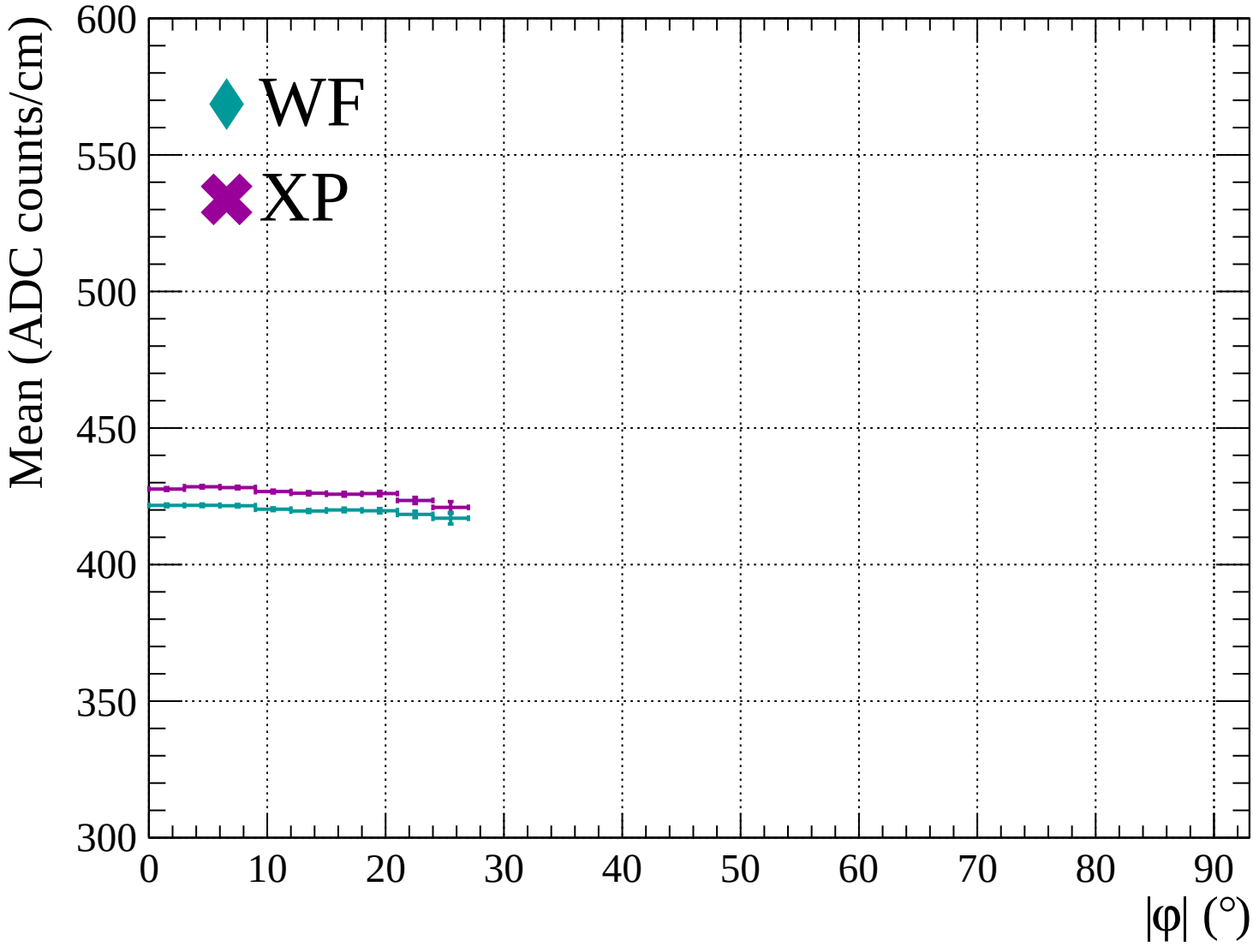


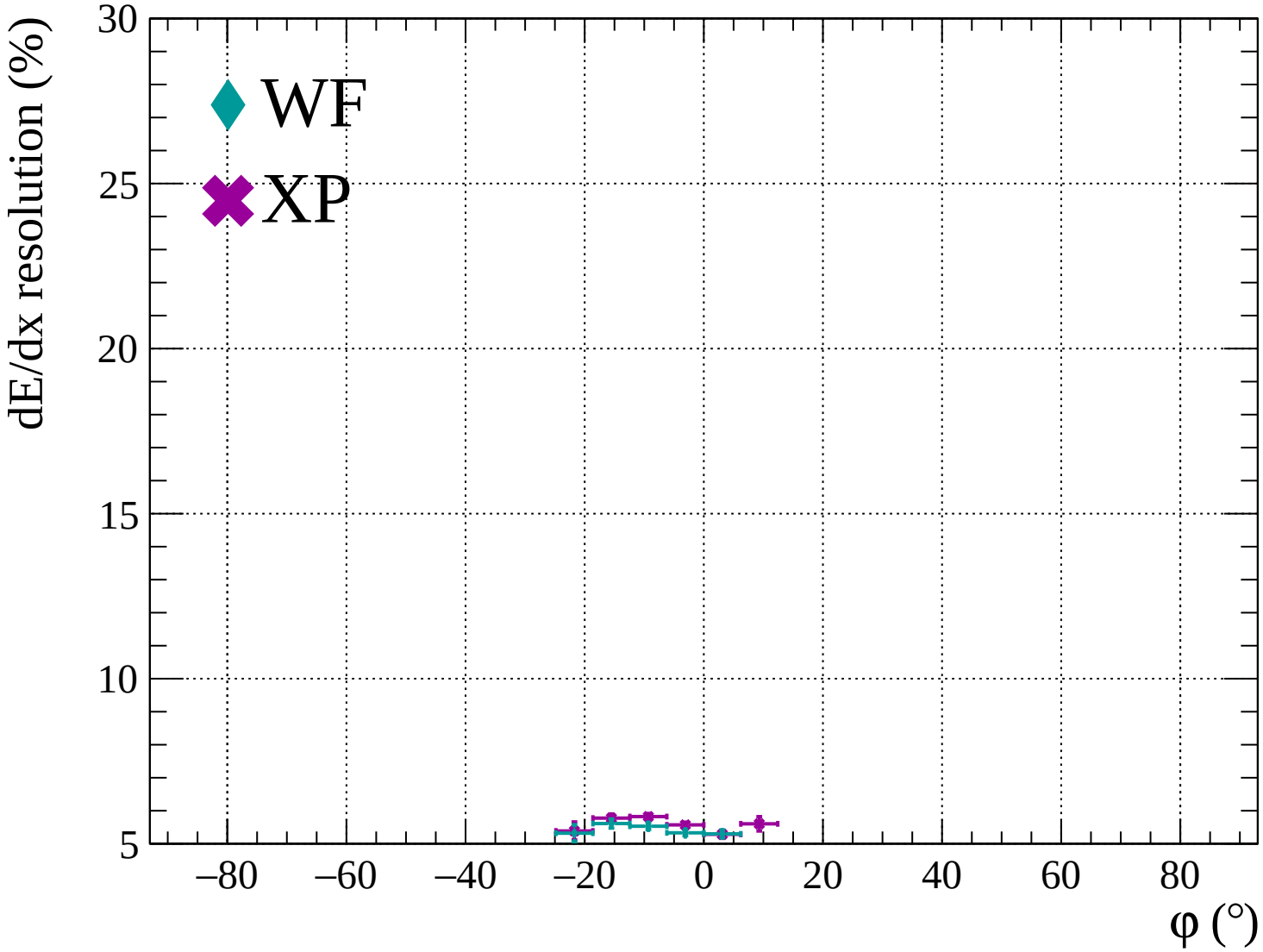


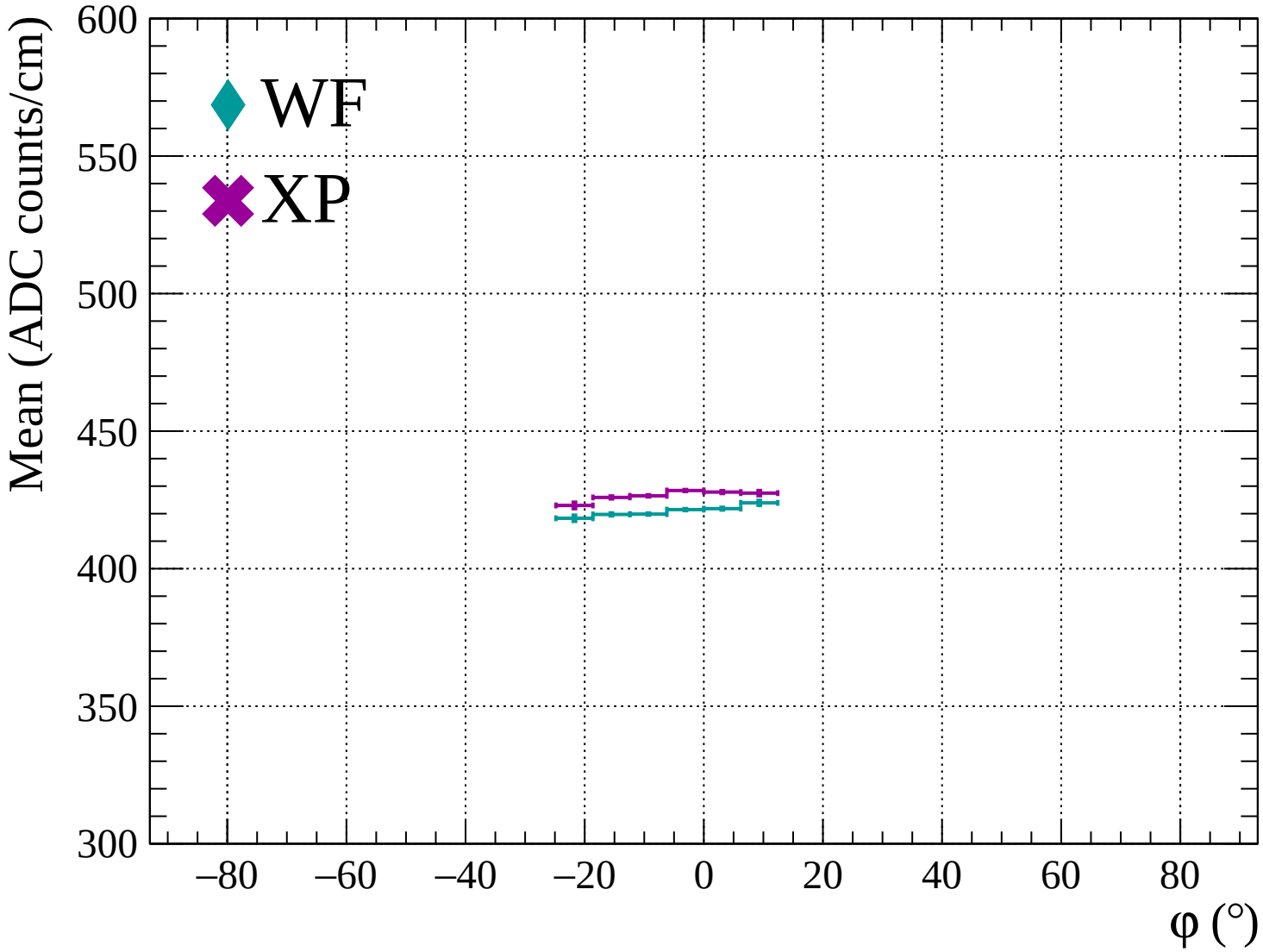


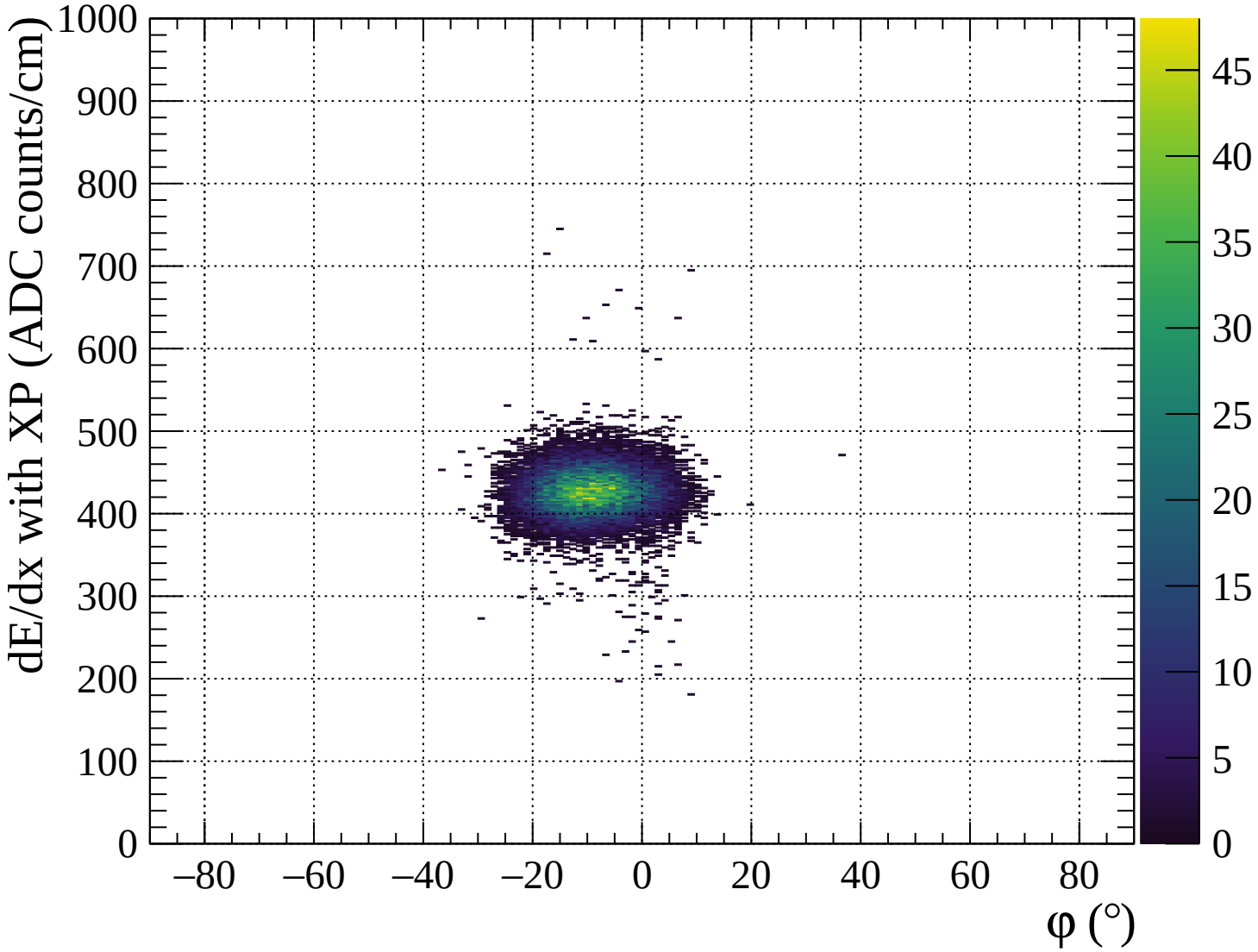


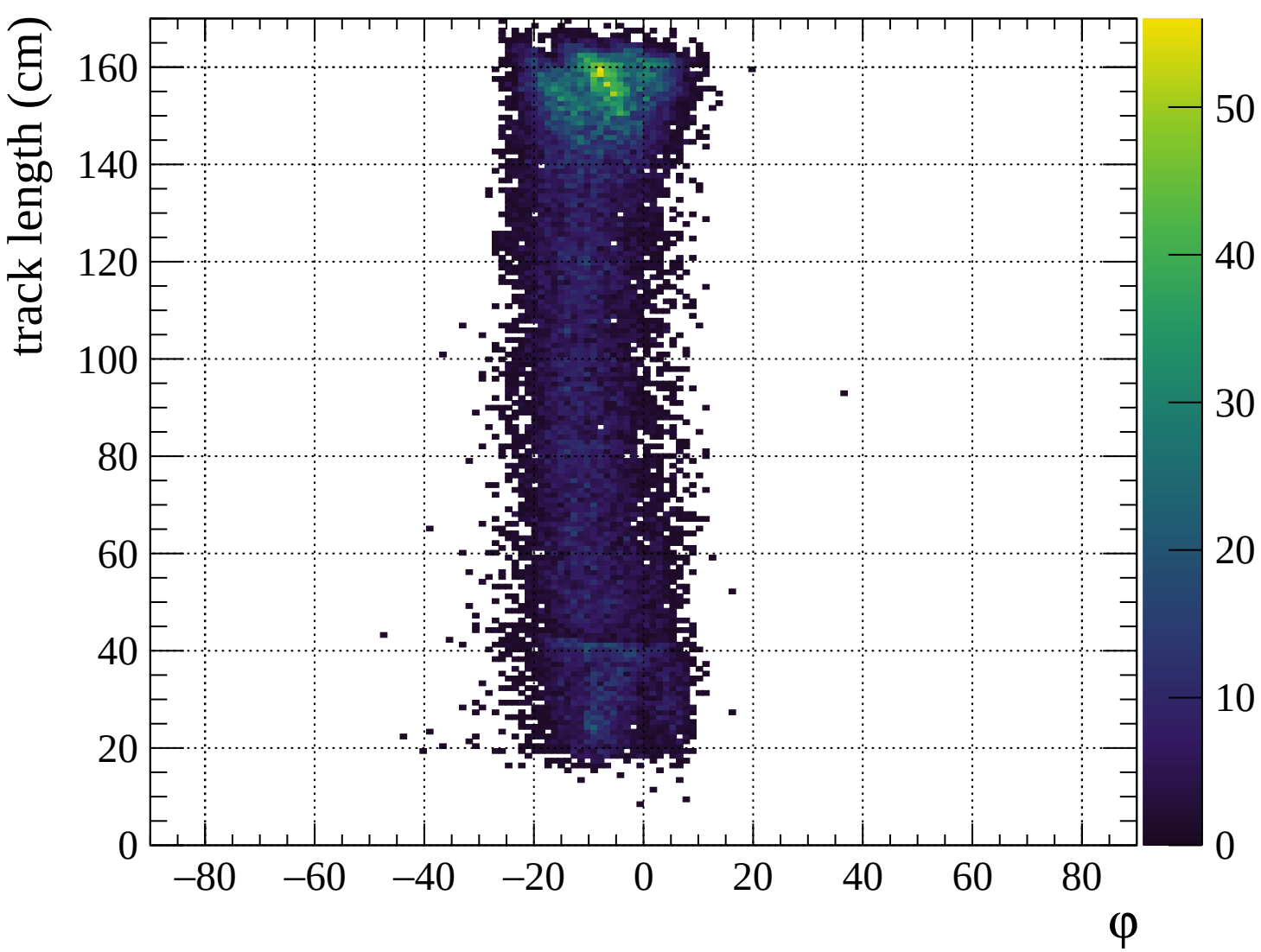


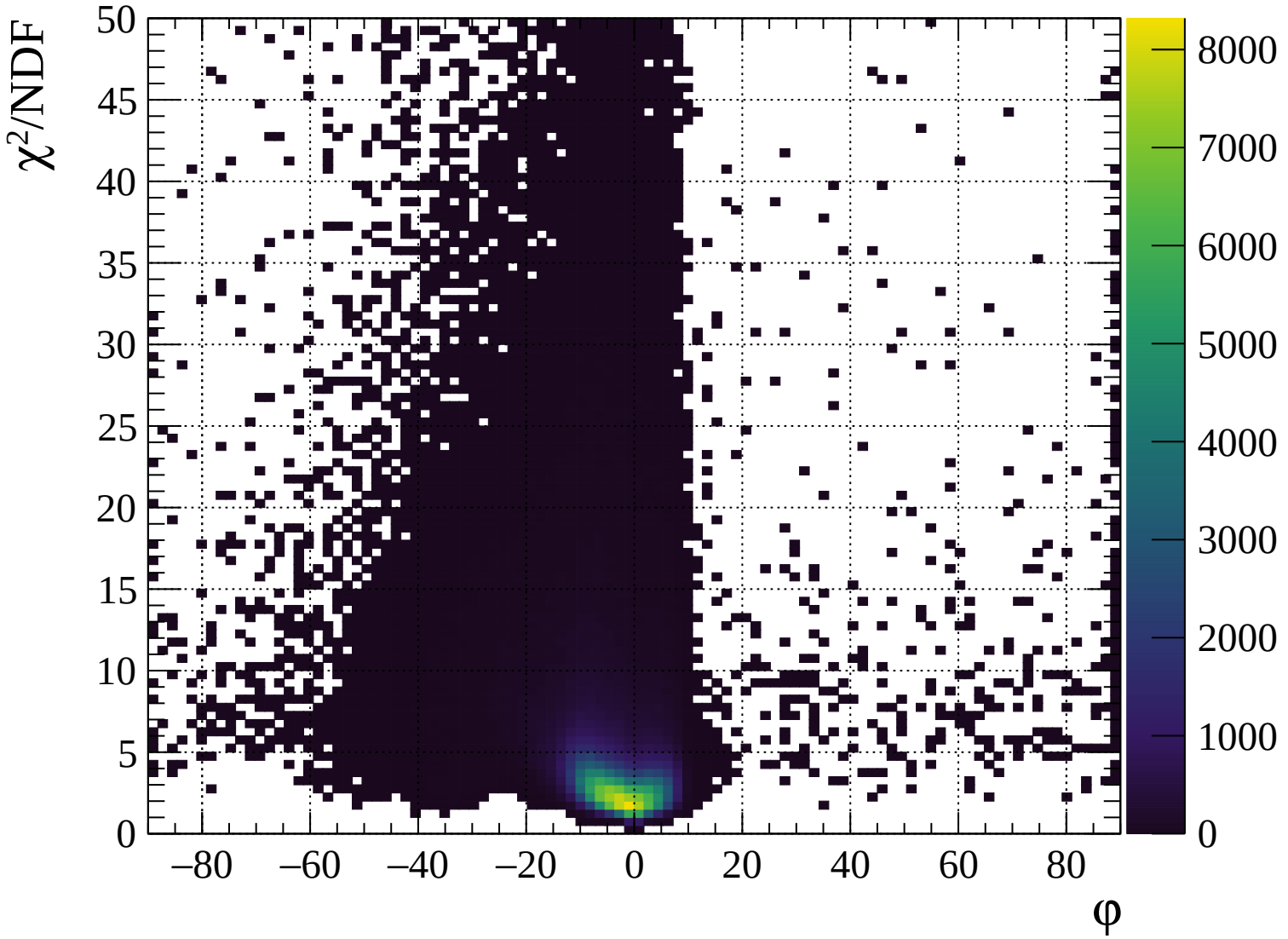


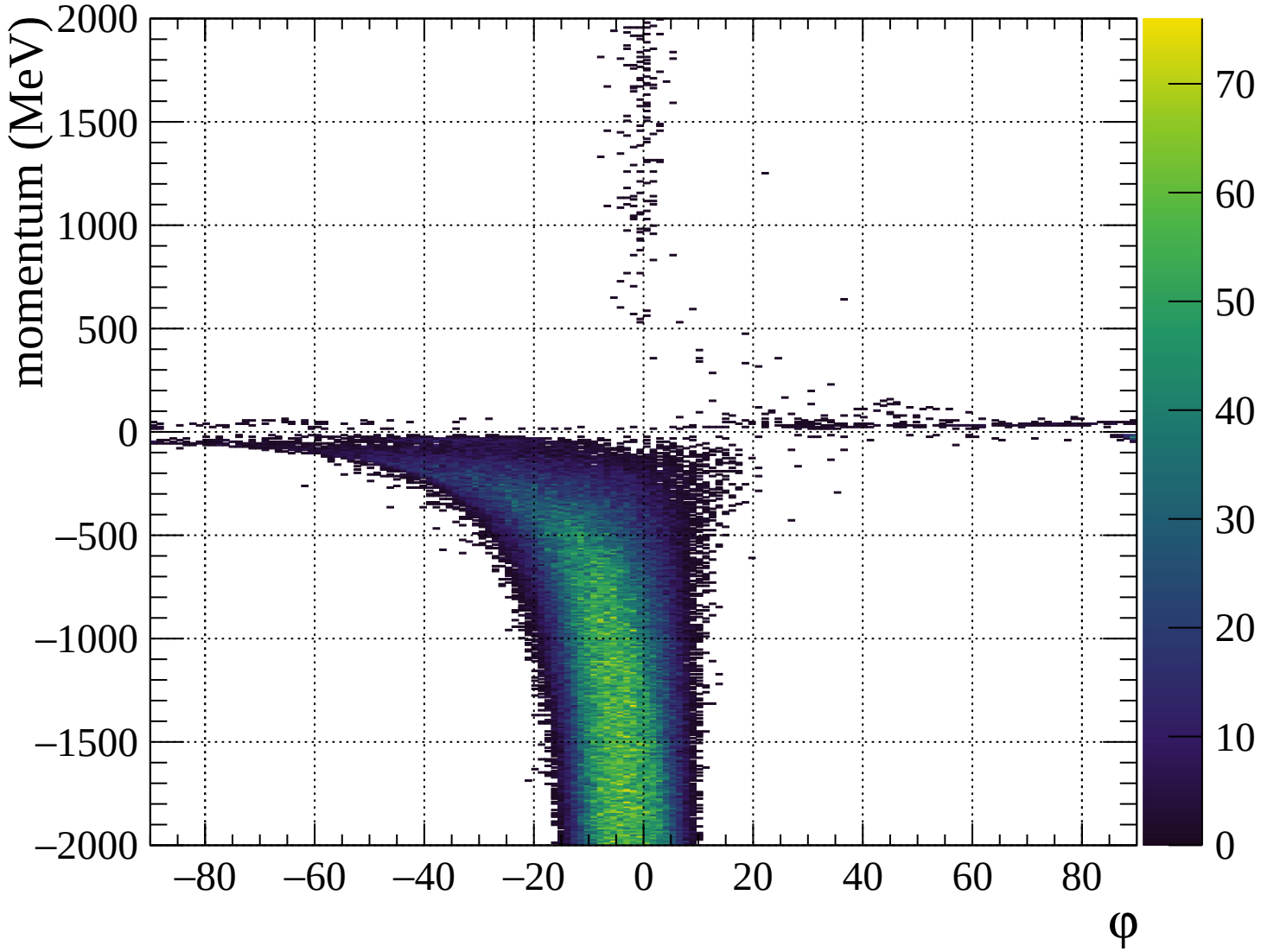


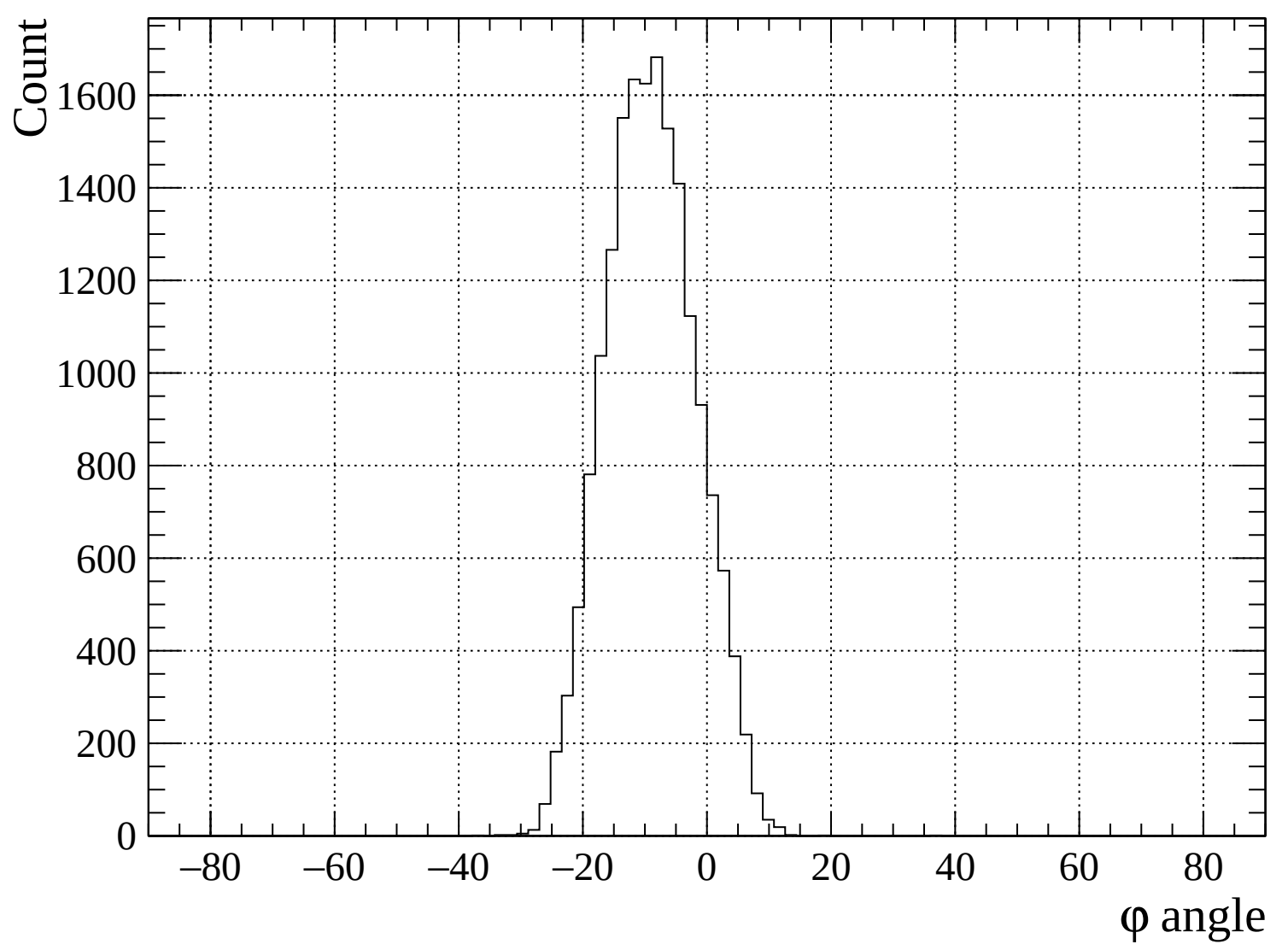


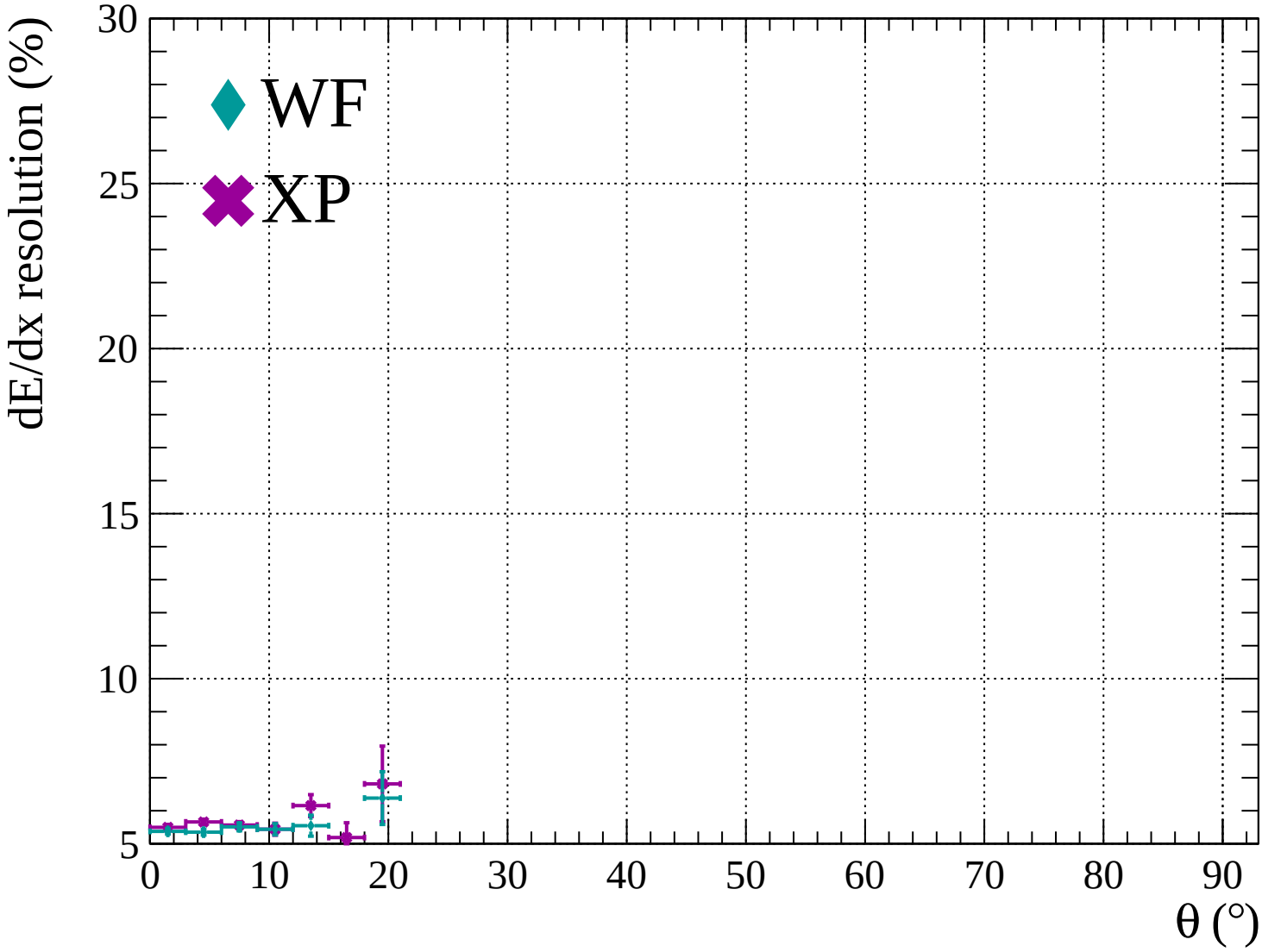


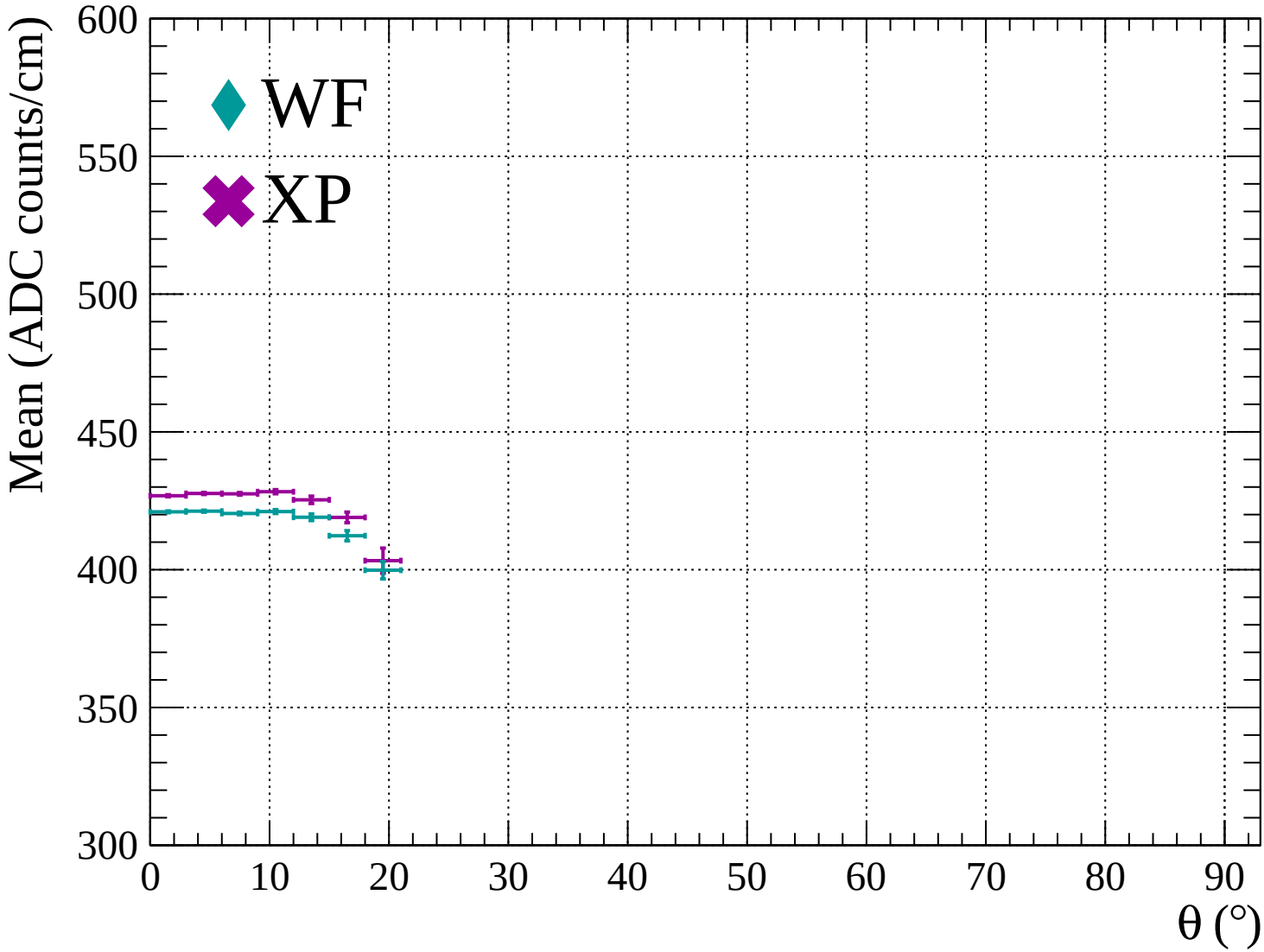


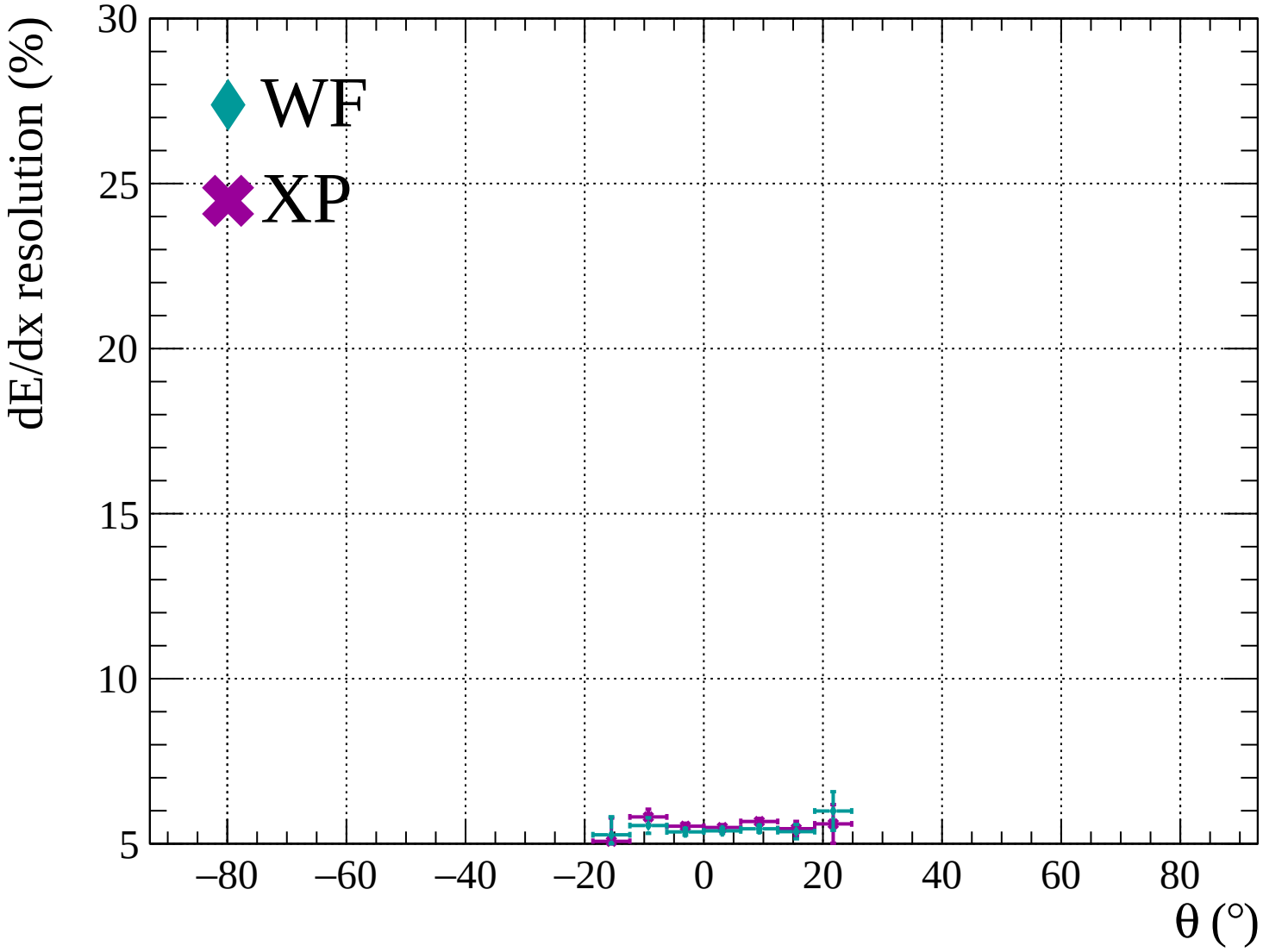


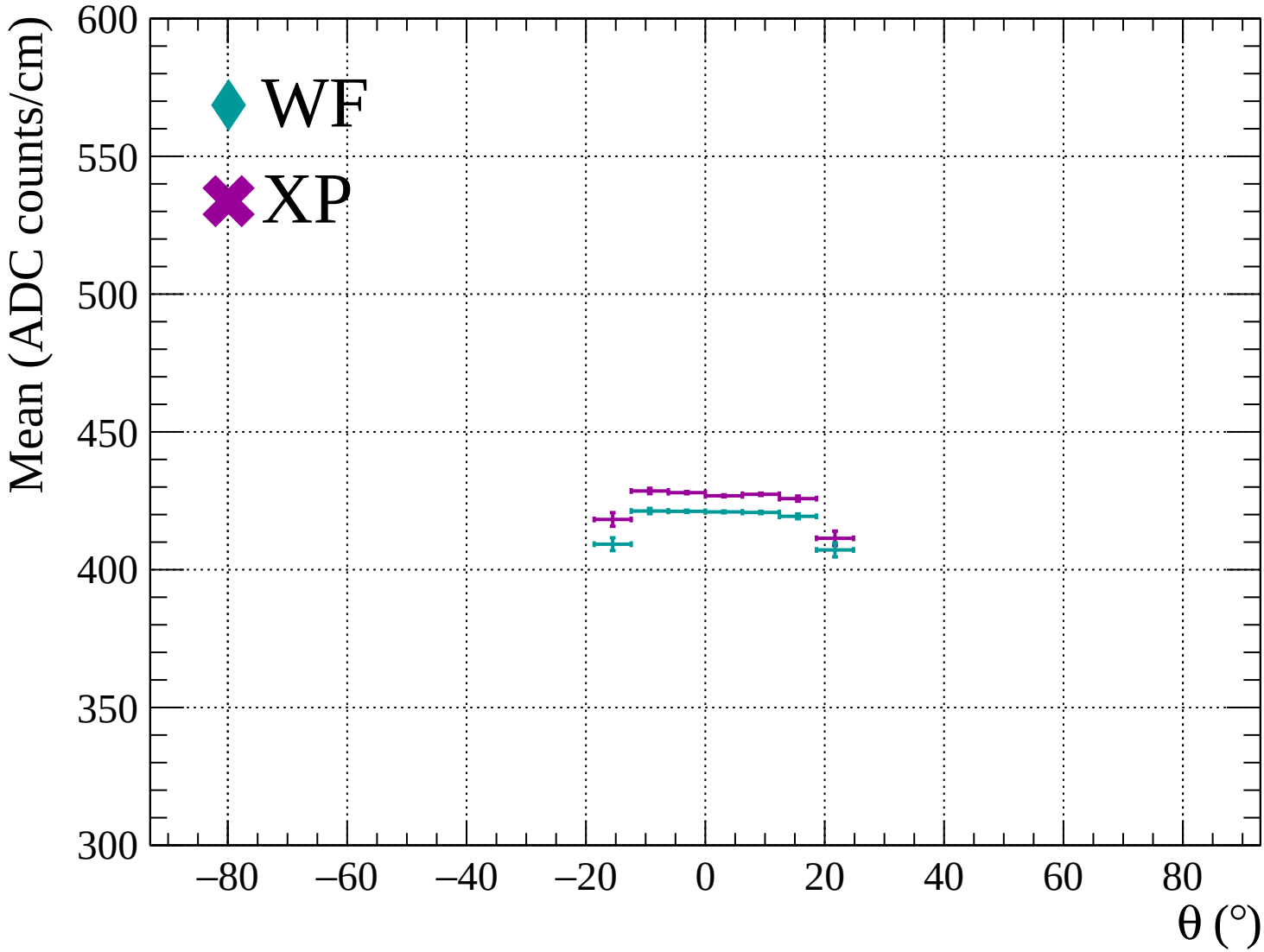


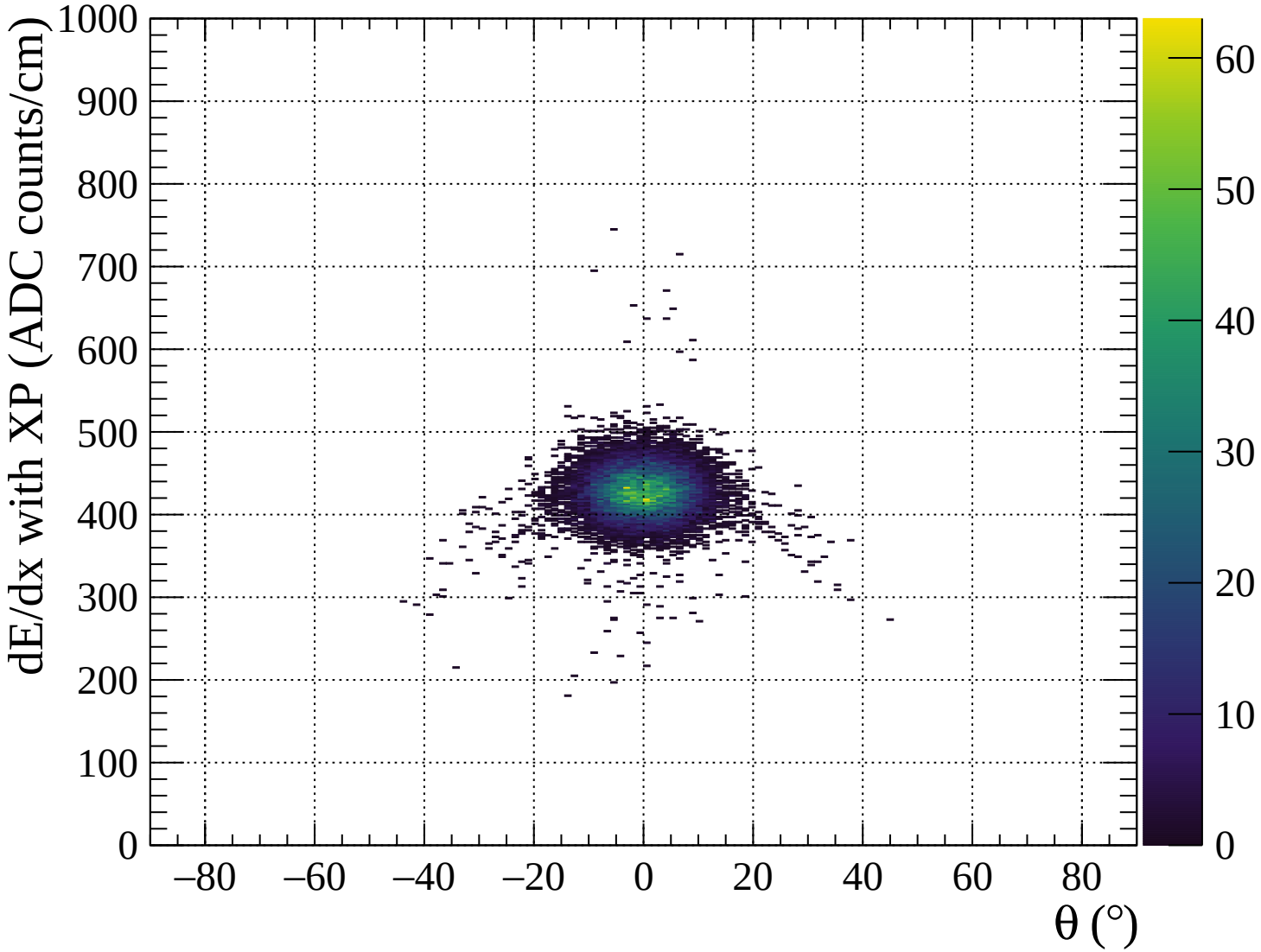


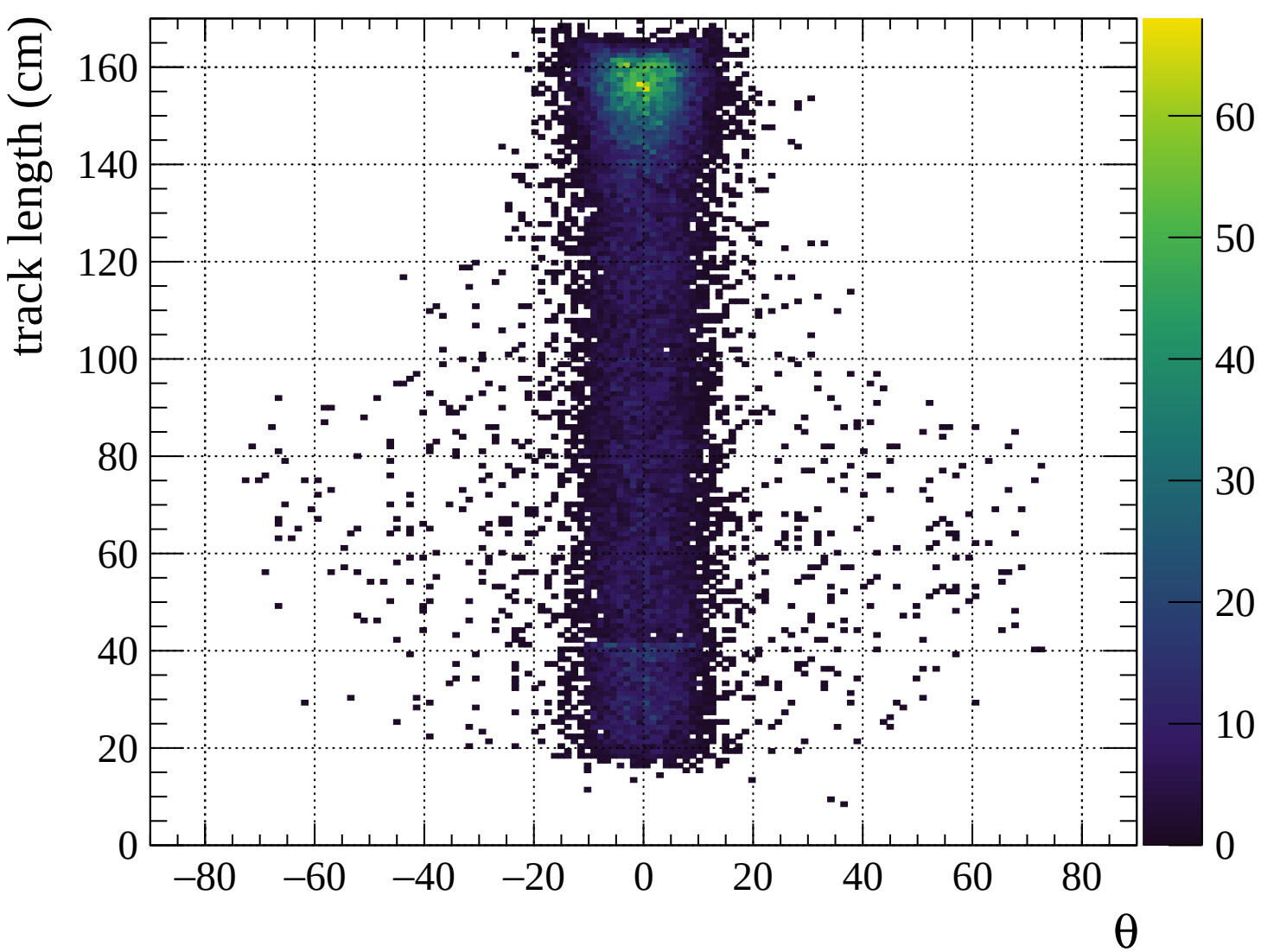


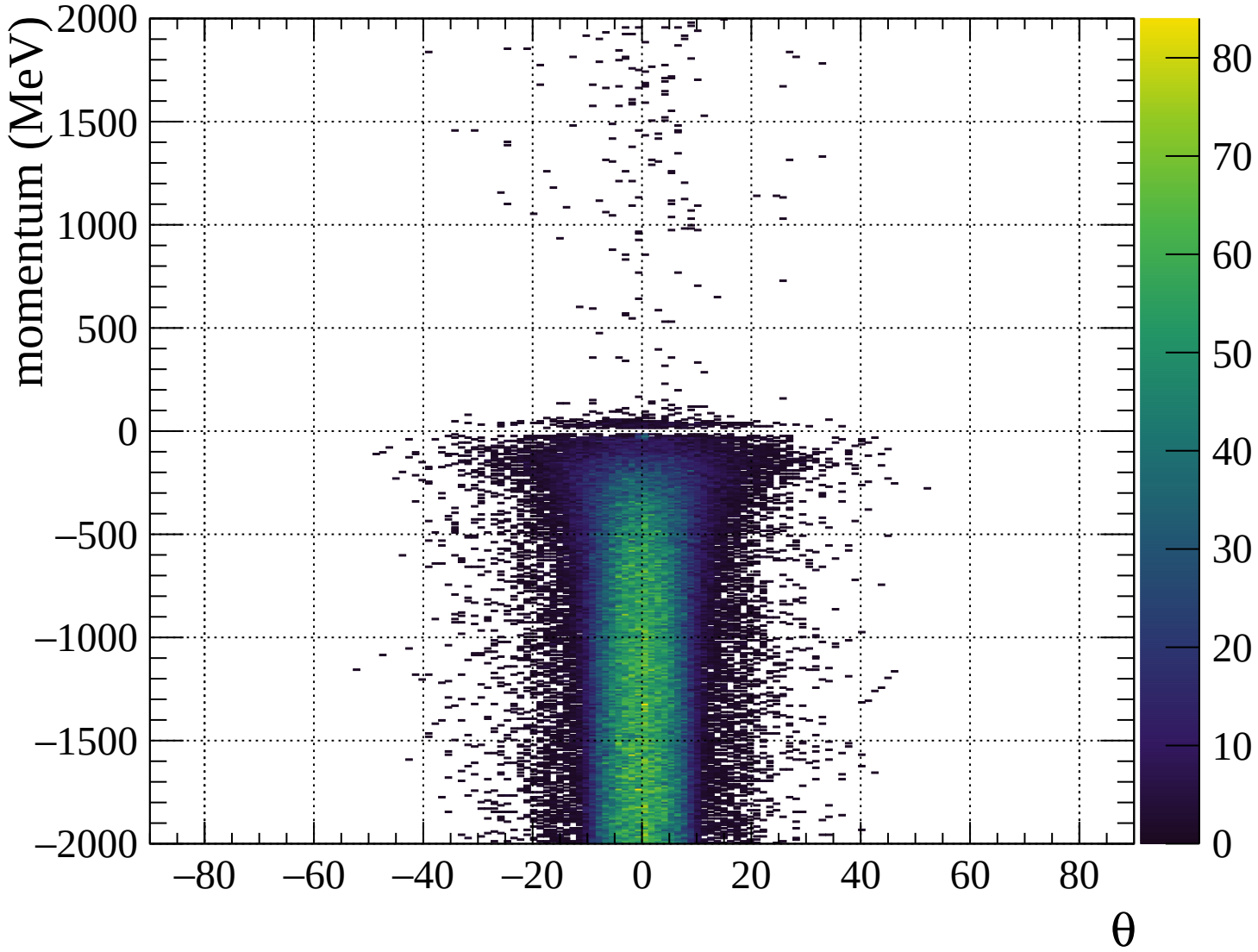


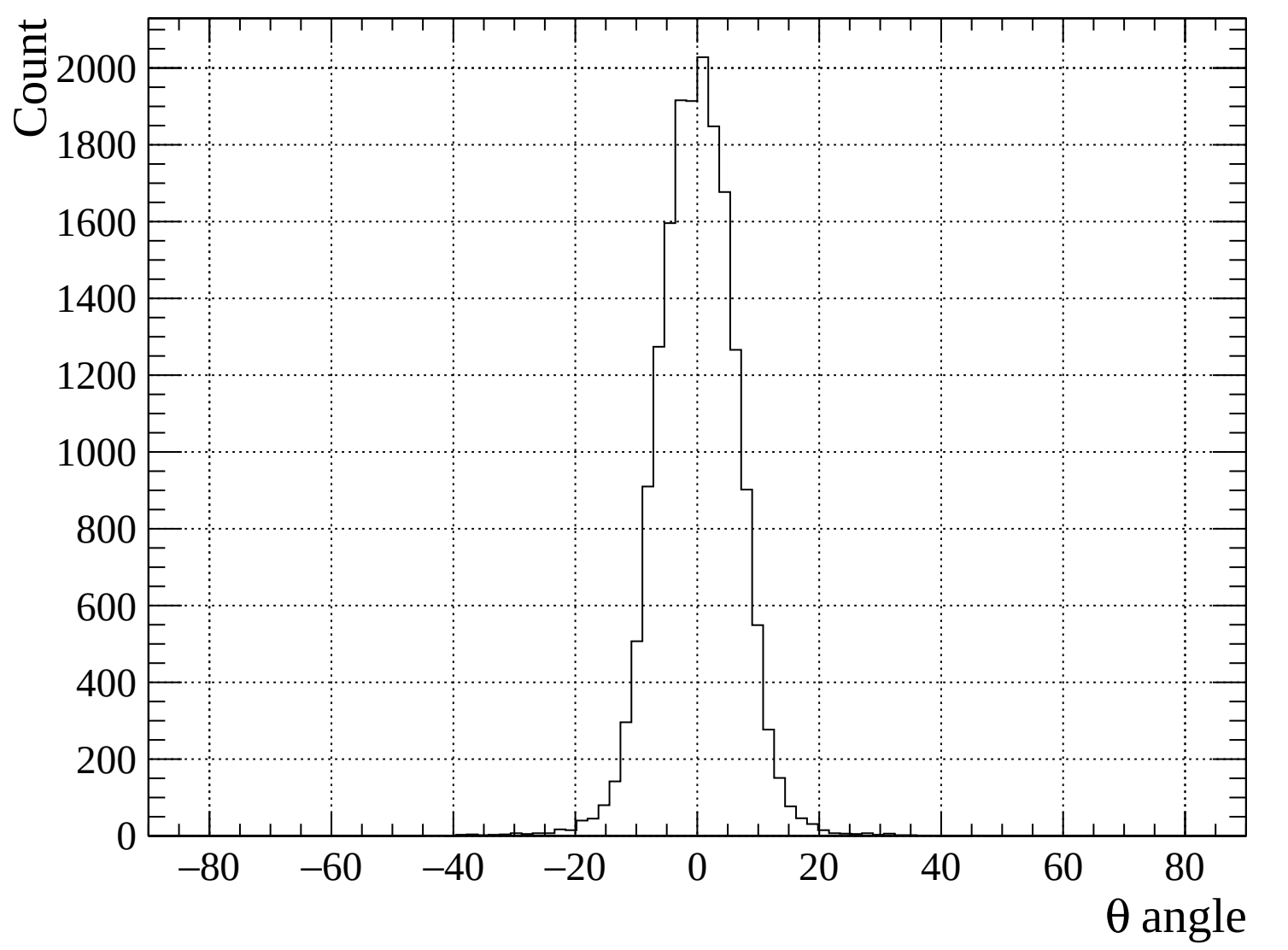


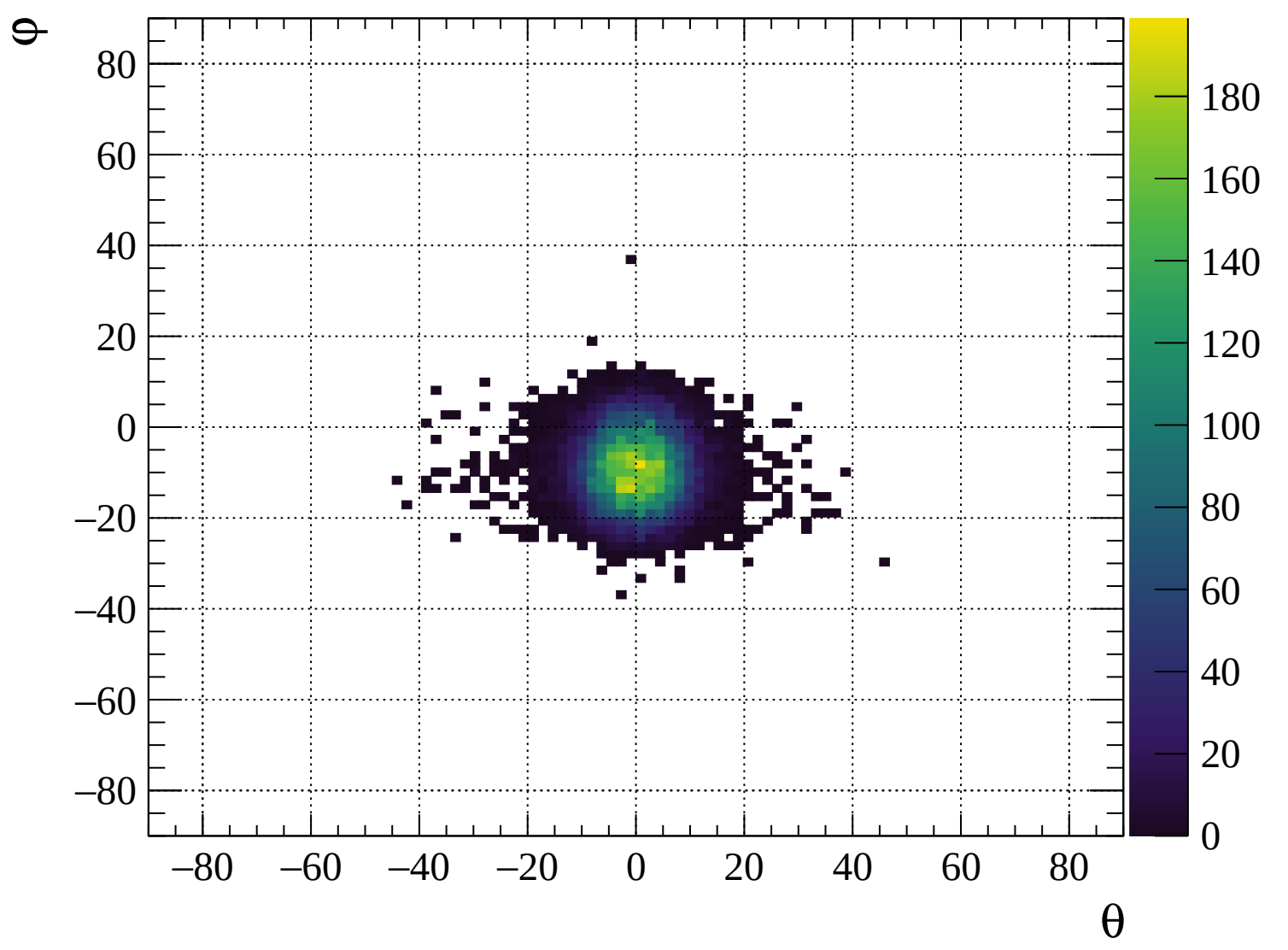


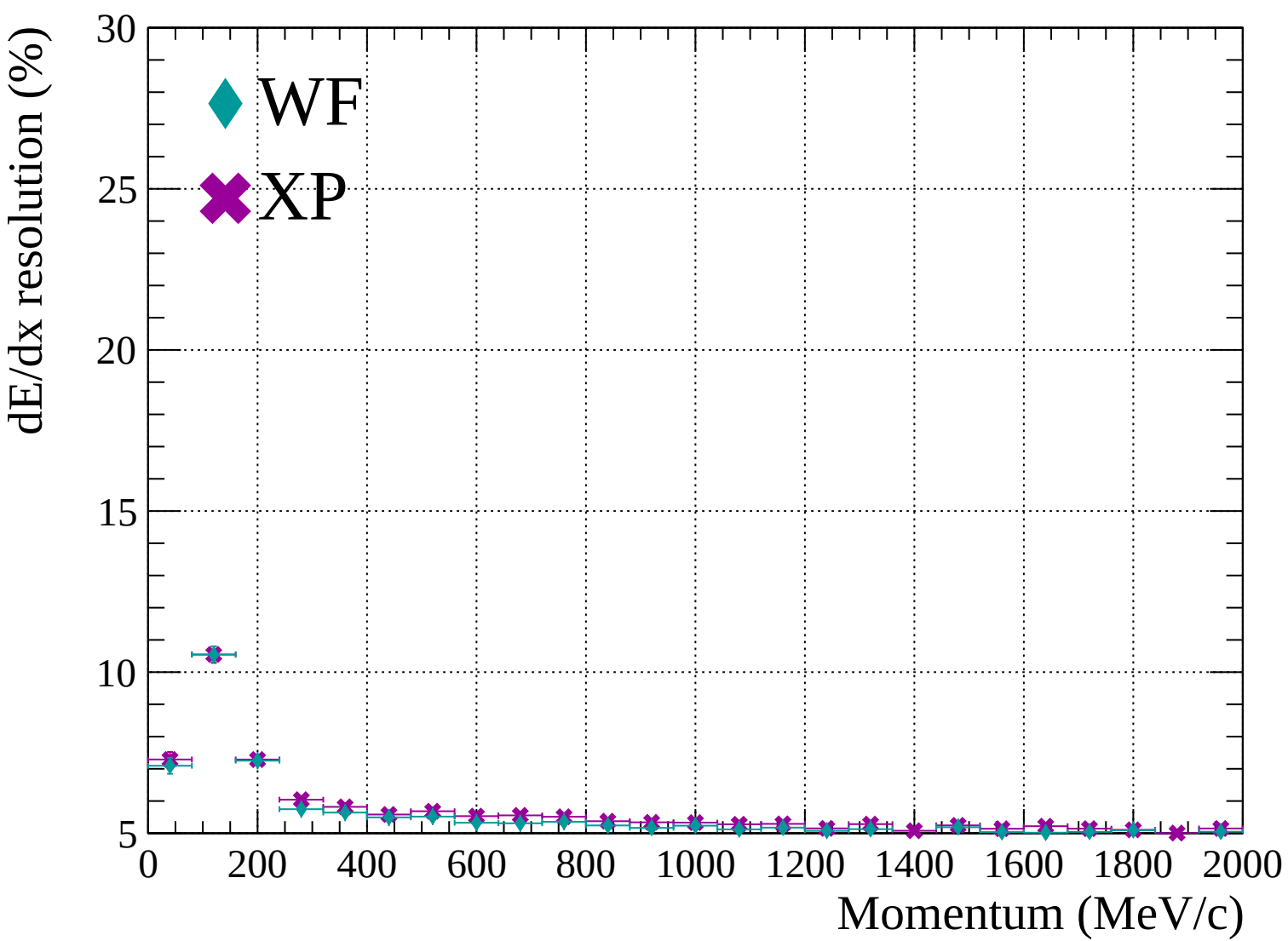


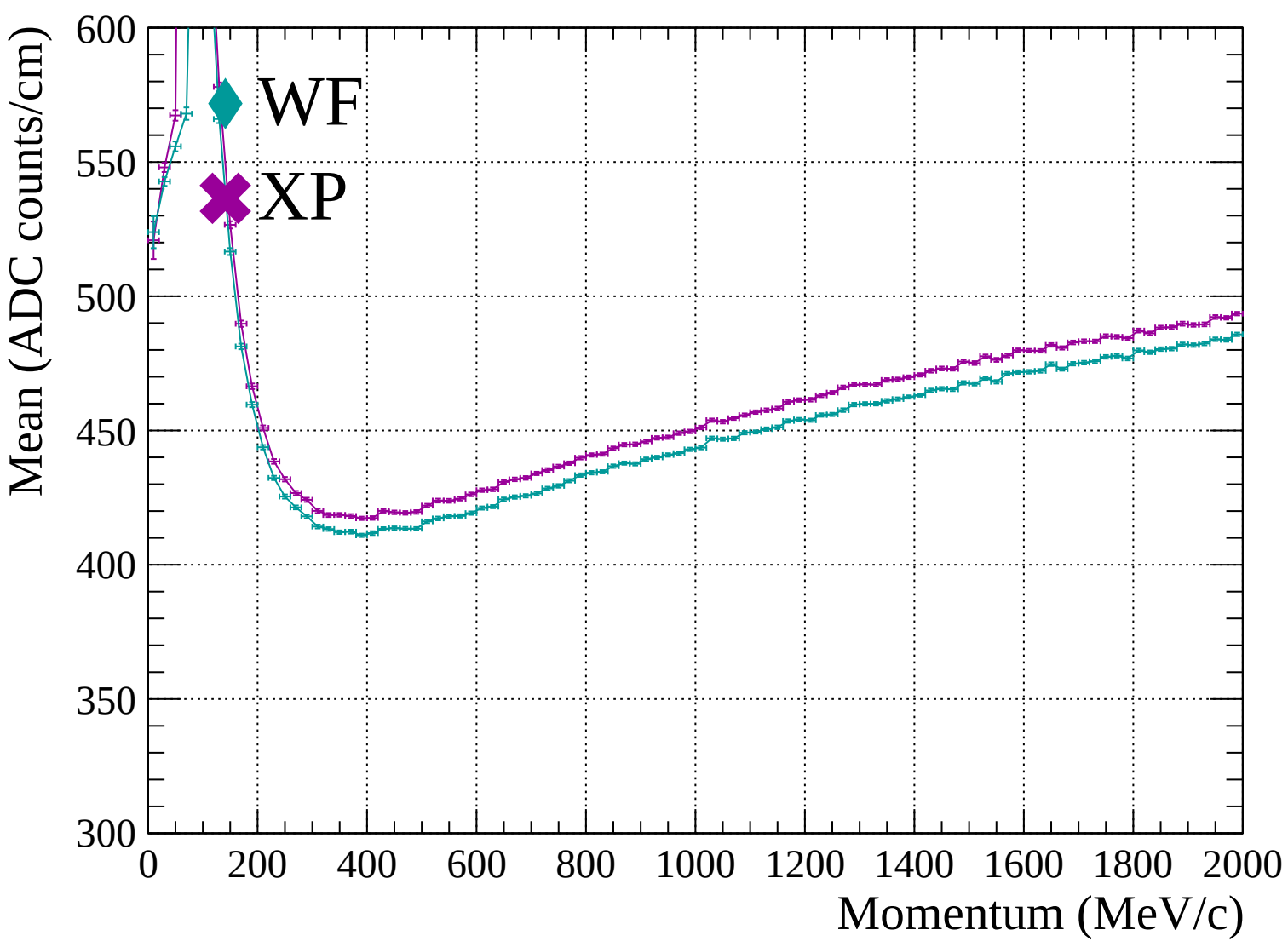


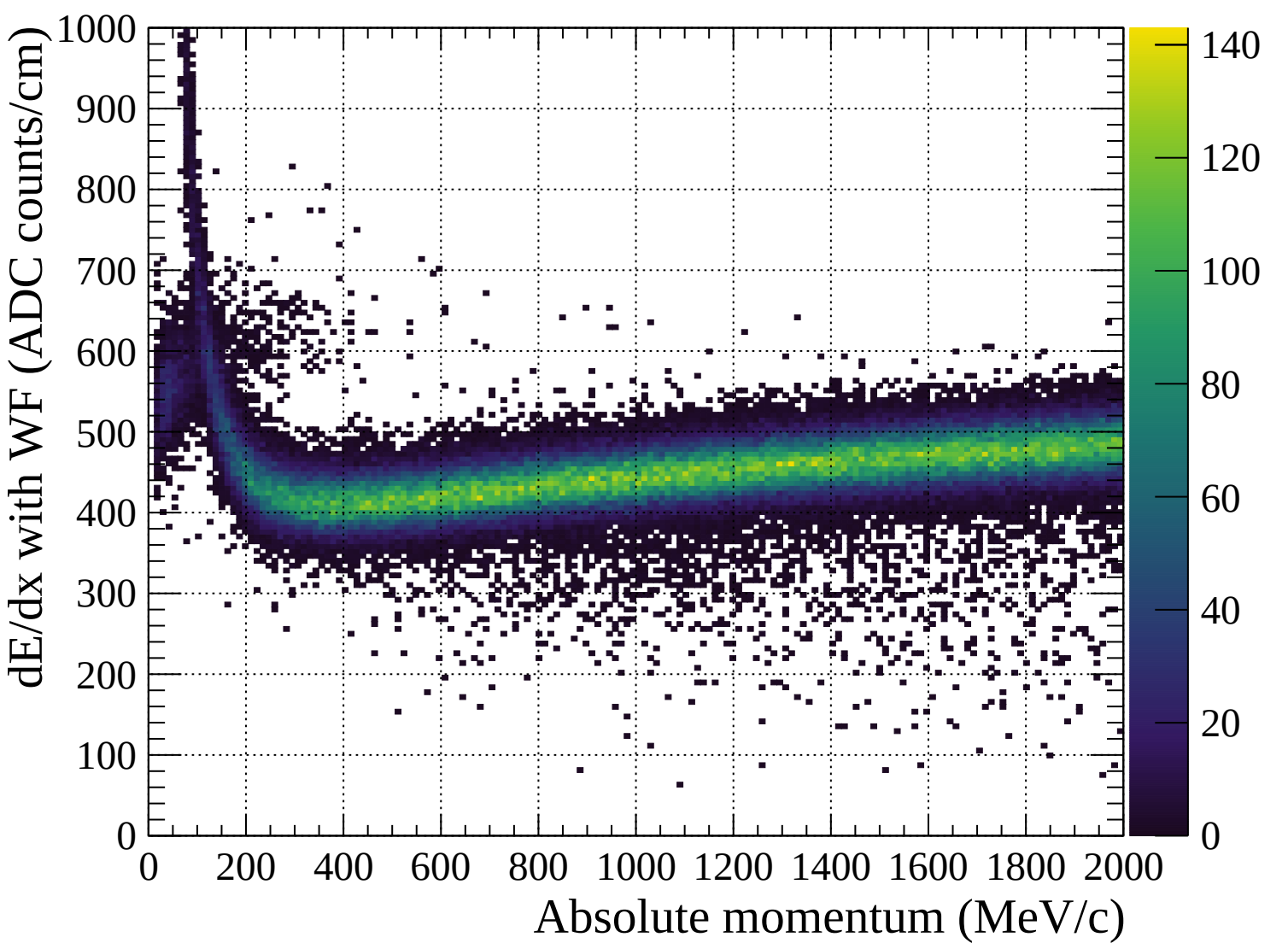


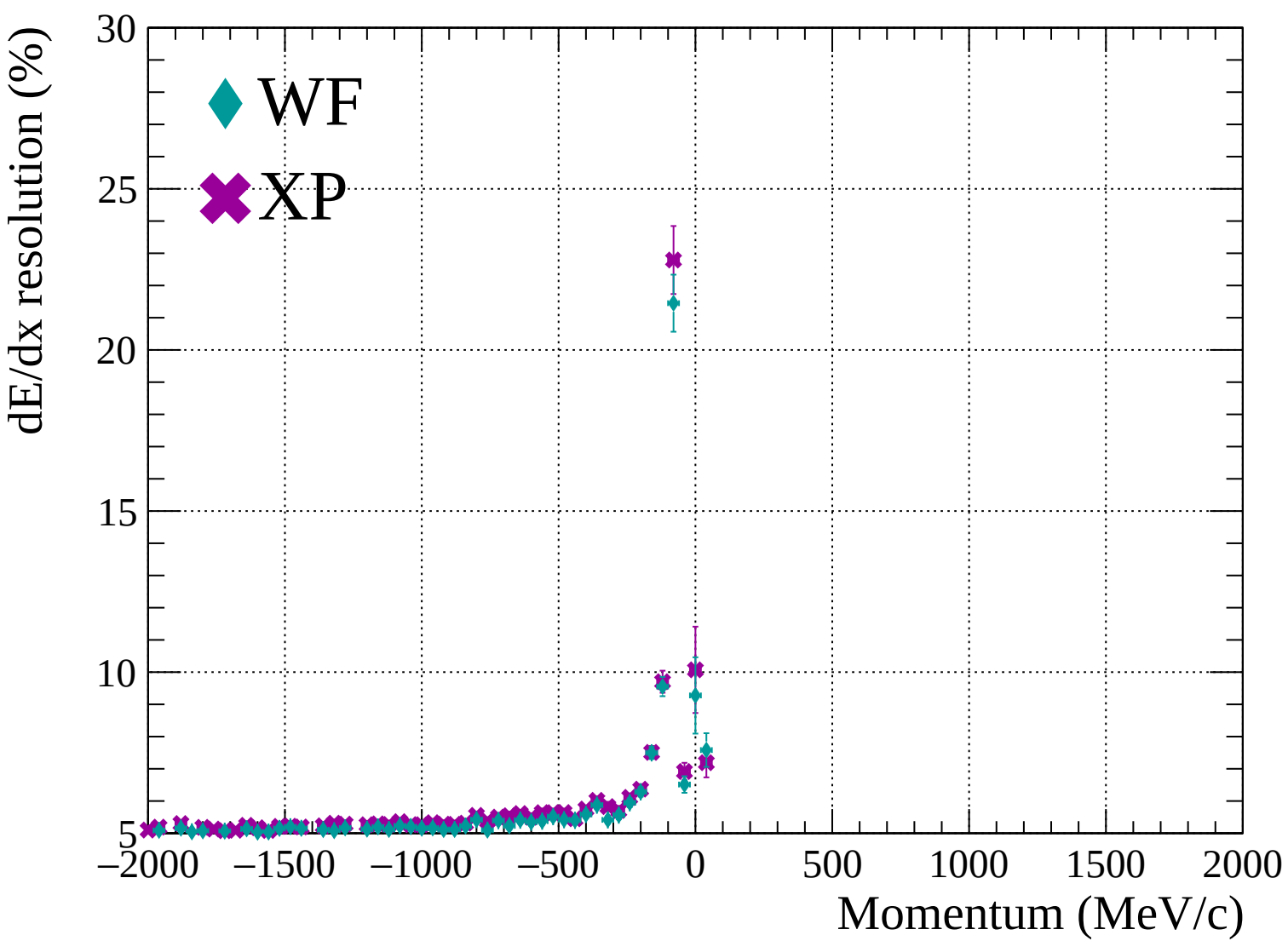


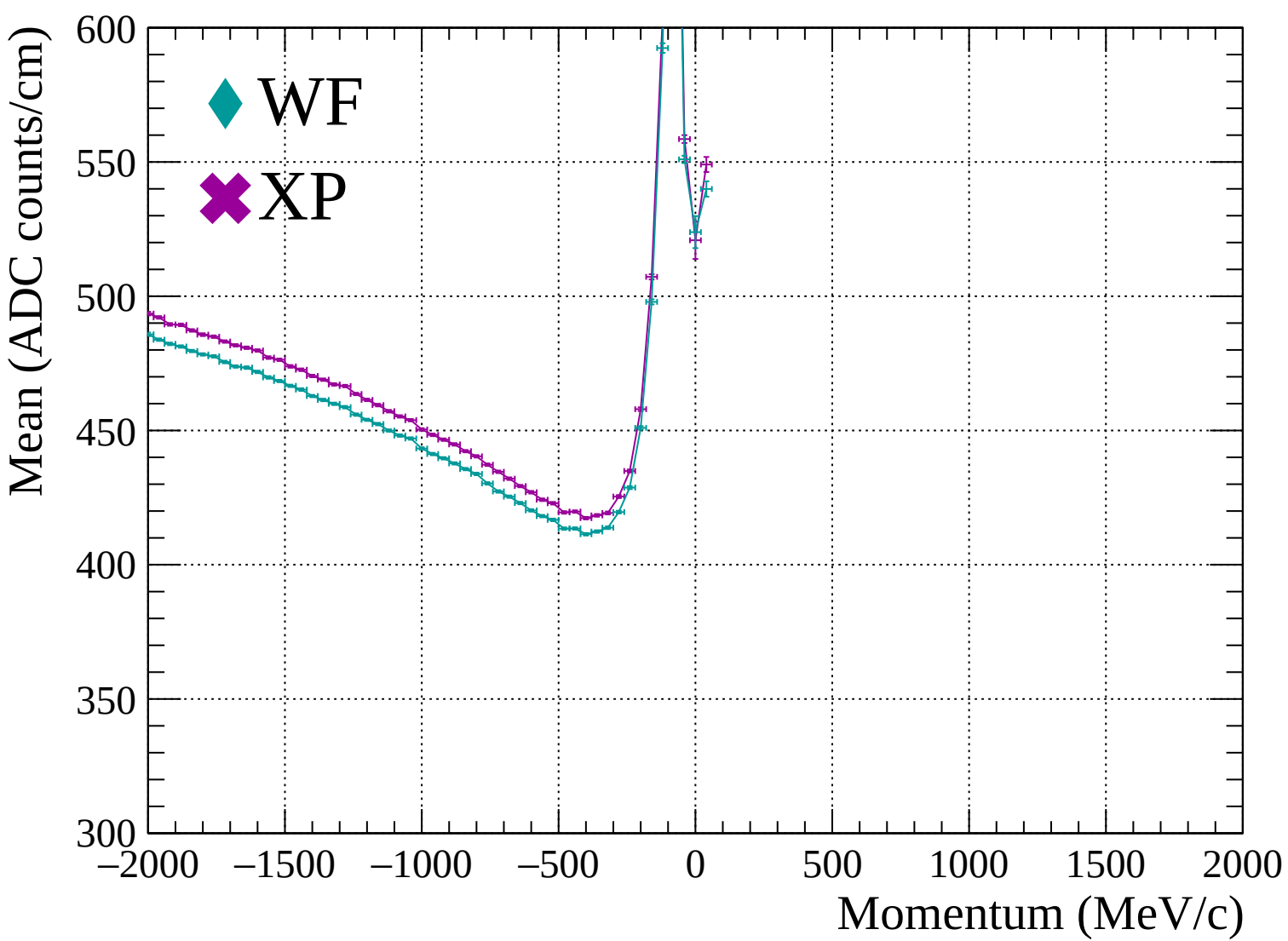


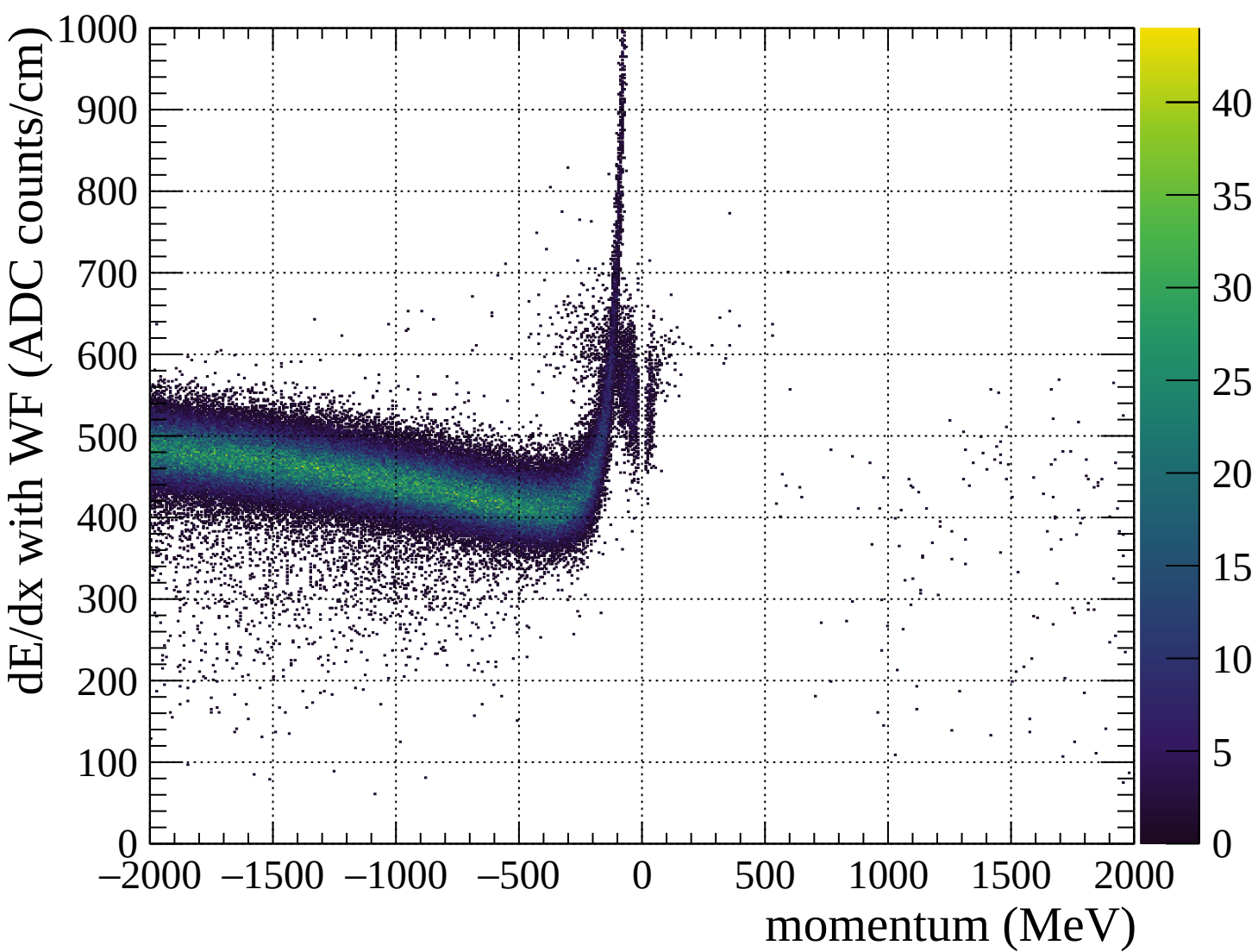


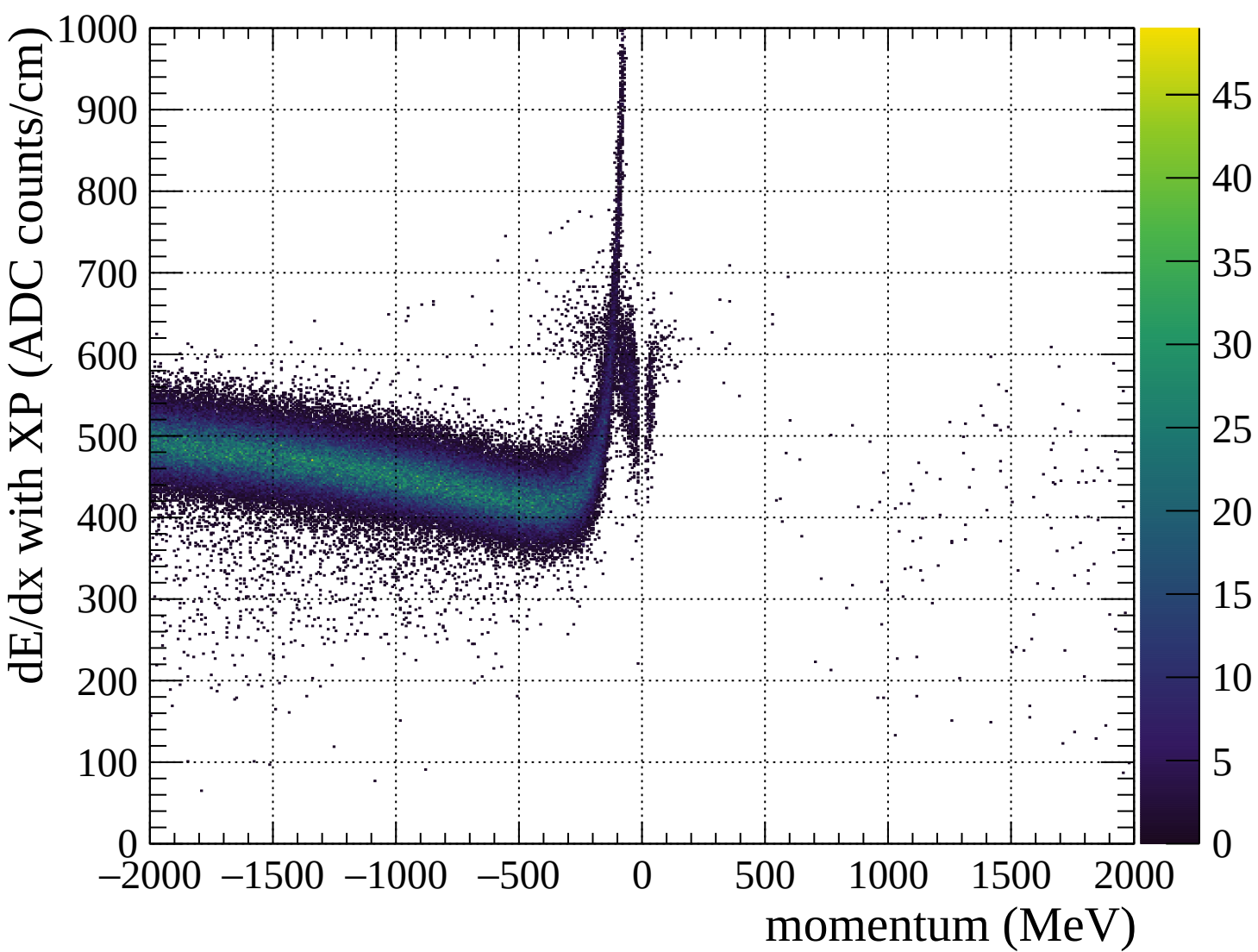


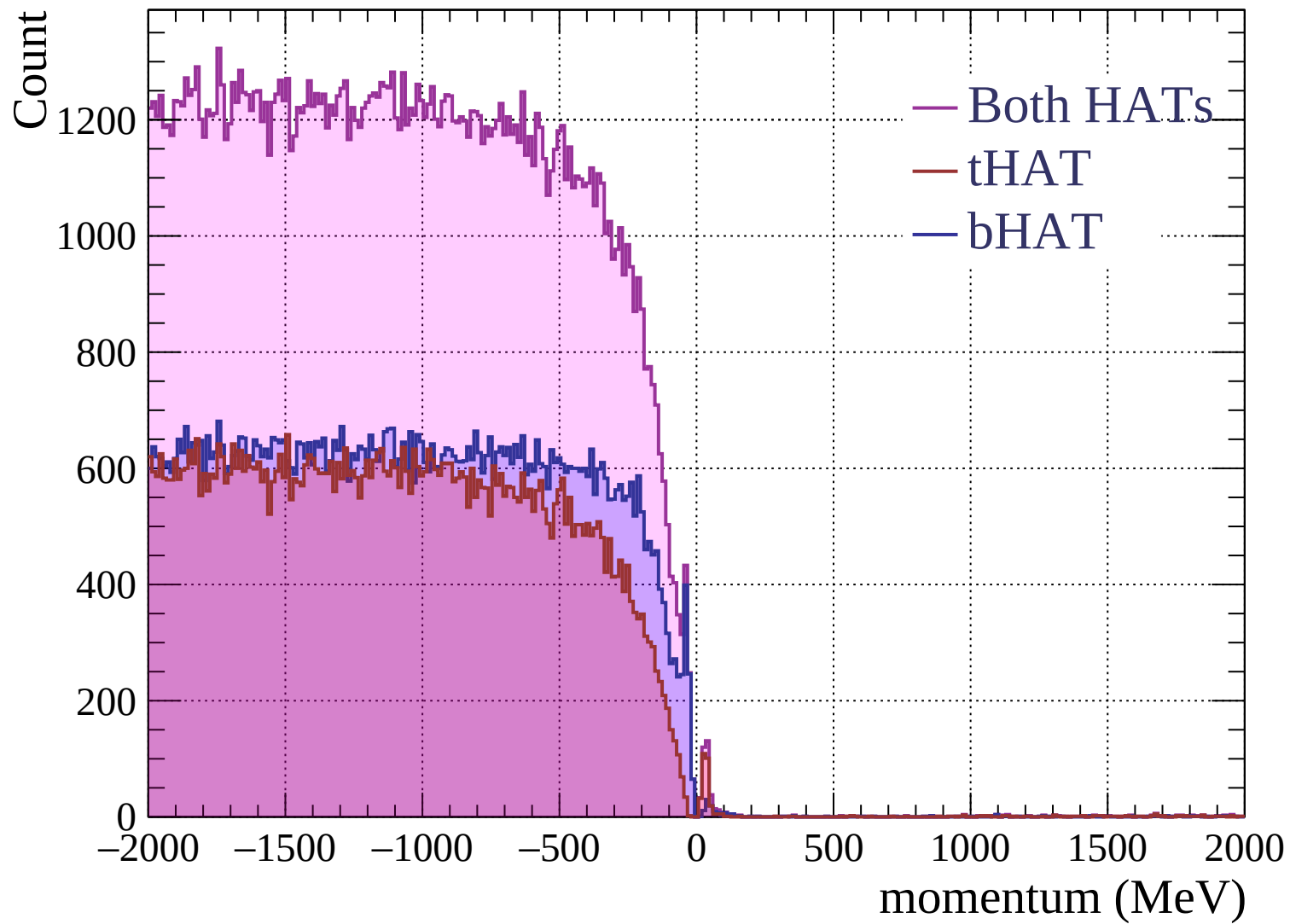


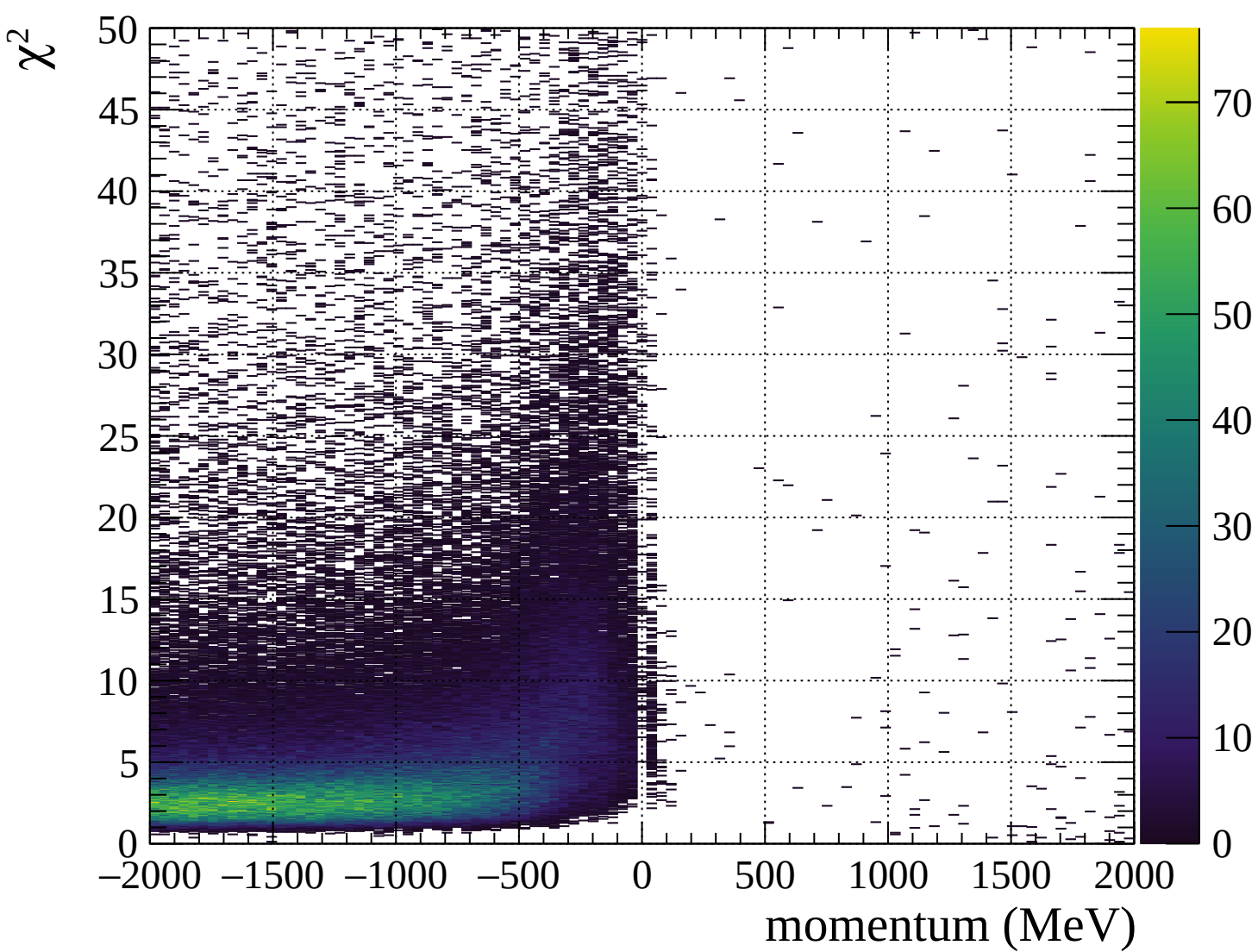




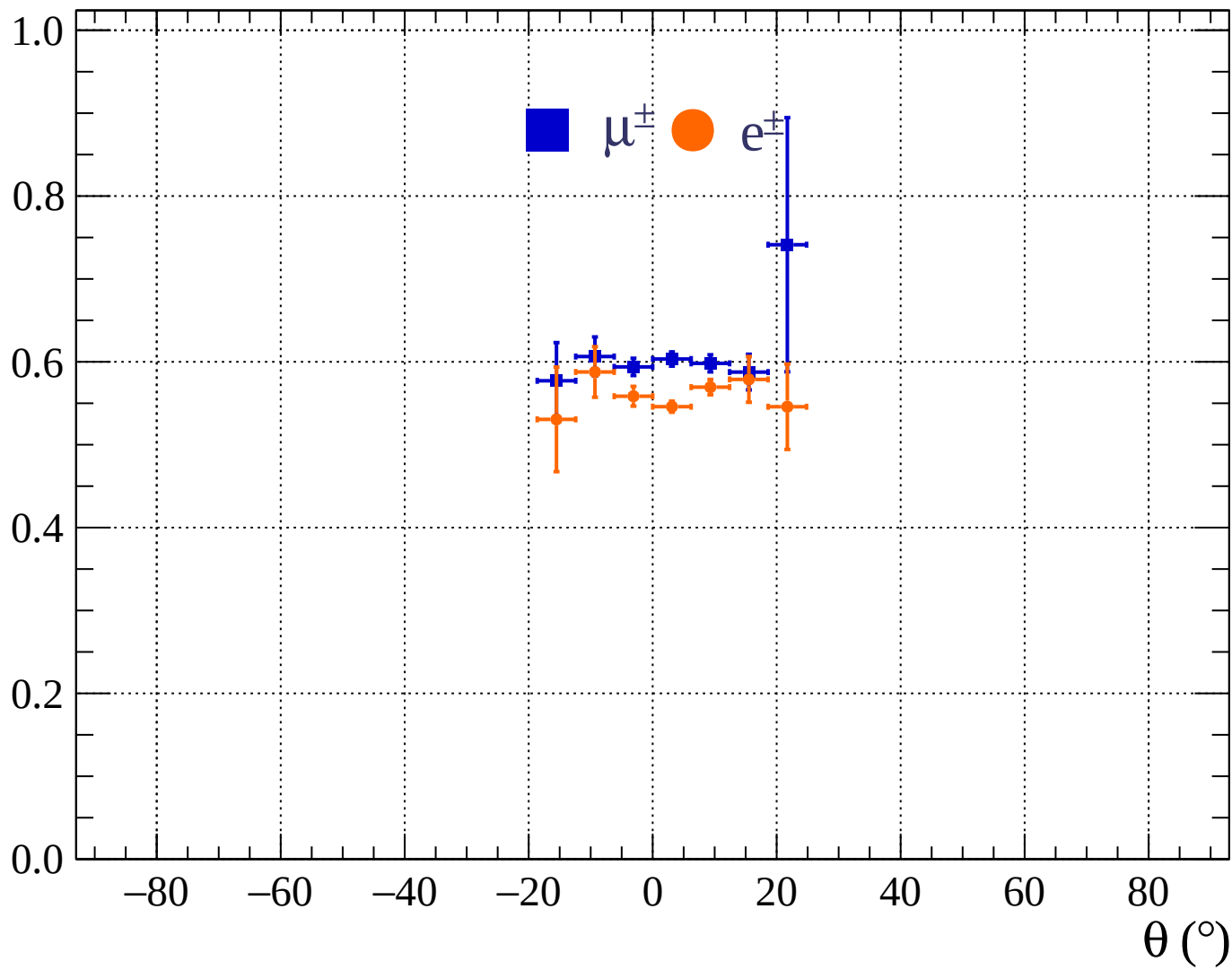


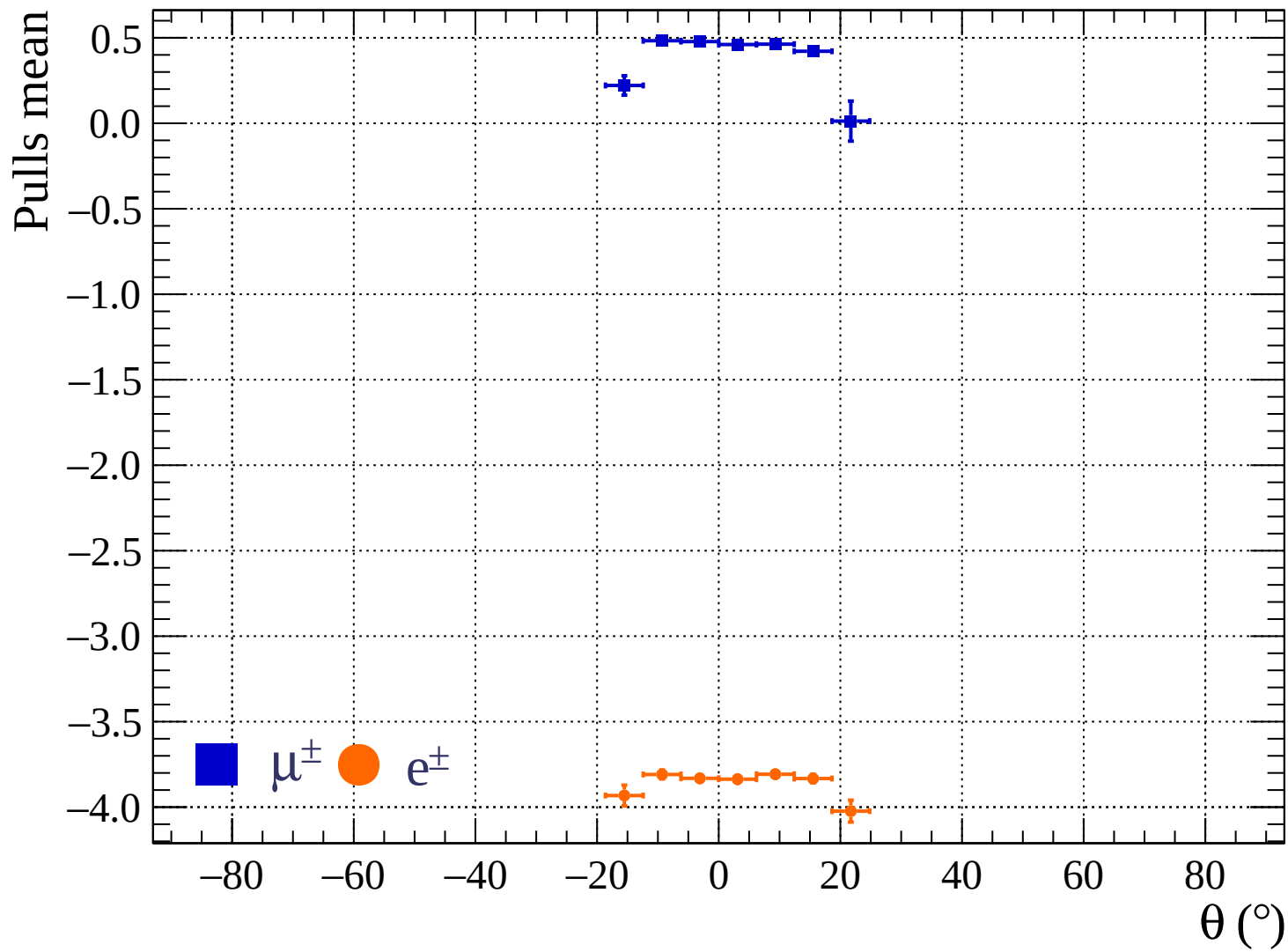




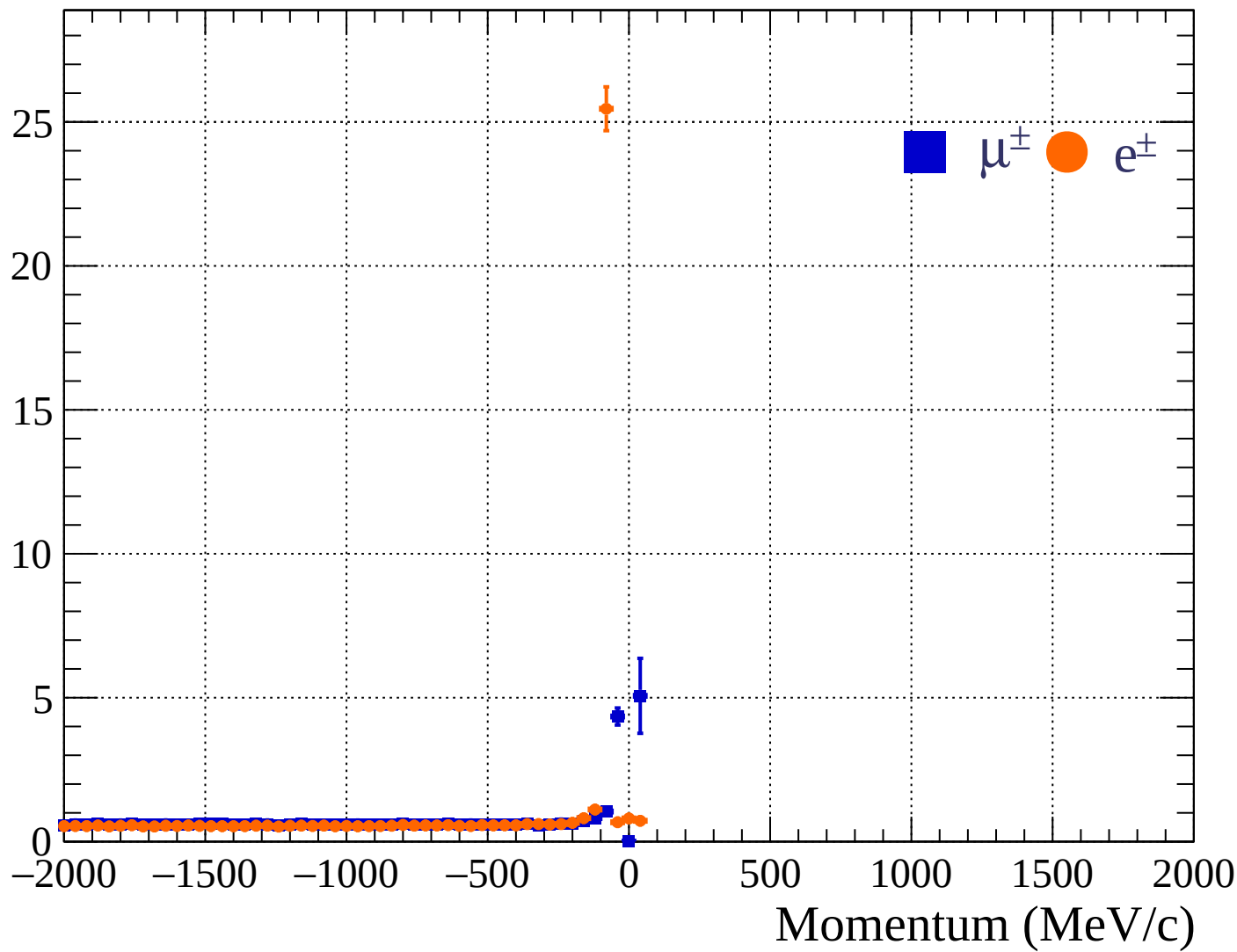


Pulls standard deviation

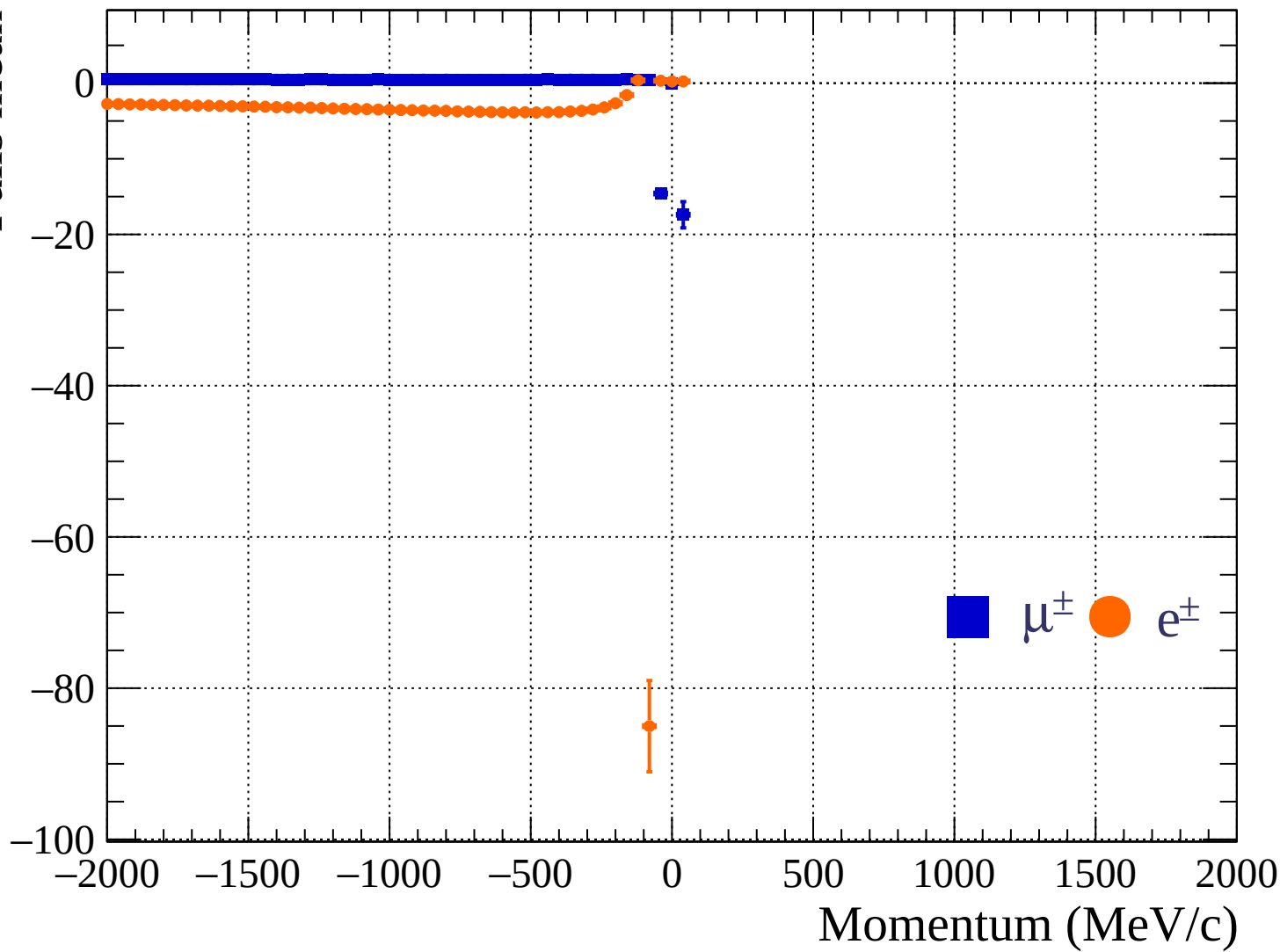


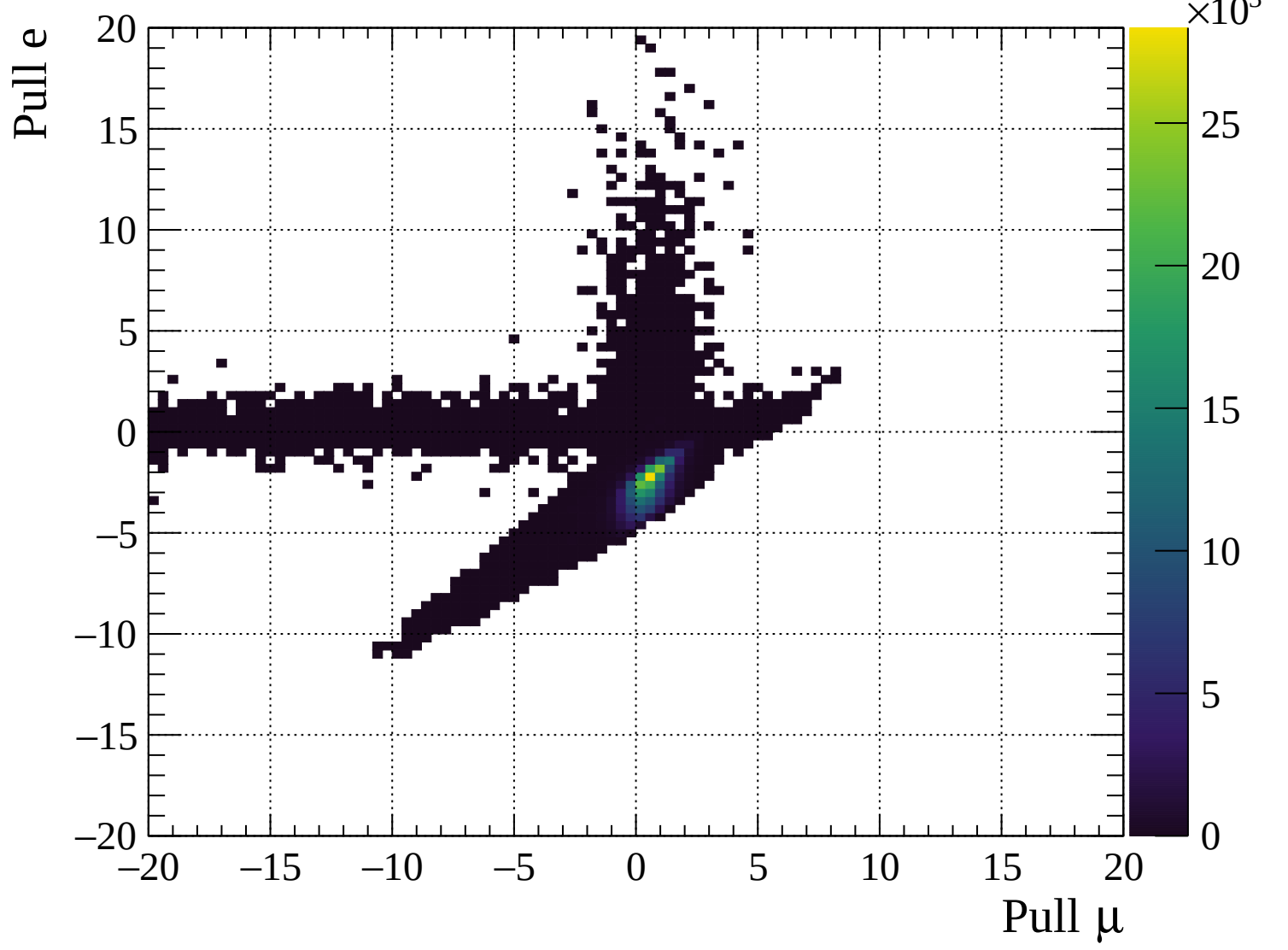


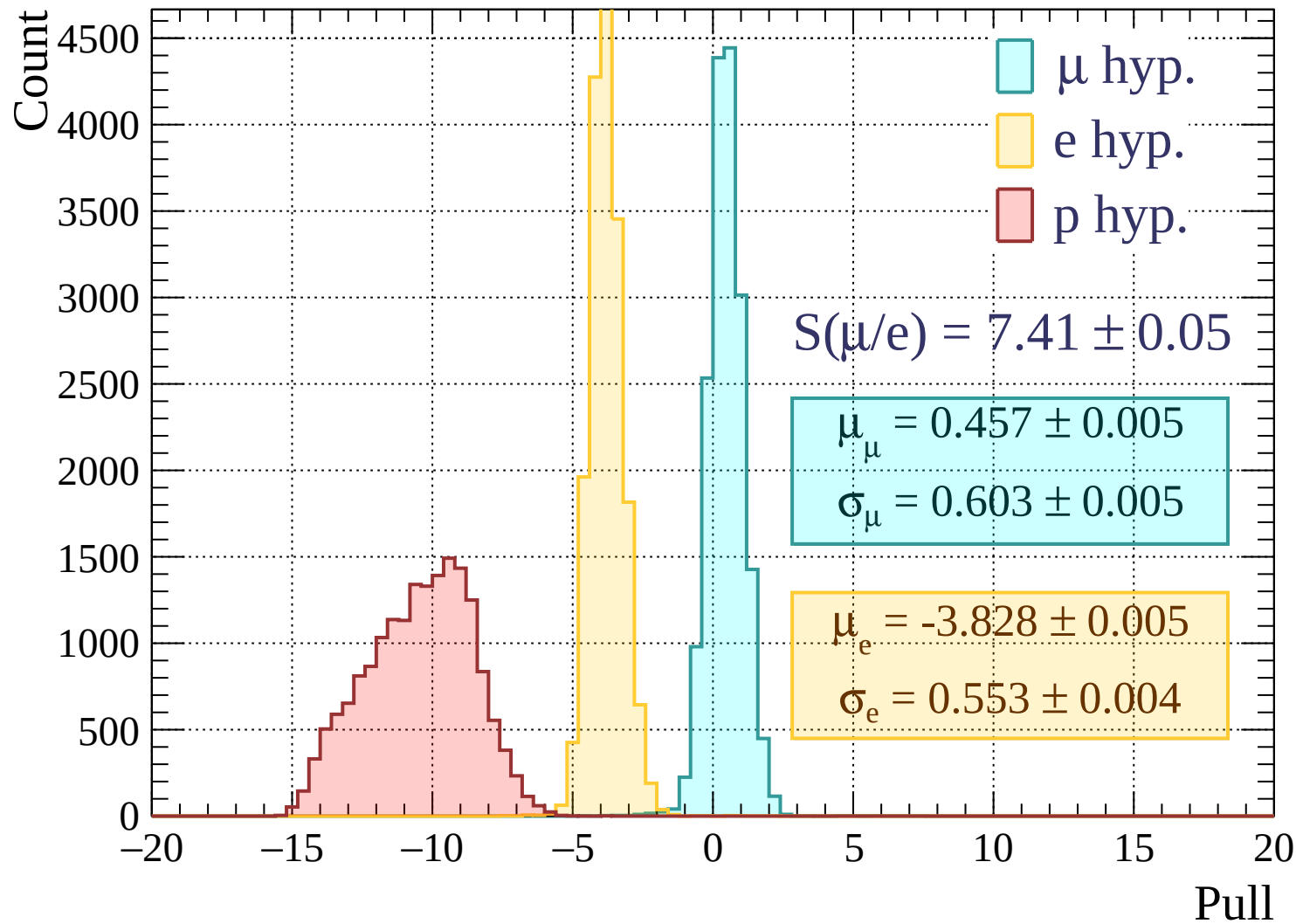
Pulls standard deviation

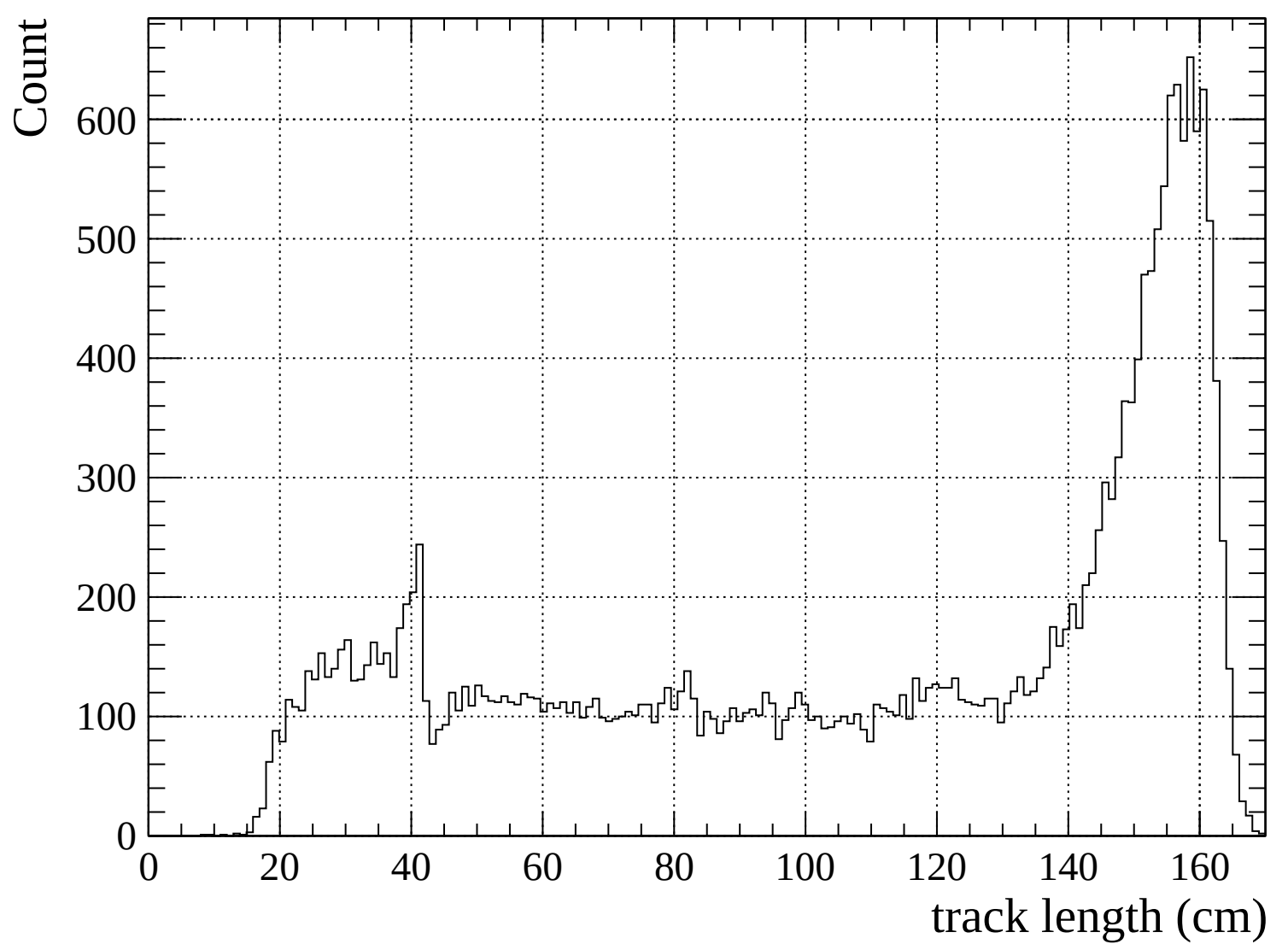


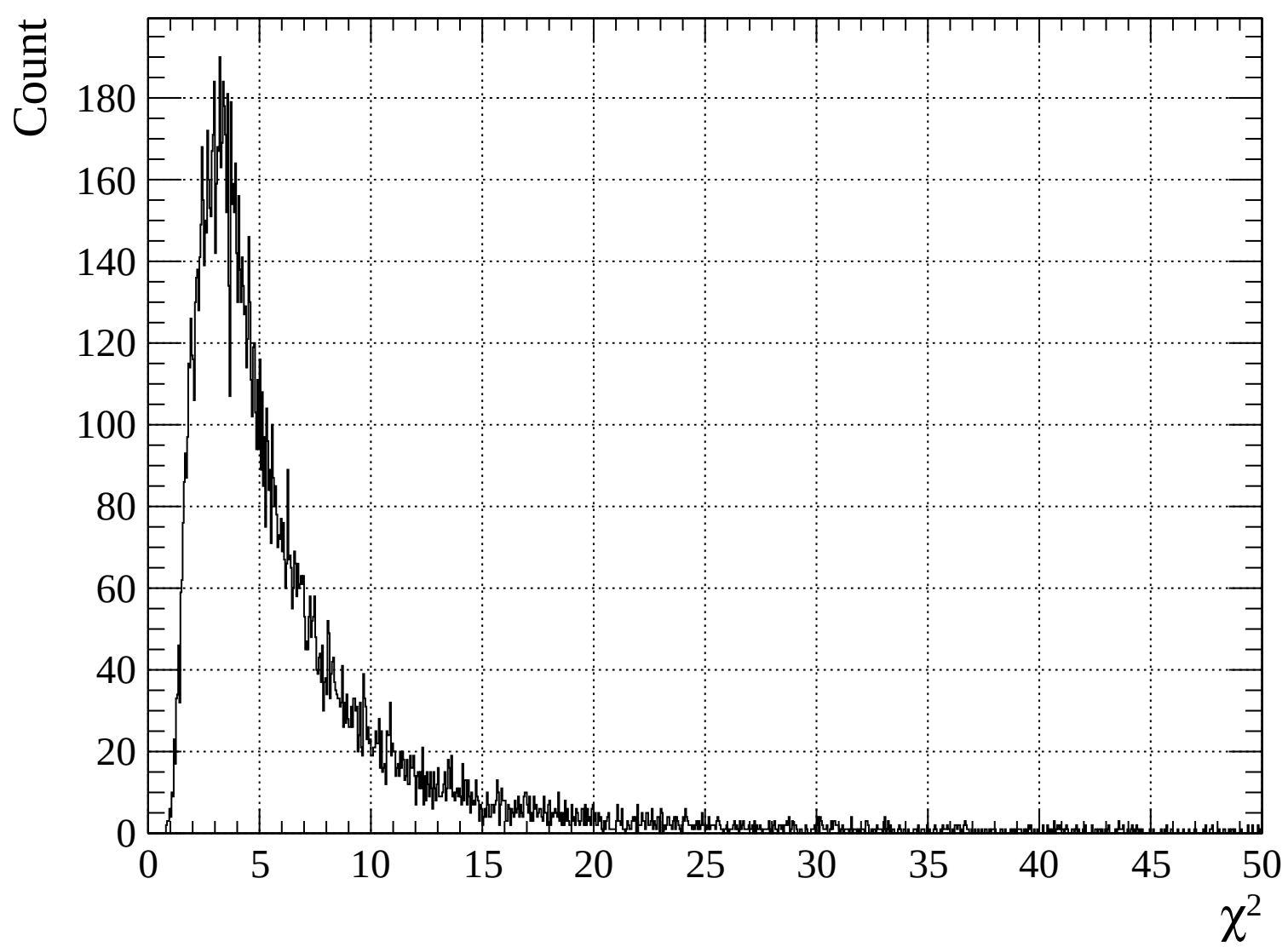
Pulls mean

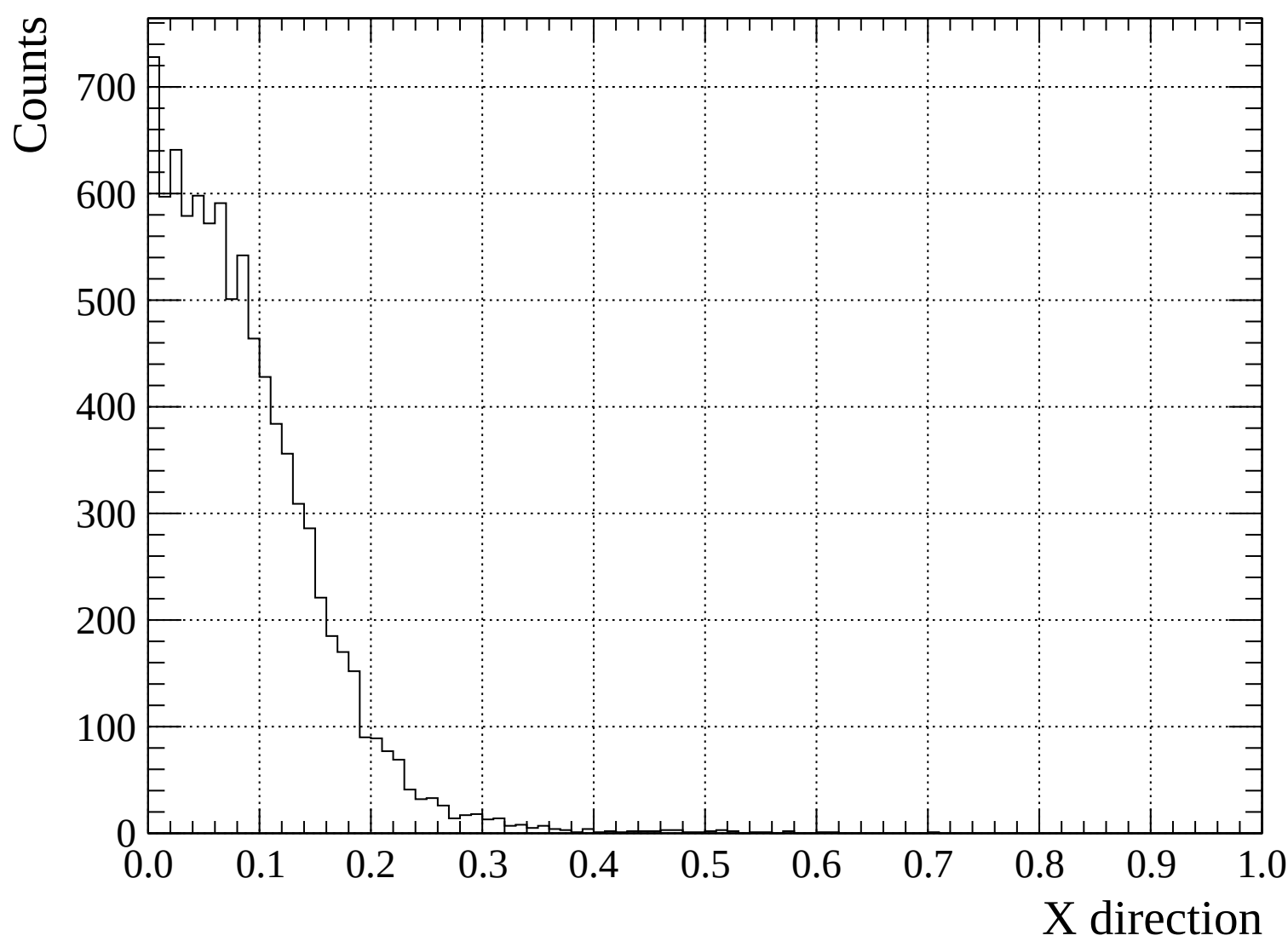


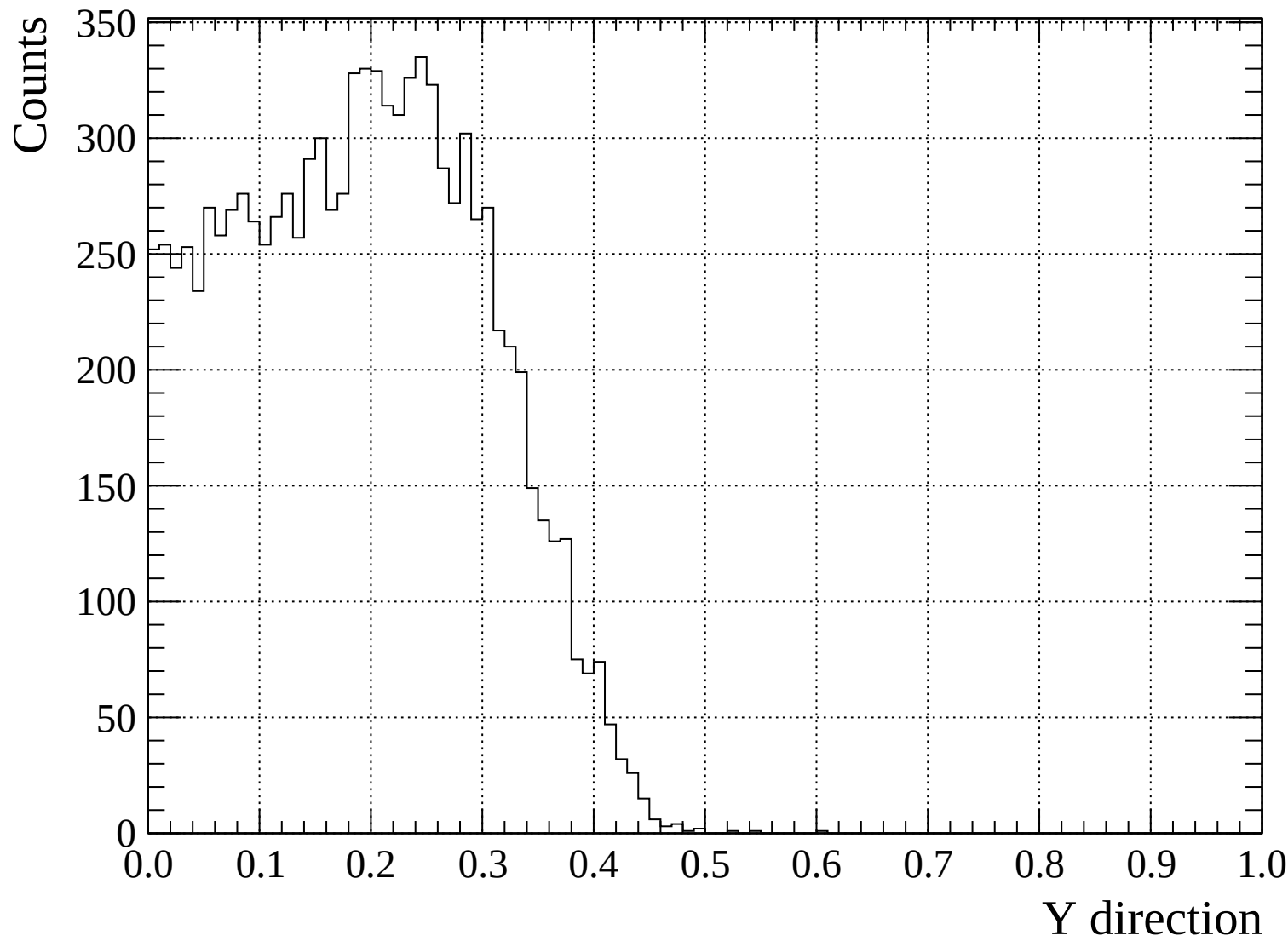


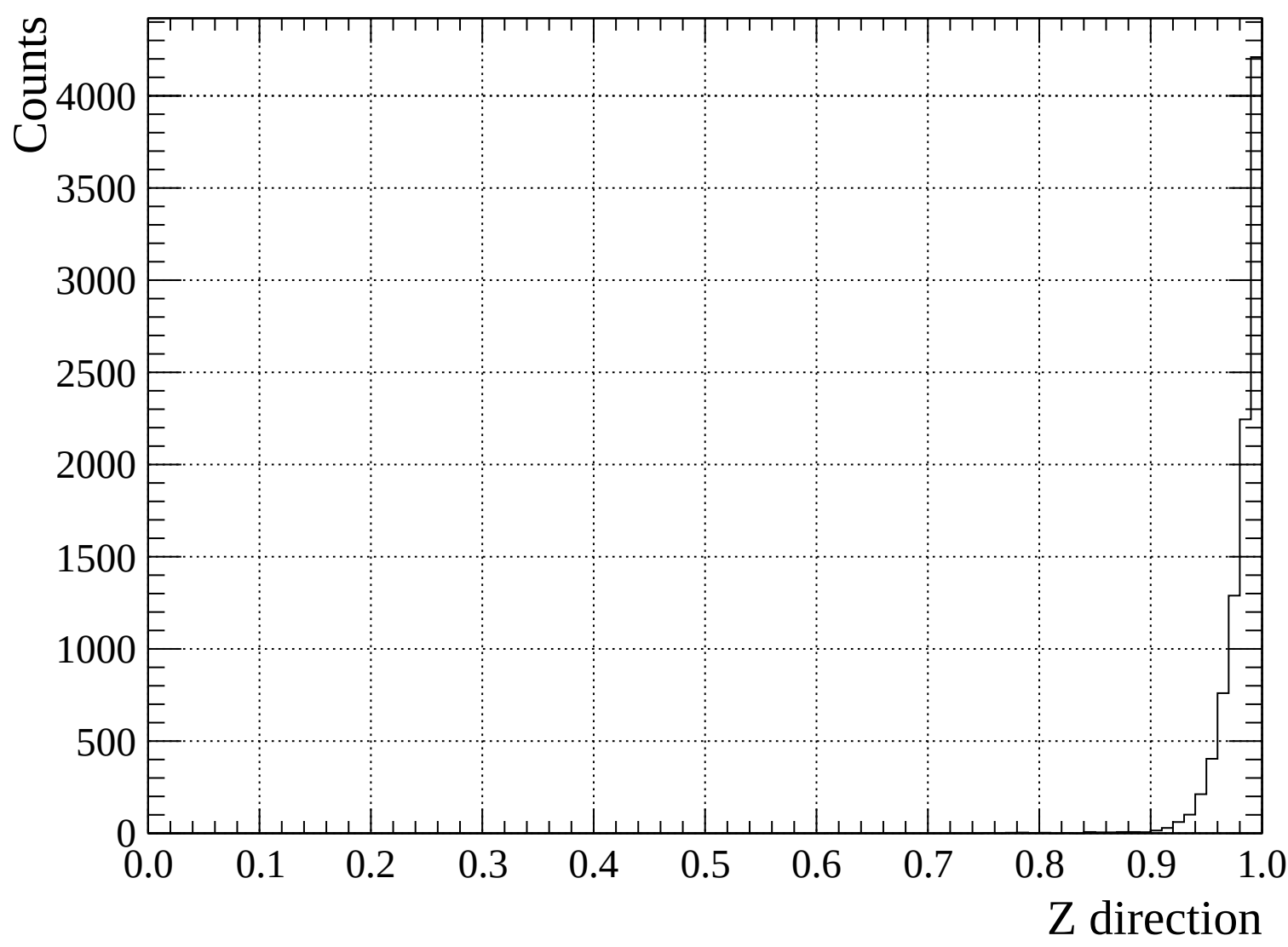


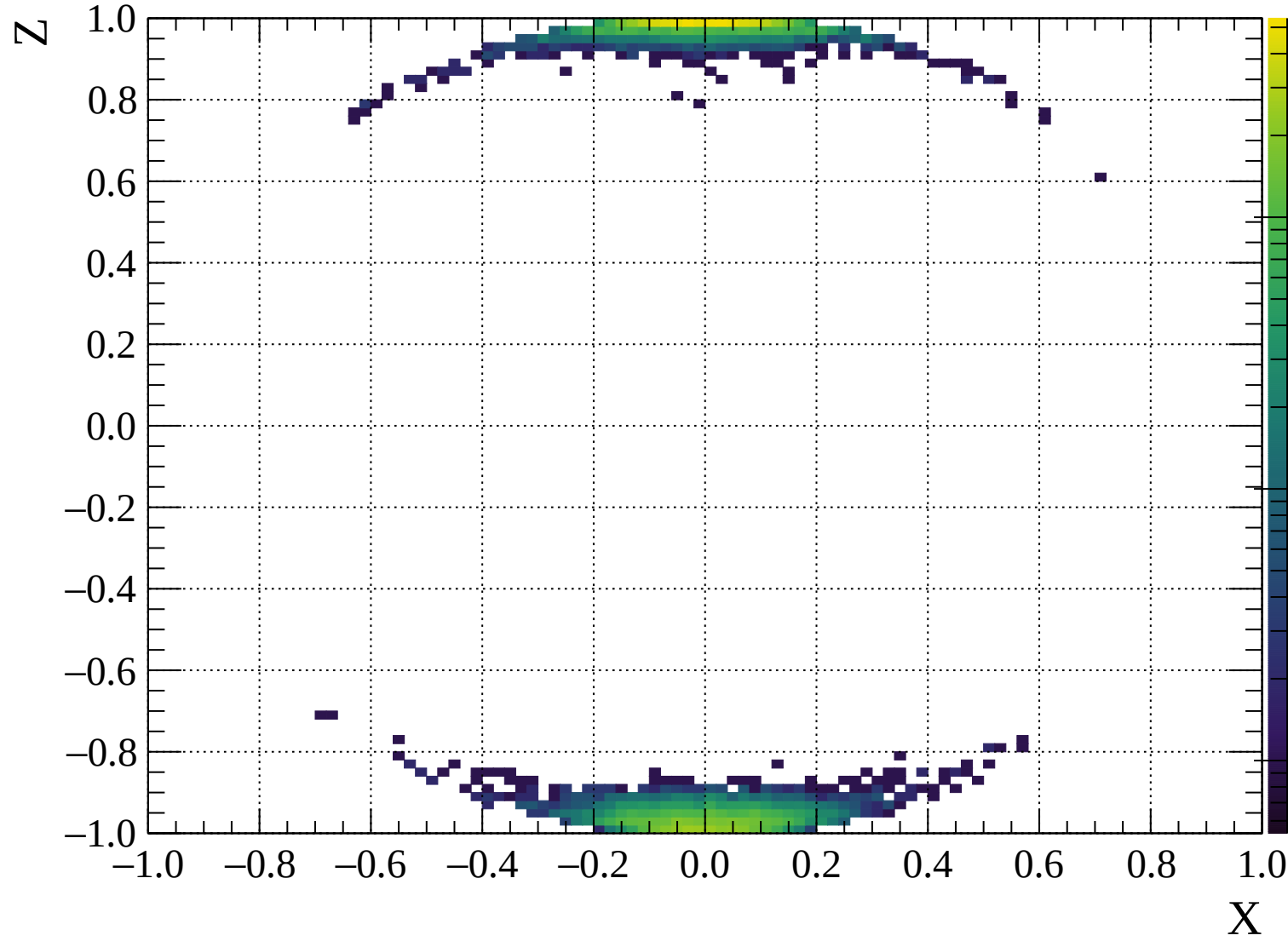


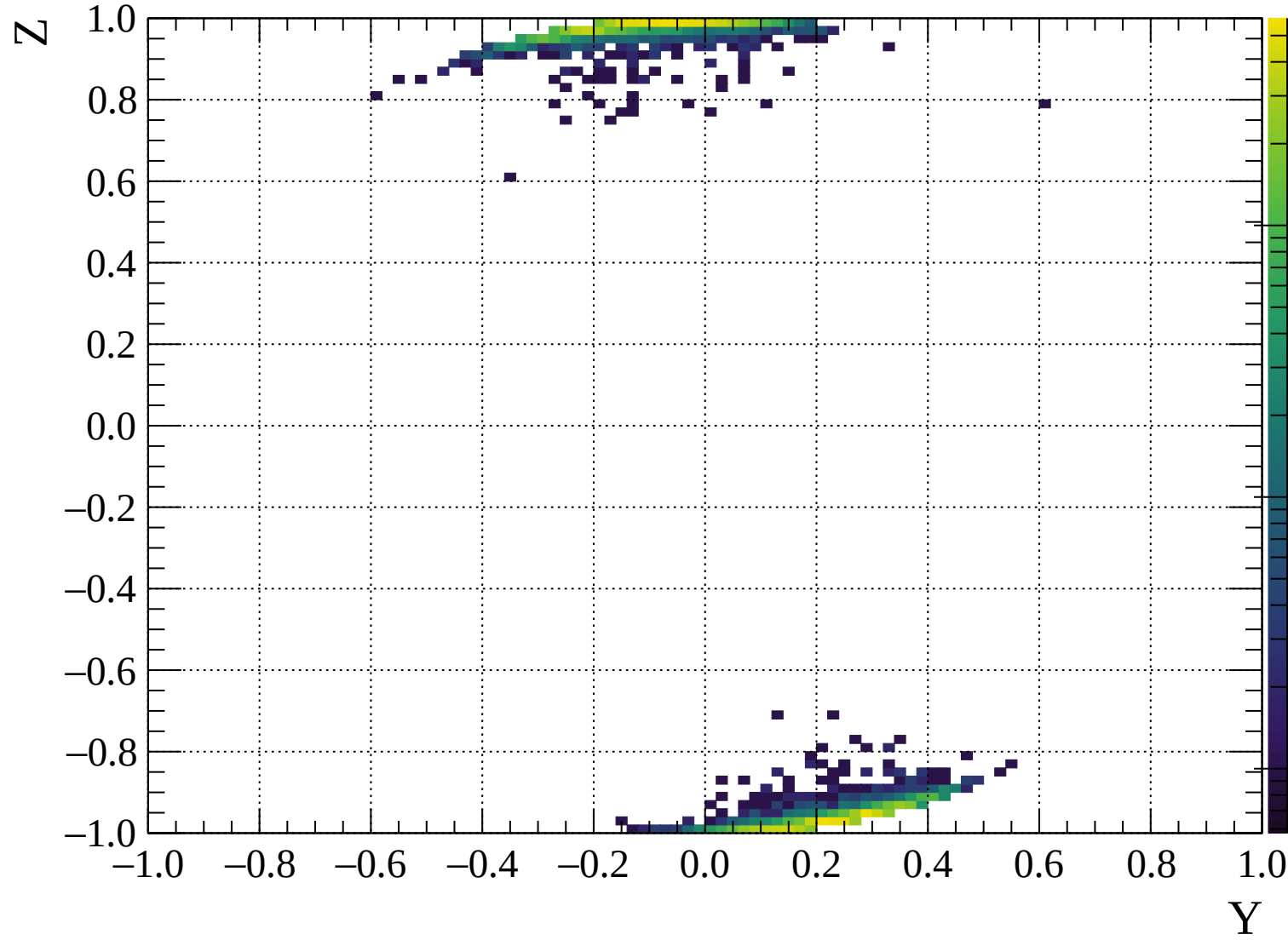


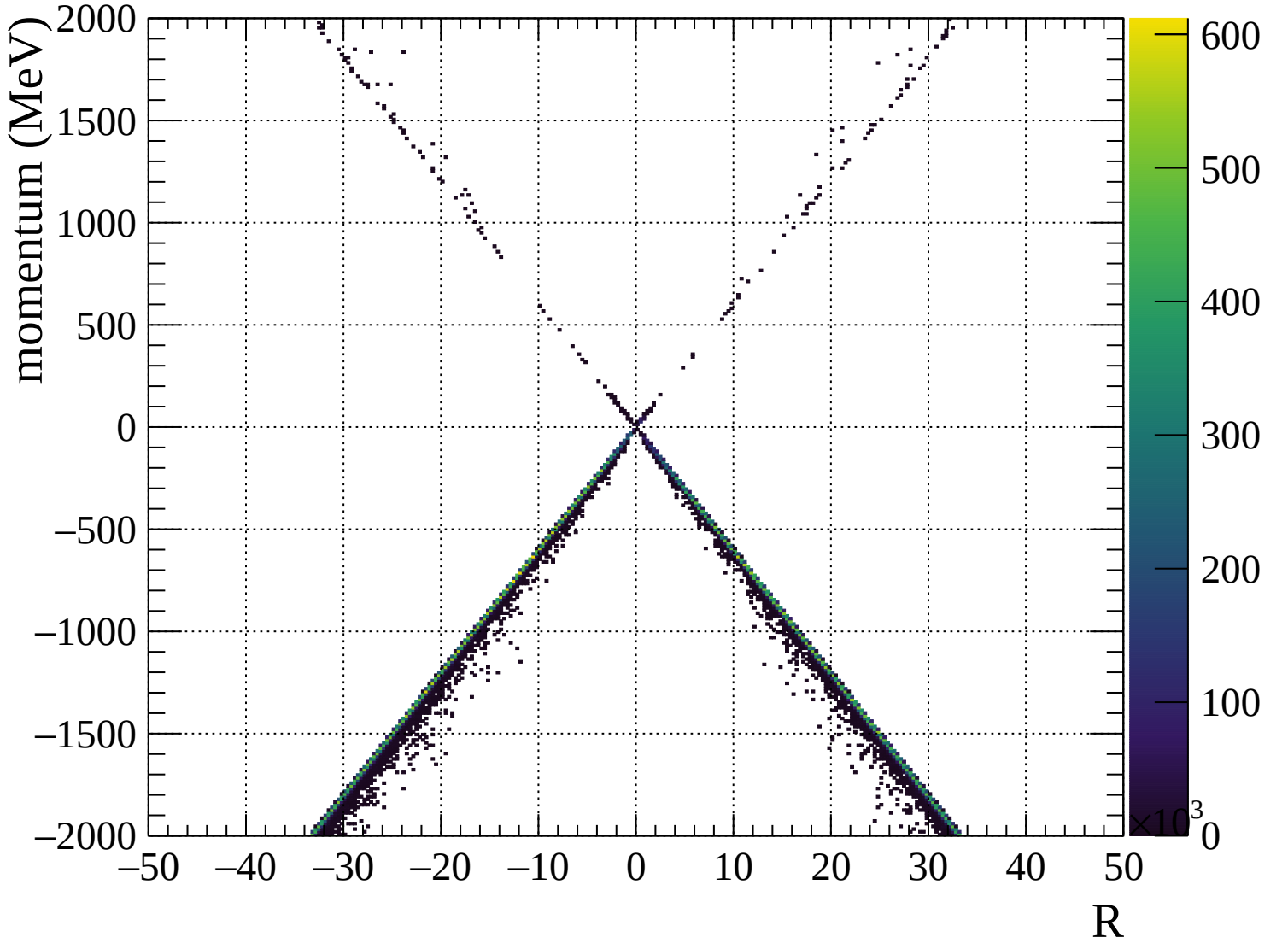


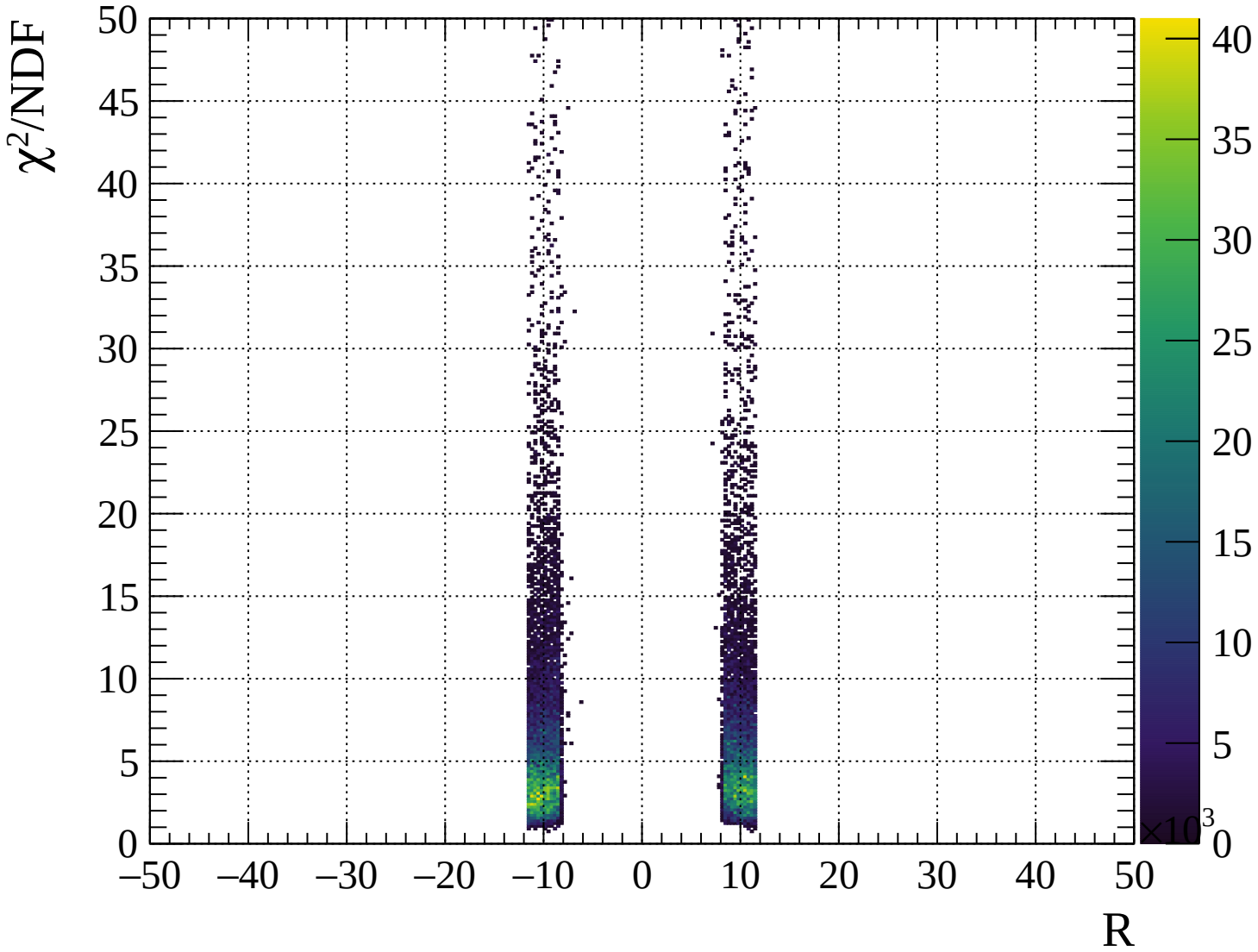




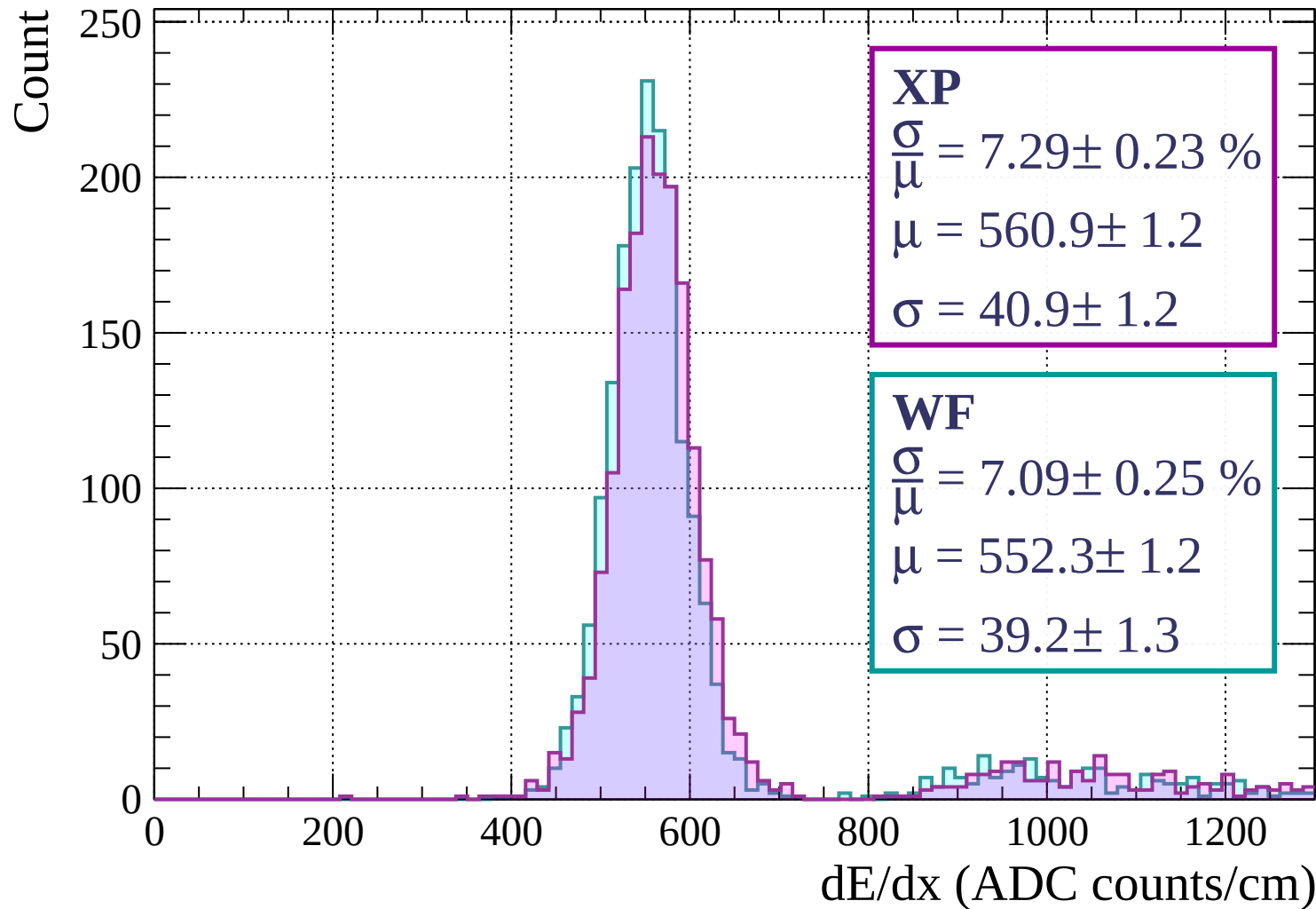






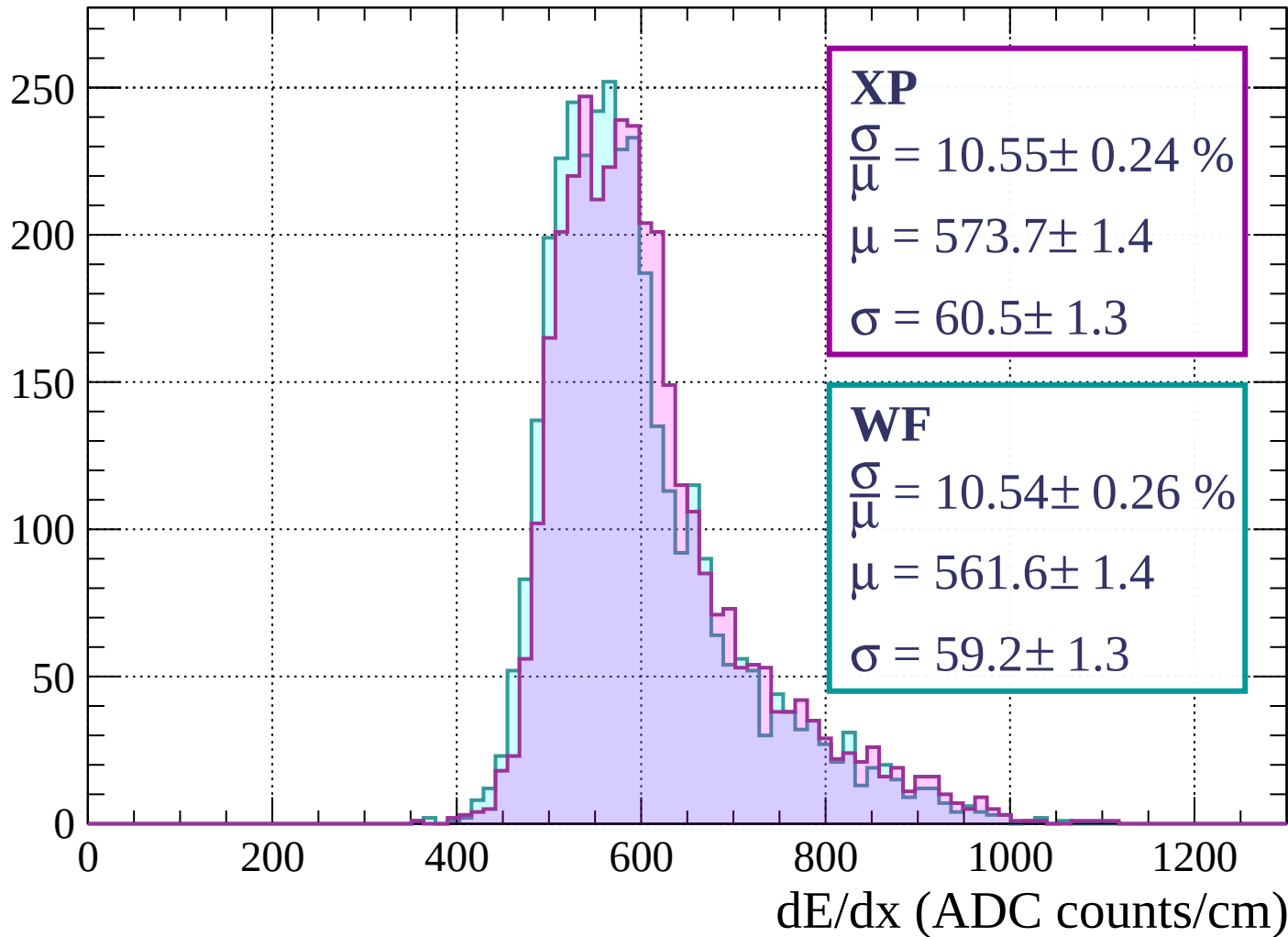


Energy loss | $0 < p < 80$

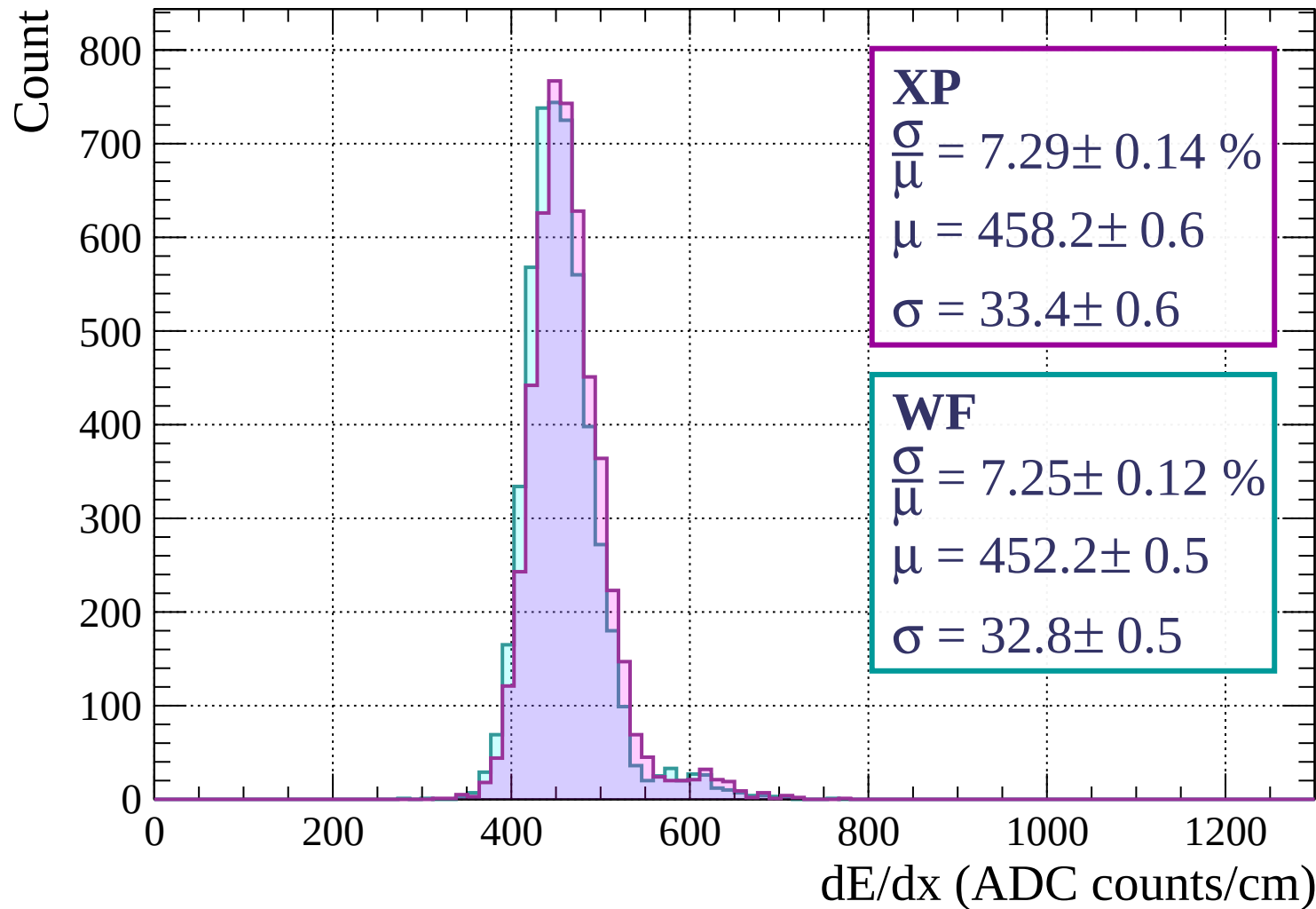


Energy loss | $80 < p < 160$

Count

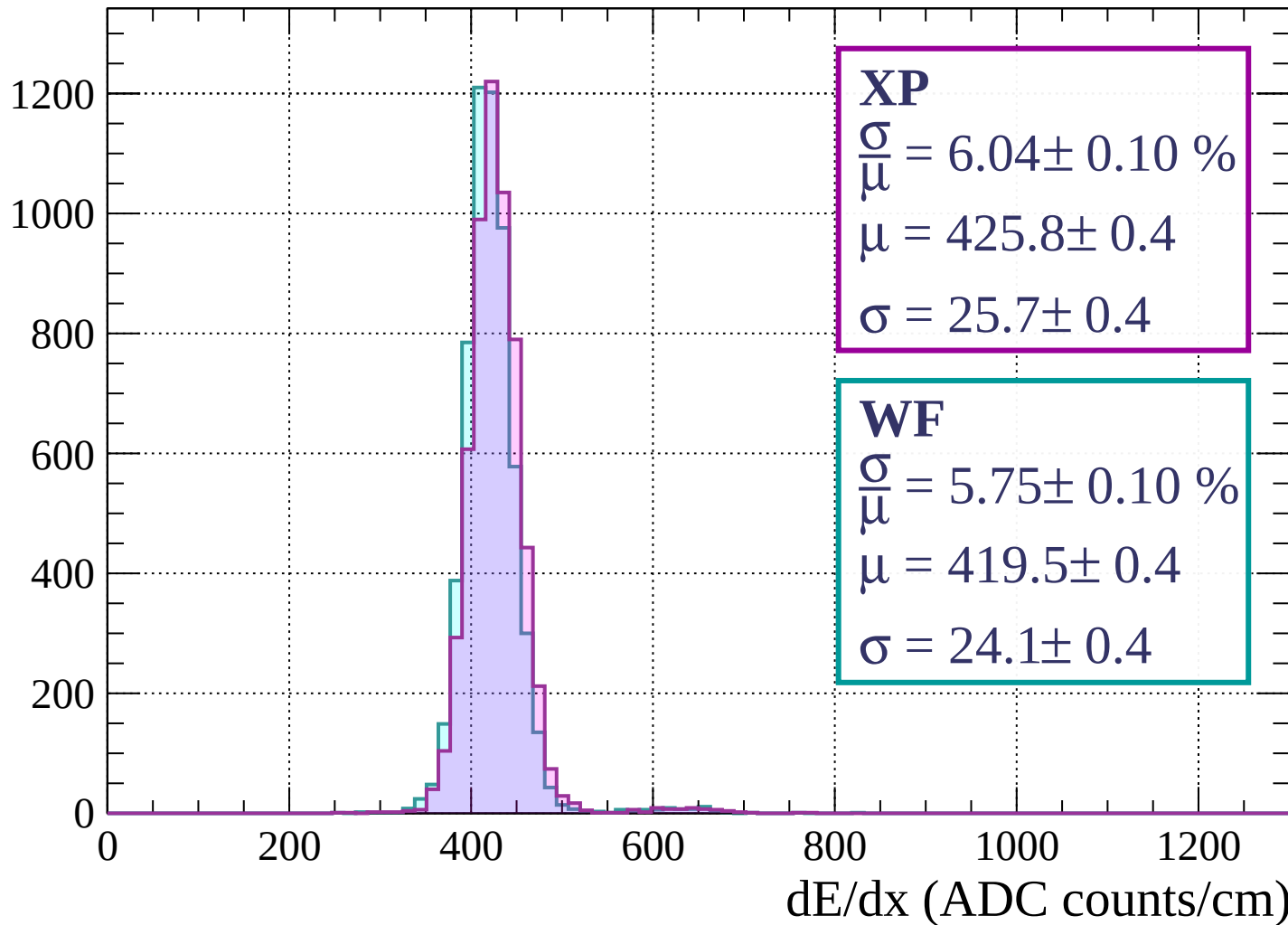


Energy loss | $160 < p < 240$

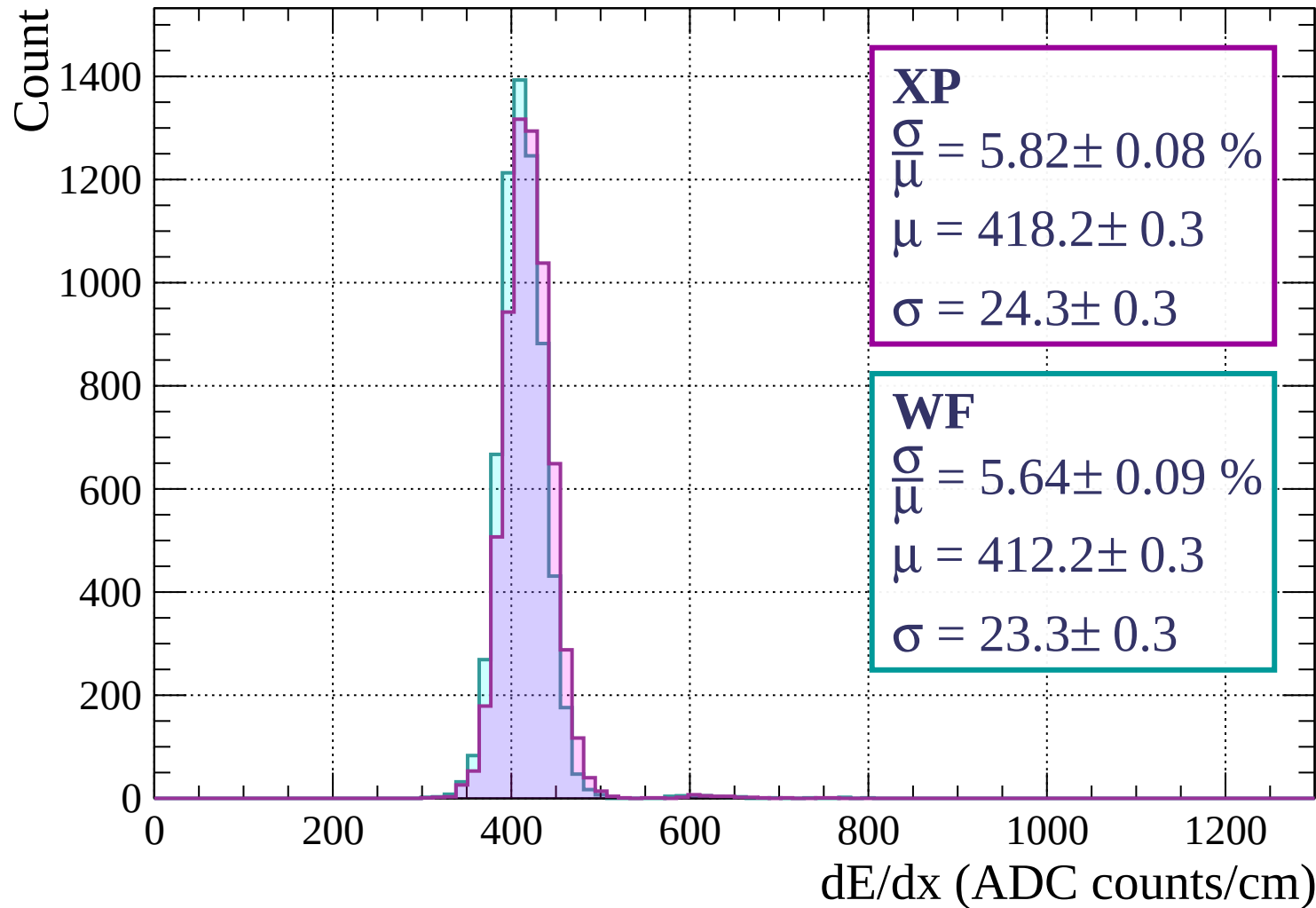


Energy loss | $240 < p < 320$

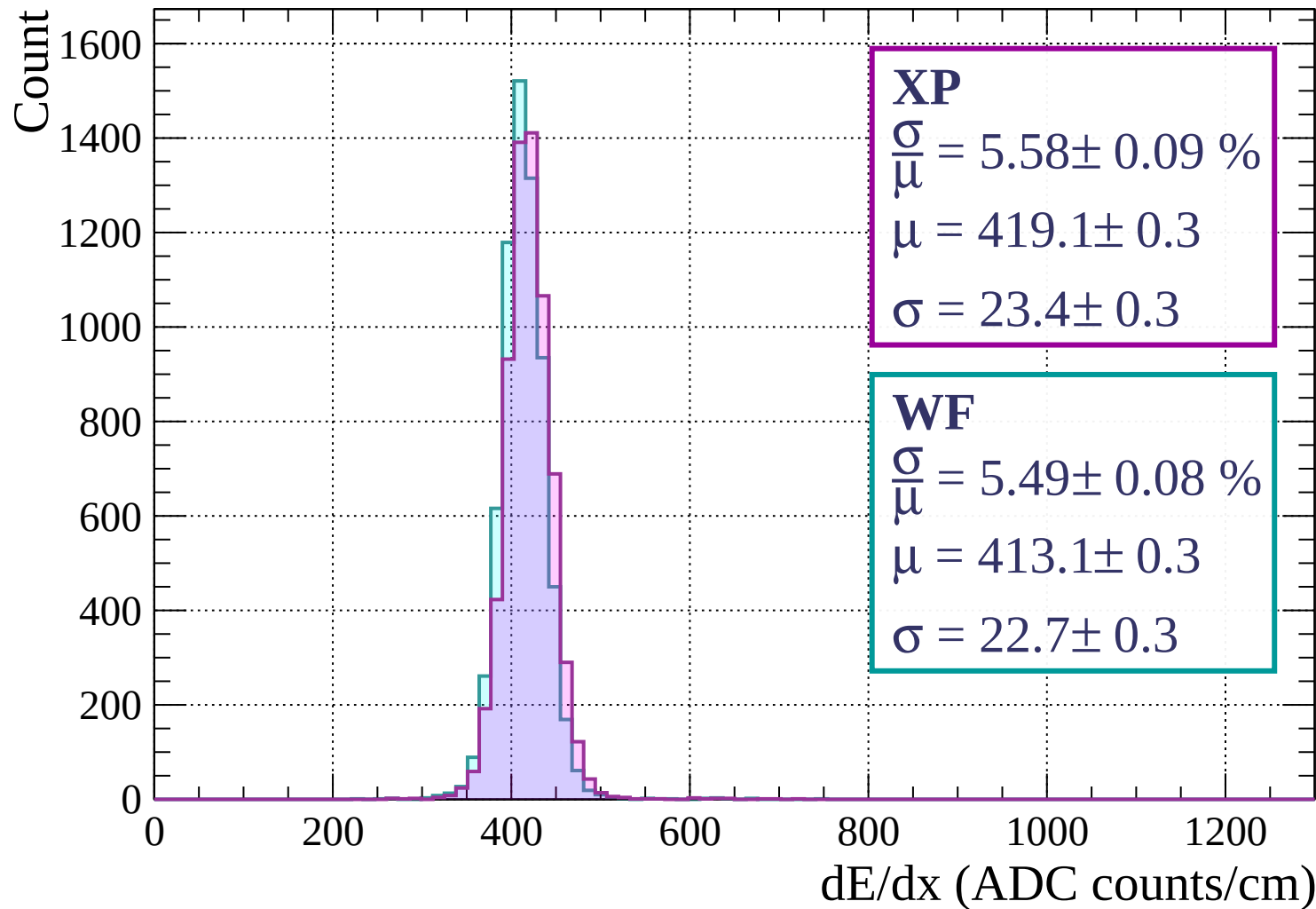
Count



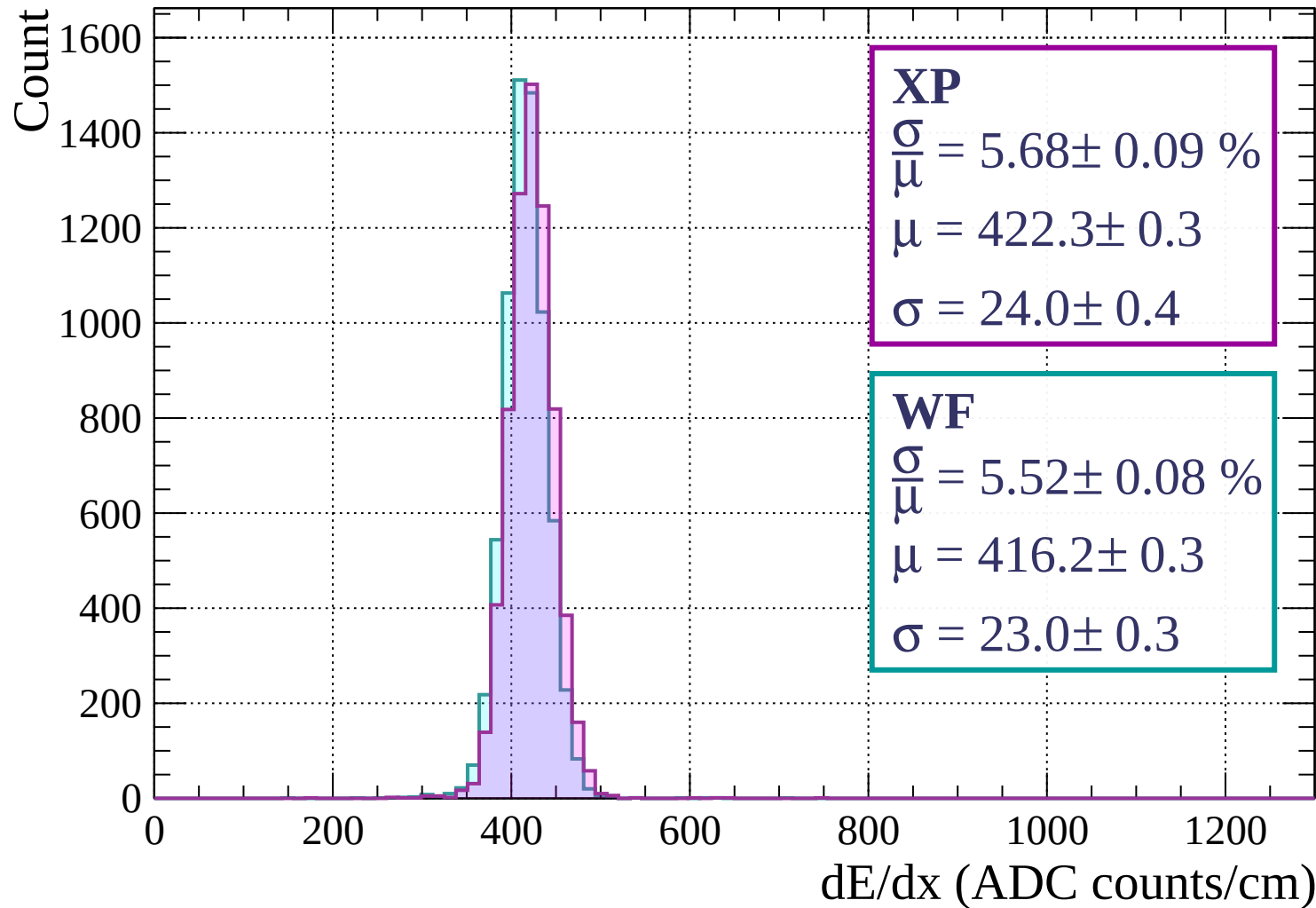
Energy loss | $320 < p < 400$



Energy loss | $400 < p < 480$

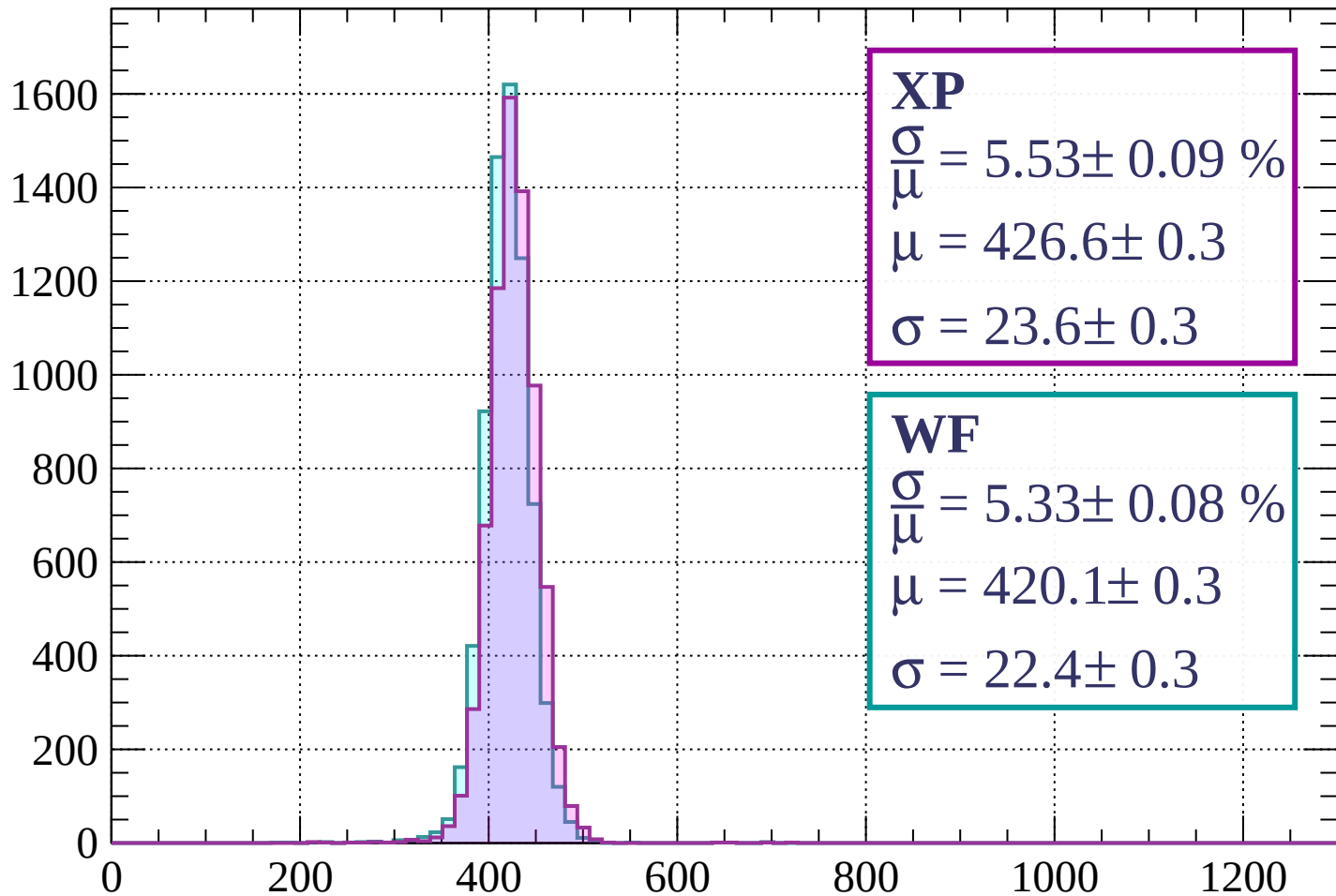


Energy loss | $480 < p < 560$



Energy loss | $560 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 5.53 \pm 0.09 \%$$

$$\mu = 426.6 \pm 0.3$$

$$\sigma = 23.6 \pm 0.3$$

WF

$$\frac{\sigma}{\mu} = 5.33 \pm 0.08 \%$$

$$\mu = 420.1 \pm 0.3$$

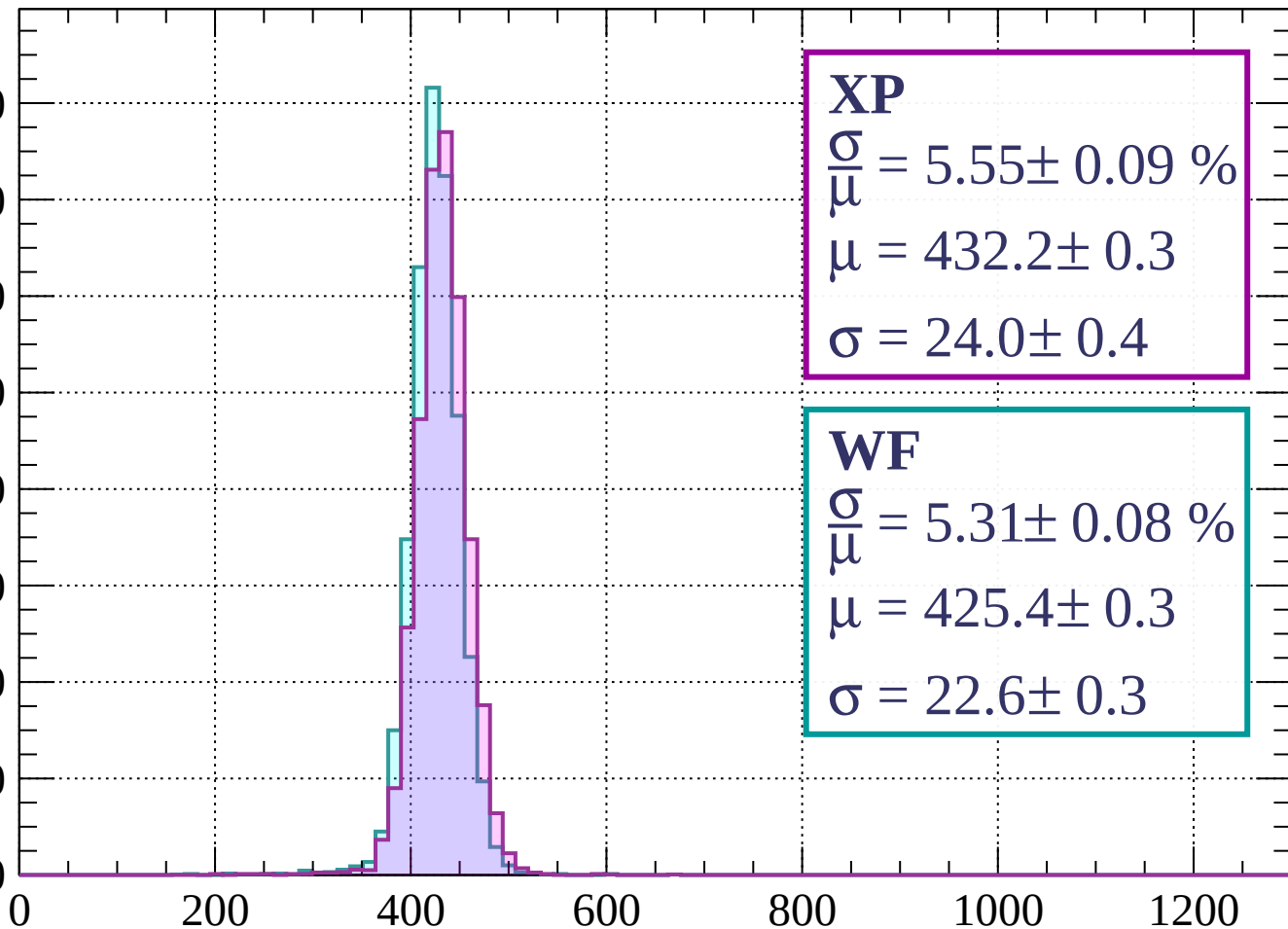
$$\sigma = 22.4 \pm 0.3$$

dE/dx (ADC counts/cm)

Energy loss | $640 < p < 720$

Count

1600
1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 5.55 \pm 0.09 \%$$

$$\mu = 432.2 \pm 0.3$$

$$\sigma = 24.0 \pm 0.4$$

WF

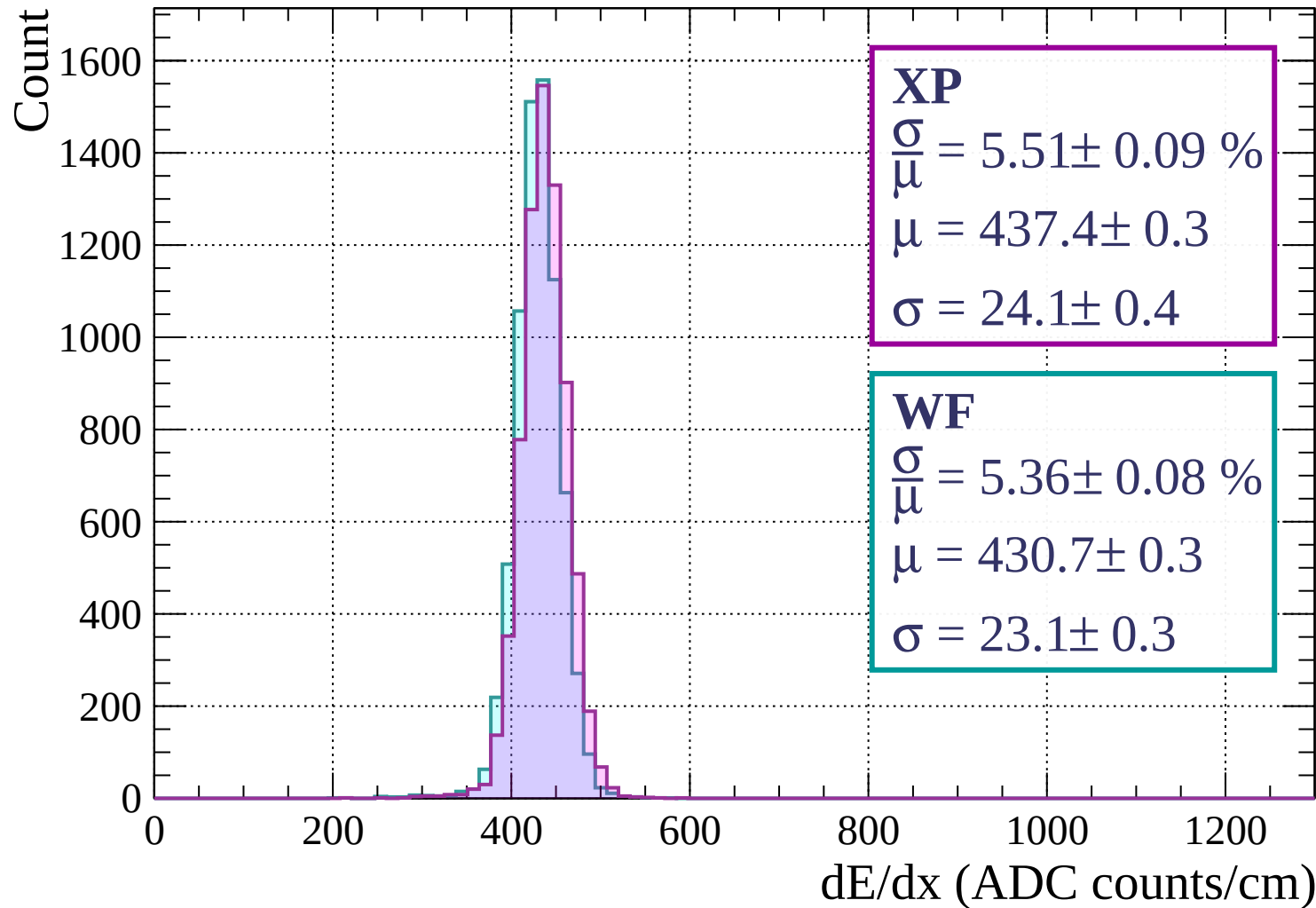
$$\frac{\sigma}{\mu} = 5.31 \pm 0.08 \%$$

$$\mu = 425.4 \pm 0.3$$

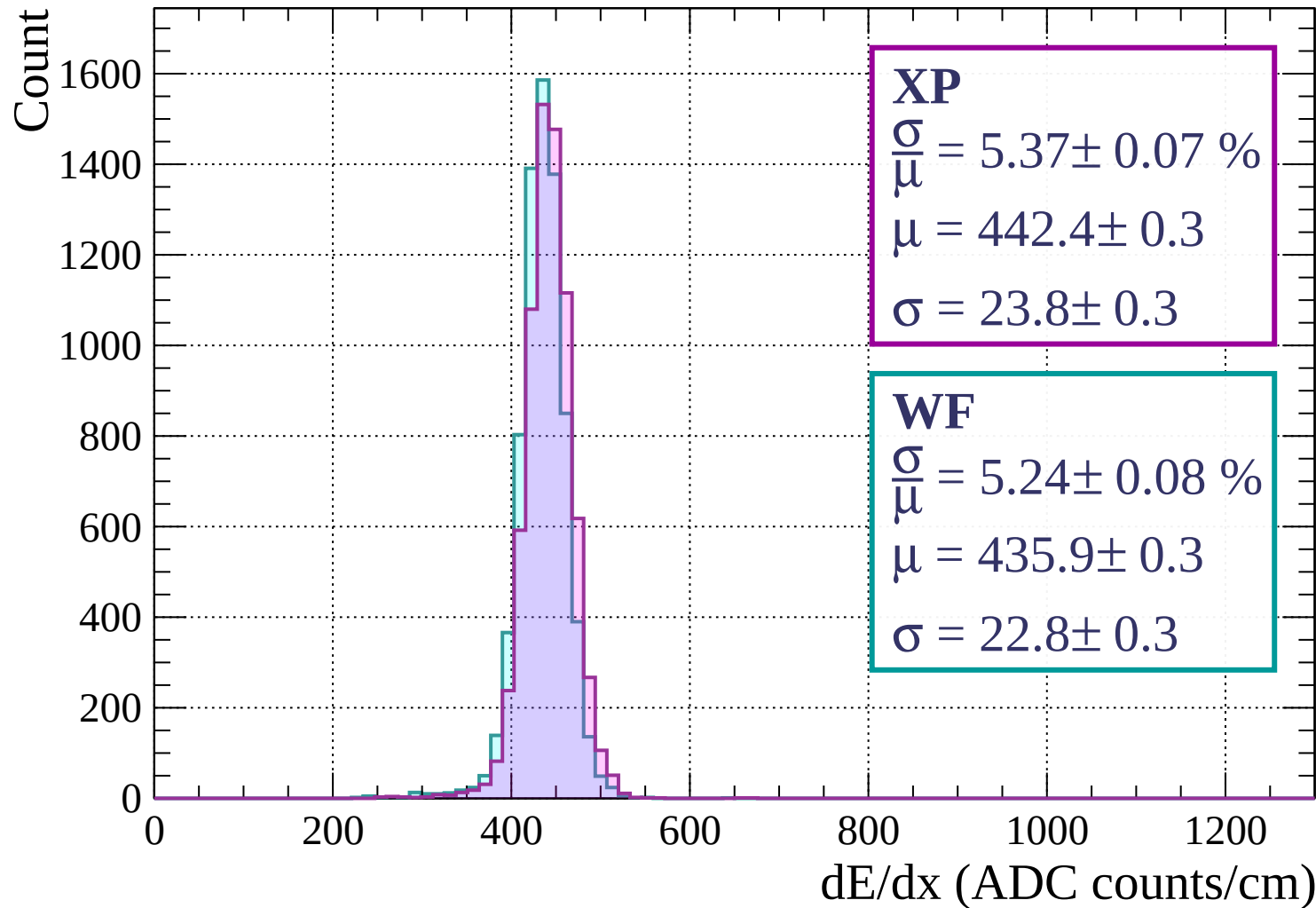
$$\sigma = 22.6 \pm 0.3$$

dE/dx (ADC counts/cm)

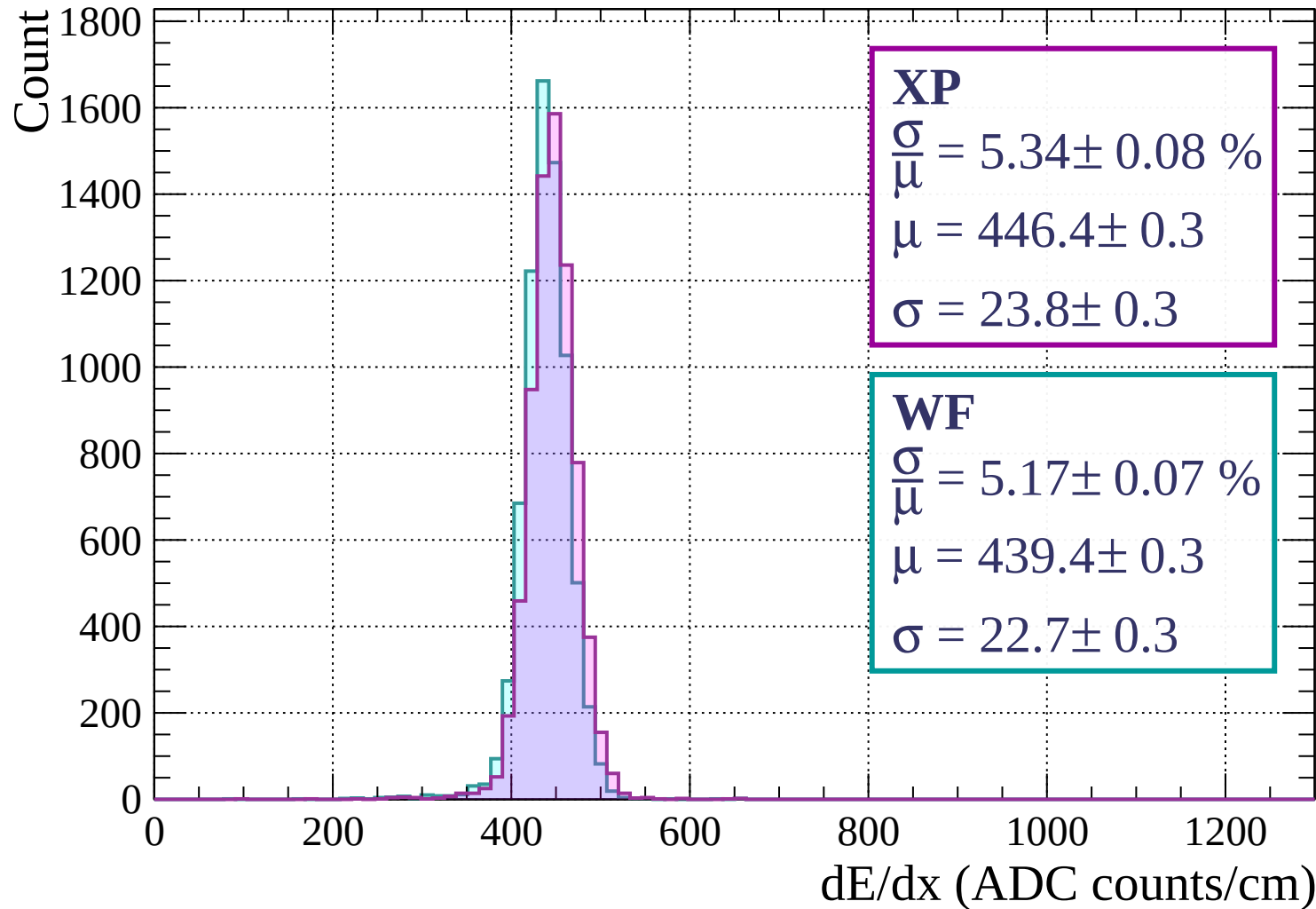
Energy loss | $720 < p < 800$



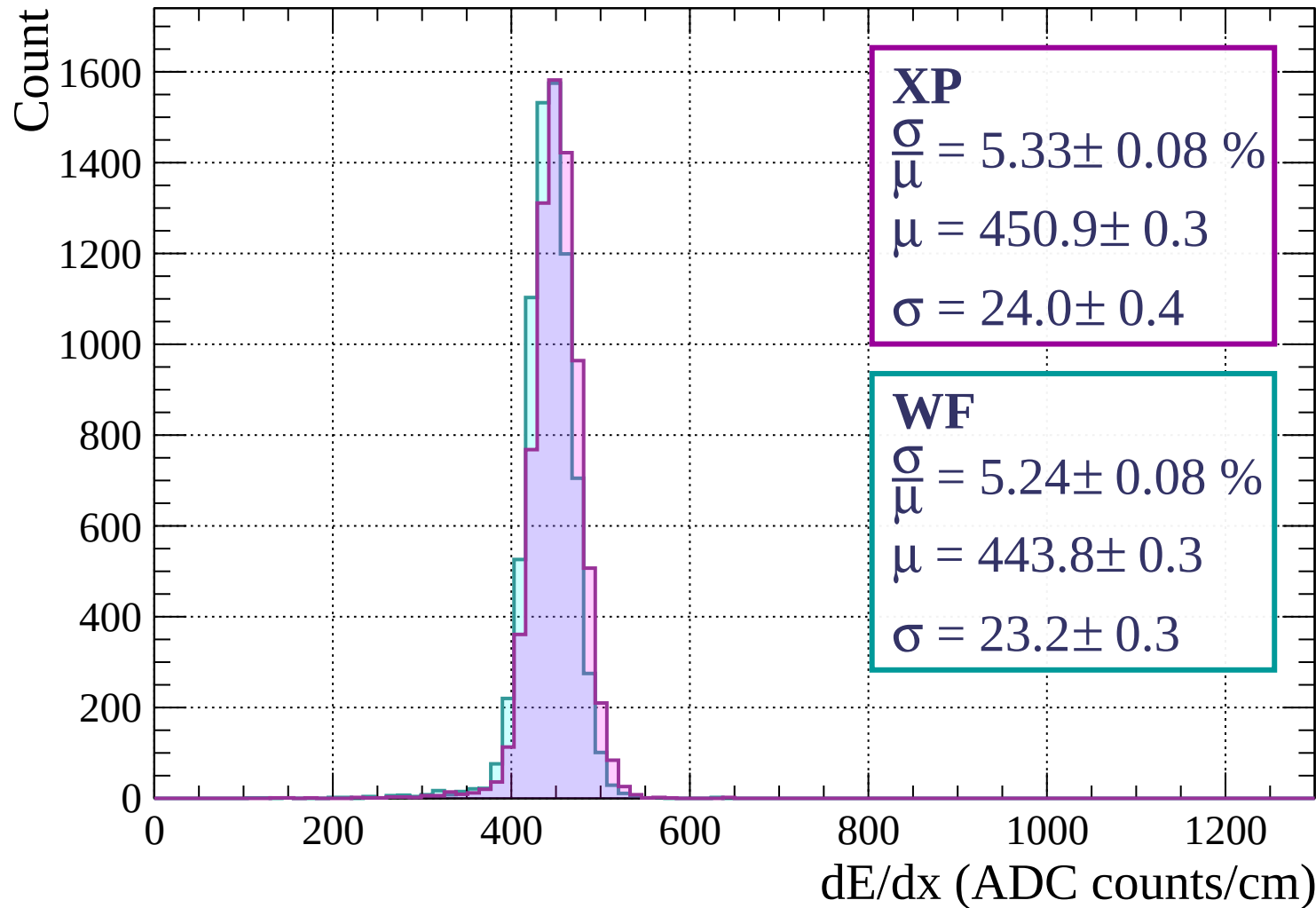
Energy loss | $800 < p < 880$



Energy loss | $880 < p < 960$



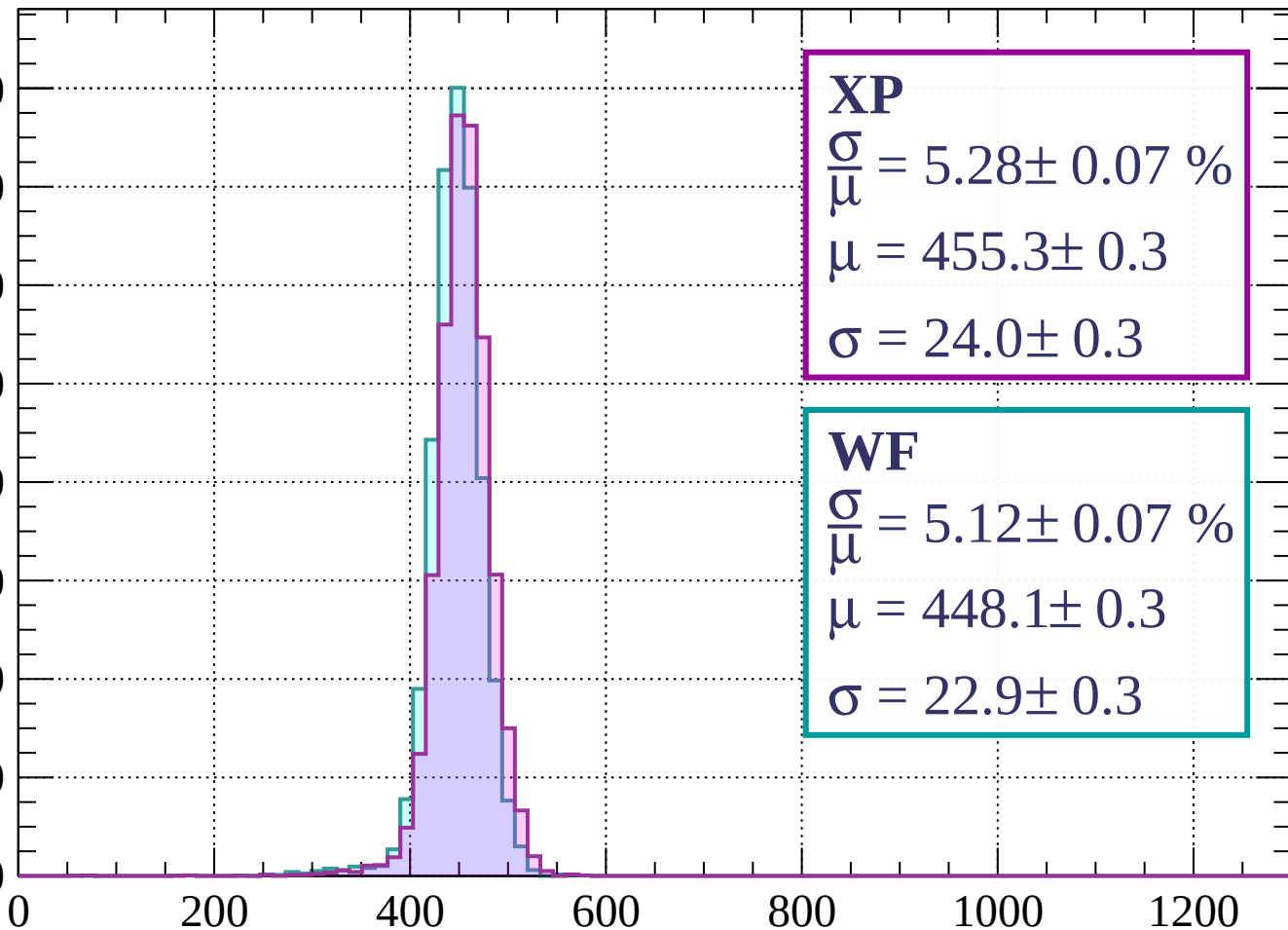
Energy loss | $960 < p < 1040$



Energy loss | $1040 < p < 1120$

Count

1600
1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 5.28 \pm 0.07 \%$$

$$\mu = 455.3 \pm 0.3$$

$$\sigma = 24.0 \pm 0.3$$

WF

$$\frac{\sigma}{\mu} = 5.12 \pm 0.07 \%$$

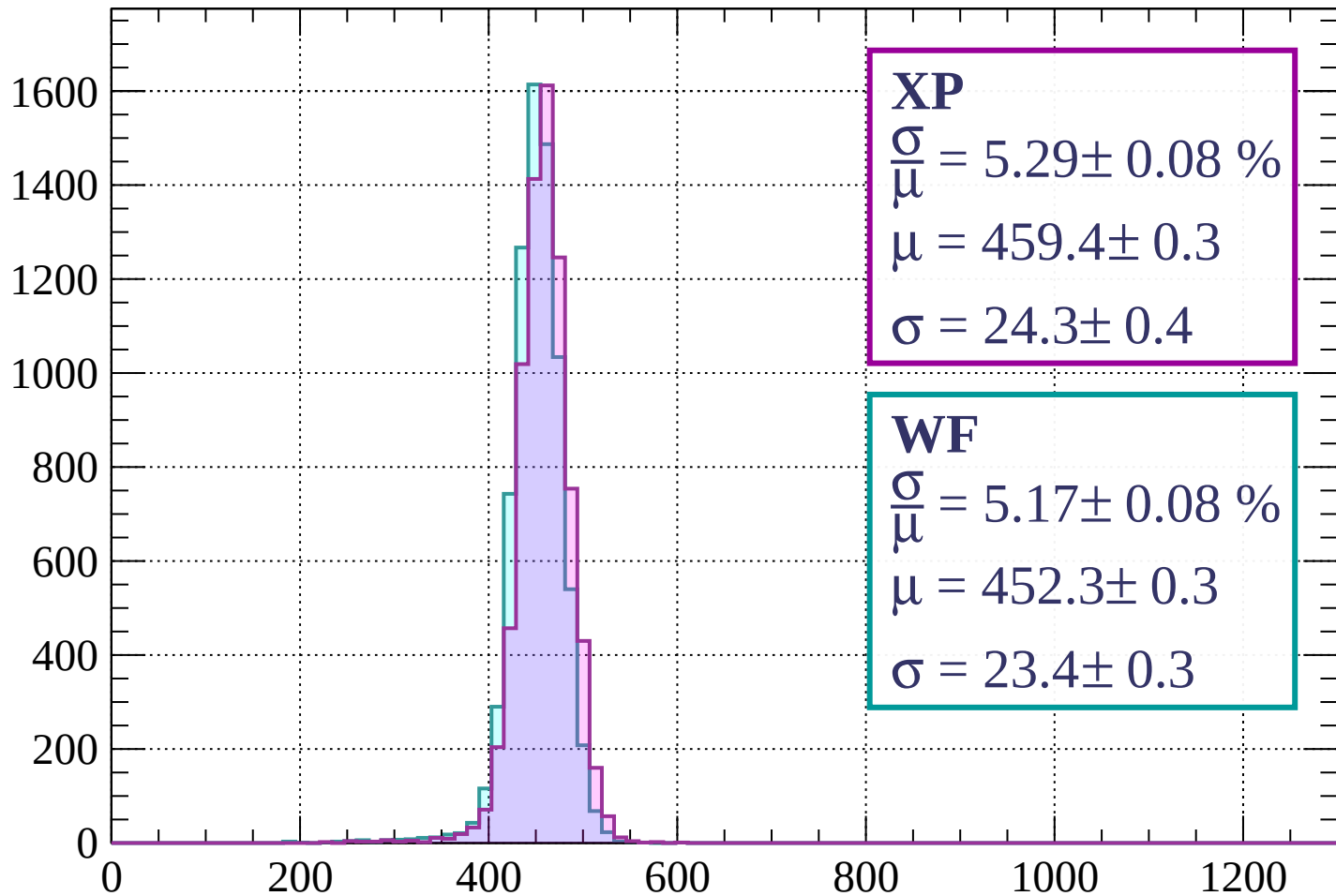
$$\mu = 448.1 \pm 0.3$$

$$\sigma = 22.9 \pm 0.3$$

dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 5.29 \pm 0.08 \%$$

$$\mu = 459.4 \pm 0.3$$

$$\sigma = 24.3 \pm 0.4$$

WF

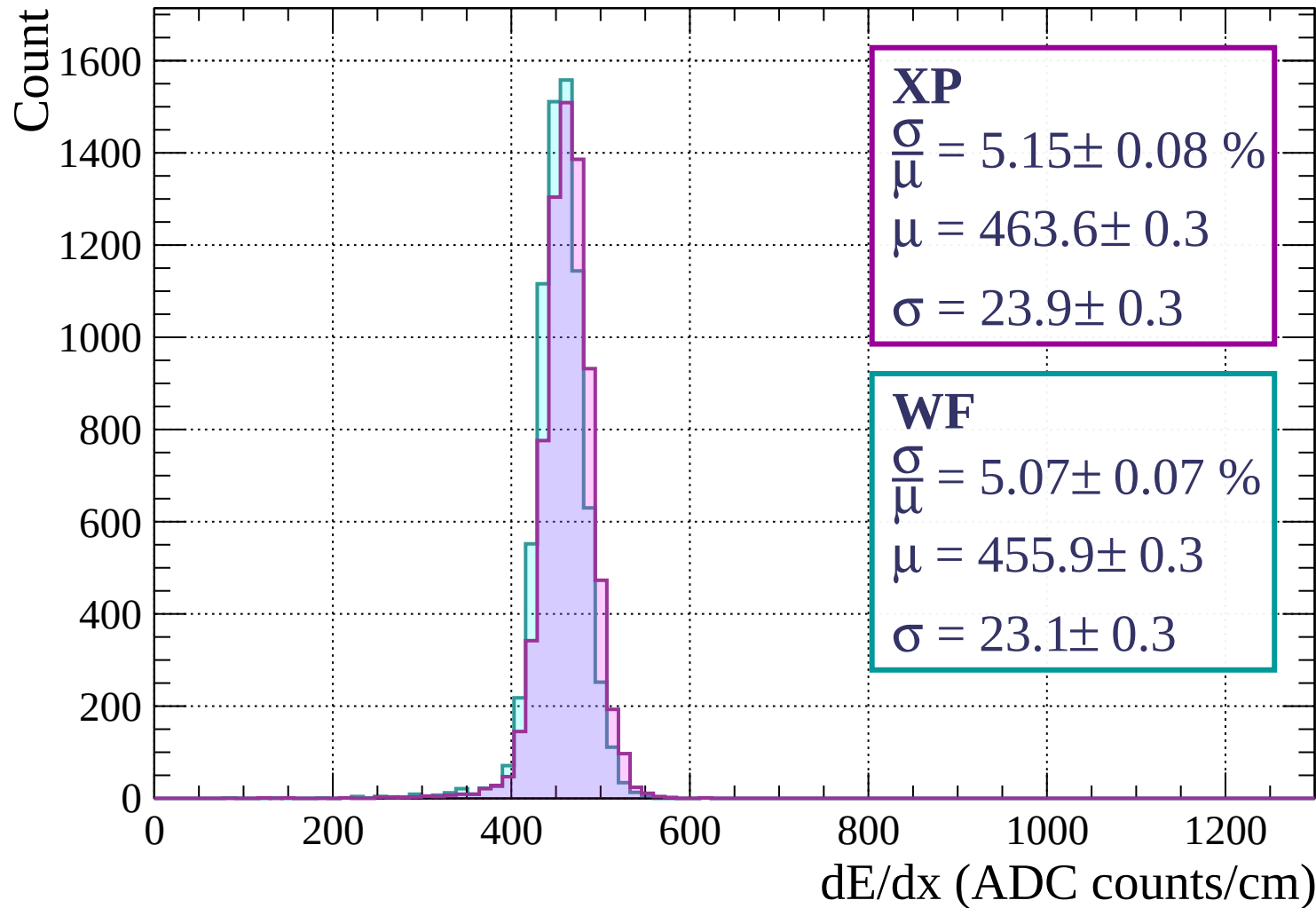
$$\frac{\sigma}{\mu} = 5.17 \pm 0.08 \%$$

$$\mu = 452.3 \pm 0.3$$

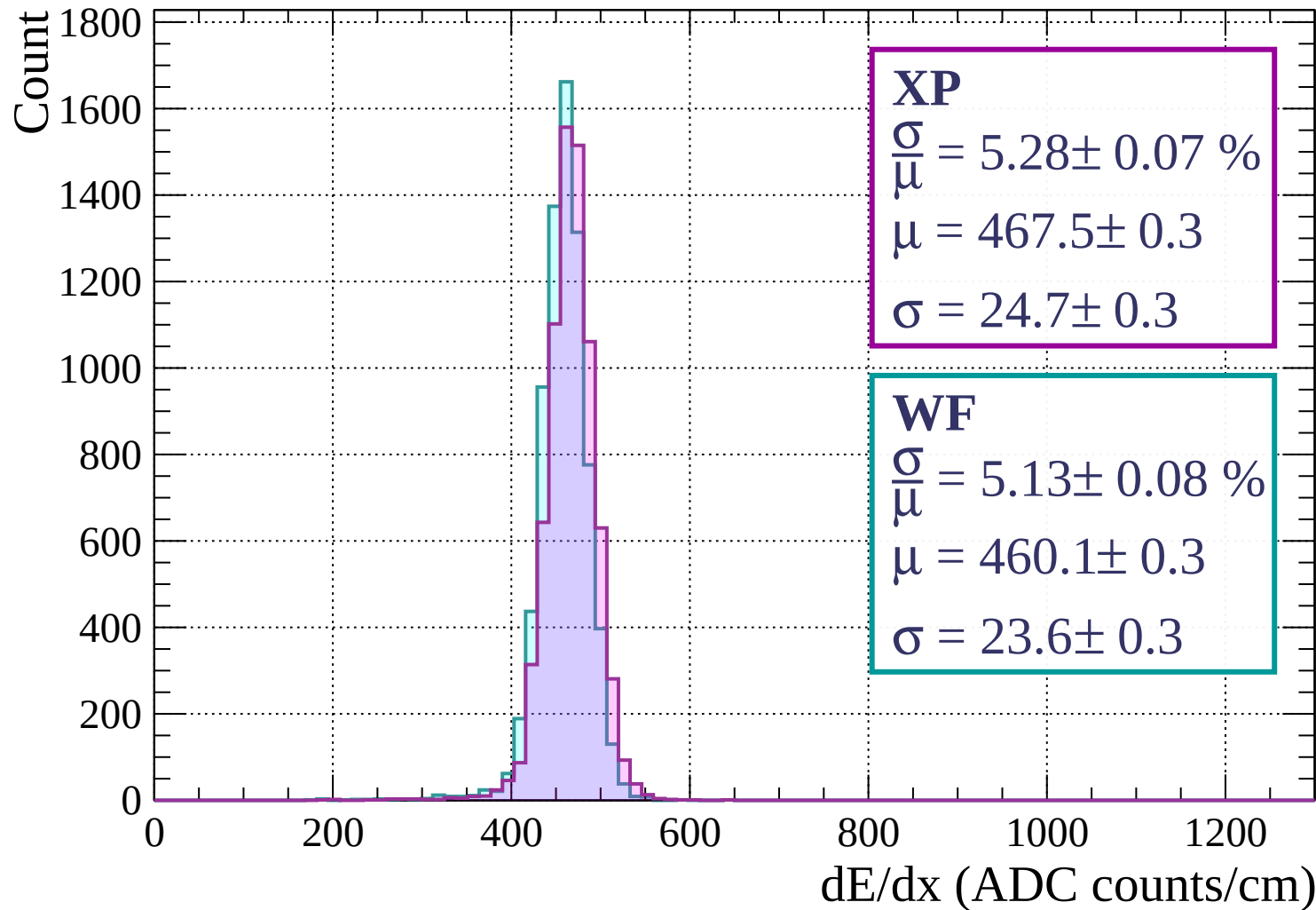
$$\sigma = 23.4 \pm 0.3$$

dE/dx (ADC counts/cm)

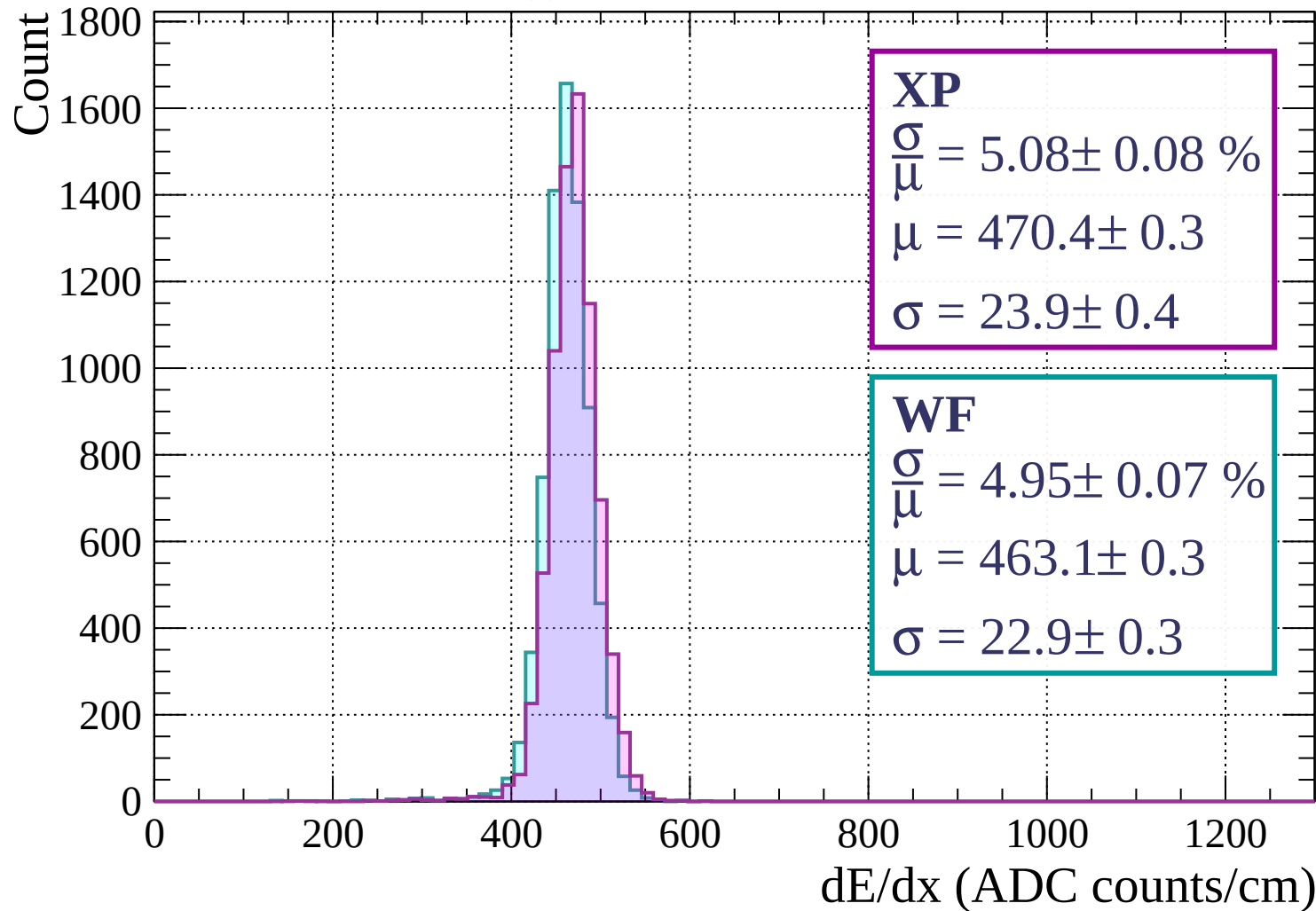
Energy loss | $1200 < p < 1280$



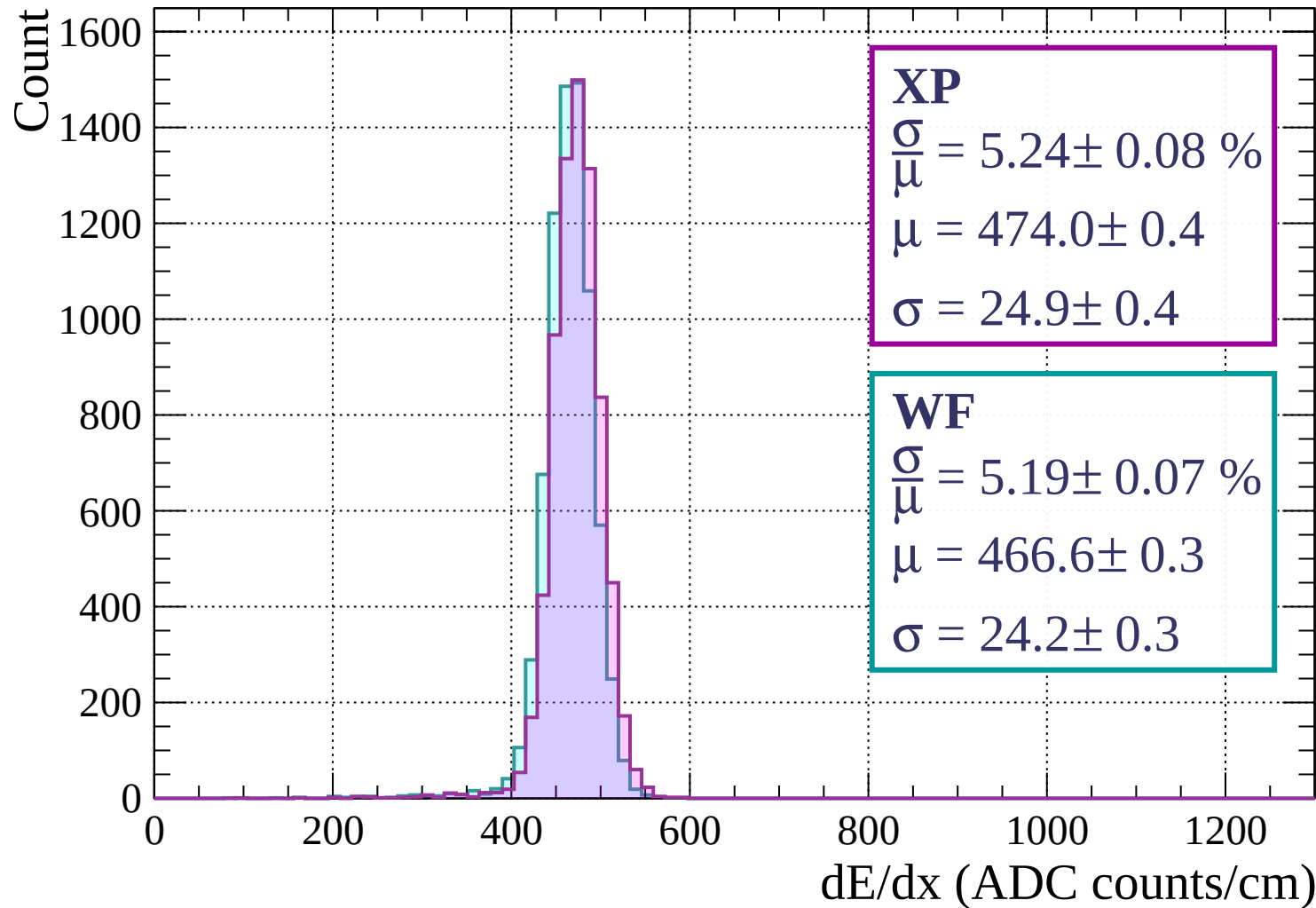
Energy loss | $1280 < p < 1360$



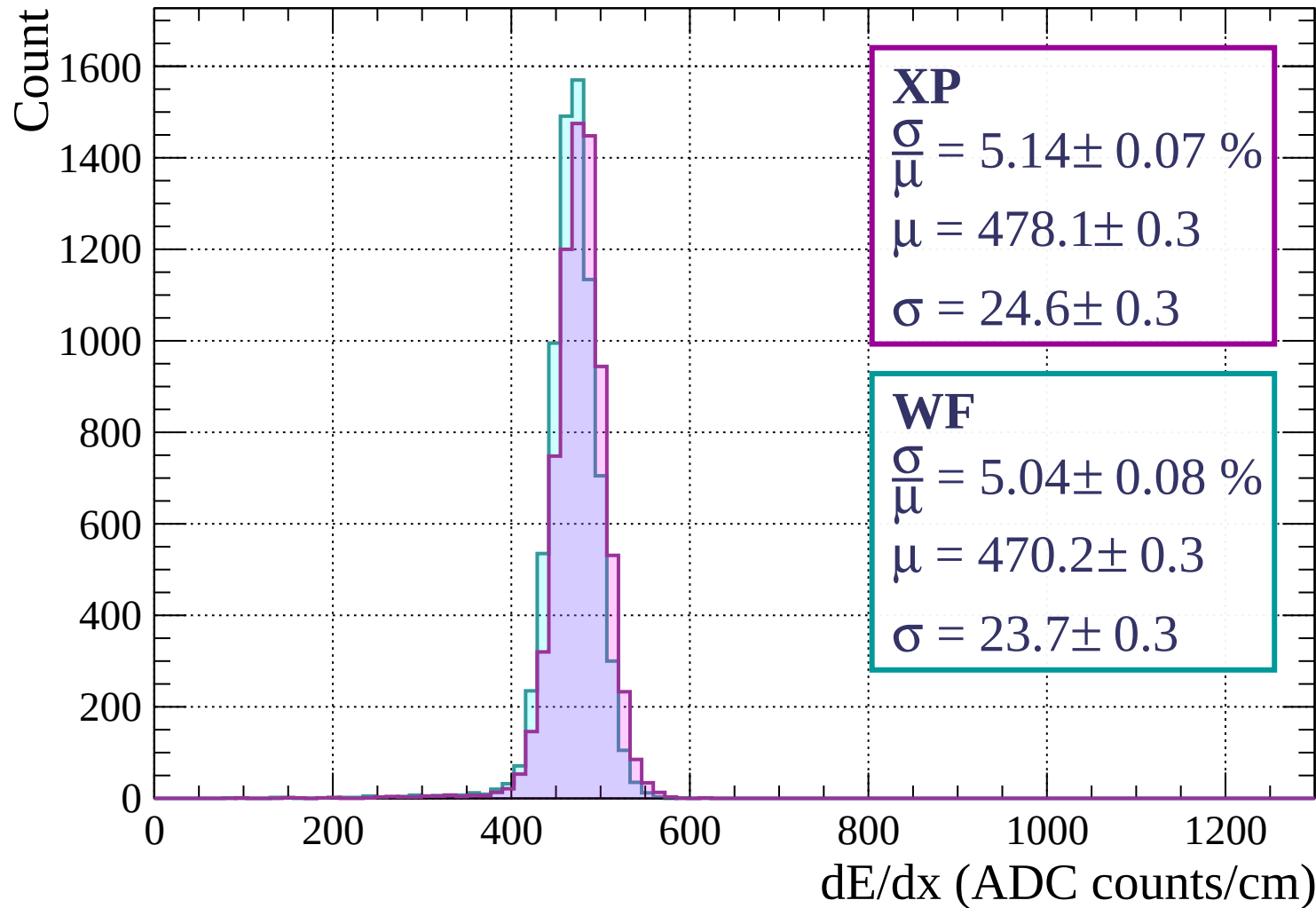
Energy loss | $1360 < p < 1440$



Energy loss | $1440 < p < 1520$

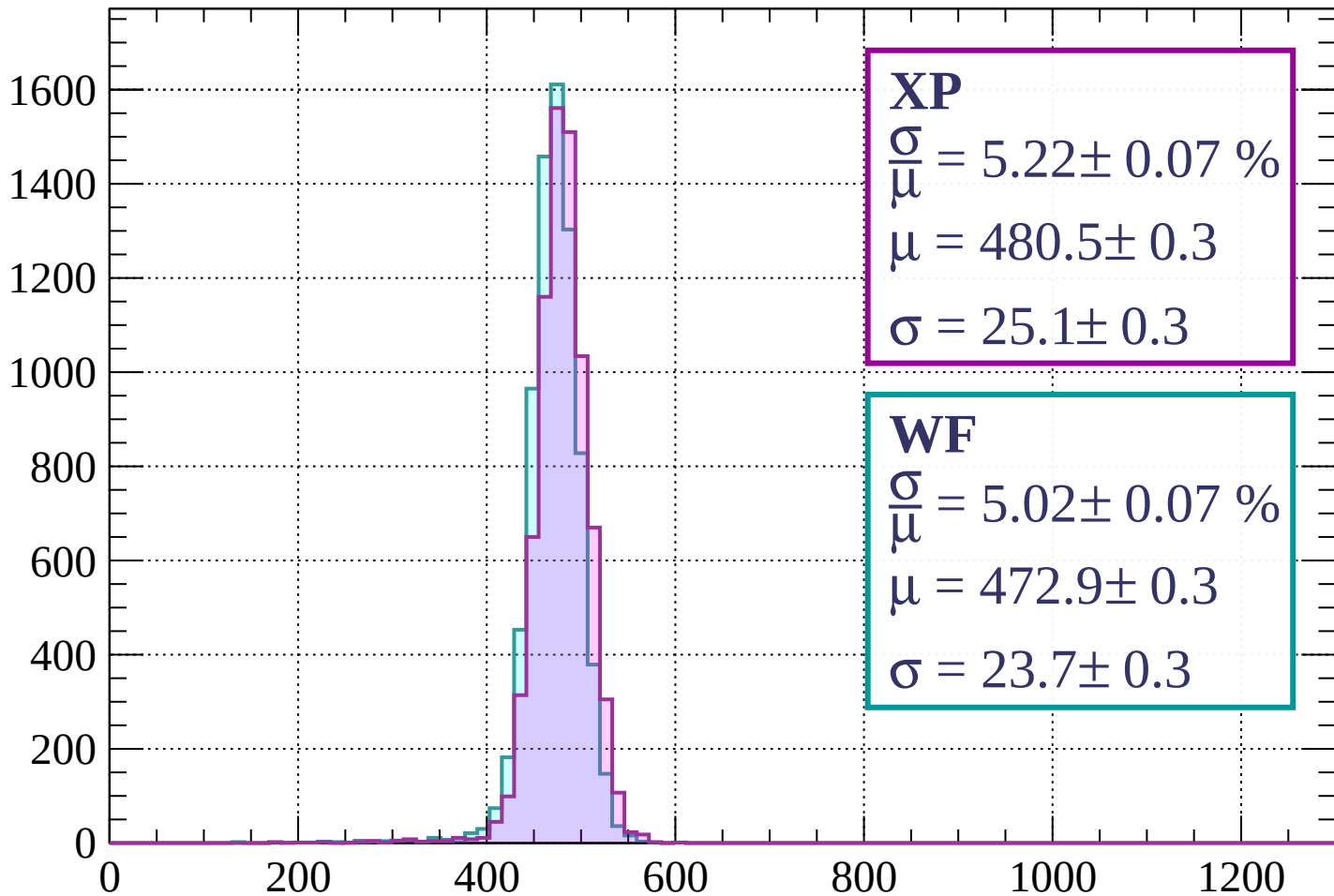


Energy loss | $1520 < p < 1600$



Energy loss | $1600 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 5.22 \pm 0.07 \%$$

$$\mu = 480.5 \pm 0.3$$

$$\sigma = 25.1 \pm 0.3$$

WF

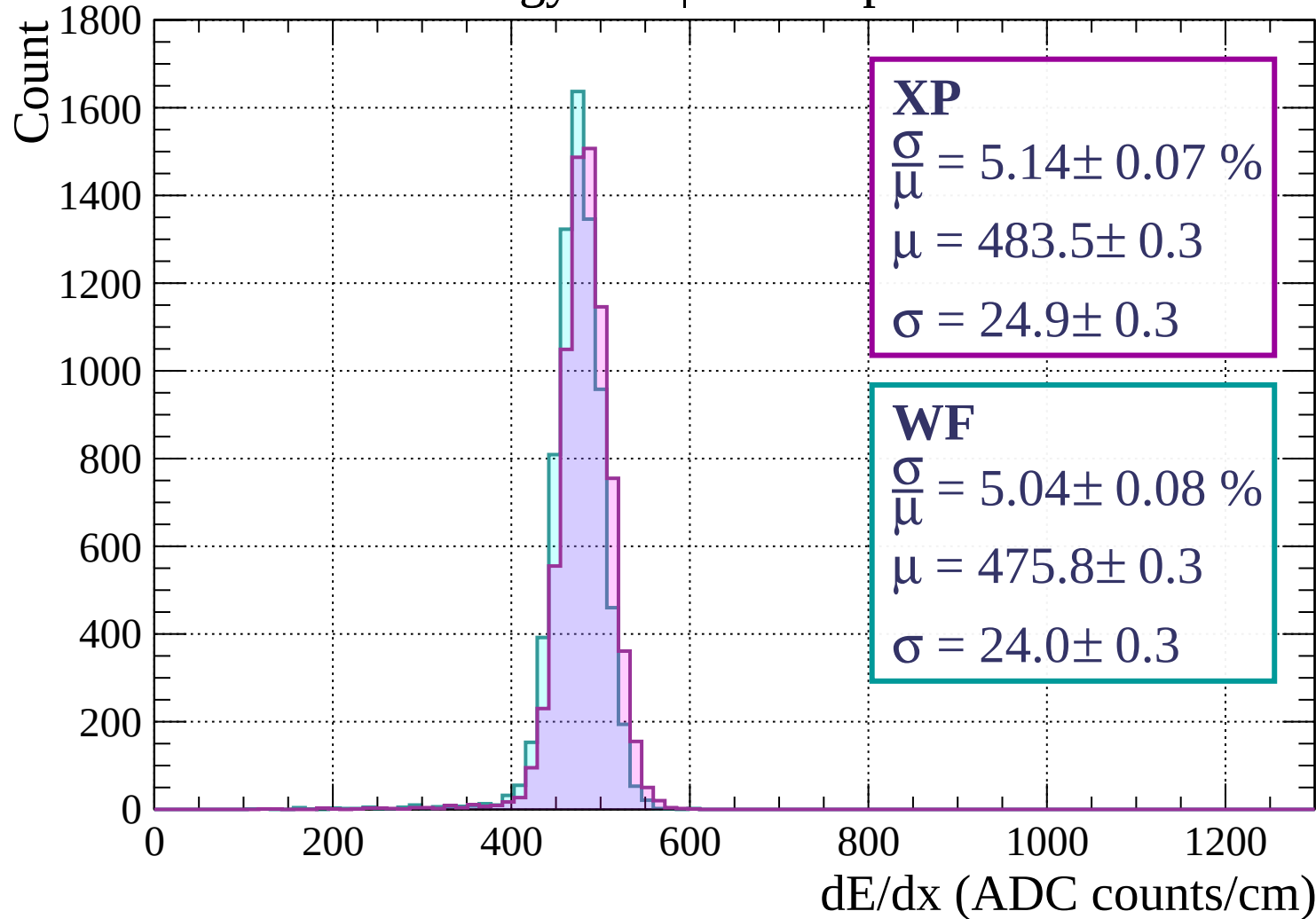
$$\frac{\sigma}{\mu} = 5.02 \pm 0.07 \%$$

$$\mu = 472.9 \pm 0.3$$

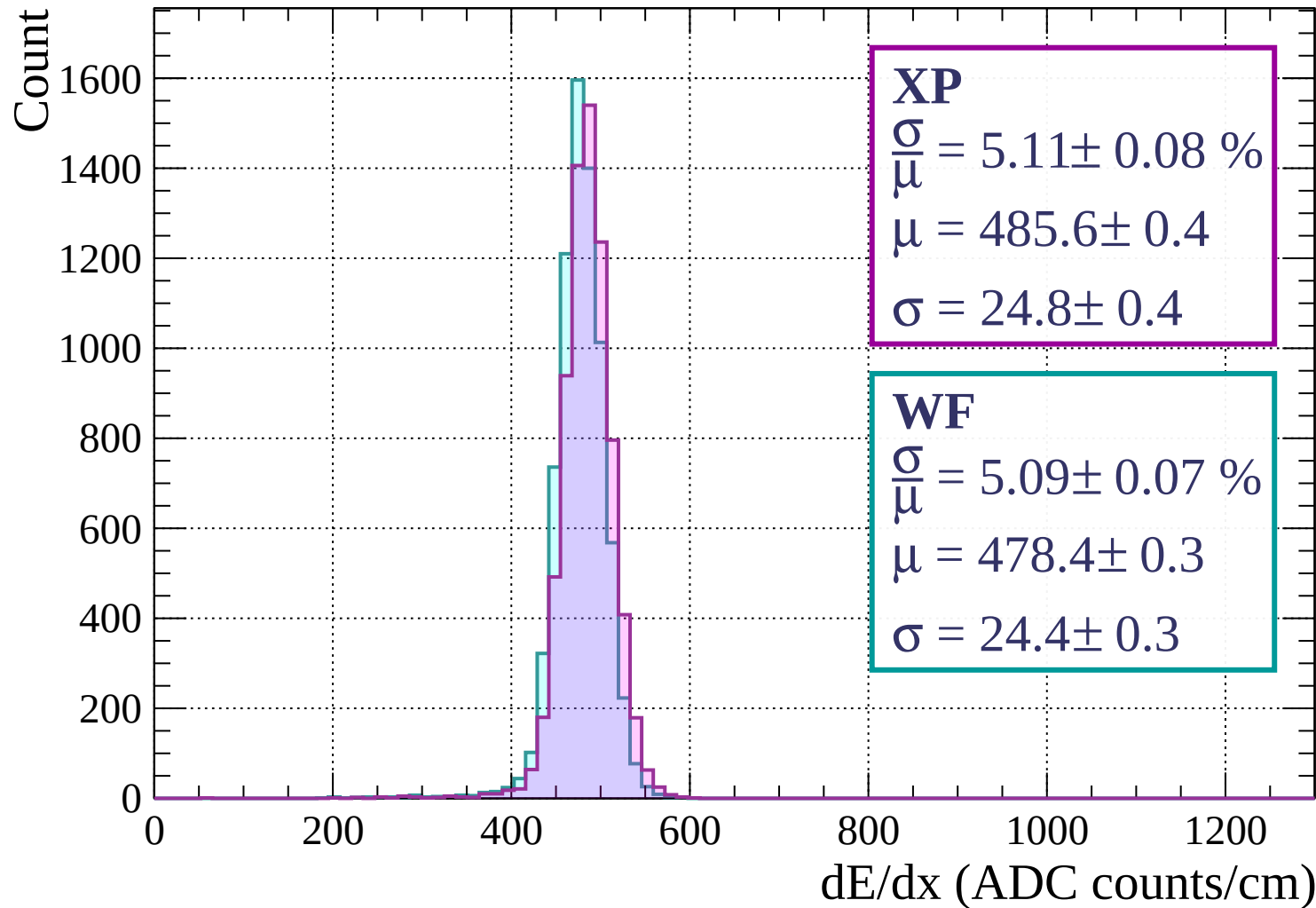
$$\sigma = 23.7 \pm 0.3$$

dE/dx (ADC counts/cm)

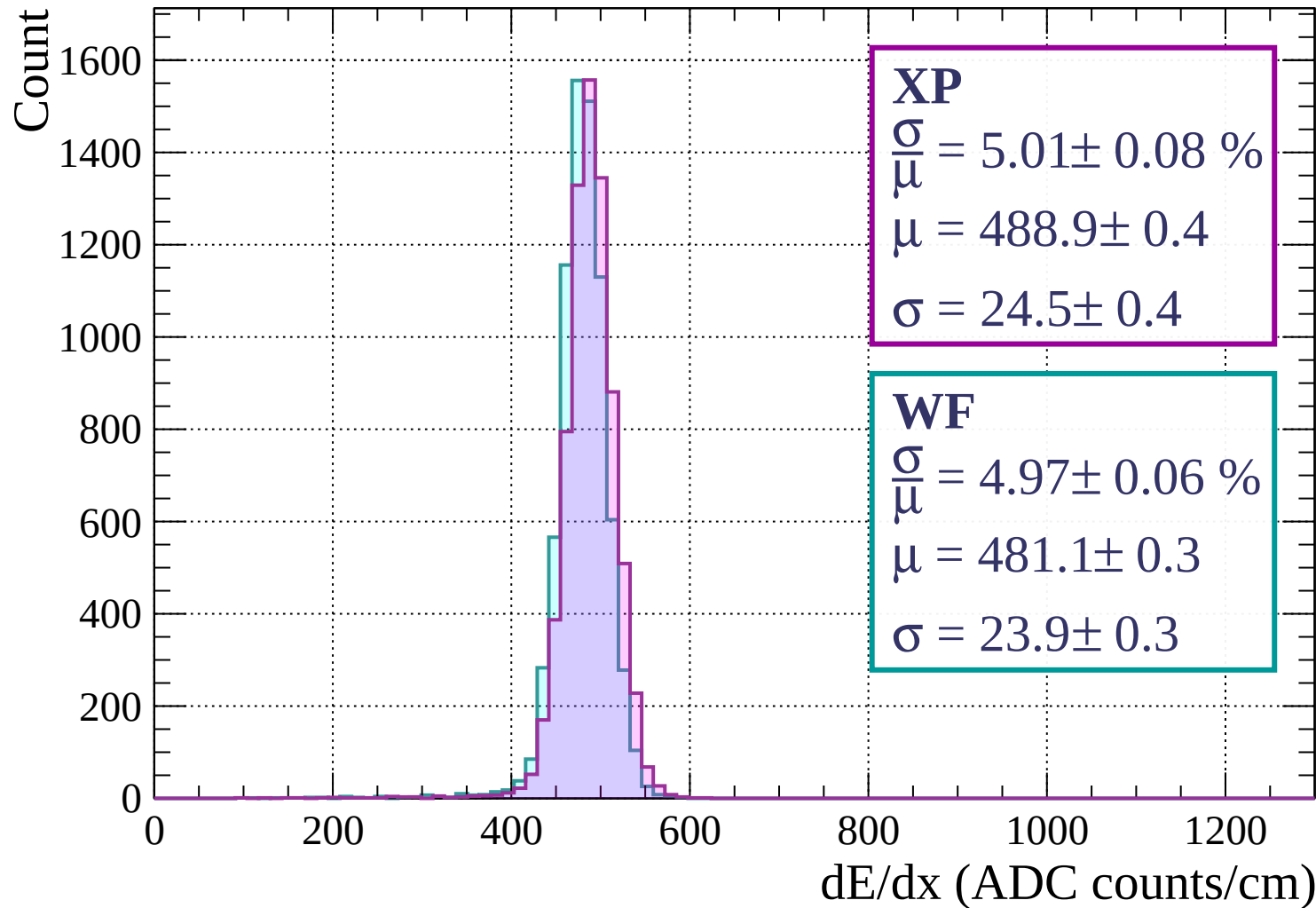
Energy loss | $1680 < p < 1760$



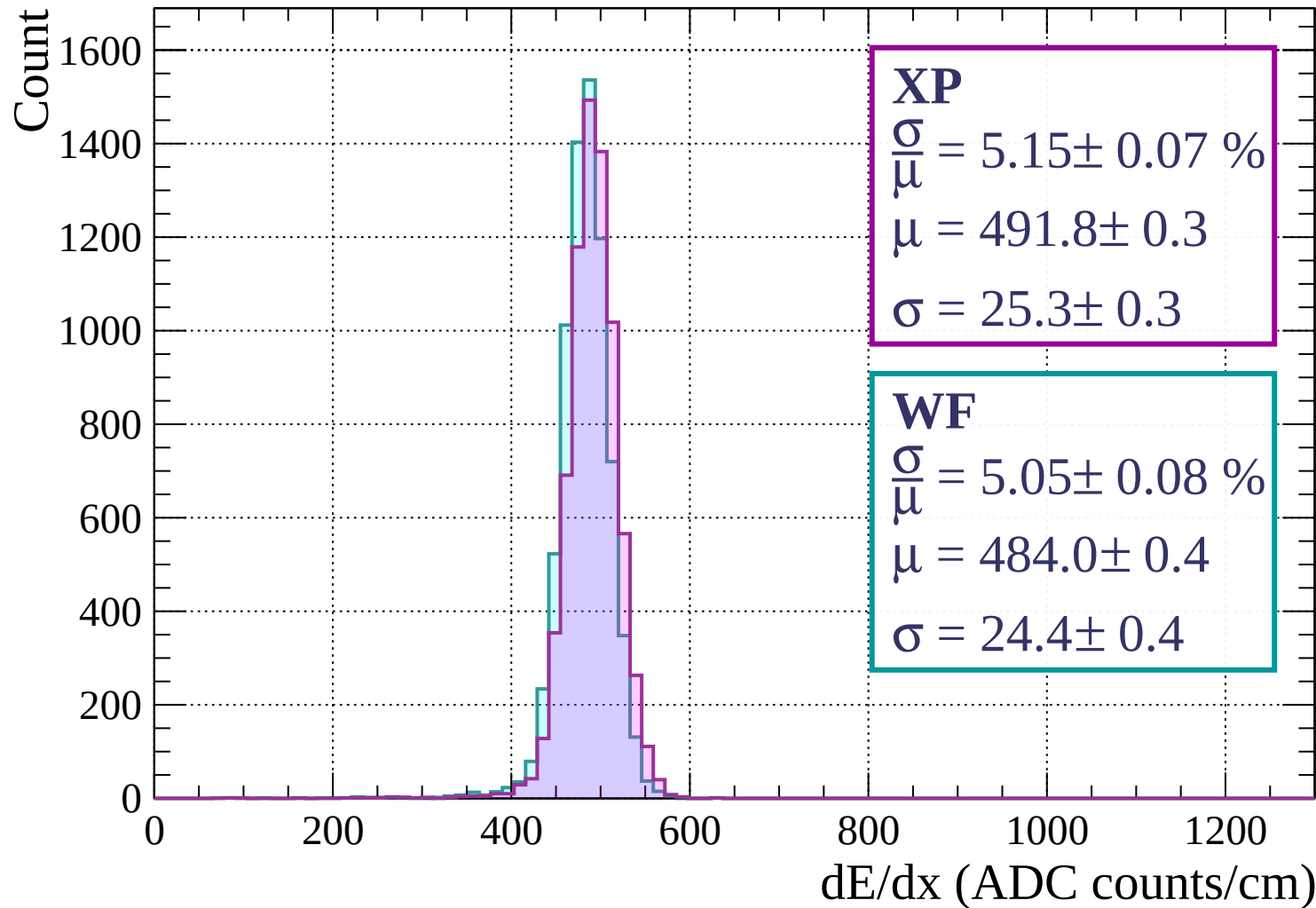
Energy loss | $1760 < p < 1840$



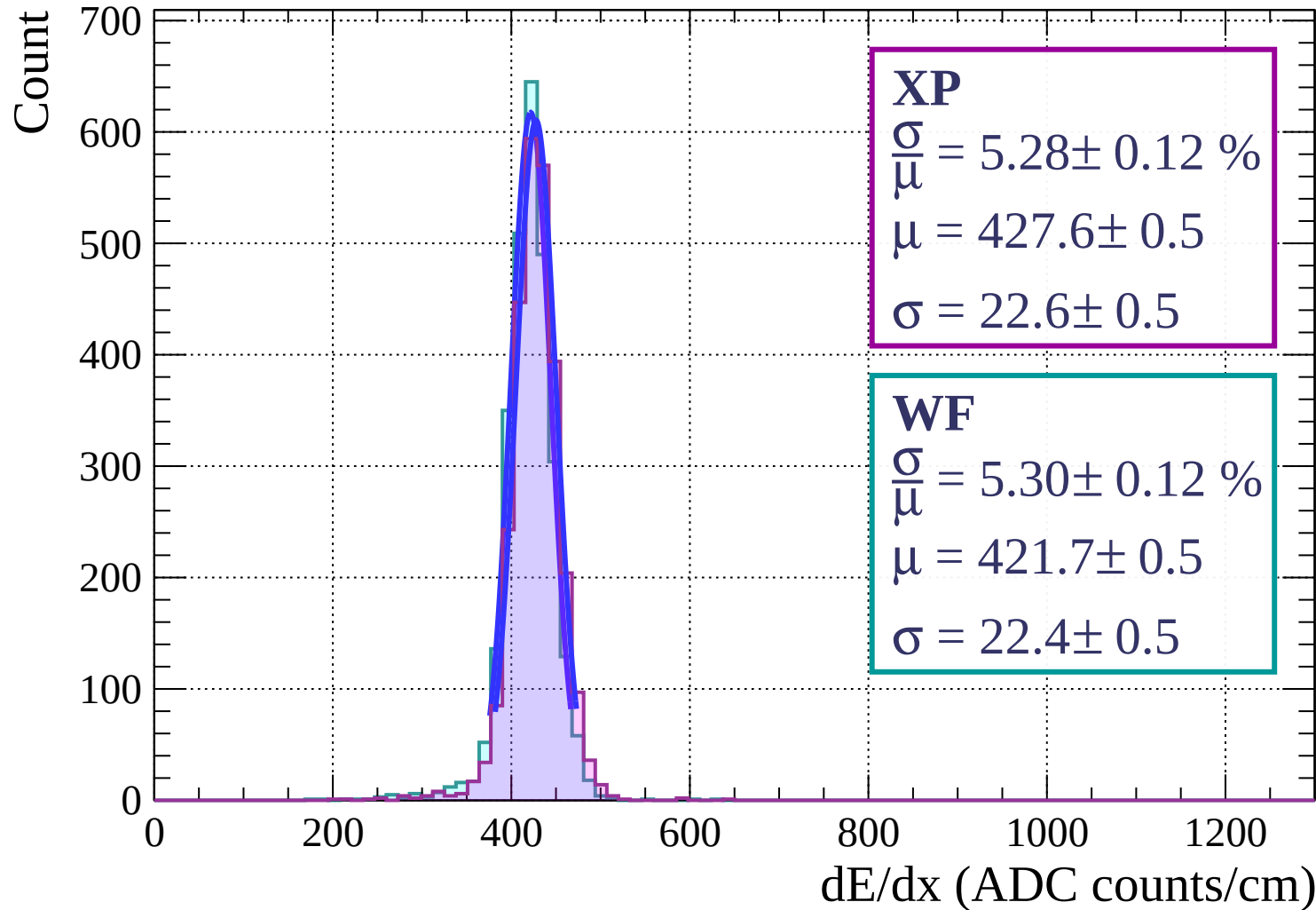
Energy loss | $1840 < p < 1920$



Energy loss | $1920 < p < 2000$

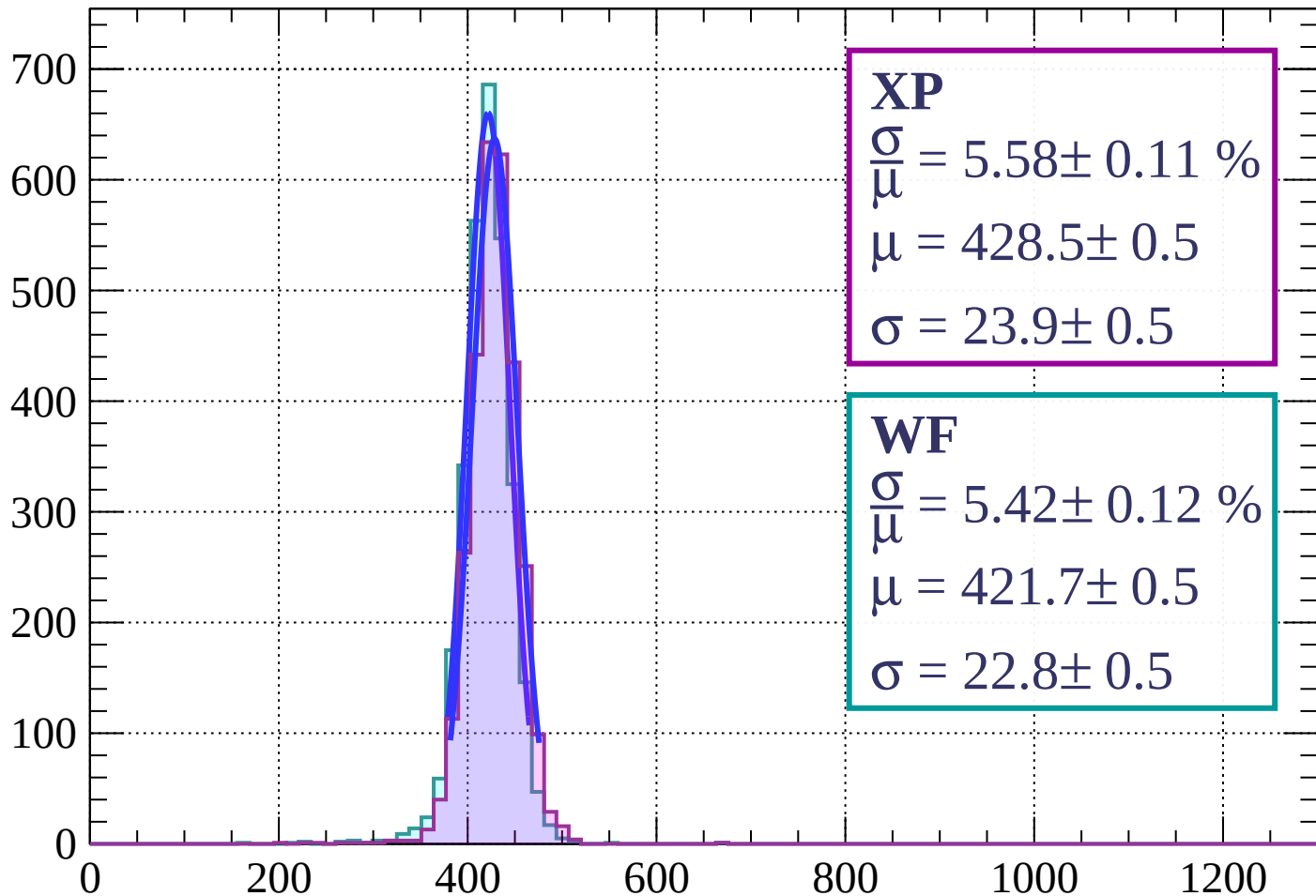


Energy loss | $0 < \phi < 3$



Energy loss | $3 < \phi < 6$

Count



XP

$$\frac{\sigma}{\mu} = 5.58 \pm 0.11 \%$$

$$\mu = 428.5 \pm 0.5$$

$$\sigma = 23.9 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.42 \pm 0.12 \%$$

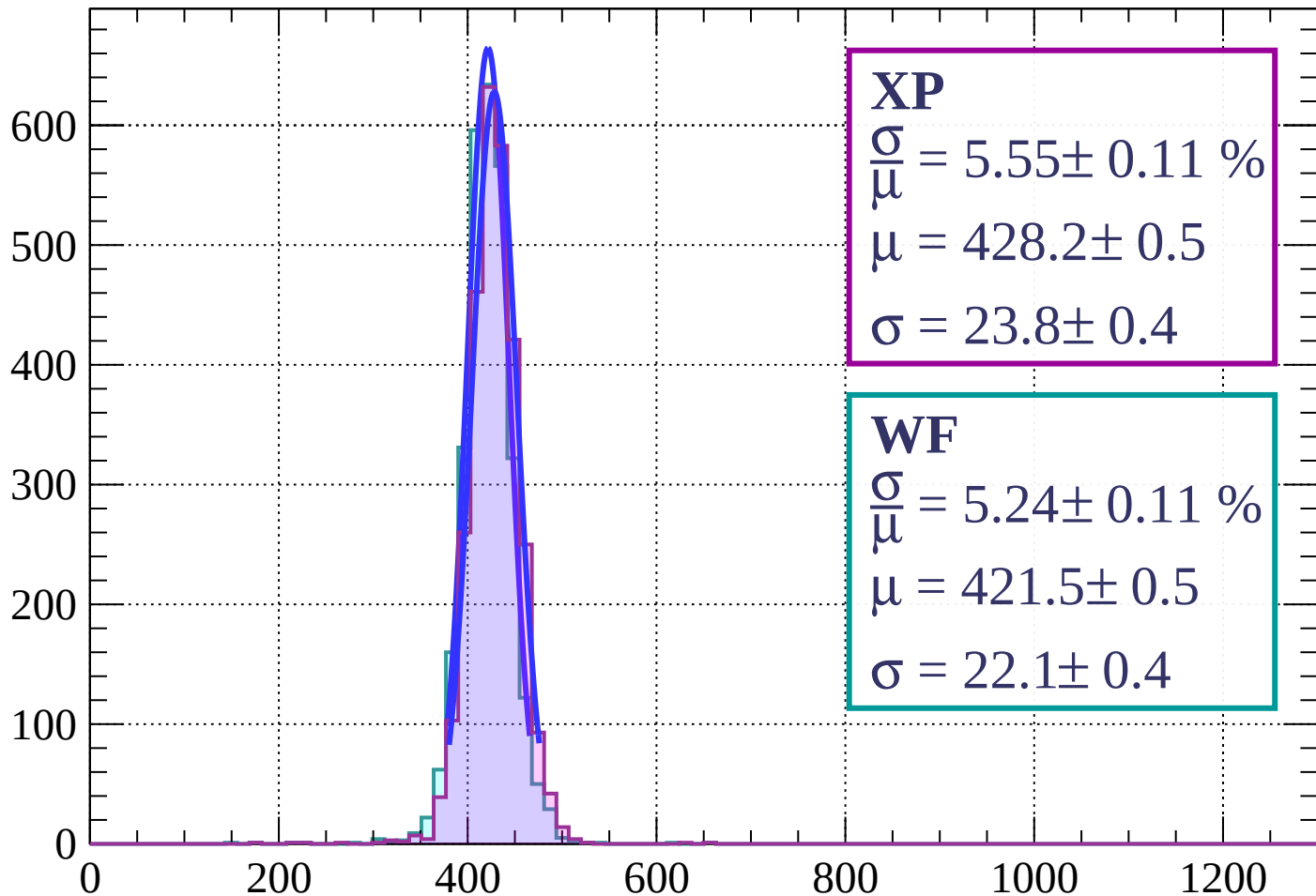
$$\mu = 421.7 \pm 0.5$$

$$\sigma = 22.8 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $6 < \phi < 9$

Count



XP

$$\frac{\sigma}{\mu} = 5.55 \pm 0.11 \%$$

$$\mu = 428.2 \pm 0.5$$

$$\sigma = 23.8 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 5.24 \pm 0.11 \%$$

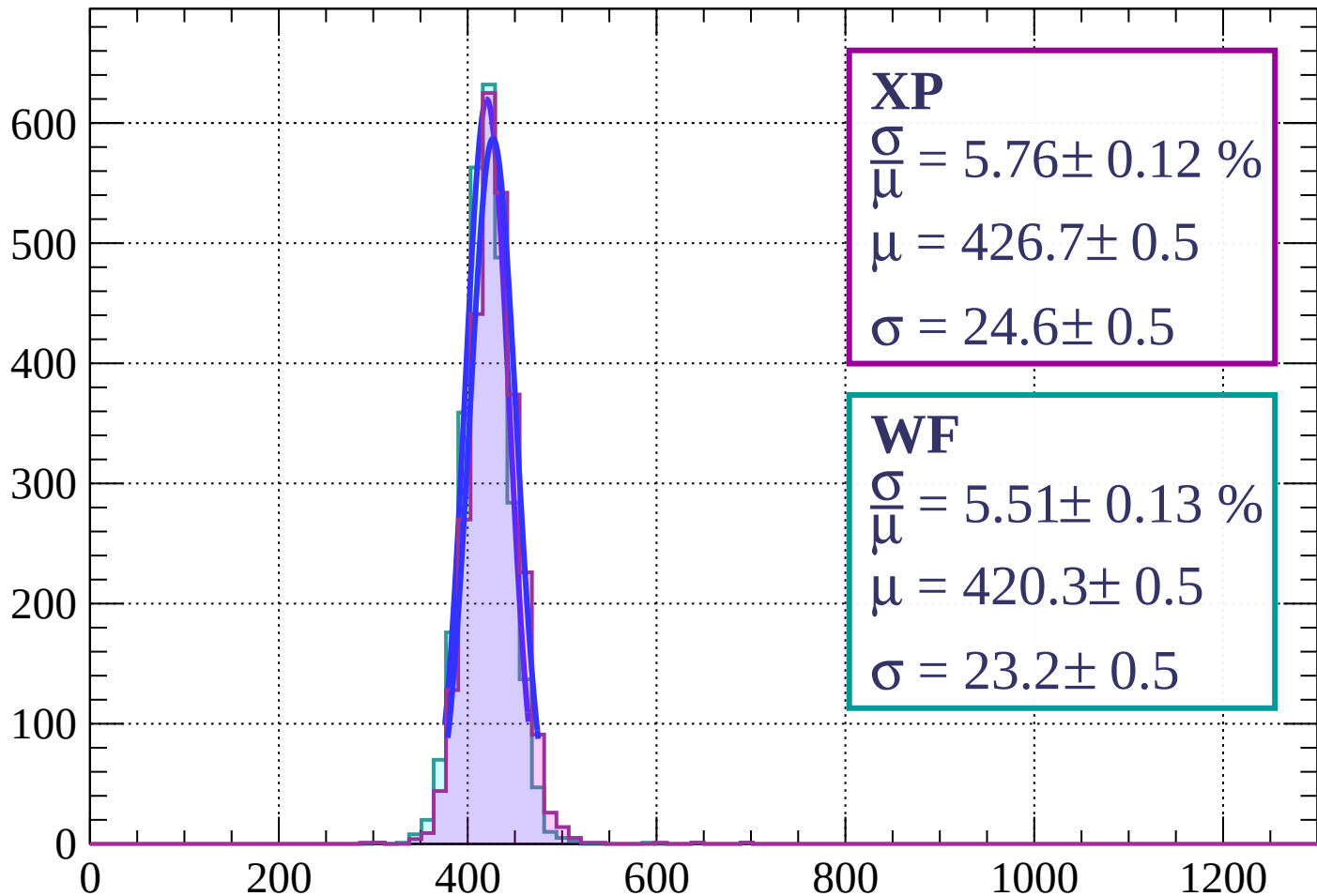
$$\mu = 421.5 \pm 0.5$$

$$\sigma = 22.1 \pm 0.4$$

dE/dx (ADC counts/cm)

Energy loss | $9 < \phi < 12$

Count



XP

$$\frac{\sigma}{\mu} = 5.76 \pm 0.12 \%$$

$$\mu = 426.7 \pm 0.5$$

$$\sigma = 24.6 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.51 \pm 0.13 \%$$

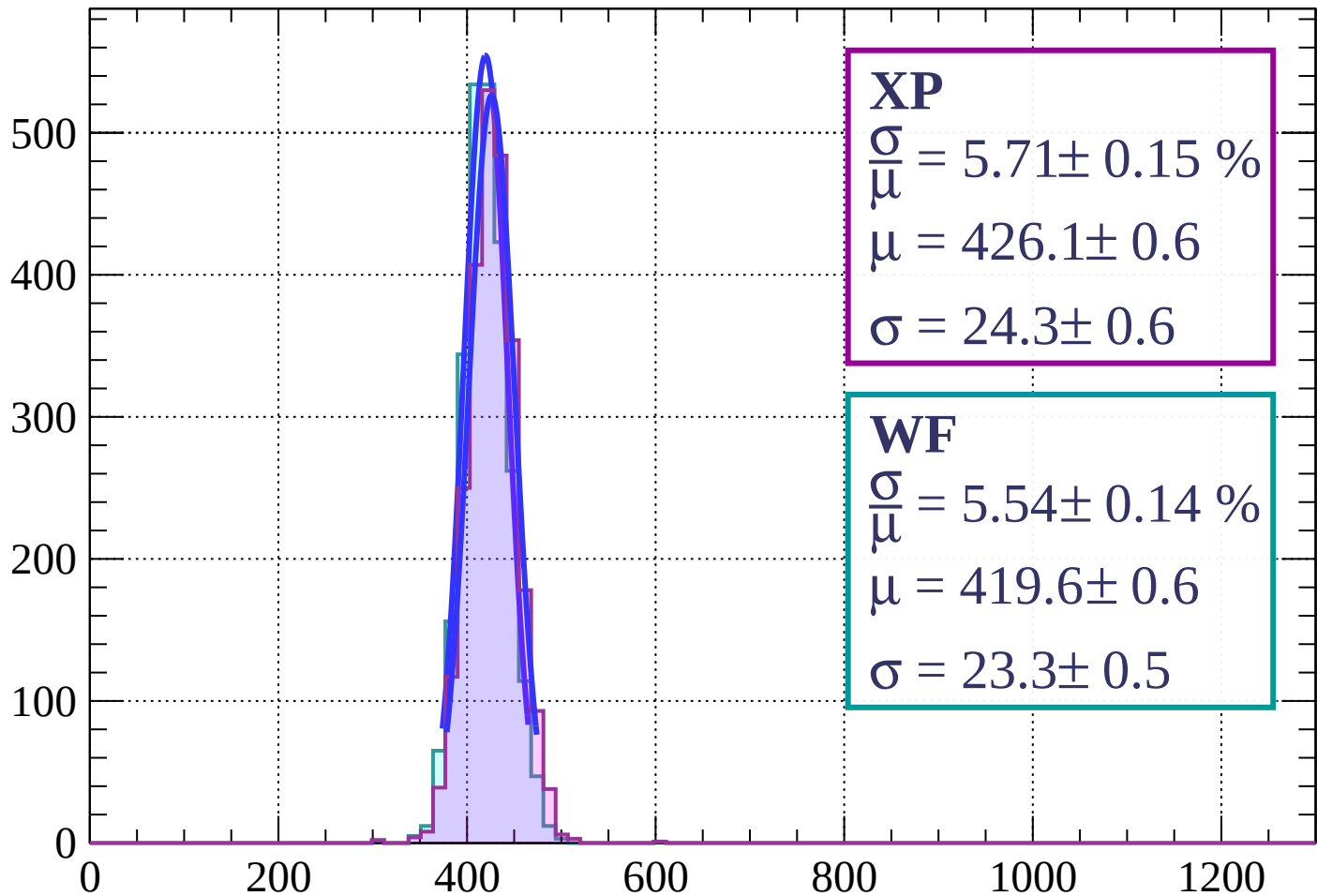
$$\mu = 420.3 \pm 0.5$$

$$\sigma = 23.2 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $12 < \phi < 15$

Count



XP

$$\frac{\sigma}{\mu} = 5.71 \pm 0.15 \%$$

$$\mu = 426.1 \pm 0.6$$

$$\sigma = 24.3 \pm 0.6$$

WF

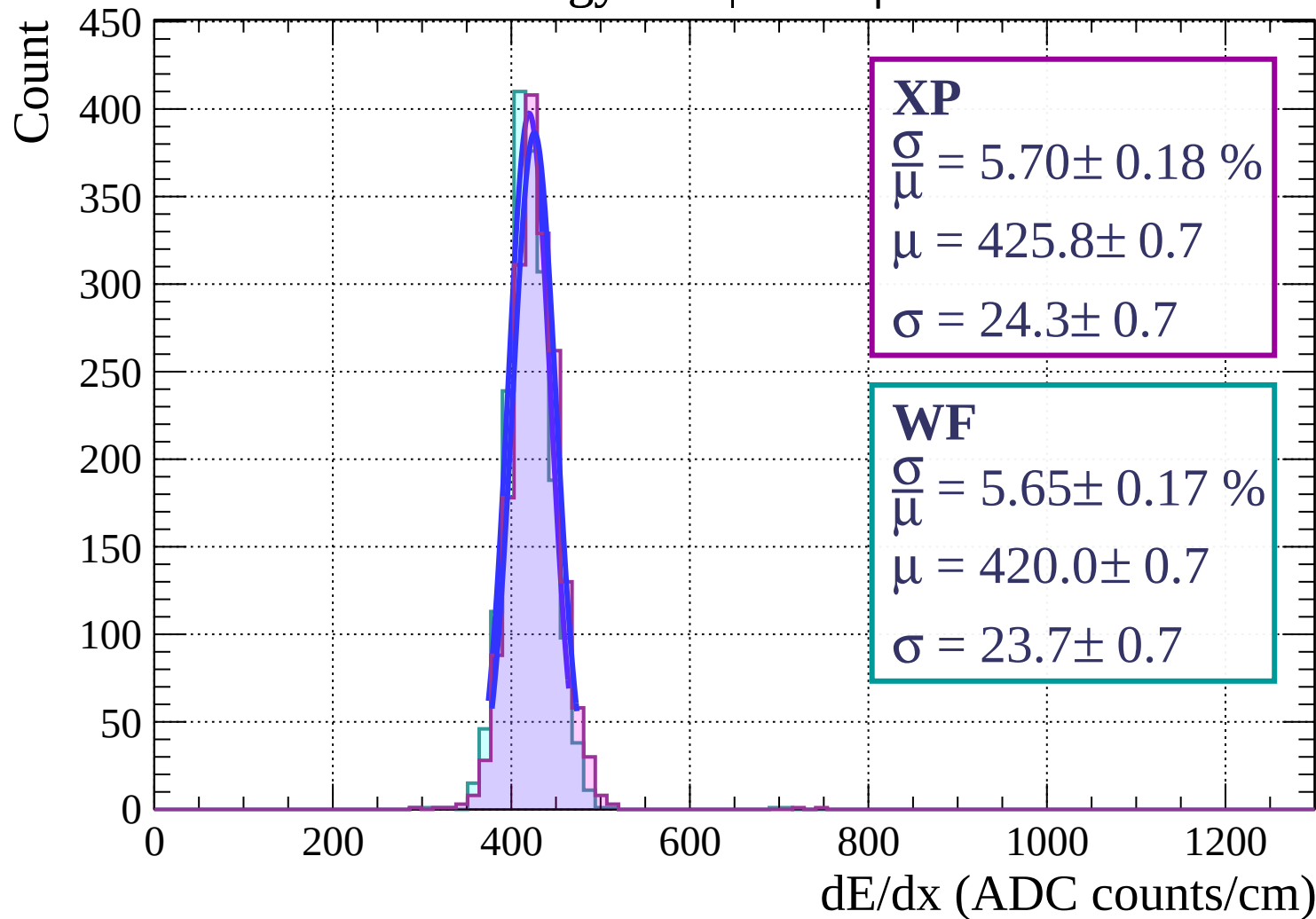
$$\frac{\sigma}{\mu} = 5.54 \pm 0.14 \%$$

$$\mu = 419.6 \pm 0.6$$

$$\sigma = 23.3 \pm 0.5$$

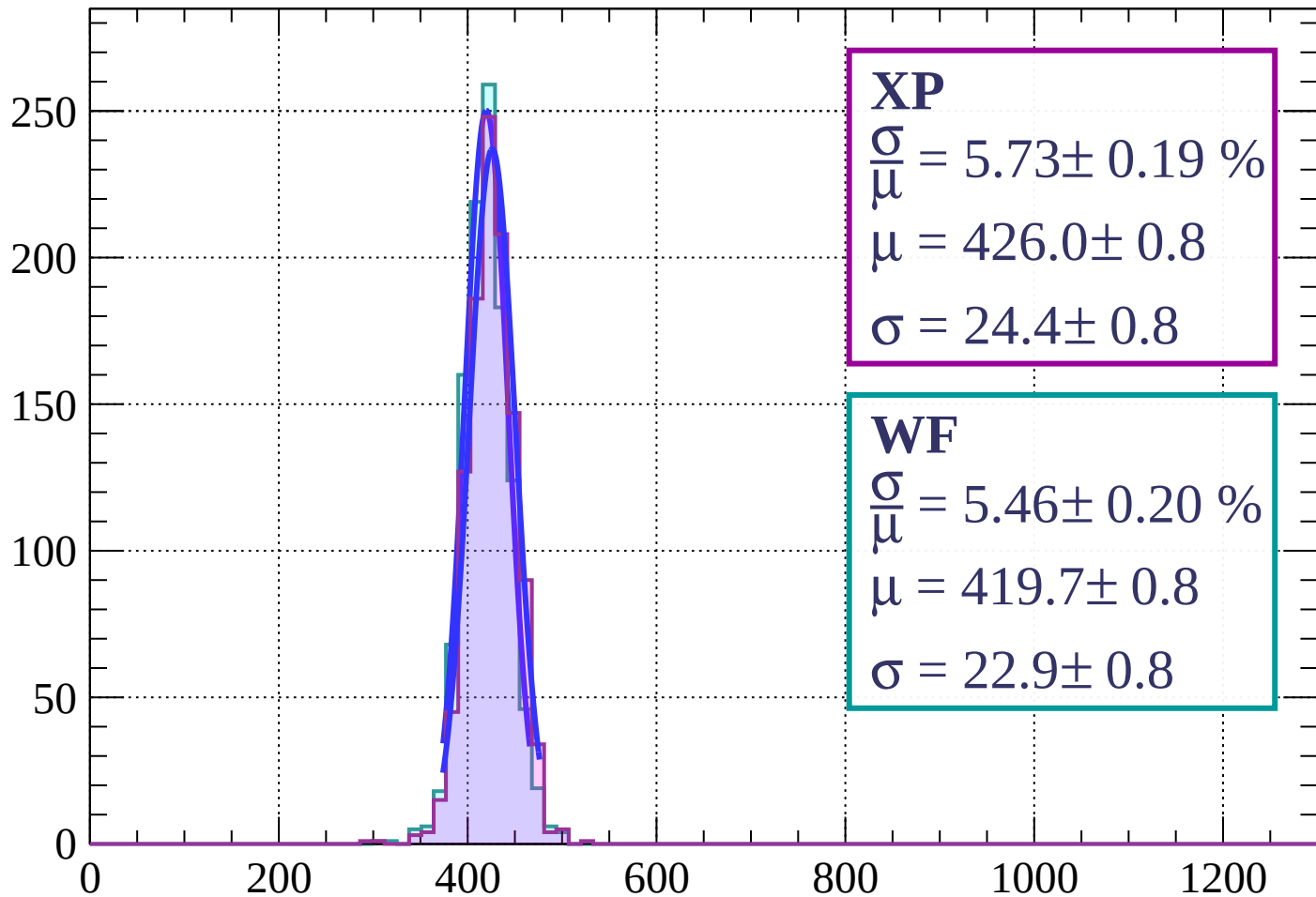
dE/dx (ADC counts/cm)

Energy loss | $15 < \phi < 18$



Energy loss | $18 < \phi < 21$

Count



XP

$$\frac{\sigma}{\mu} = 5.73 \pm 0.19 \%$$

$$\mu = 426.0 \pm 0.8$$

$$\sigma = 24.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 5.46 \pm 0.20 \%$$

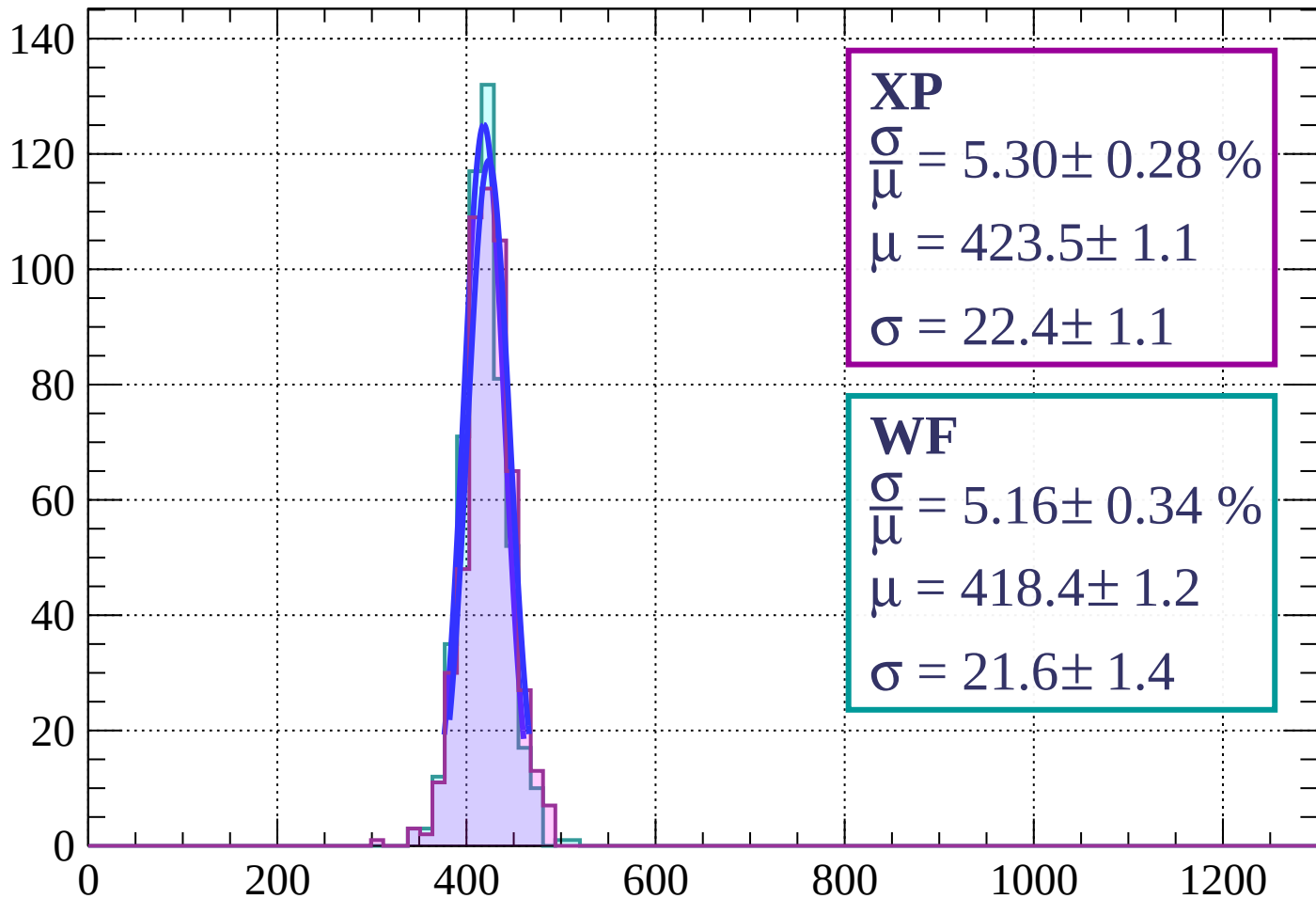
$$\mu = 419.7 \pm 0.8$$

$$\sigma = 22.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $21 < \phi < 24$

Count



XP

$$\frac{\sigma}{\mu} = 5.30 \pm 0.28 \%$$

$$\mu = 423.5 \pm 1.1$$

$$\sigma = 22.4 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 5.16 \pm 0.34 \%$$

$$\mu = 418.4 \pm 1.2$$

$$\sigma = 21.6 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $24 < \phi < 27$

Count

XP

$$\frac{\sigma}{\mu} = 6.06 \pm 0.45 \%$$

$$\mu = 420.9 \pm 2.1$$

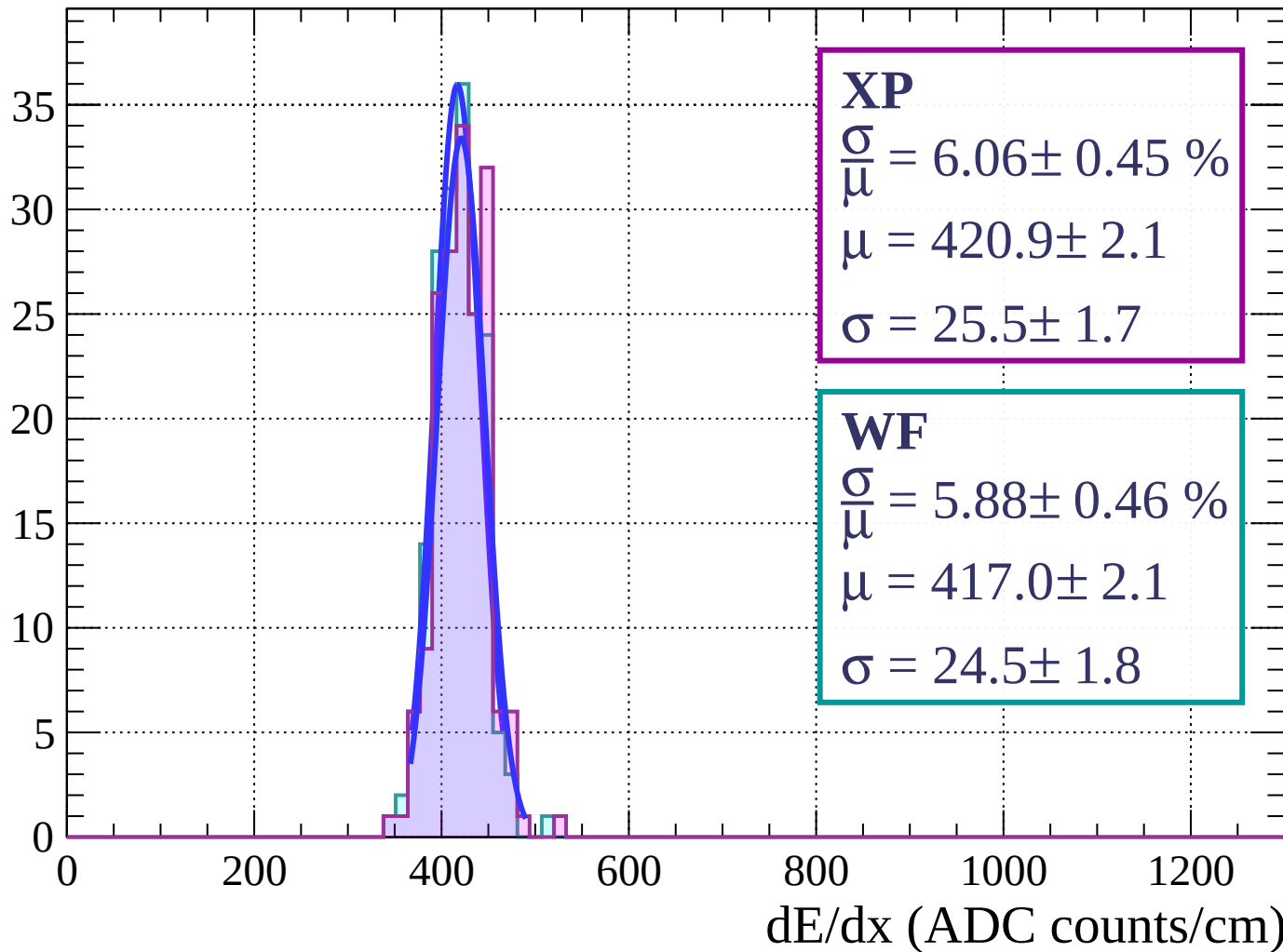
$$\sigma = 25.5 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 5.88 \pm 0.46 \%$$

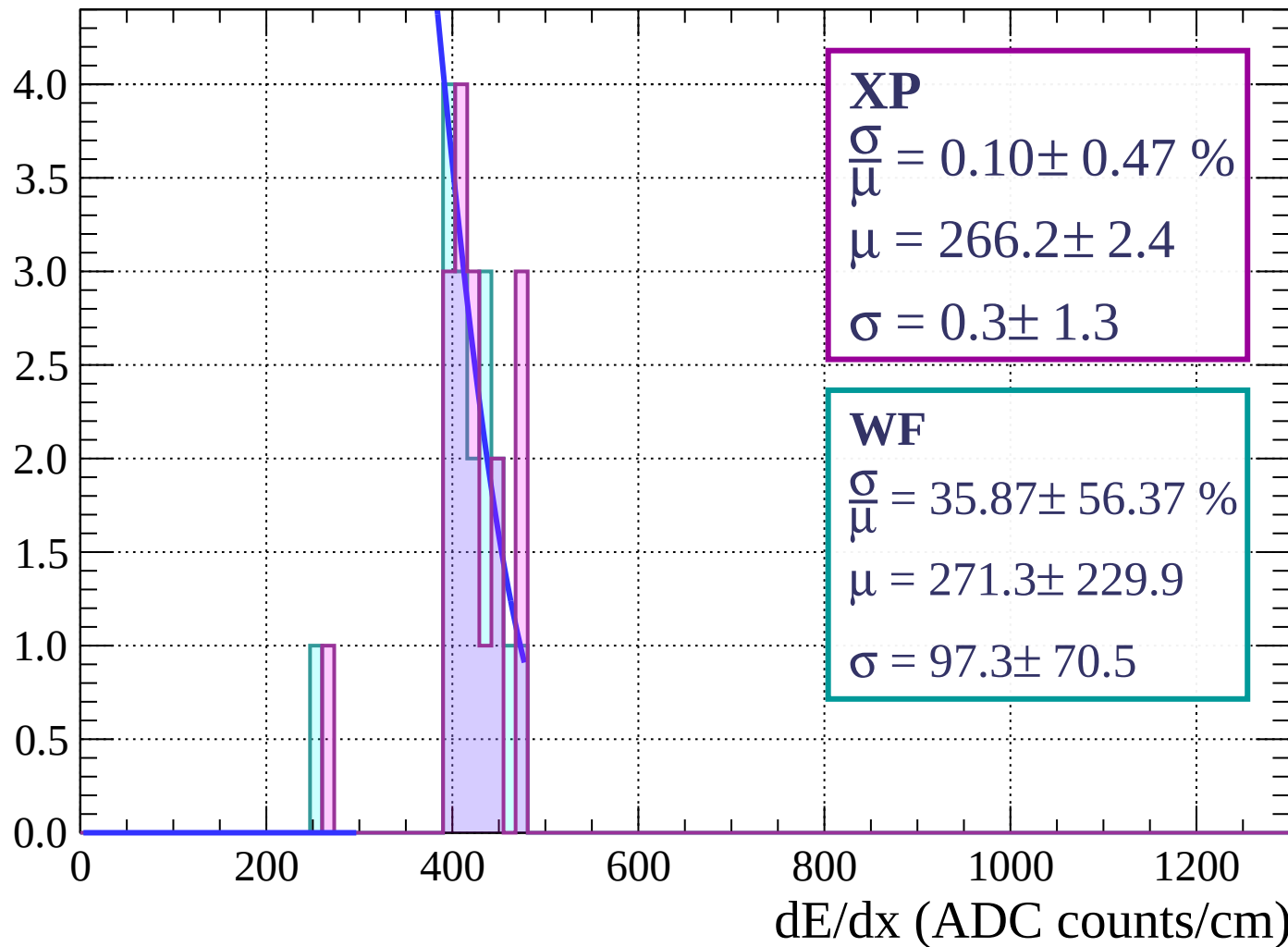
$$\mu = 417.0 \pm 2.1$$

$$\sigma = 24.5 \pm 1.8$$



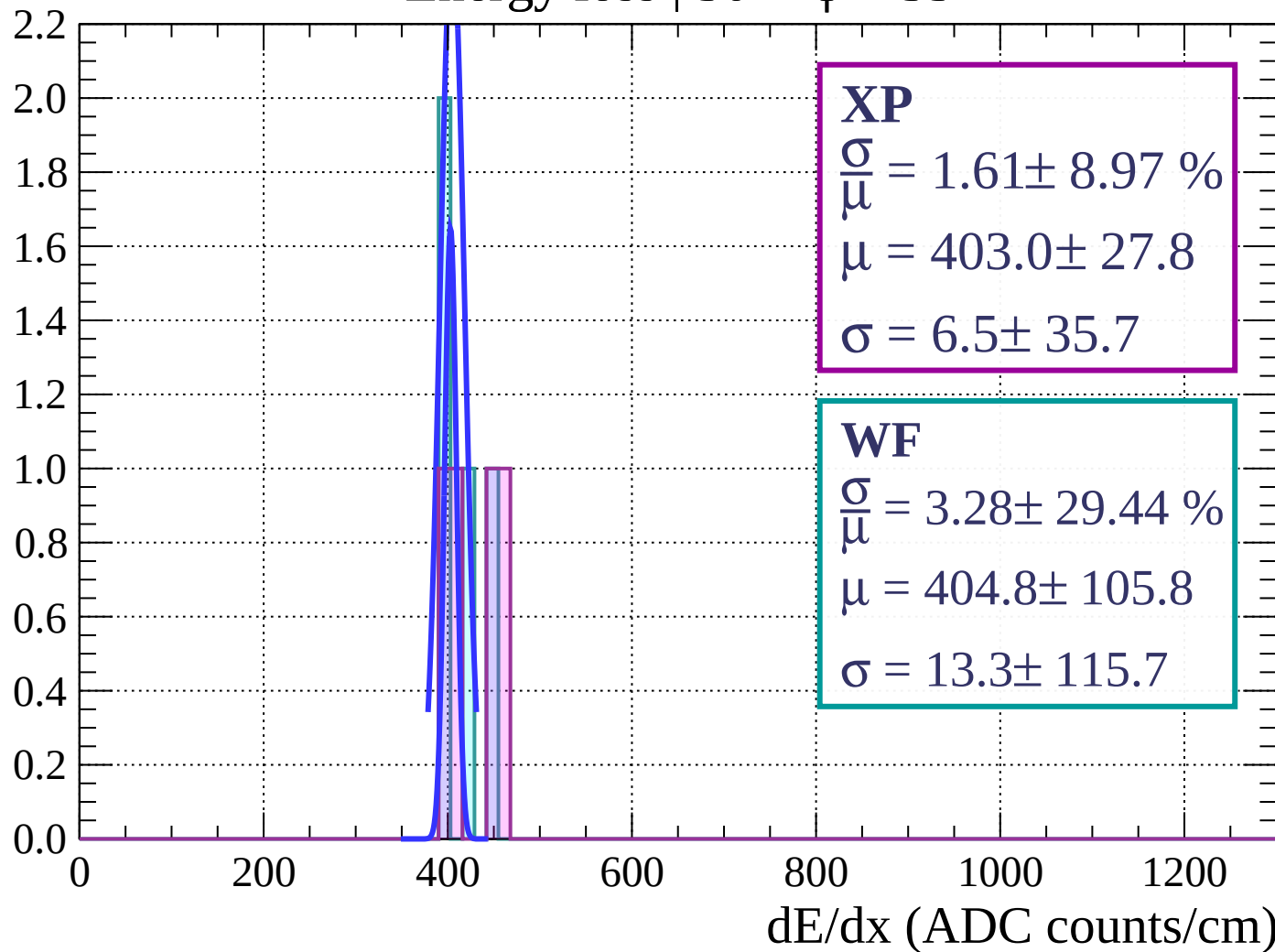
Energy loss | $27 < \phi < 30$

Count



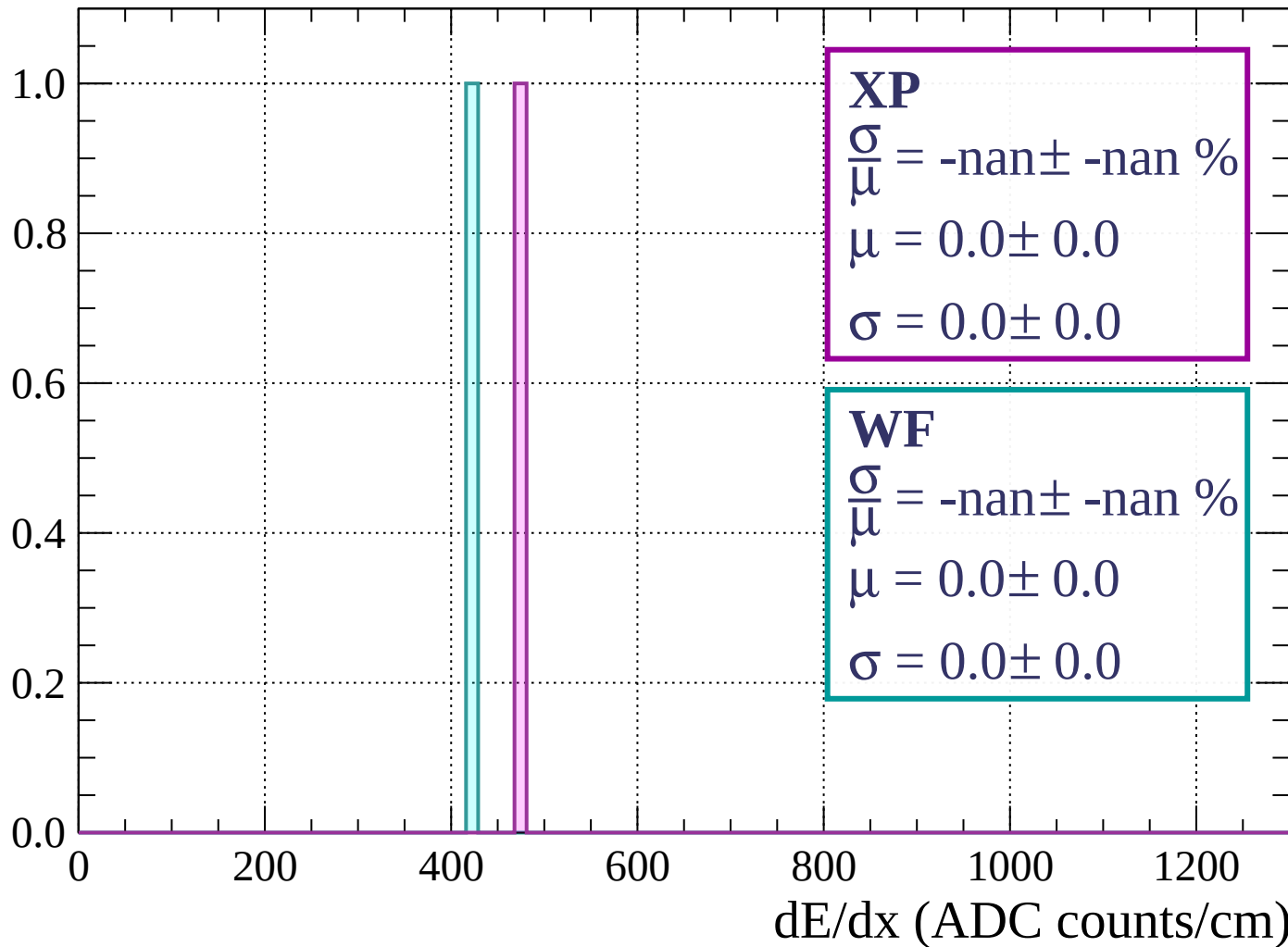
Energy loss | $30 < \phi < 33$

Count



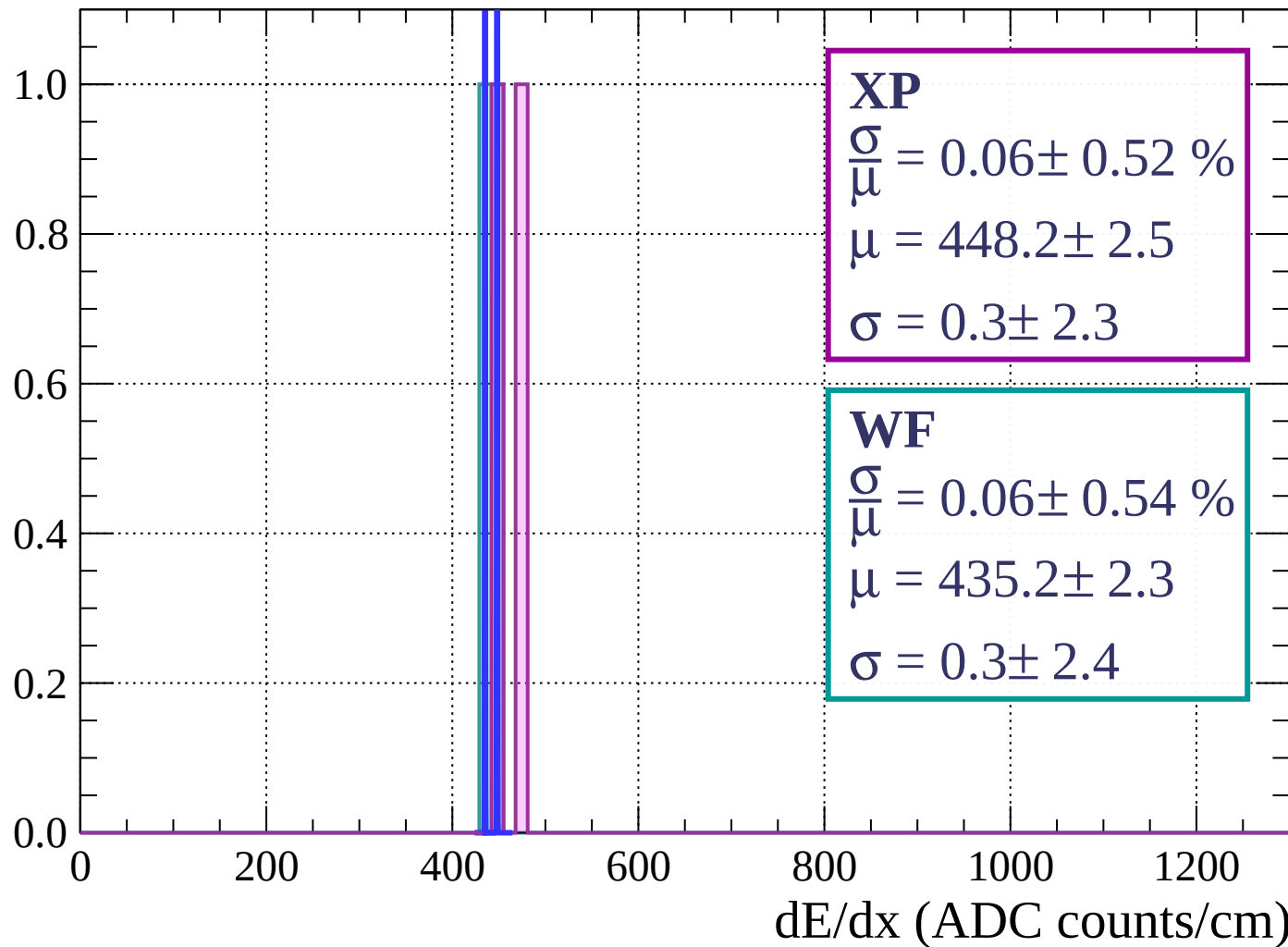
Energy loss | $33 < \phi < 36$

Count



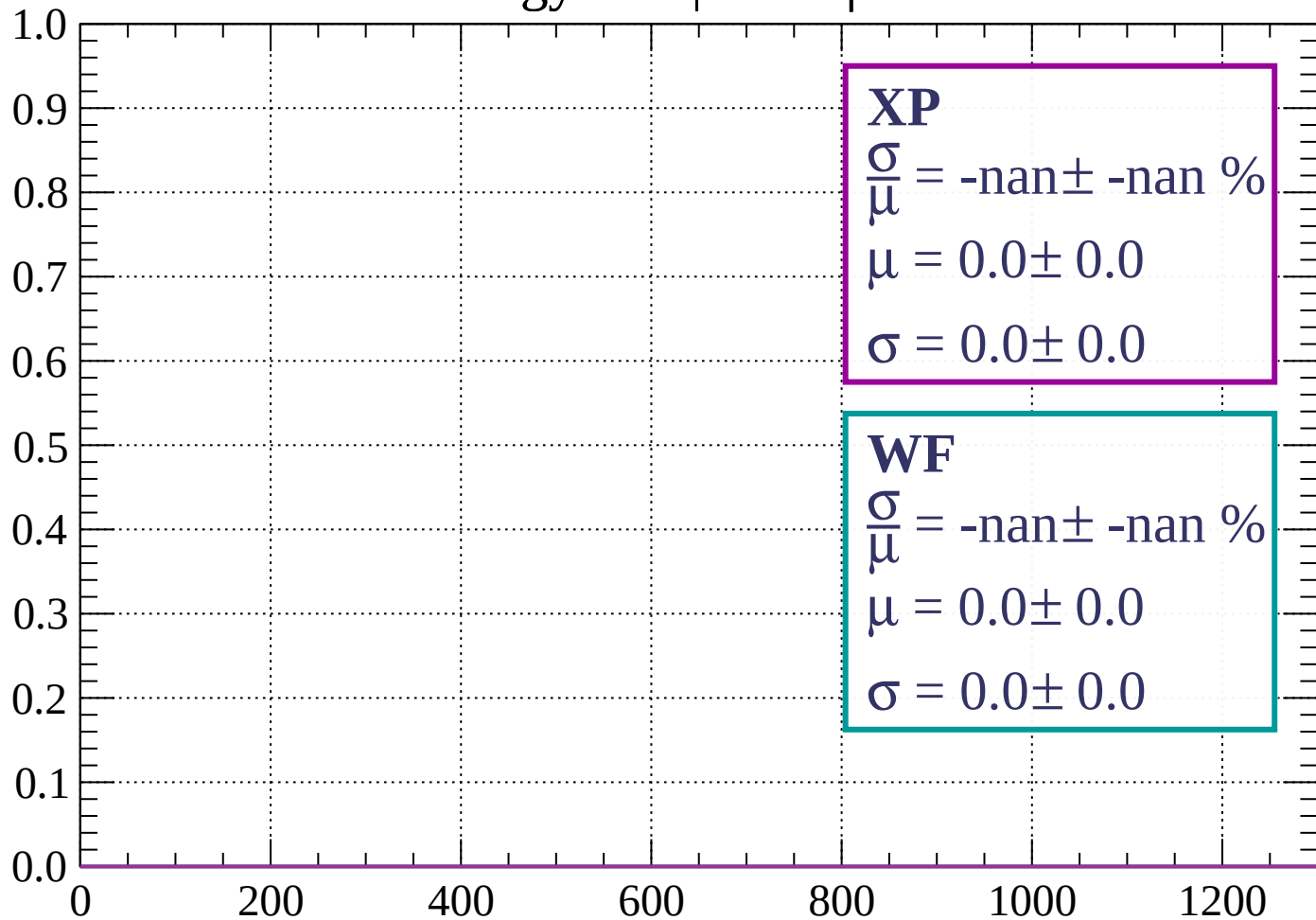
Energy loss | $36 < \phi < 39$

Count



Energy loss | $39 < \phi < 42$

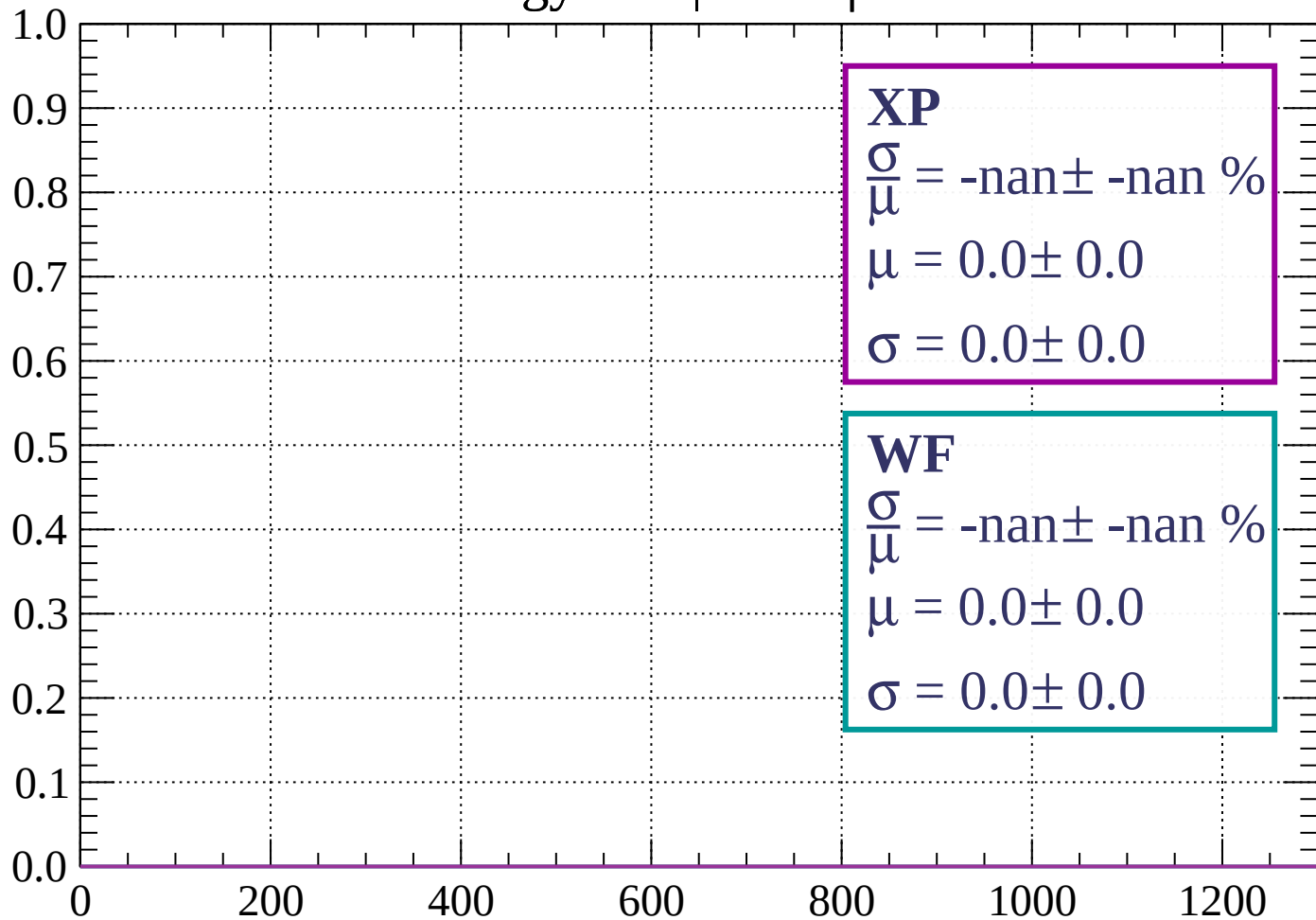
Count



dE/dx (ADC counts/cm)

Energy loss | $42 < \phi < 45$

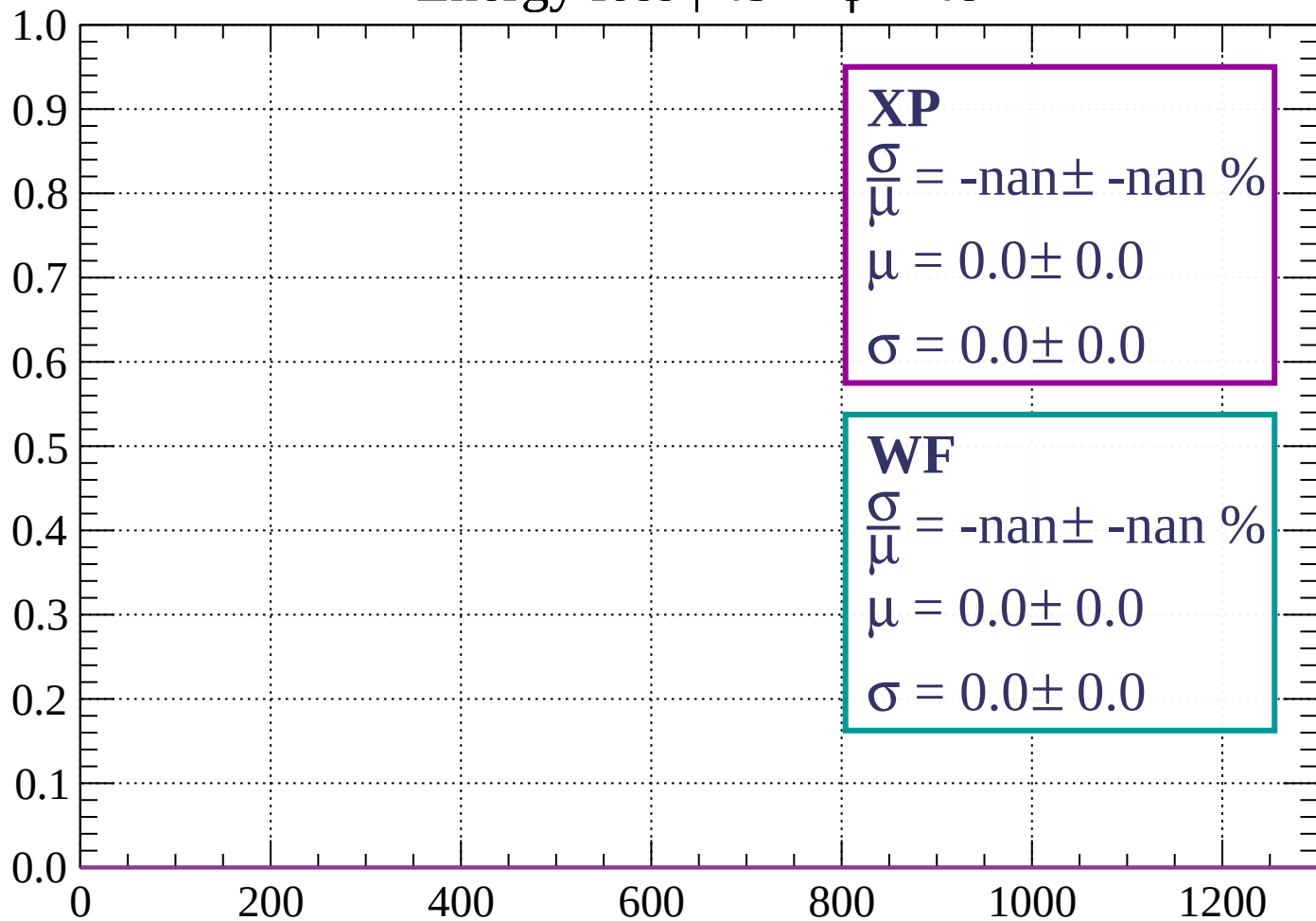
Count



dE/dx (ADC counts/cm)

Energy loss | $45 < \phi < 48$

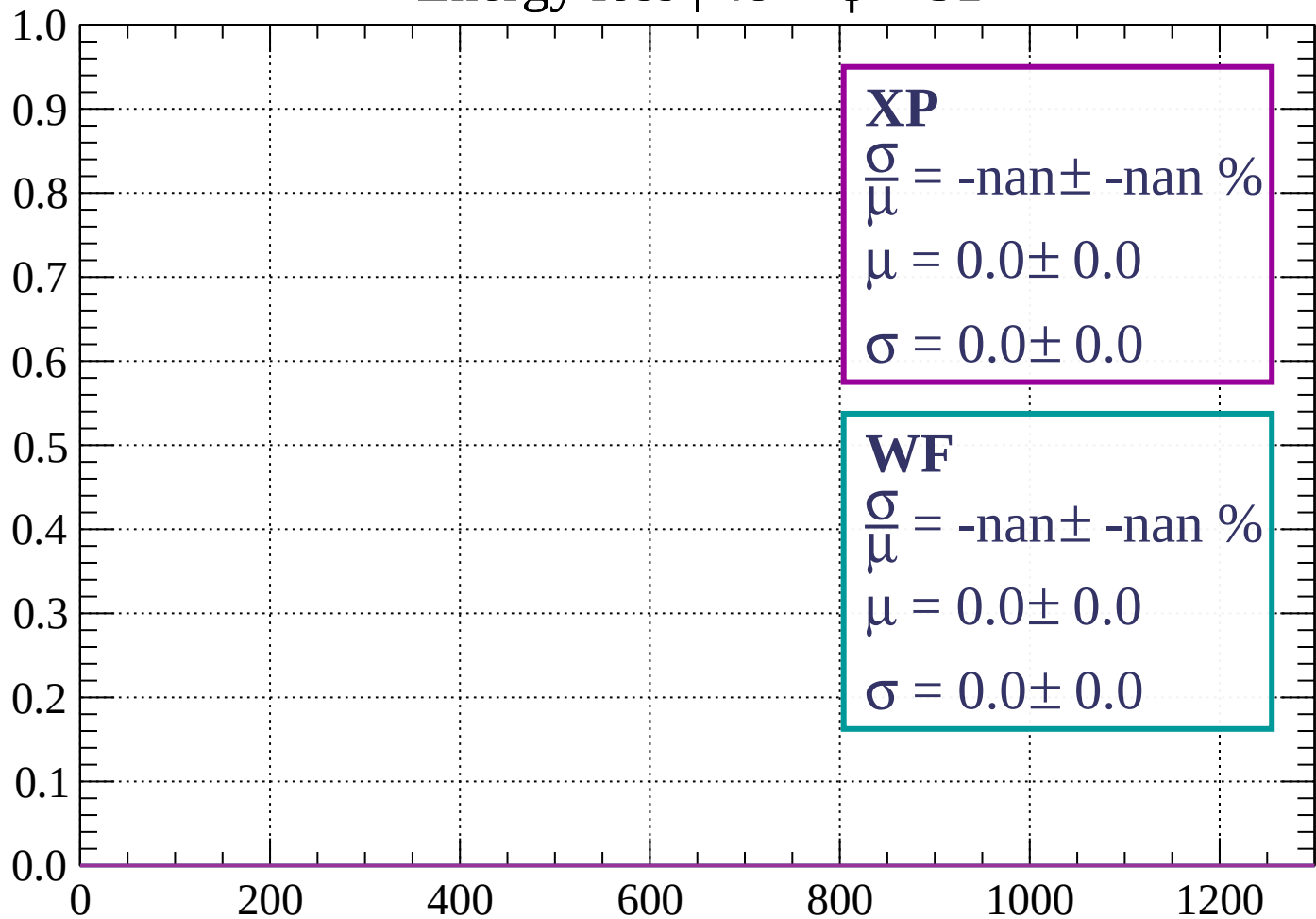
Count



dE/dx (ADC counts/cm)

Energy loss | $48 < \phi < 51$

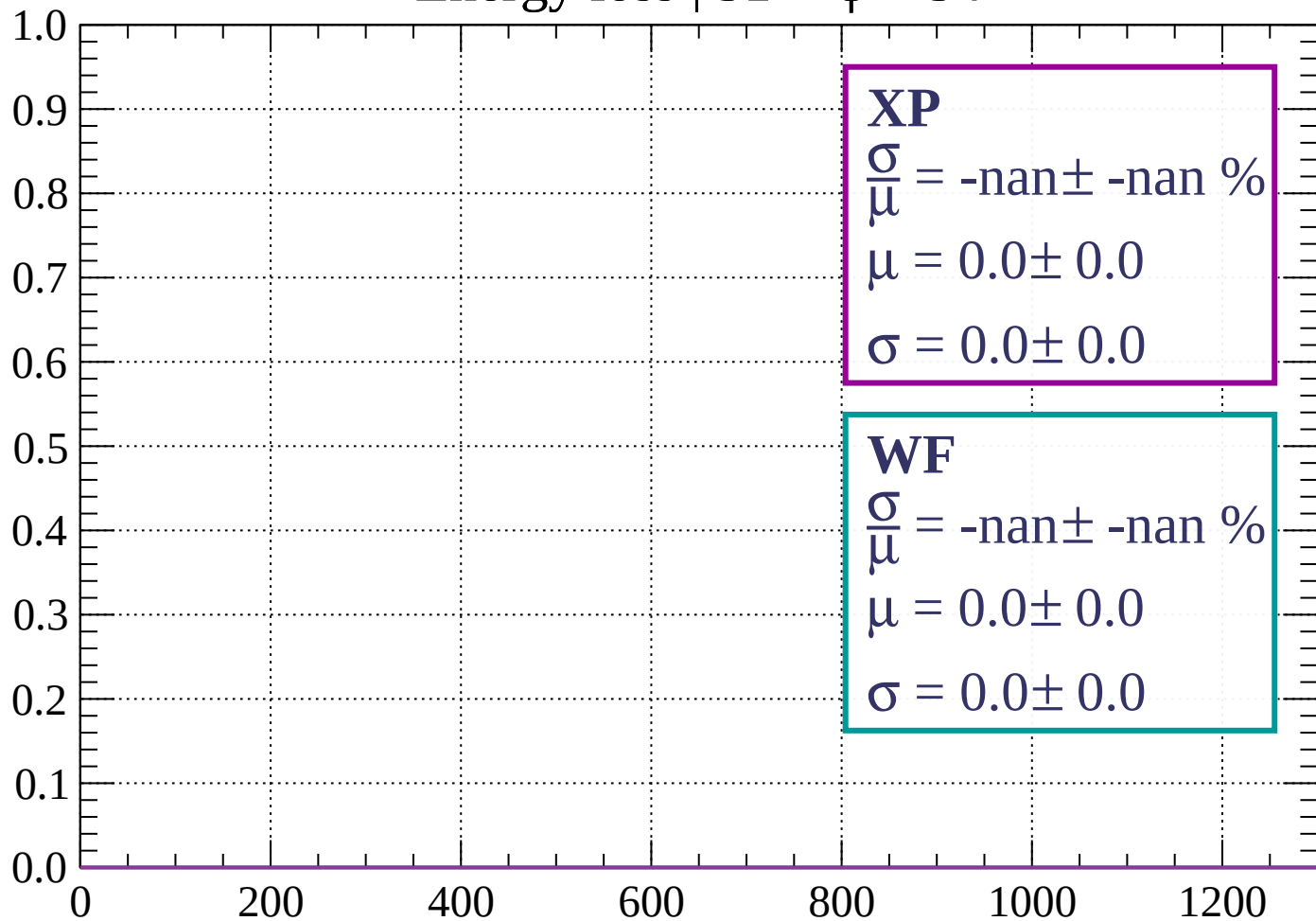
Count



dE/dx (ADC counts/cm)

Energy loss | $51 < \phi < 54$

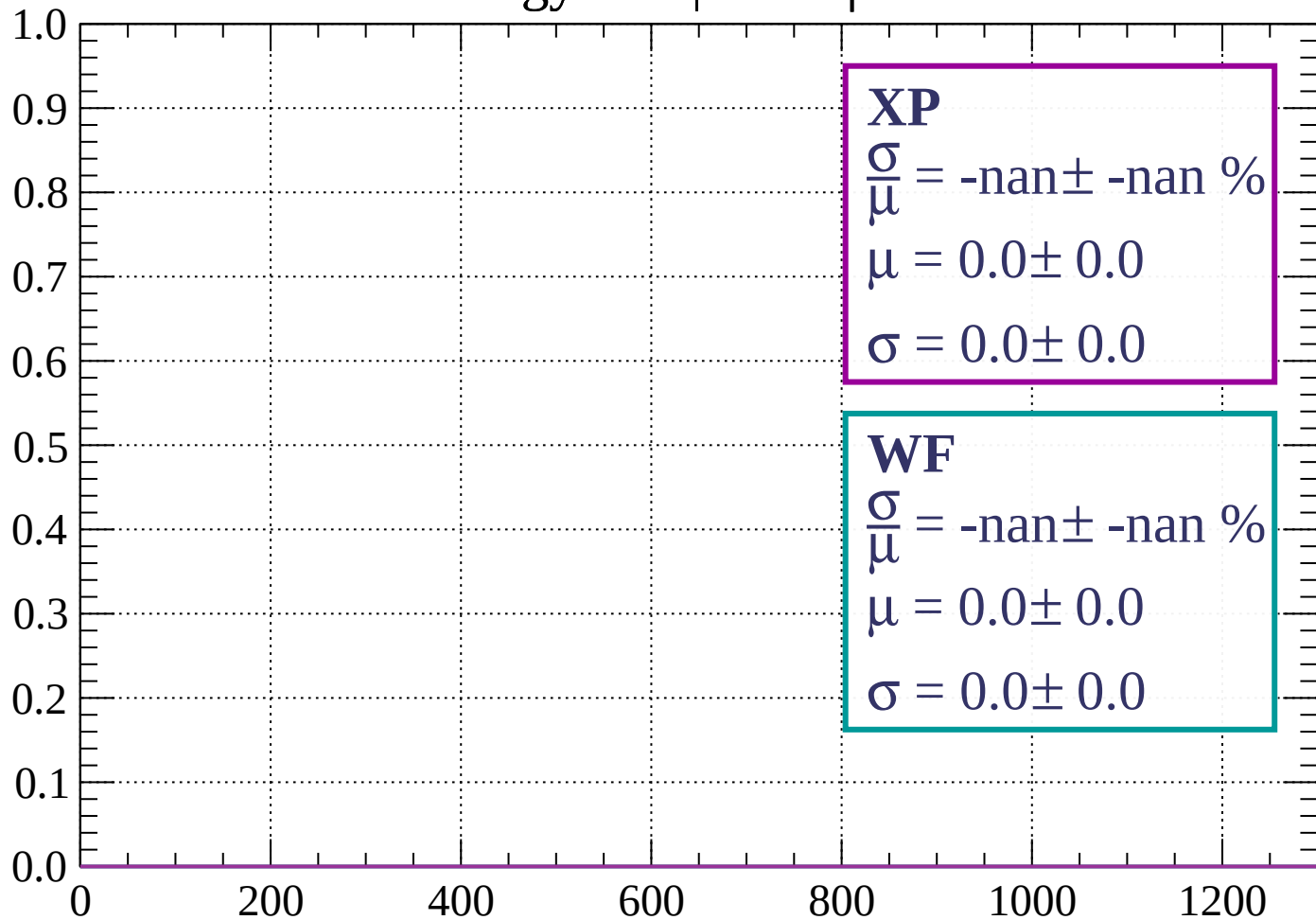
Count



dE/dx (ADC counts/cm)

Energy loss | $54 < \phi < 57$

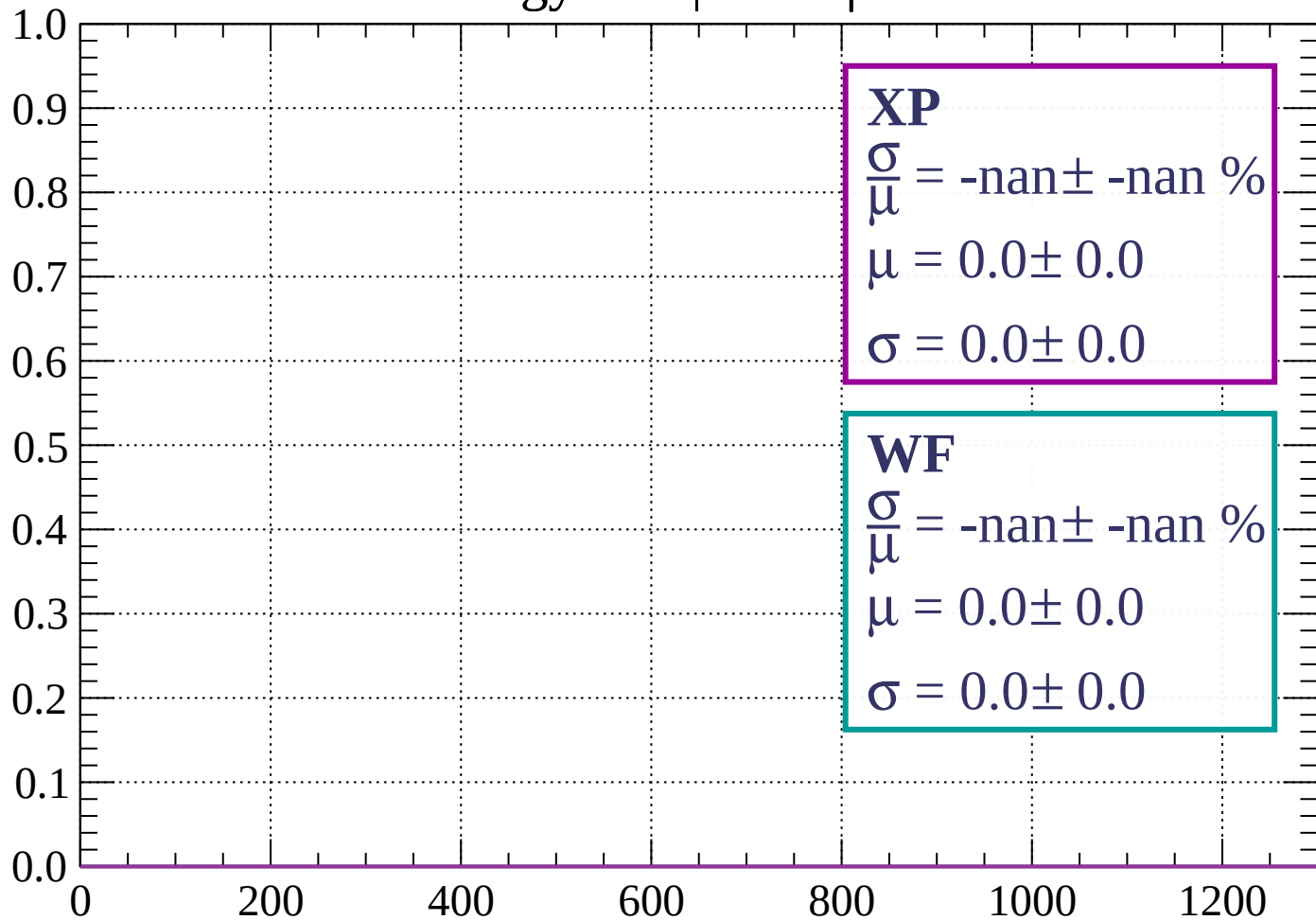
Count



dE/dx (ADC counts/cm)

Energy loss | $57 < \phi < 60$

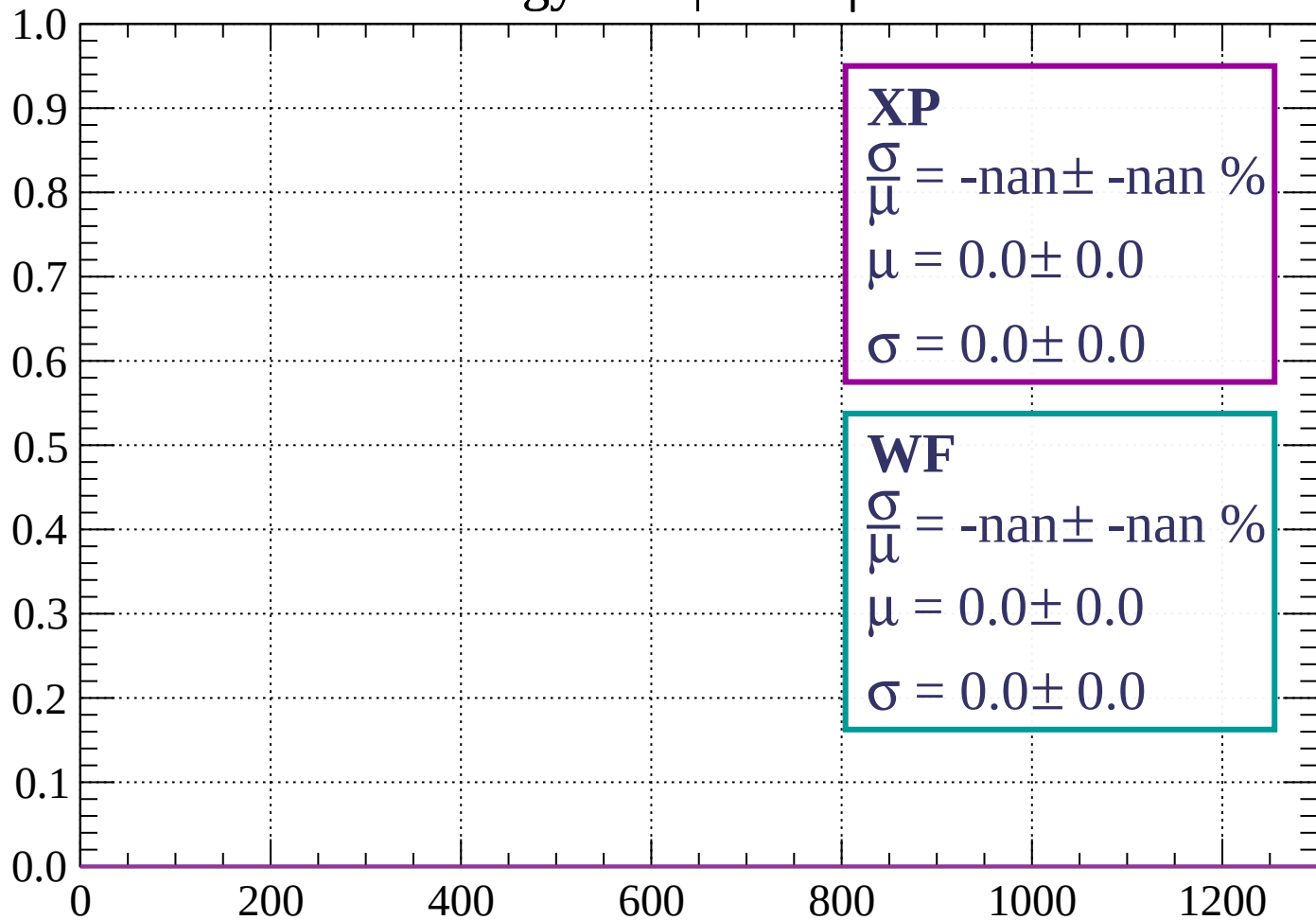
Count



dE/dx (ADC counts/cm)

Energy loss | $60 < \phi < 63$

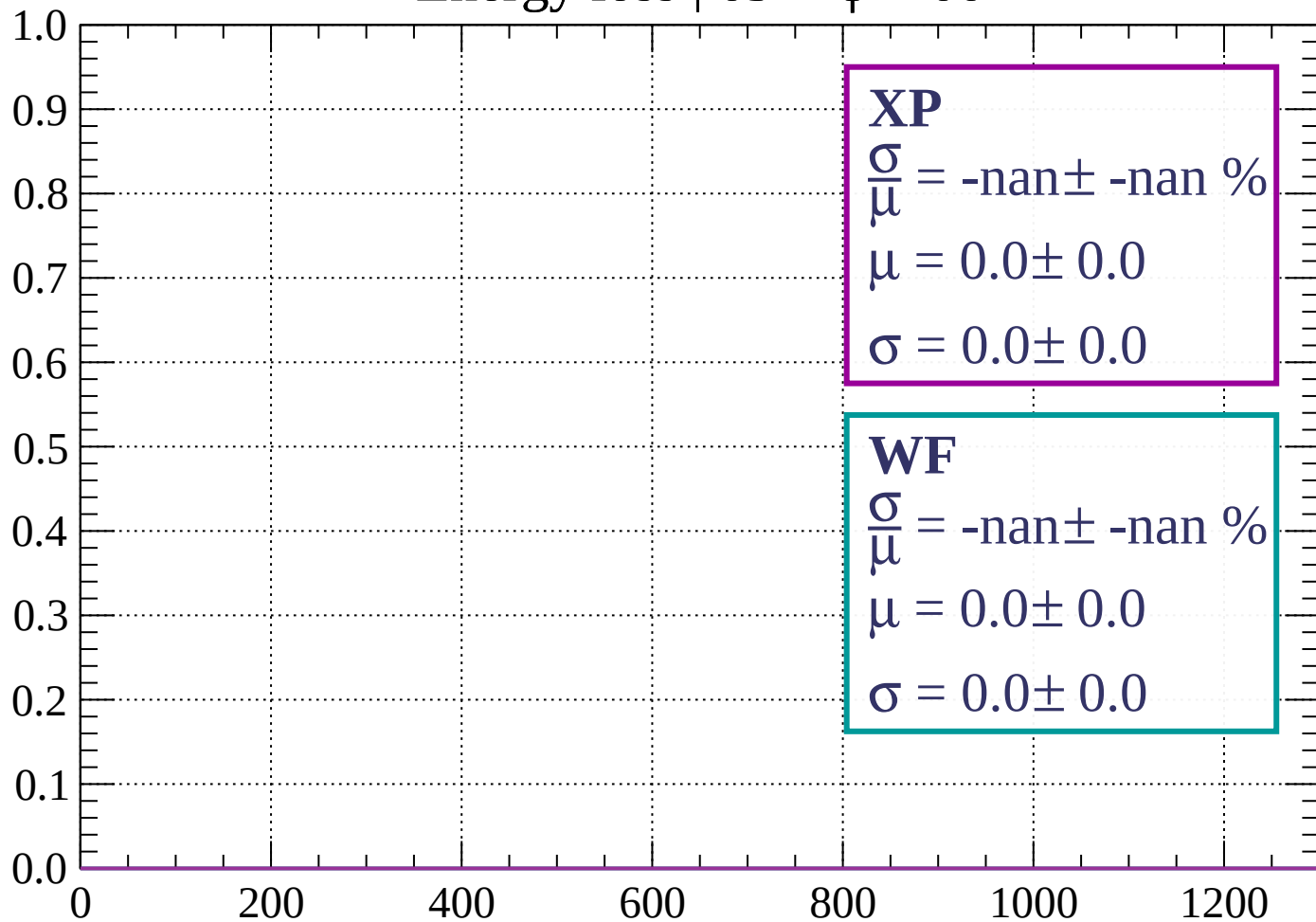
Count



dE/dx (ADC counts/cm)

Energy loss | $63 < \phi < 66$

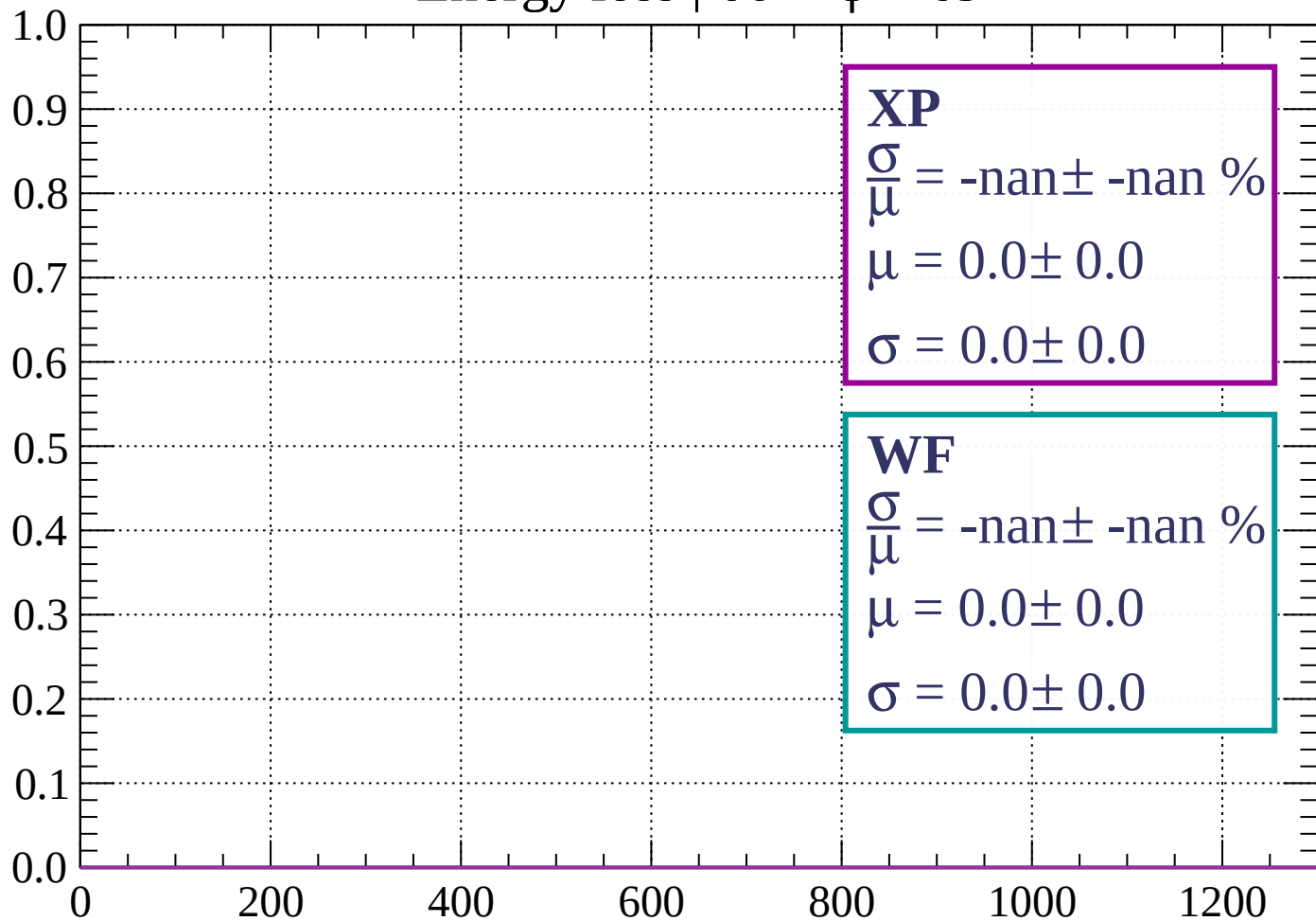
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \phi < 69$

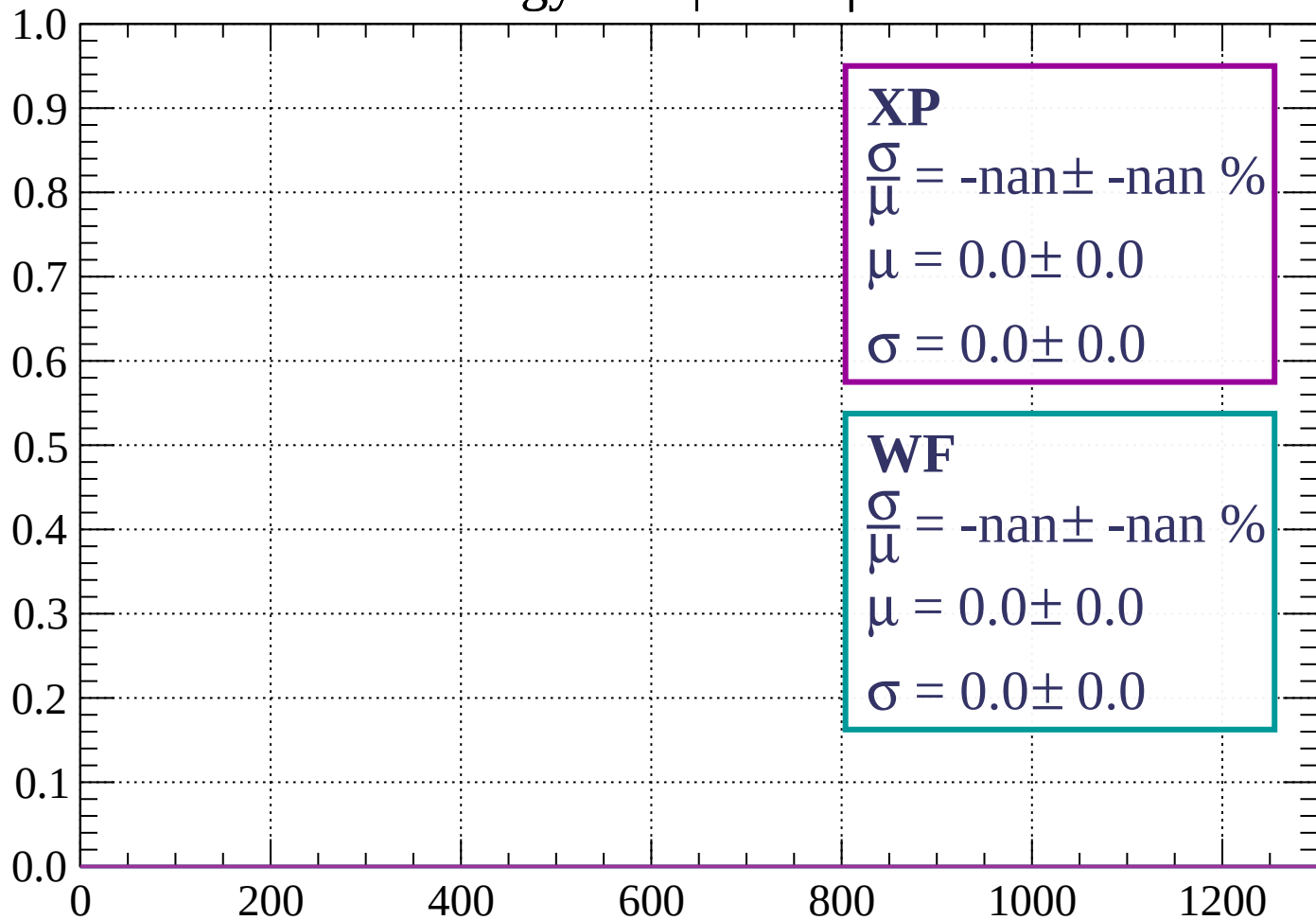
Count



dE/dx (ADC counts/cm)

Energy loss | $69 < \phi < 72$

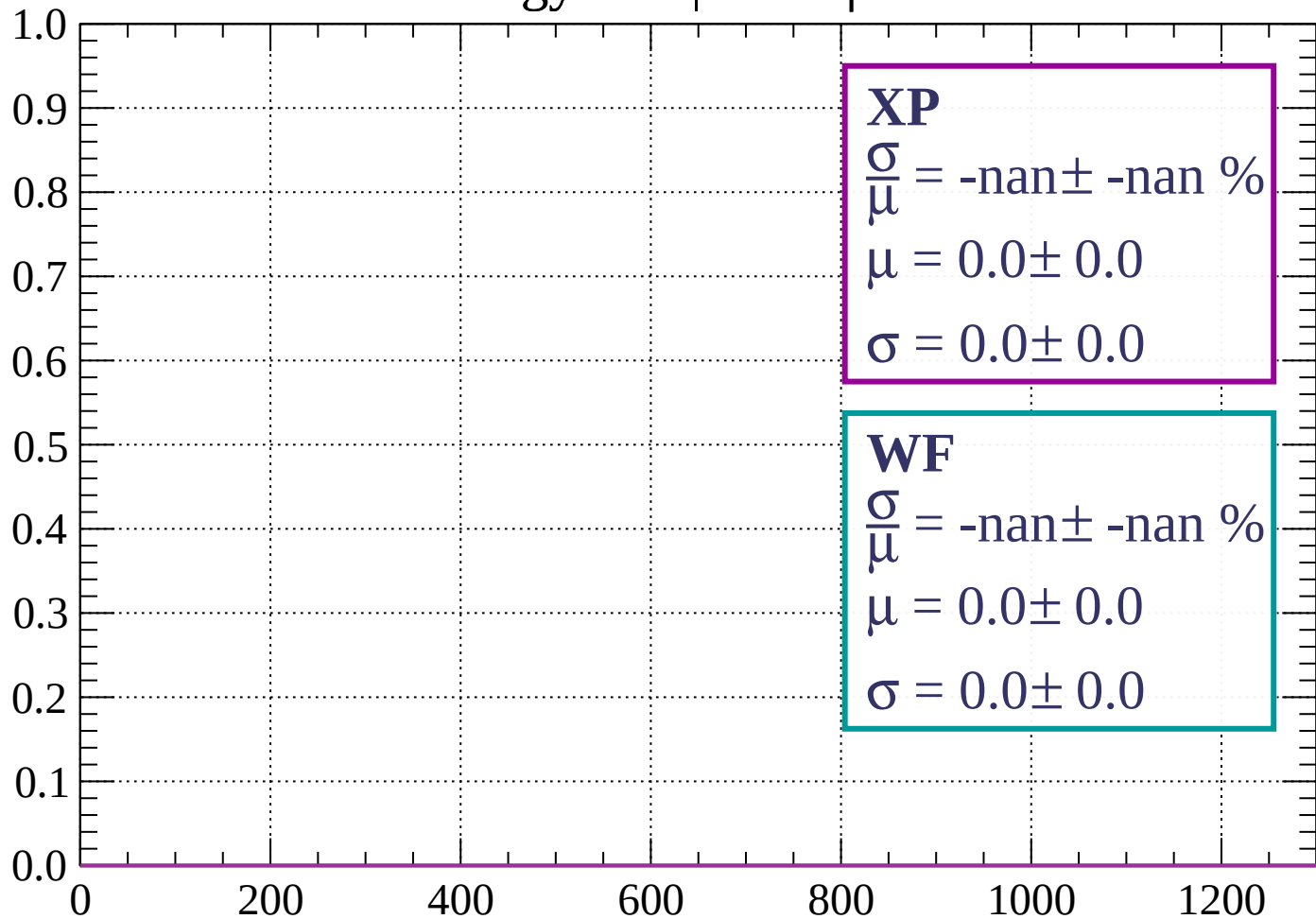
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \phi < 75$

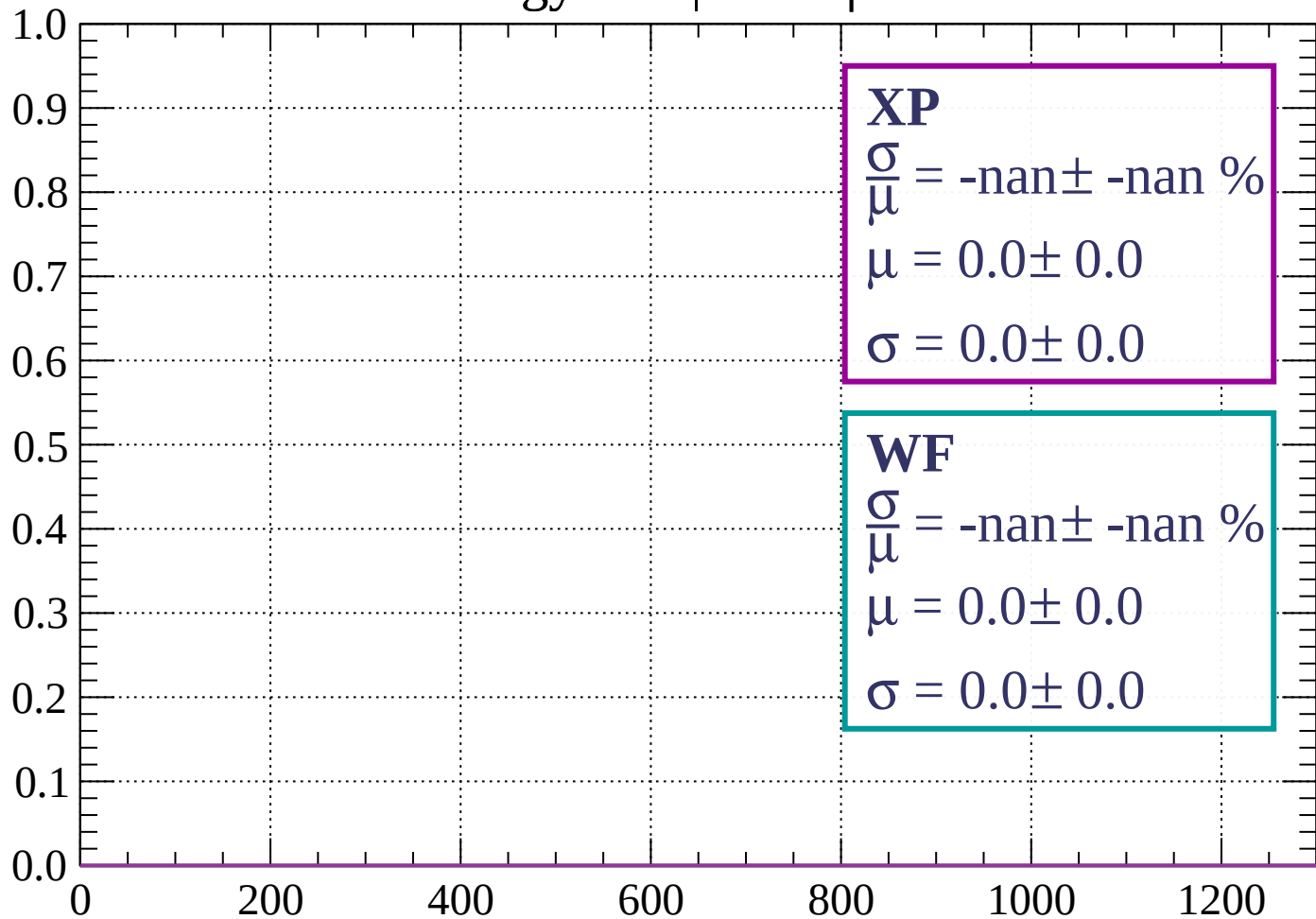
Count



dE/dx (ADC counts/cm)

Energy loss | $75 < \phi < 78$

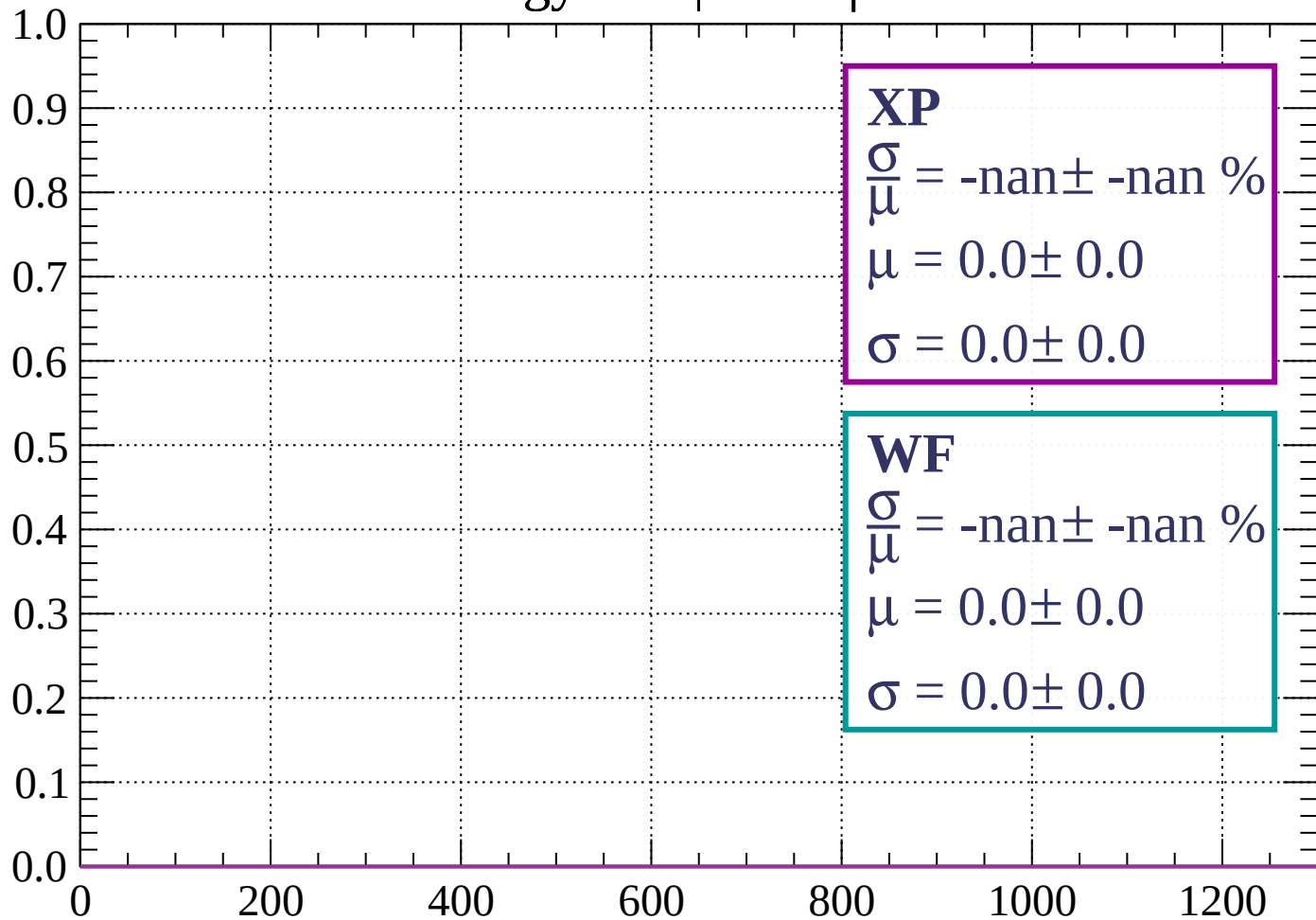
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \phi < 81$

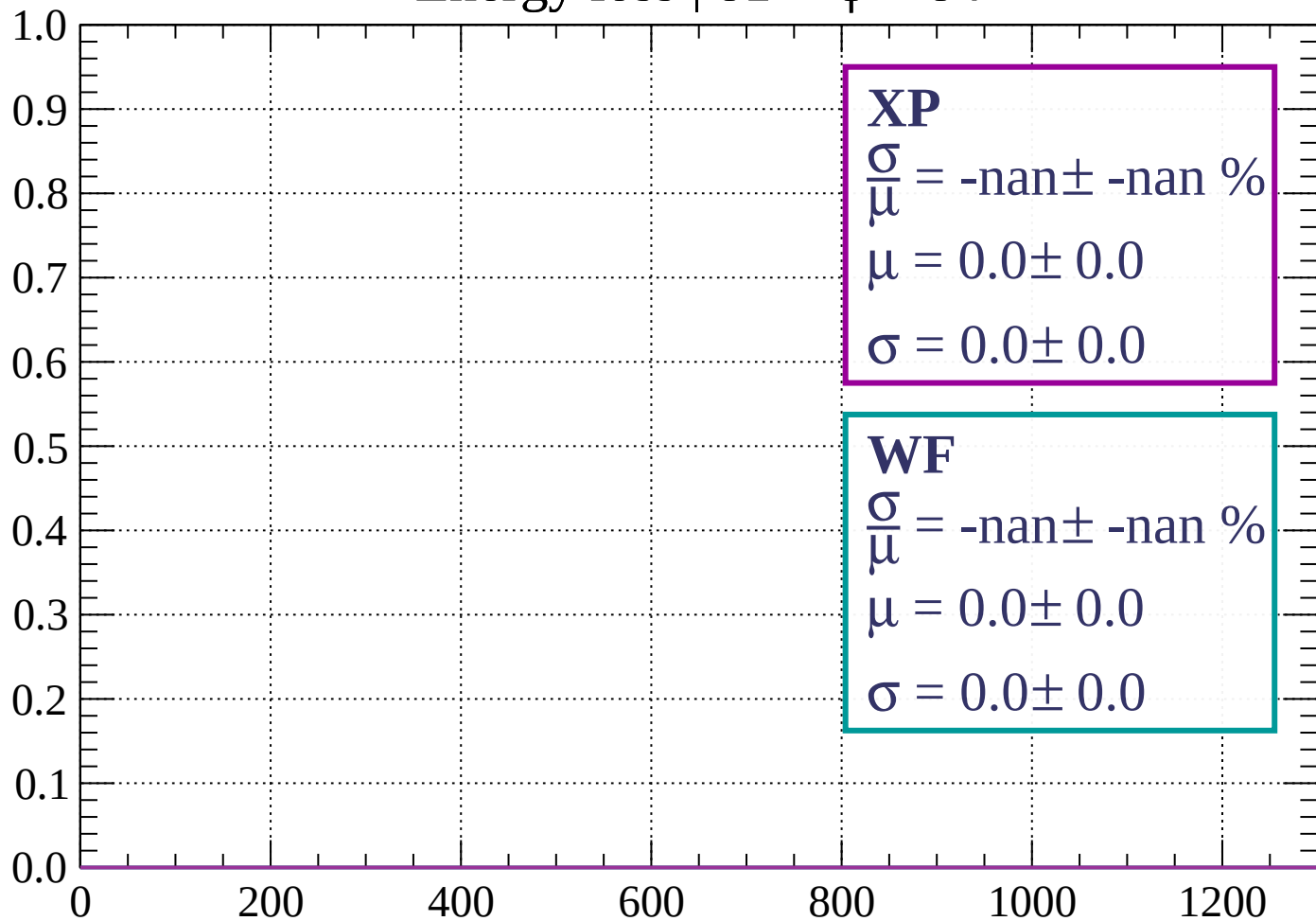
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \phi < 84$

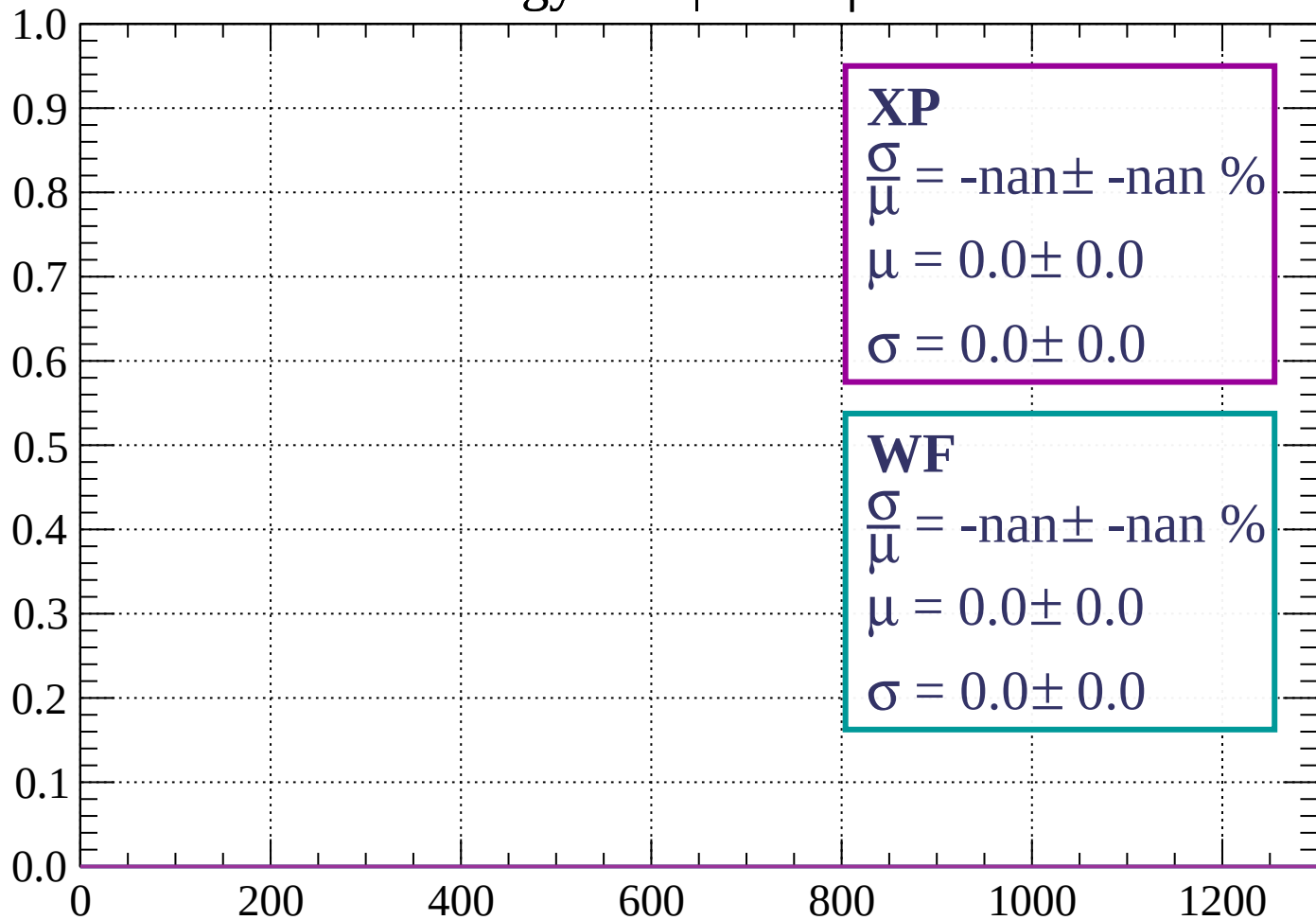
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \phi < 87$

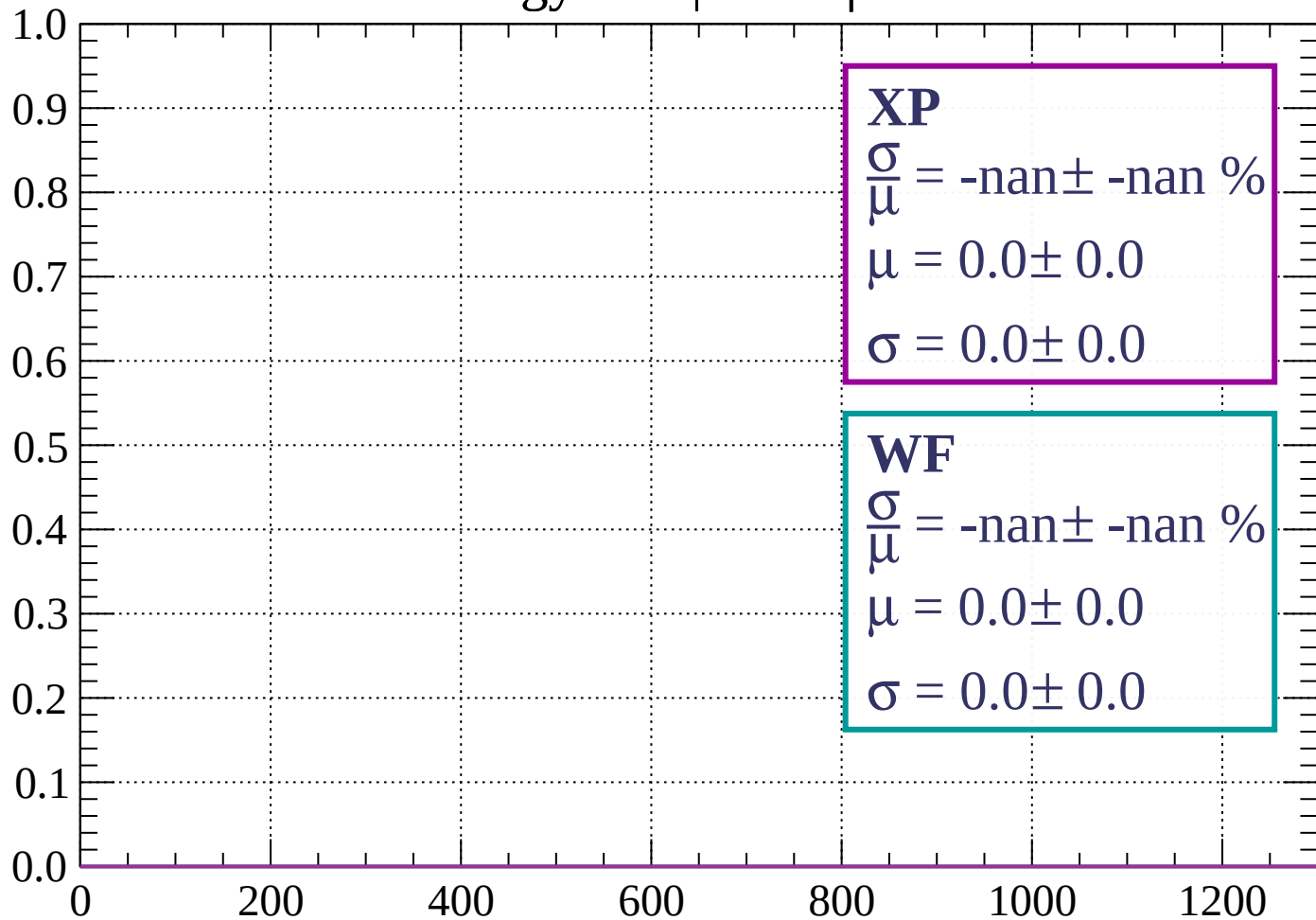
Count



dE/dx (ADC counts/cm)

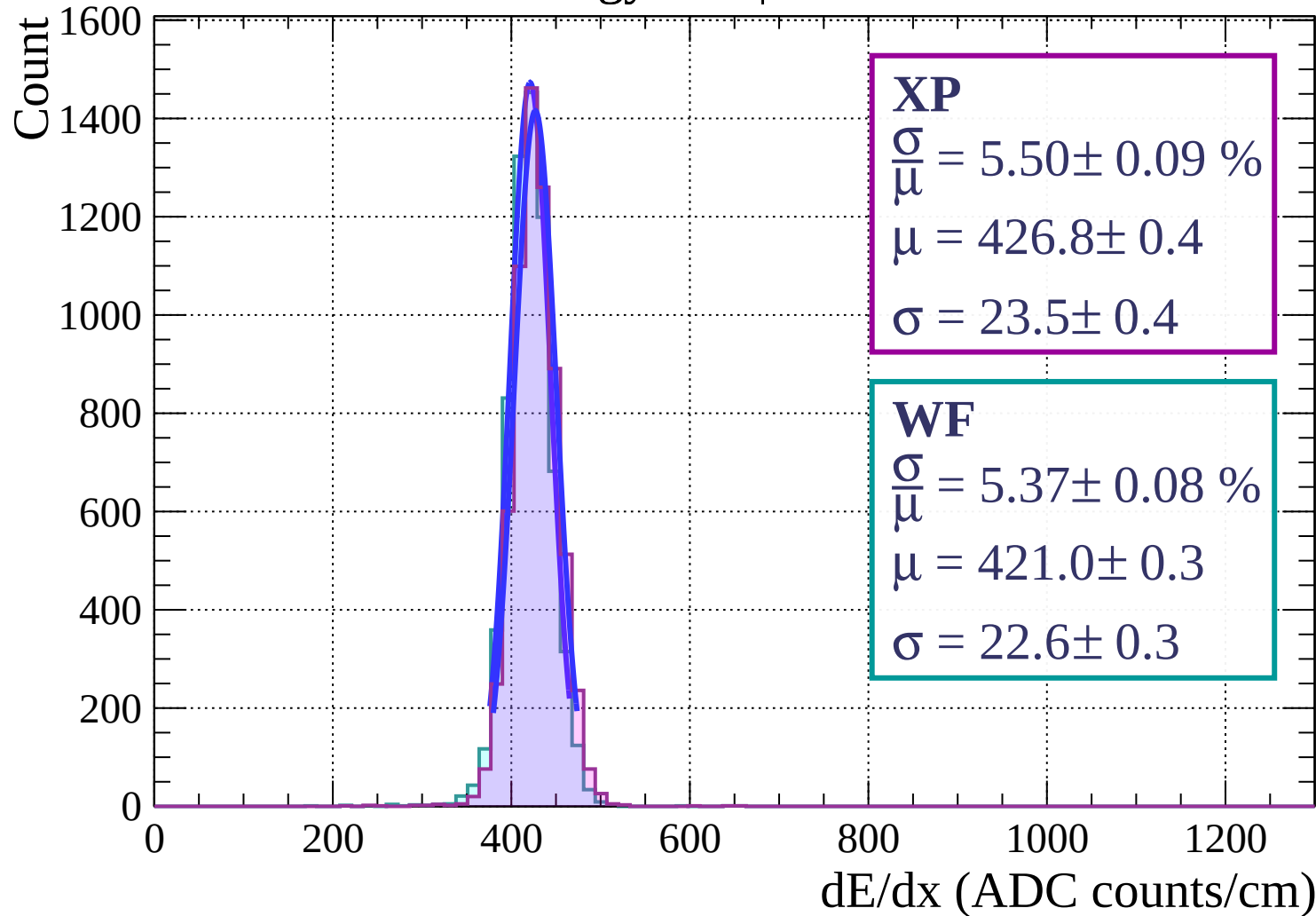
Energy loss | $87 < \phi < 90$

Count

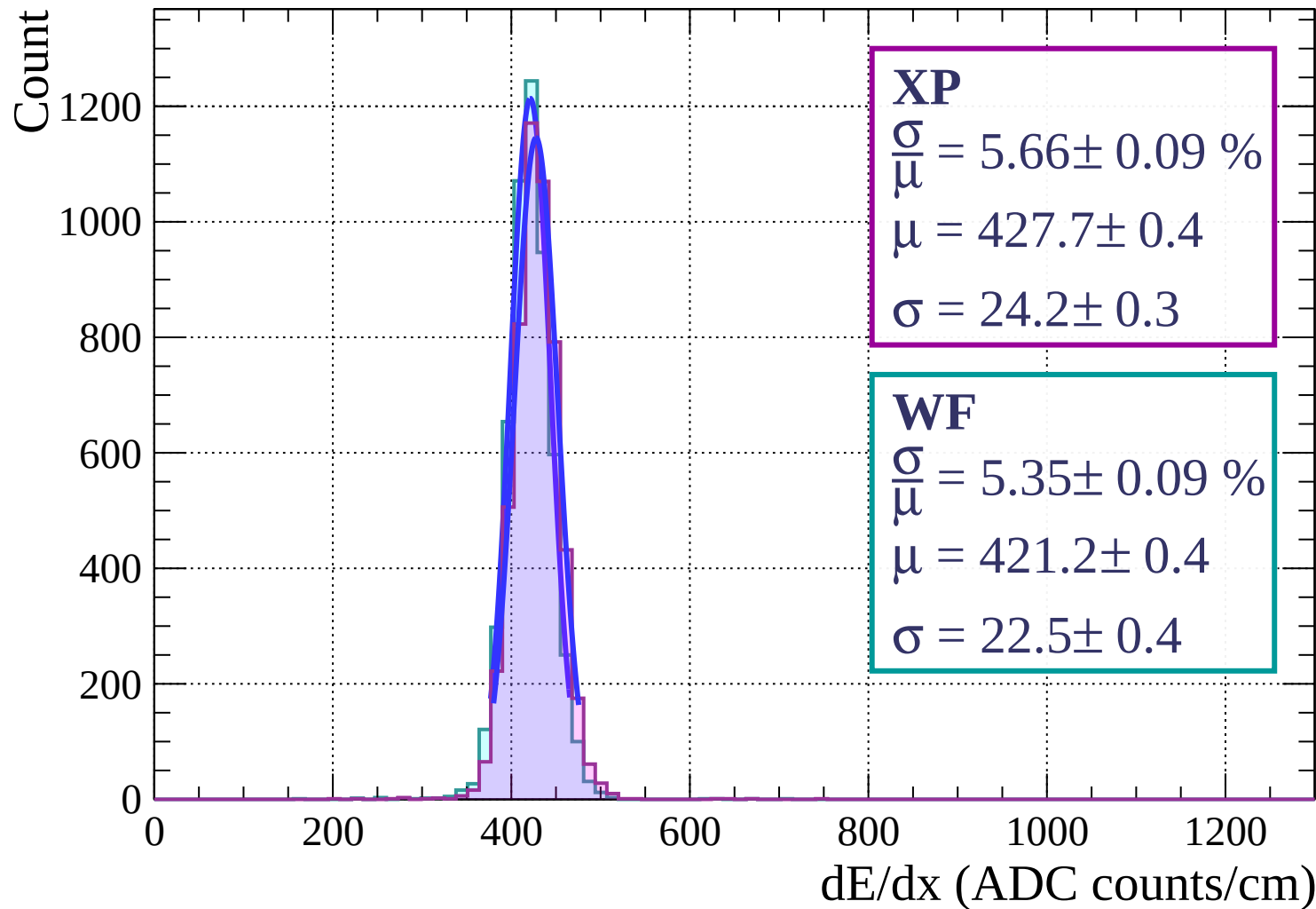


dE/dx (ADC counts/cm)

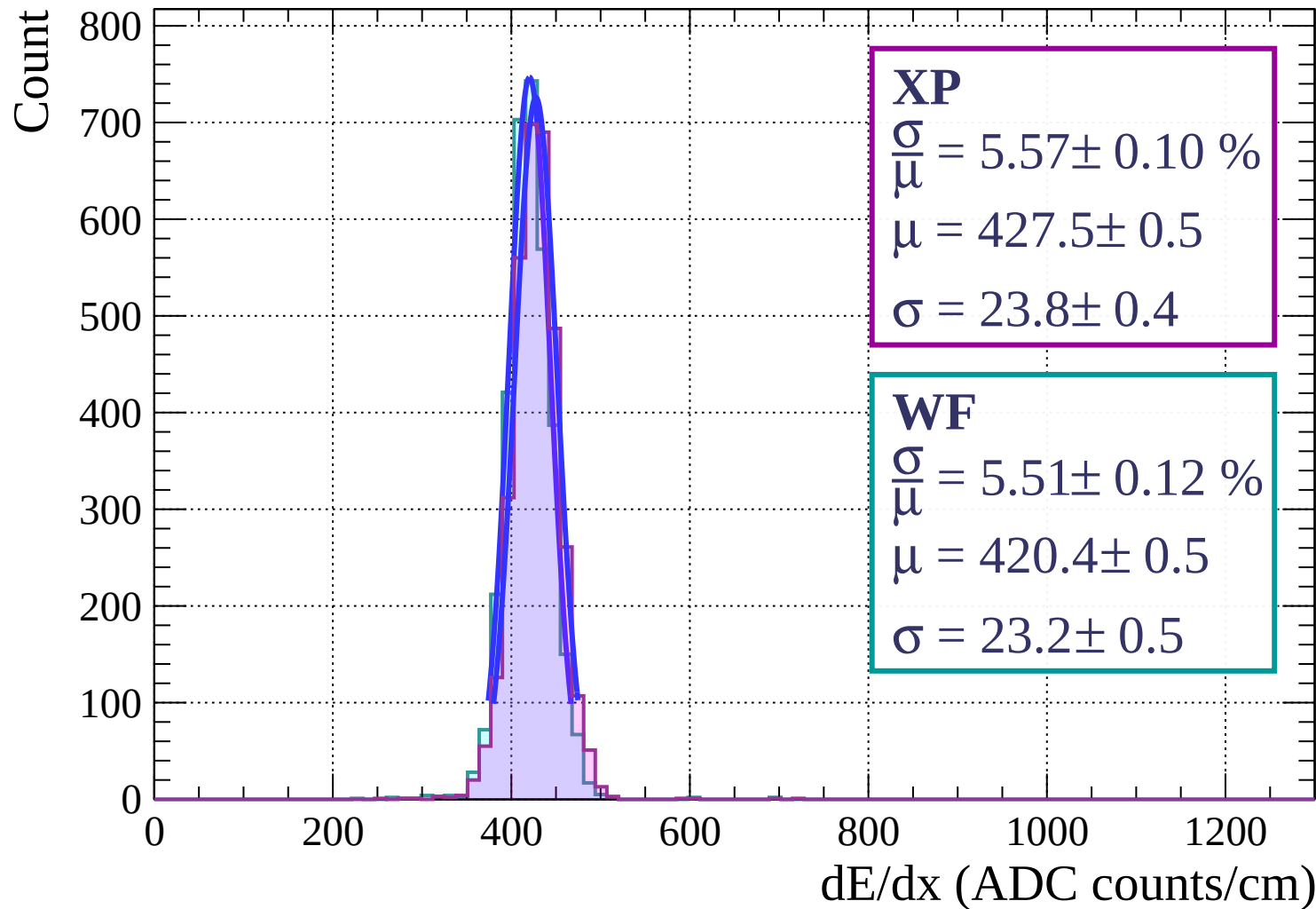
Energy loss | $0 < \theta < 3$



Energy loss | $3 < \theta < 6$

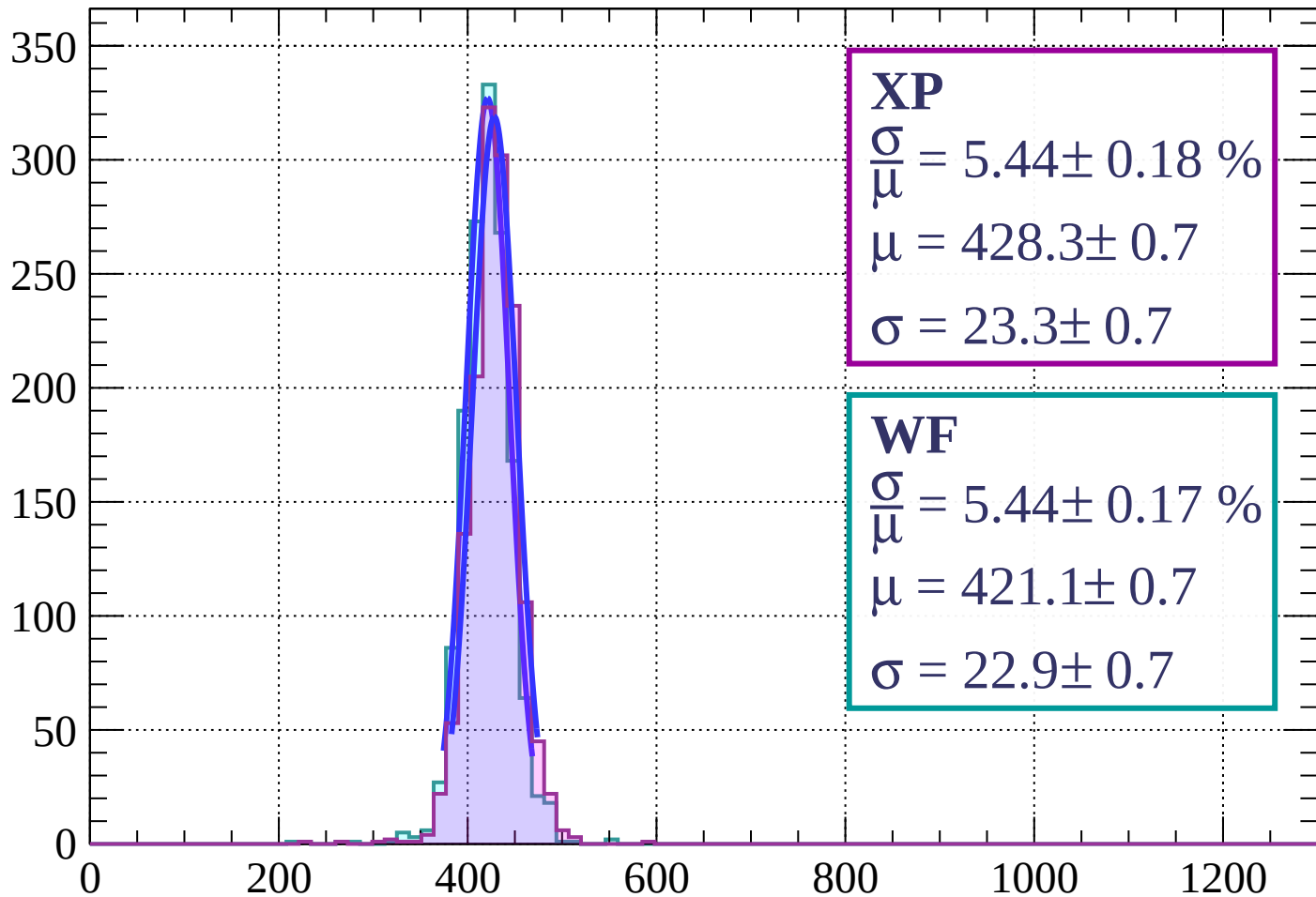


Energy loss | $6 < \theta < 9$



Energy loss | $9 < \theta < 12$

Count



XP

$$\frac{\sigma}{\mu} = 5.44 \pm 0.18 \%$$

$$\mu = 428.3 \pm 0.7$$

$$\sigma = 23.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 5.44 \pm 0.17 \%$$

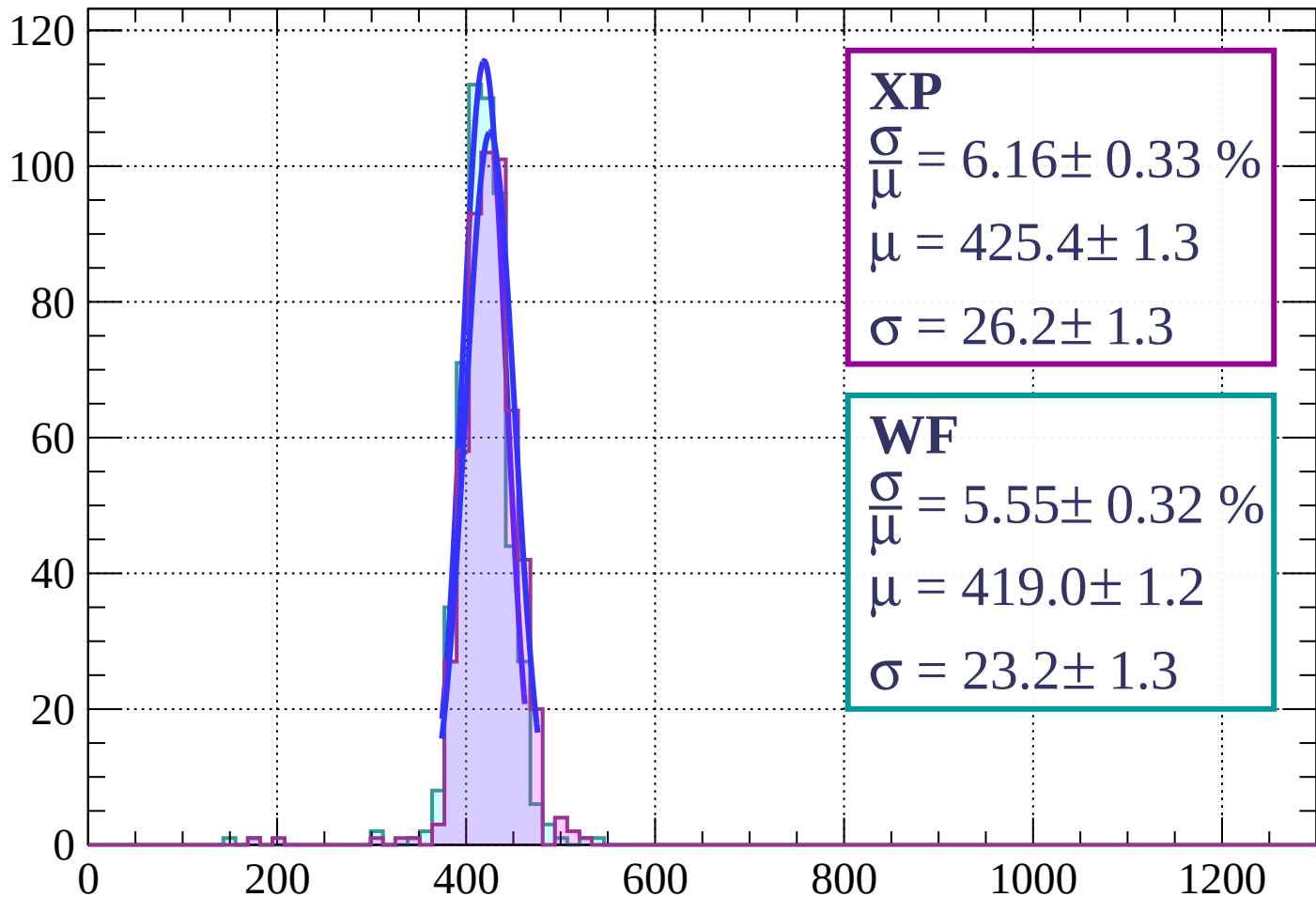
$$\mu = 421.1 \pm 0.7$$

$$\sigma = 22.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $12 < \theta < 15$

Count



XP

$$\frac{\sigma}{\mu} = 6.16 \pm 0.33 \%$$

$$\mu = 425.4 \pm 1.3$$

$$\sigma = 26.2 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 5.55 \pm 0.32 \%$$

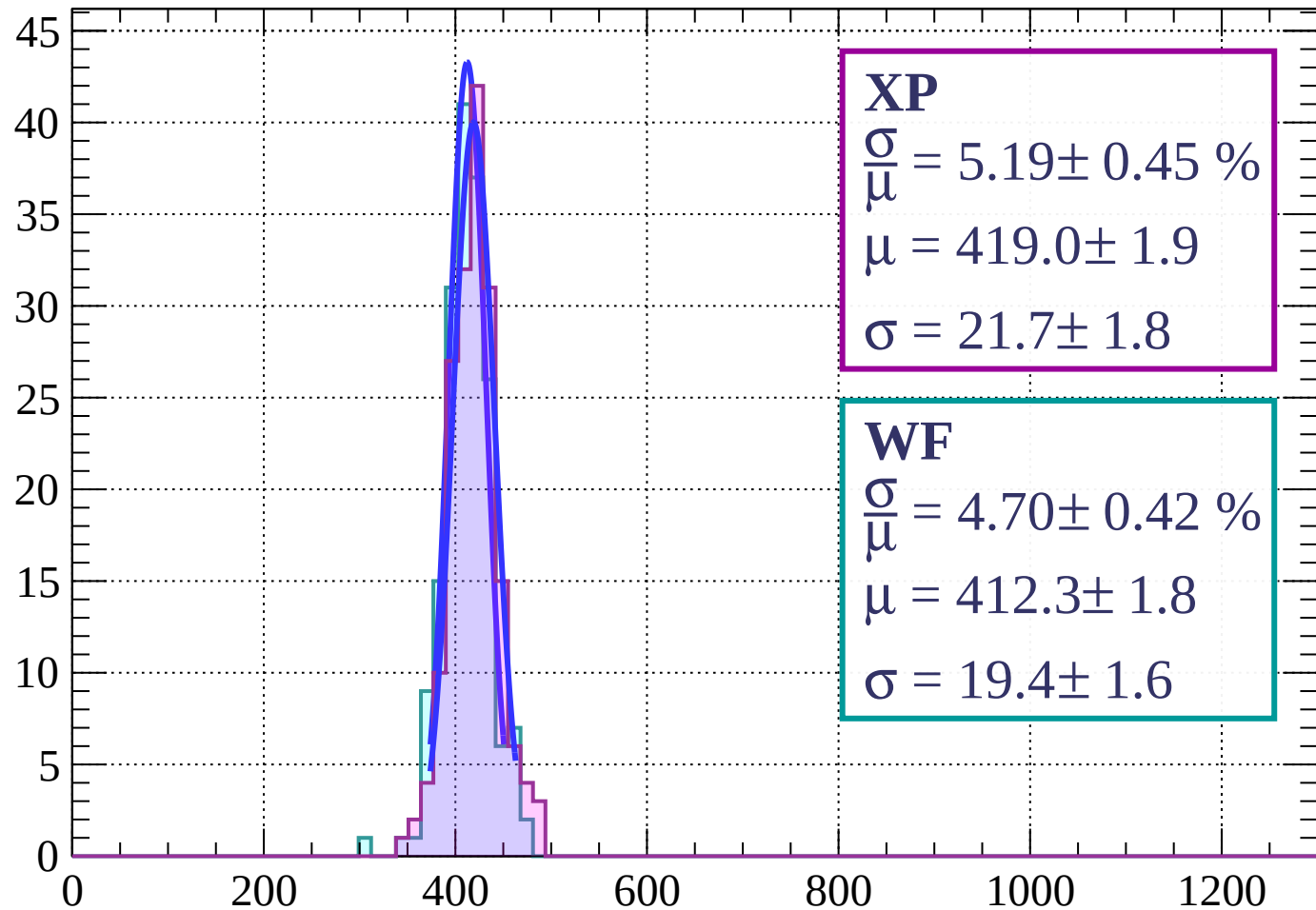
$$\mu = 419.0 \pm 1.2$$

$$\sigma = 23.2 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $15 < \theta < 18$

Count



XP

$$\frac{\sigma}{\mu} = 5.19 \pm 0.45 \%$$

$$\mu = 419.0 \pm 1.9$$

$$\sigma = 21.7 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 4.70 \pm 0.42 \%$$

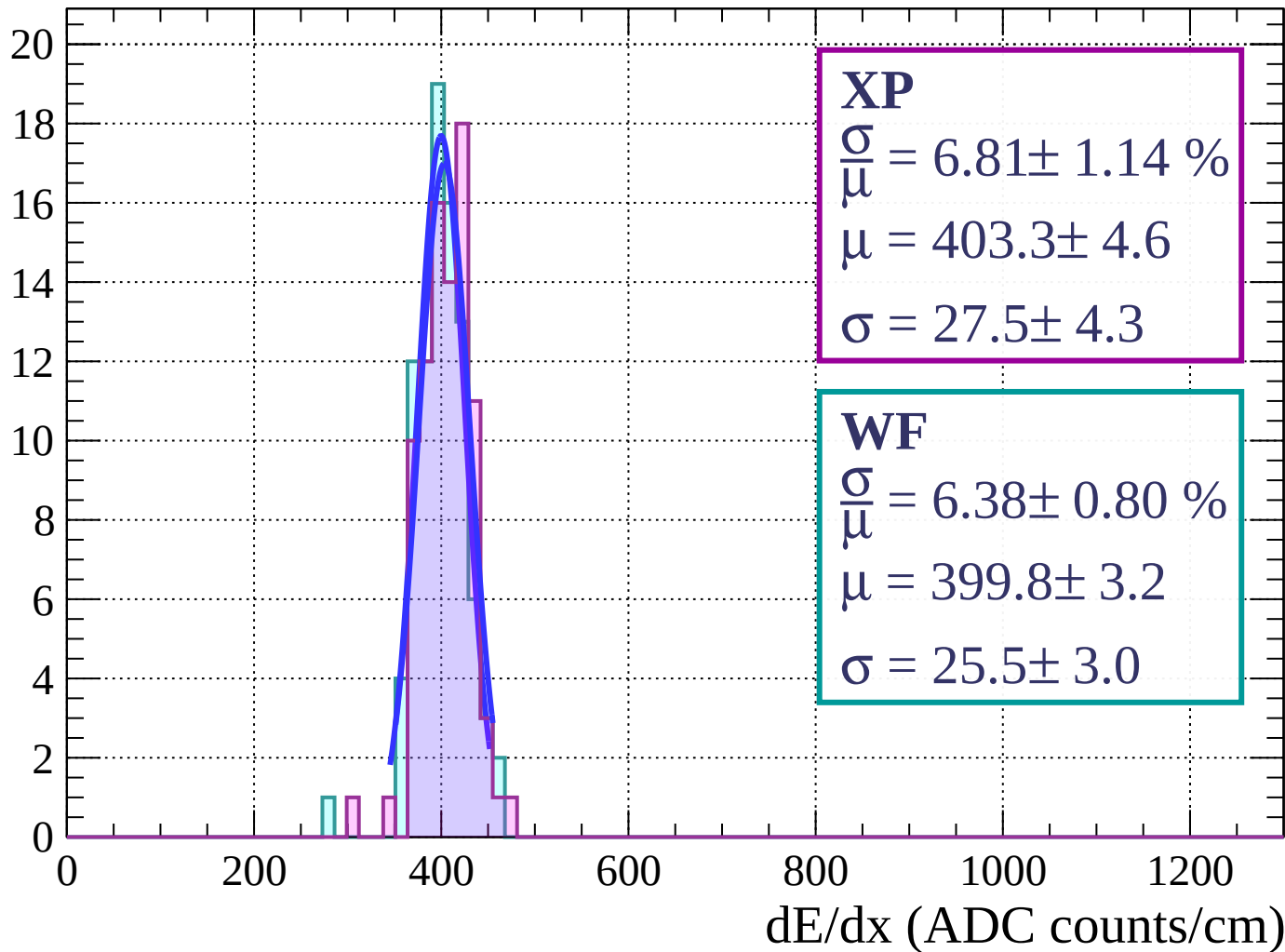
$$\mu = 412.3 \pm 1.8$$

$$\sigma = 19.4 \pm 1.6$$

dE/dx (ADC counts/cm)

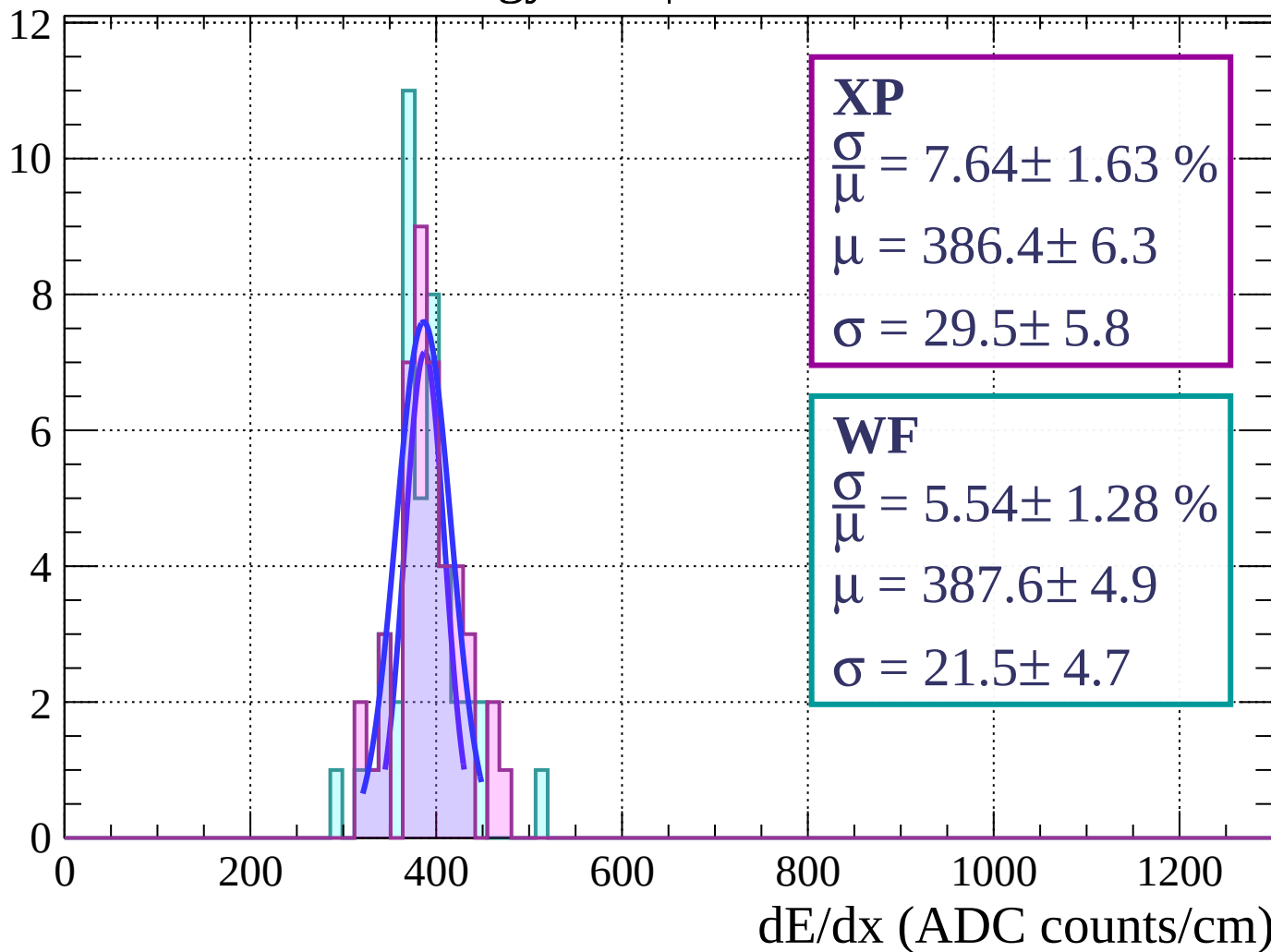
Energy loss | $18 < \theta < 21$

Count



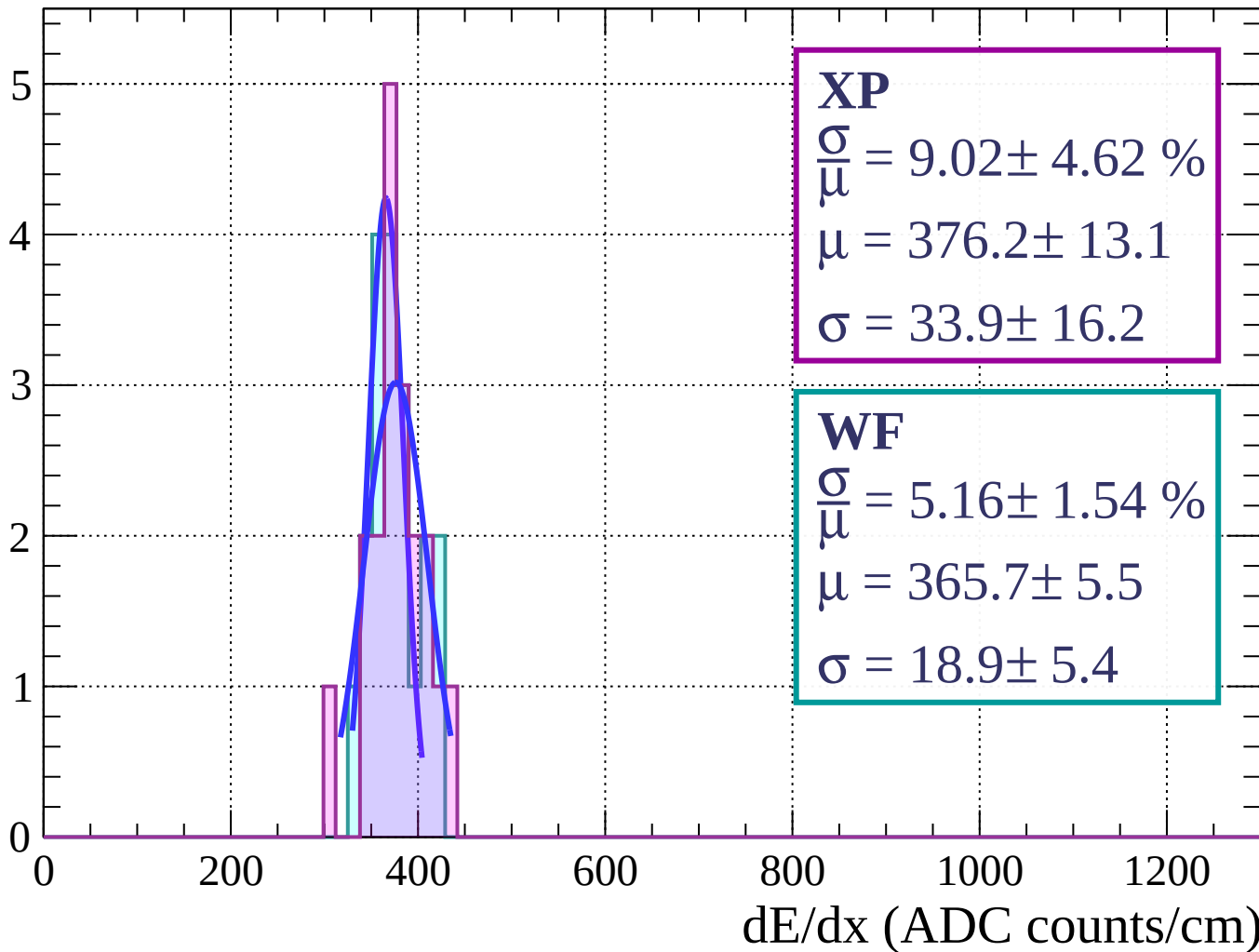
Energy loss | $21 < \theta < 24$

Count



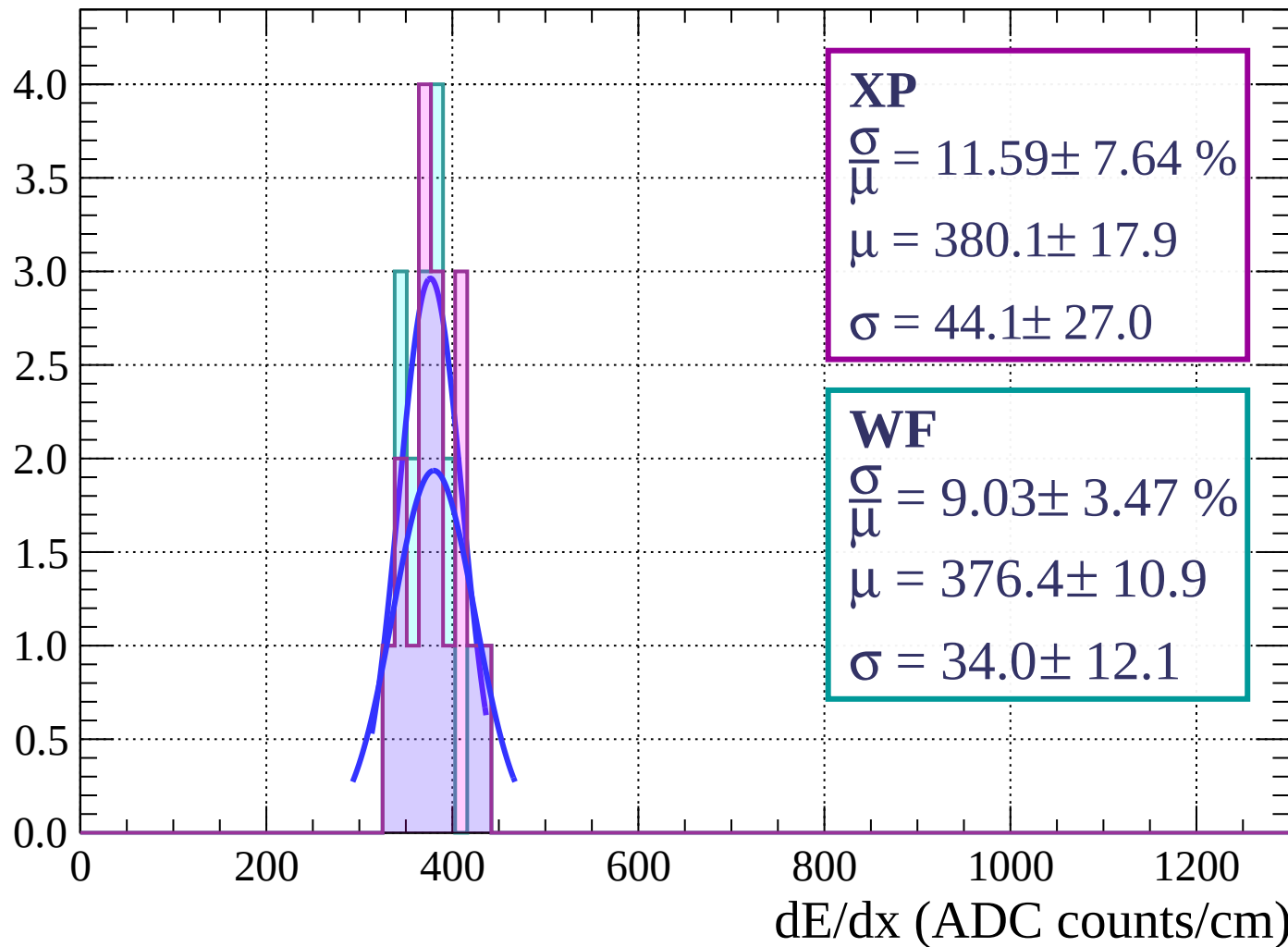
Energy loss | $24 < \theta < 27$

Count



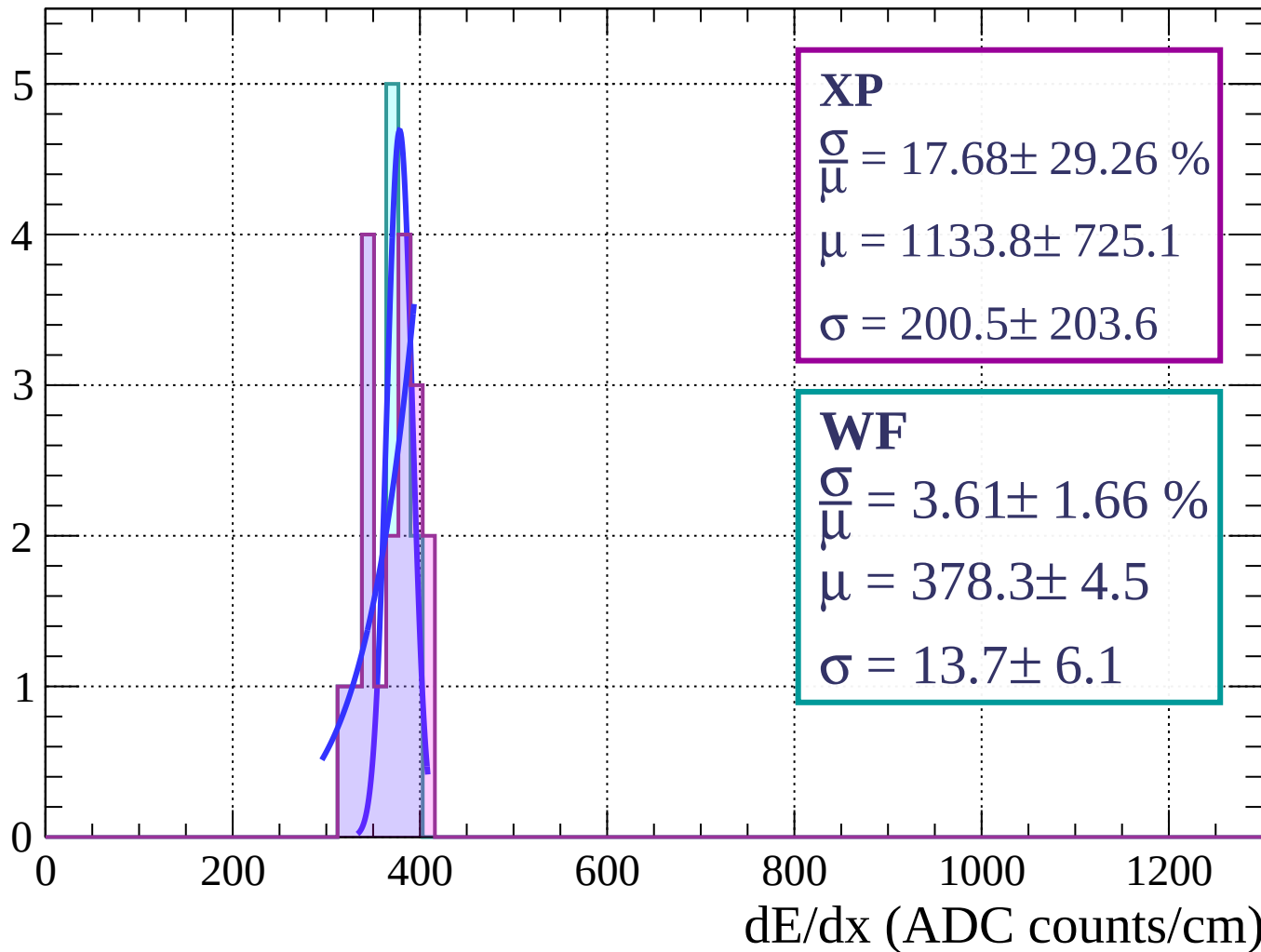
Energy loss | $27 < \theta < 30$

Count



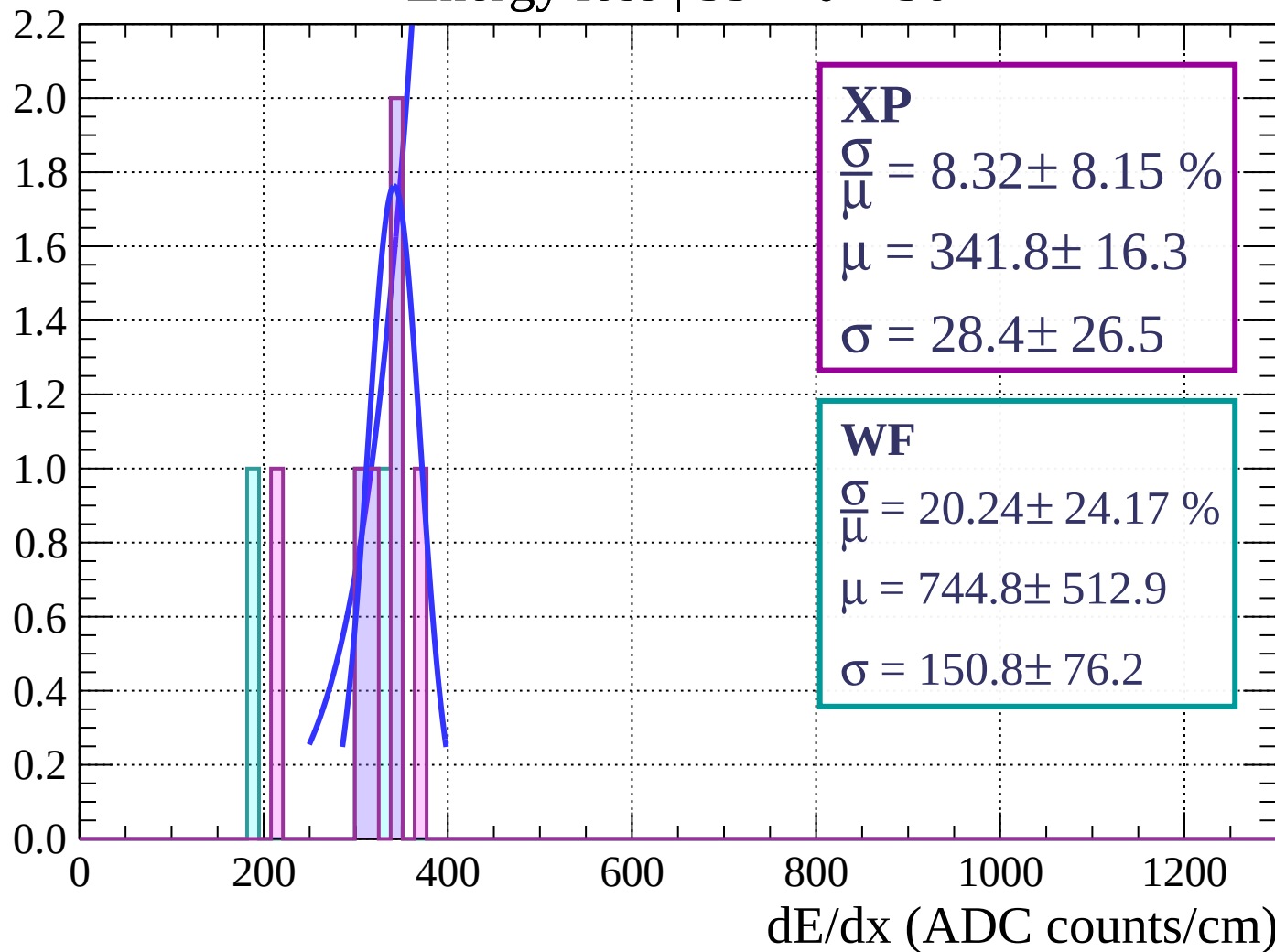
Energy loss | $30 < \theta < 33$

Count



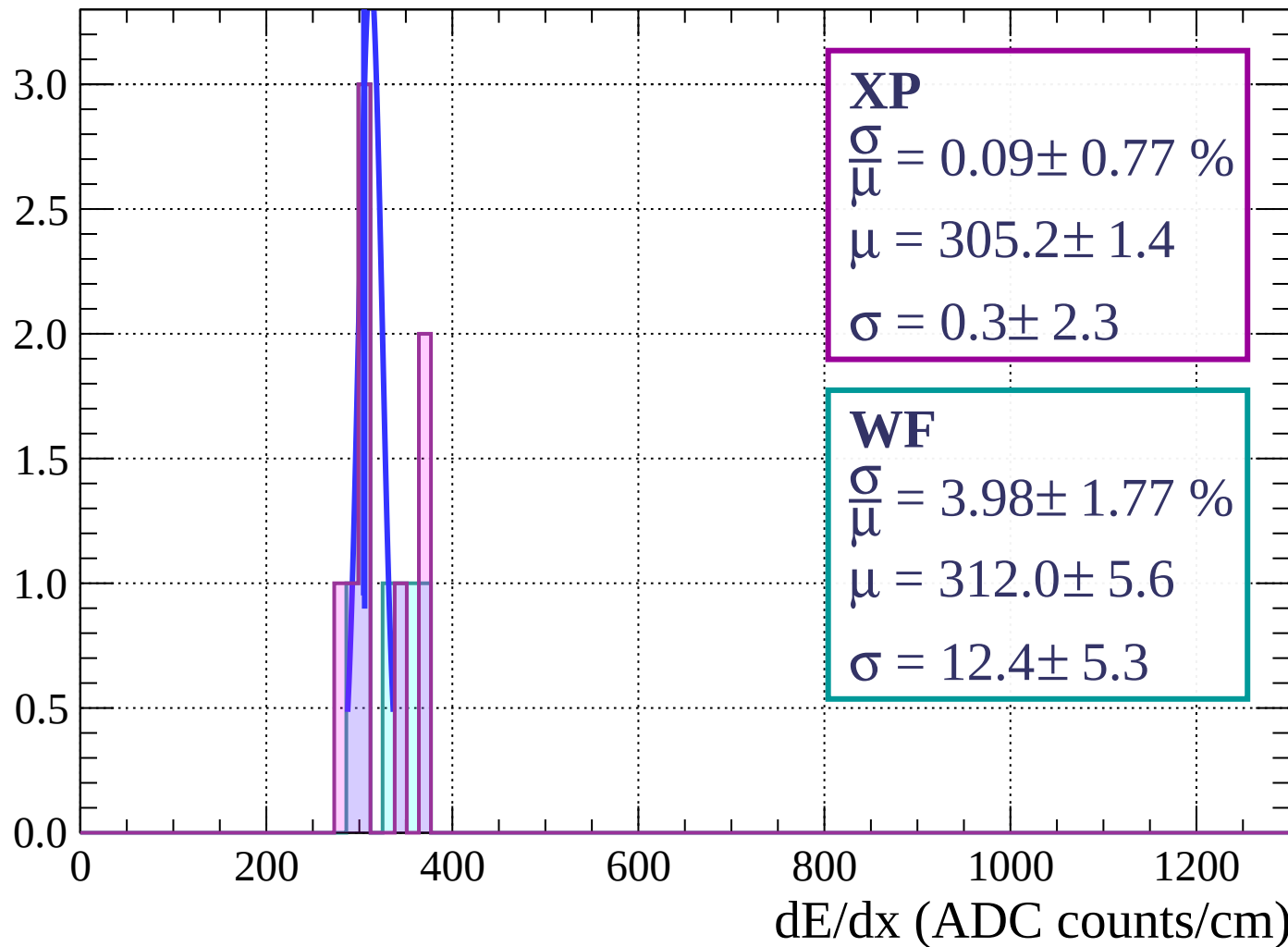
Energy loss | $33 < \theta < 36$

Count



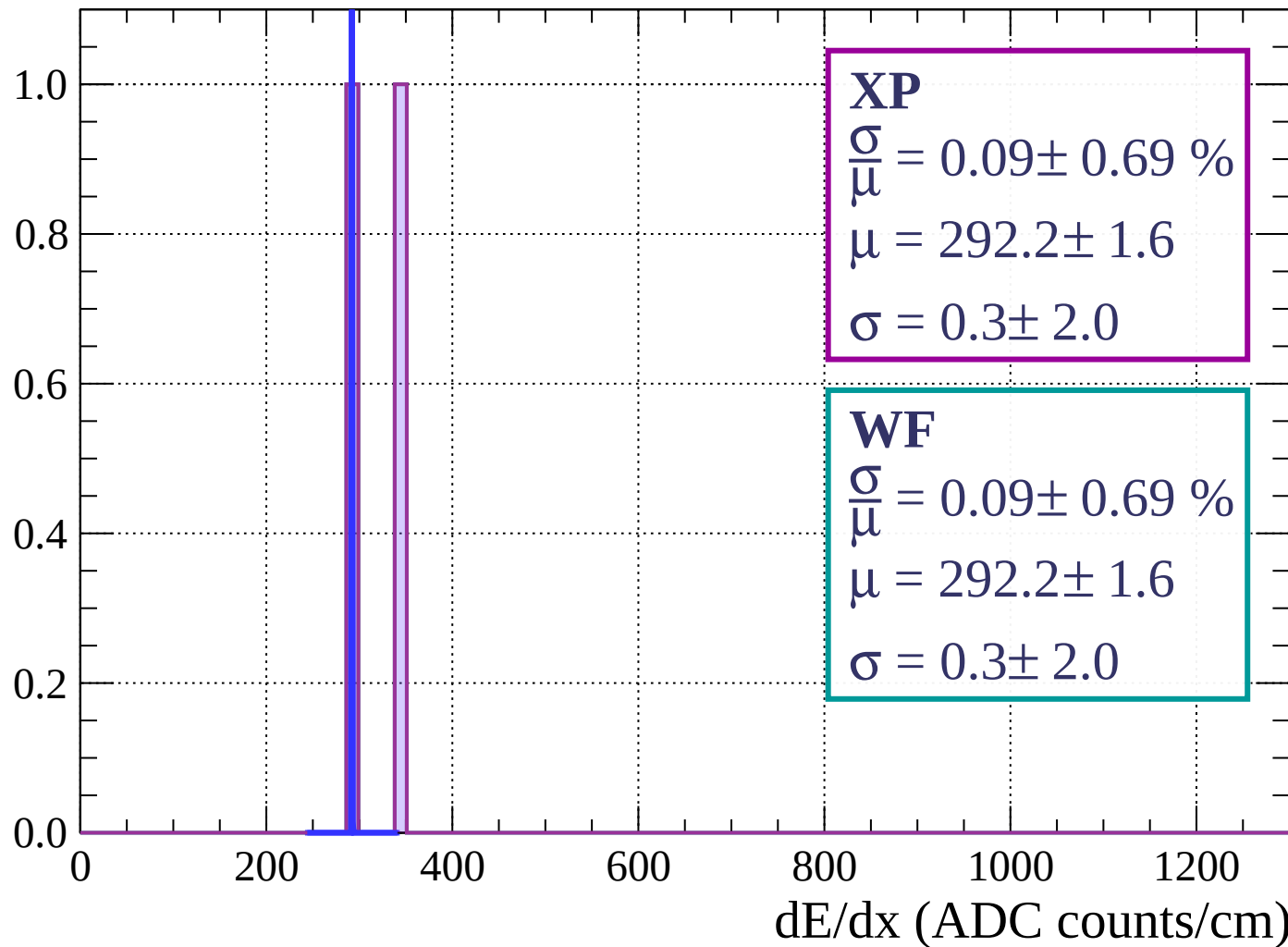
Energy loss | $36 < \theta < 39$

Count



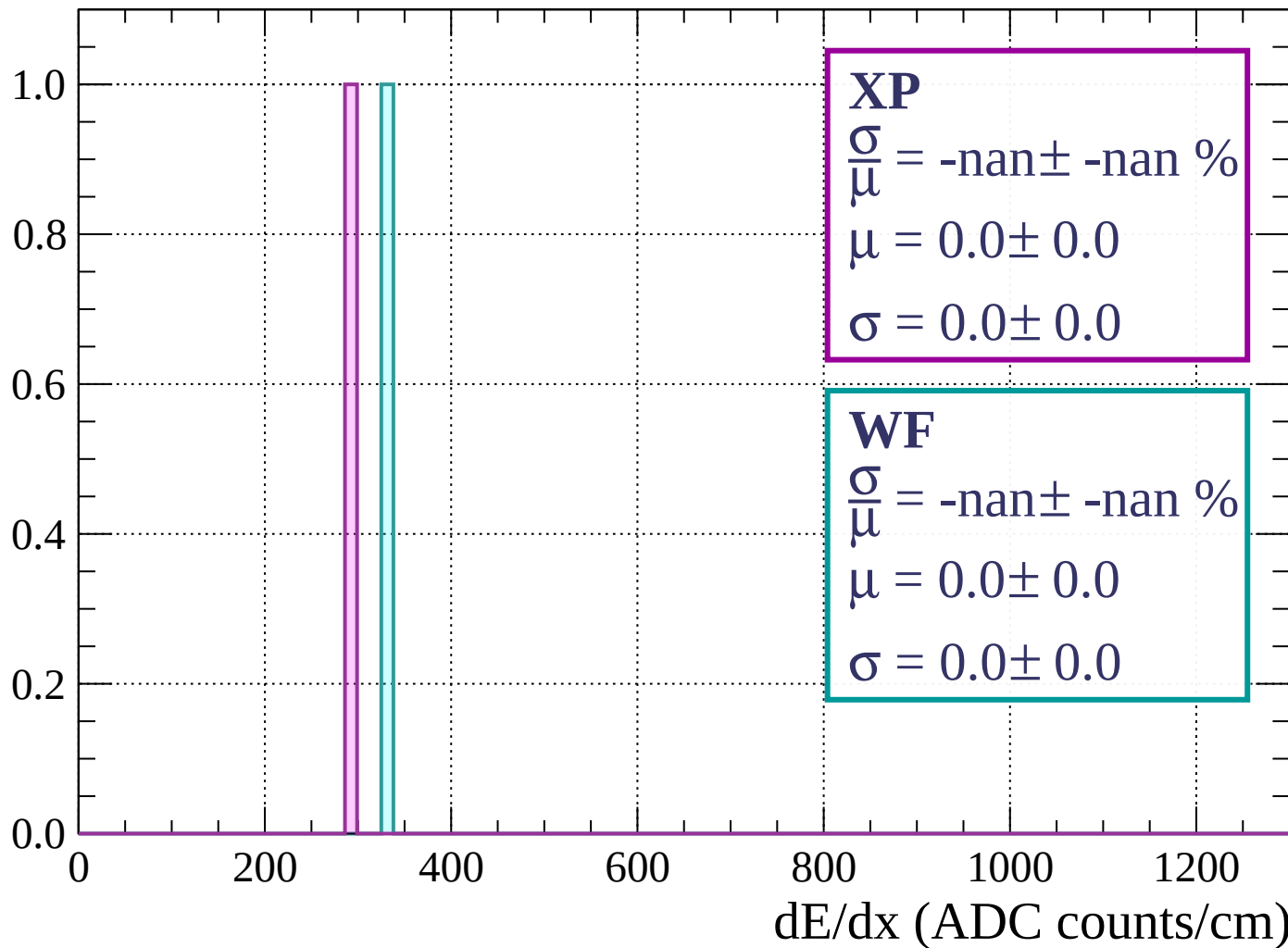
Energy loss | $39 < \theta < 42$

Count



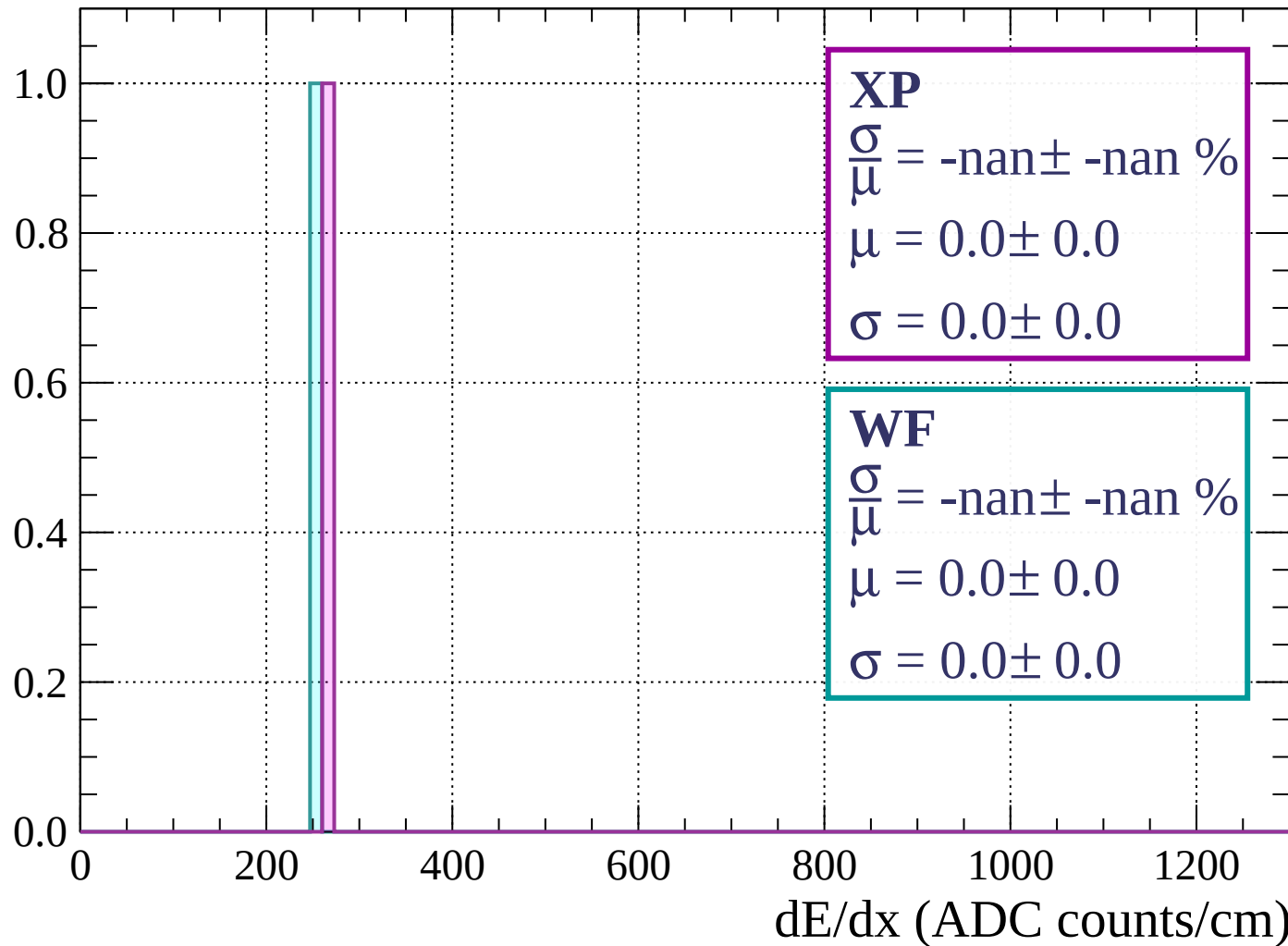
Energy loss | $42 < \theta < 45$

Count



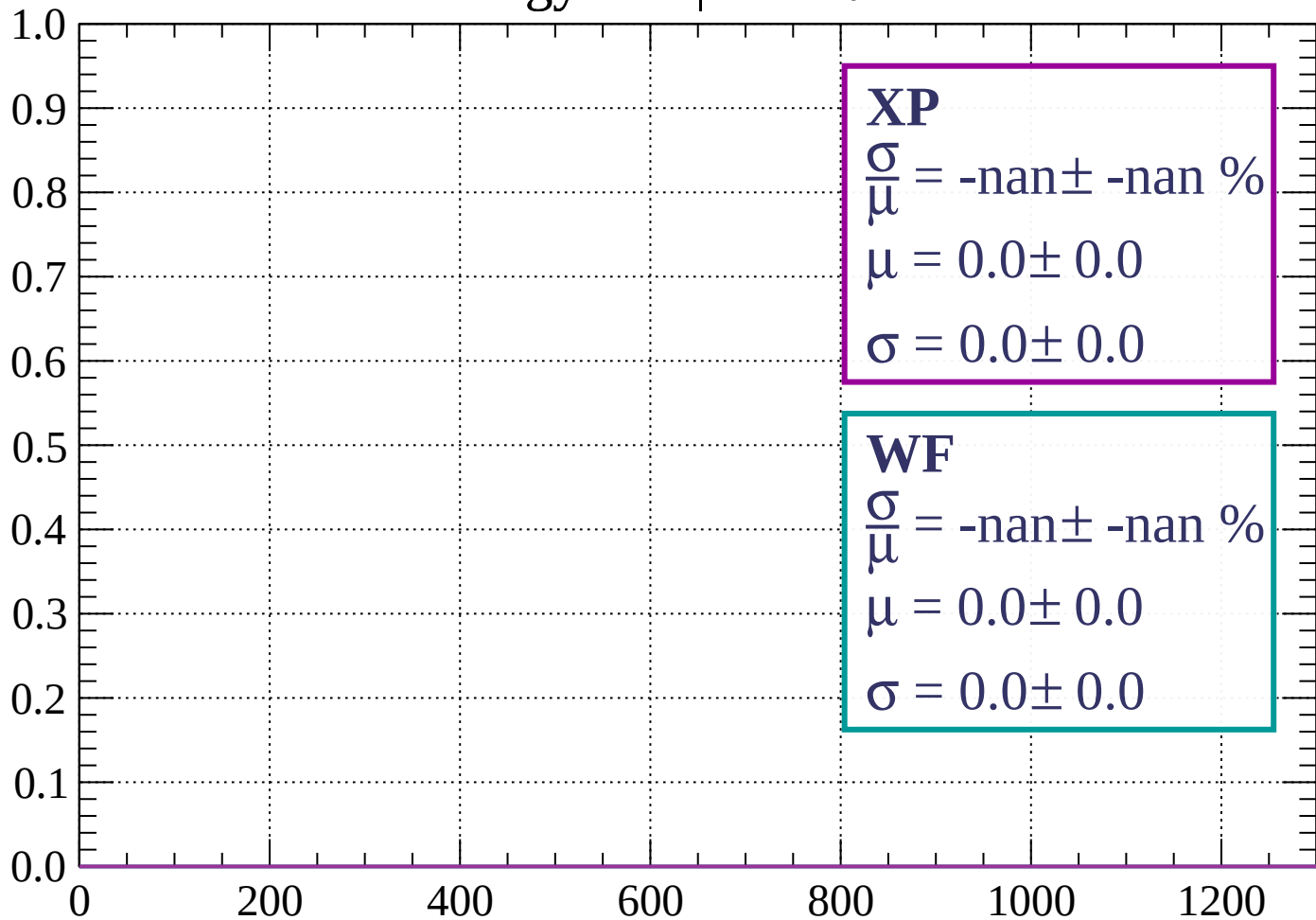
Energy loss | $45 < \theta < 48$

Count



Energy loss | $48 < \theta < 51$

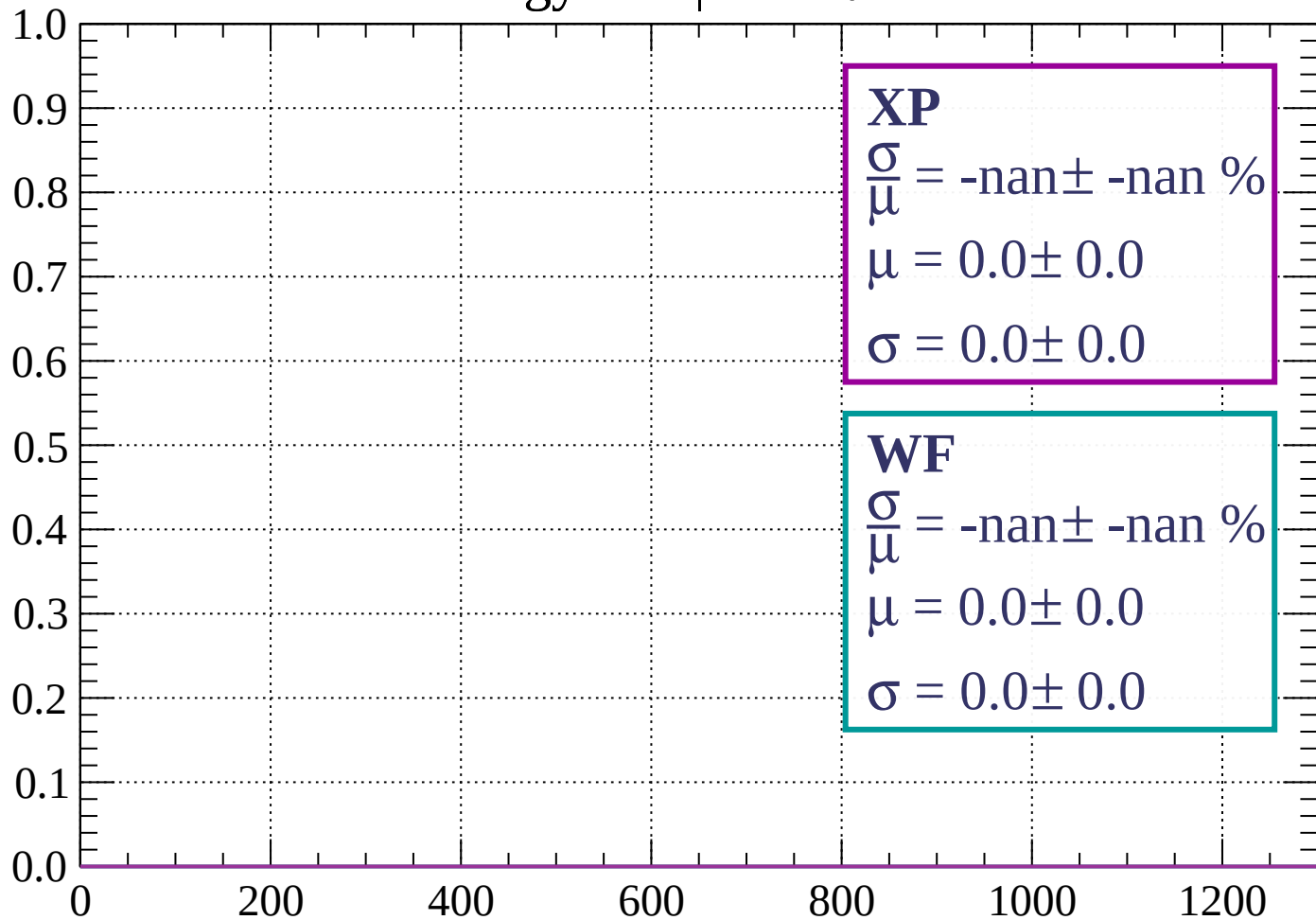
Count



dE/dx (ADC counts/cm)

Energy loss | $51 < \theta < 54$

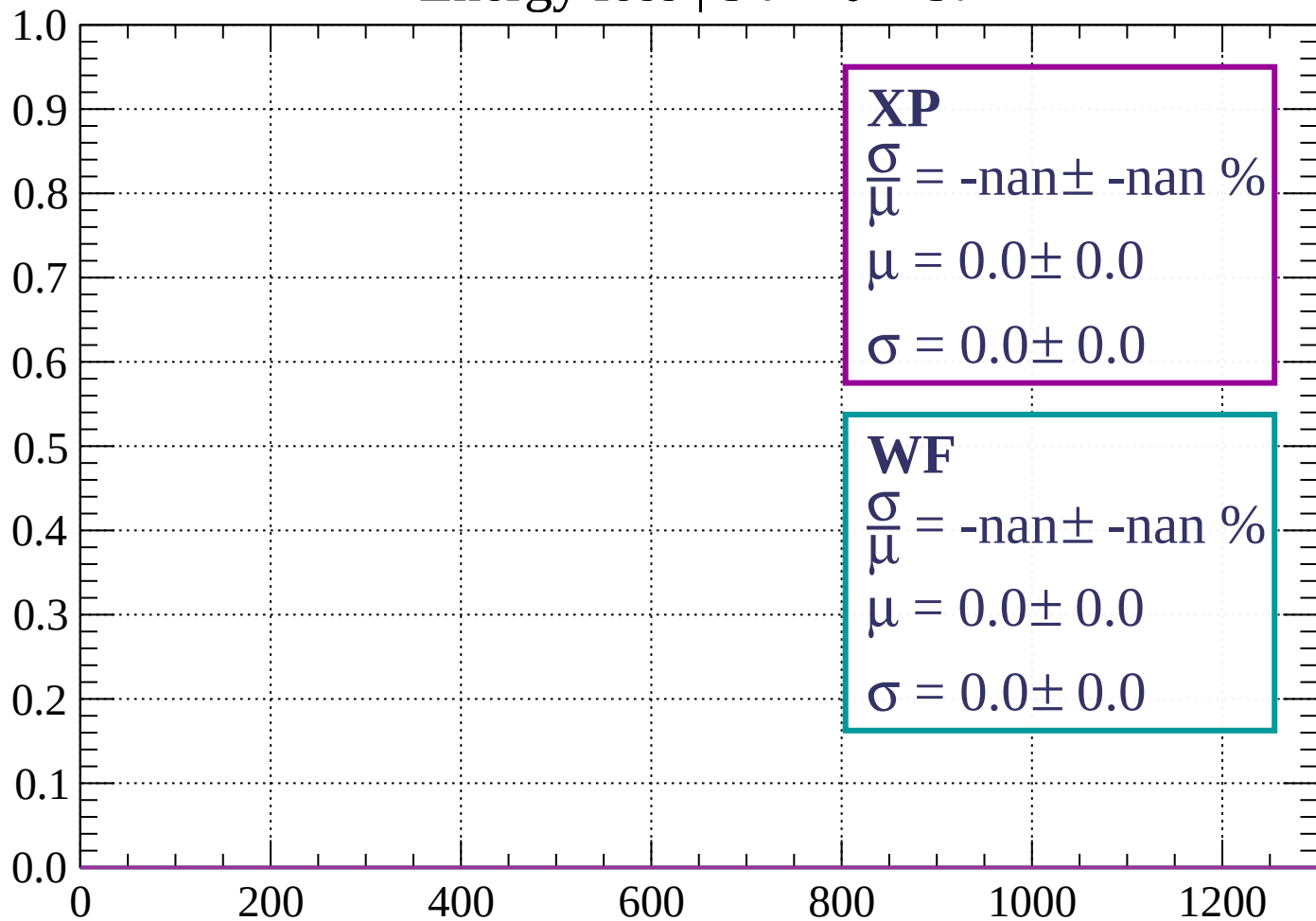
Count



dE/dx (ADC counts/cm)

Energy loss | $54 < \theta < 57$

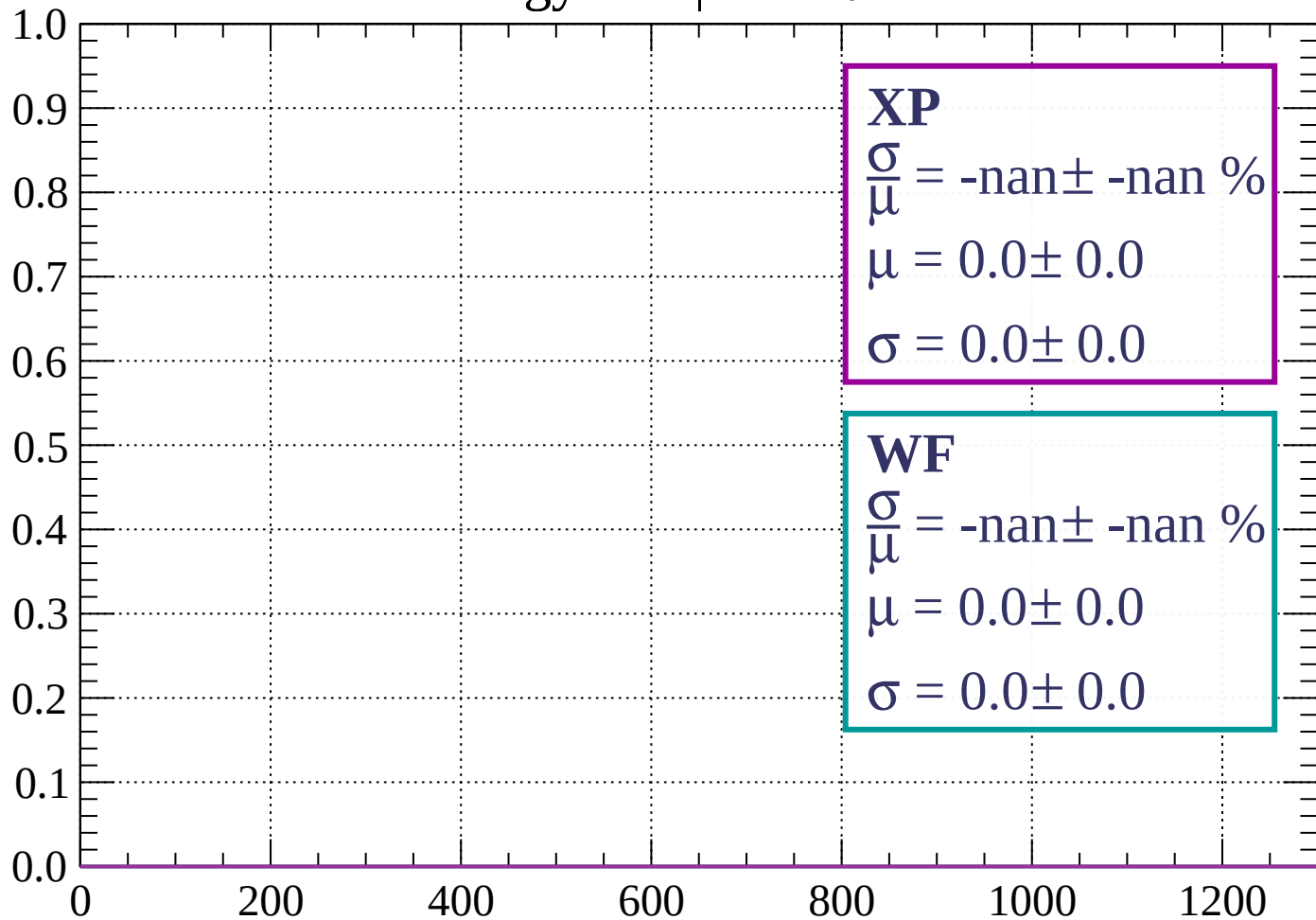
Count



dE/dx (ADC counts/cm)

Energy loss | $57 < \theta < 60$

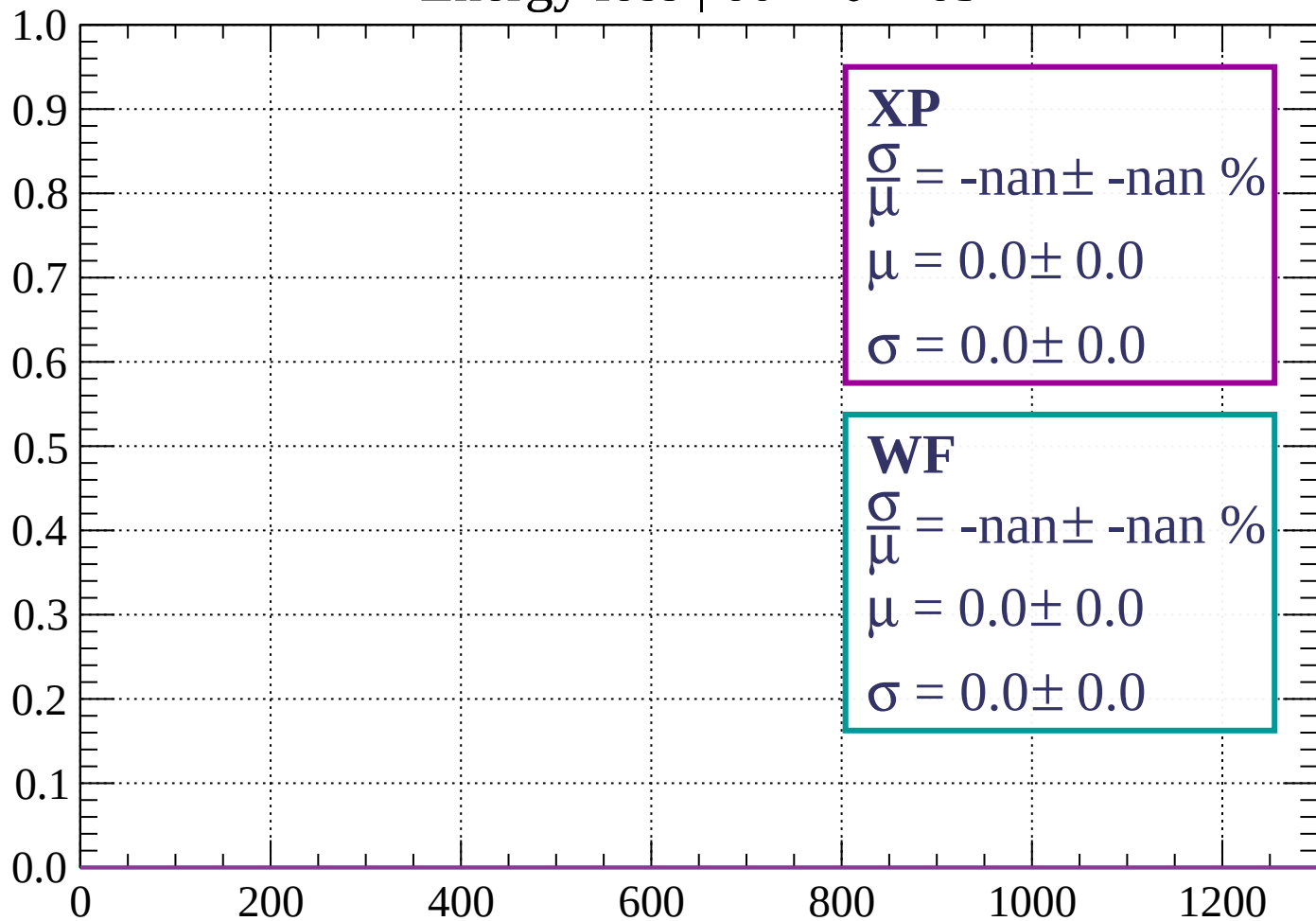
Count



dE/dx (ADC counts/cm)

Energy loss | $60 < \theta < 63$

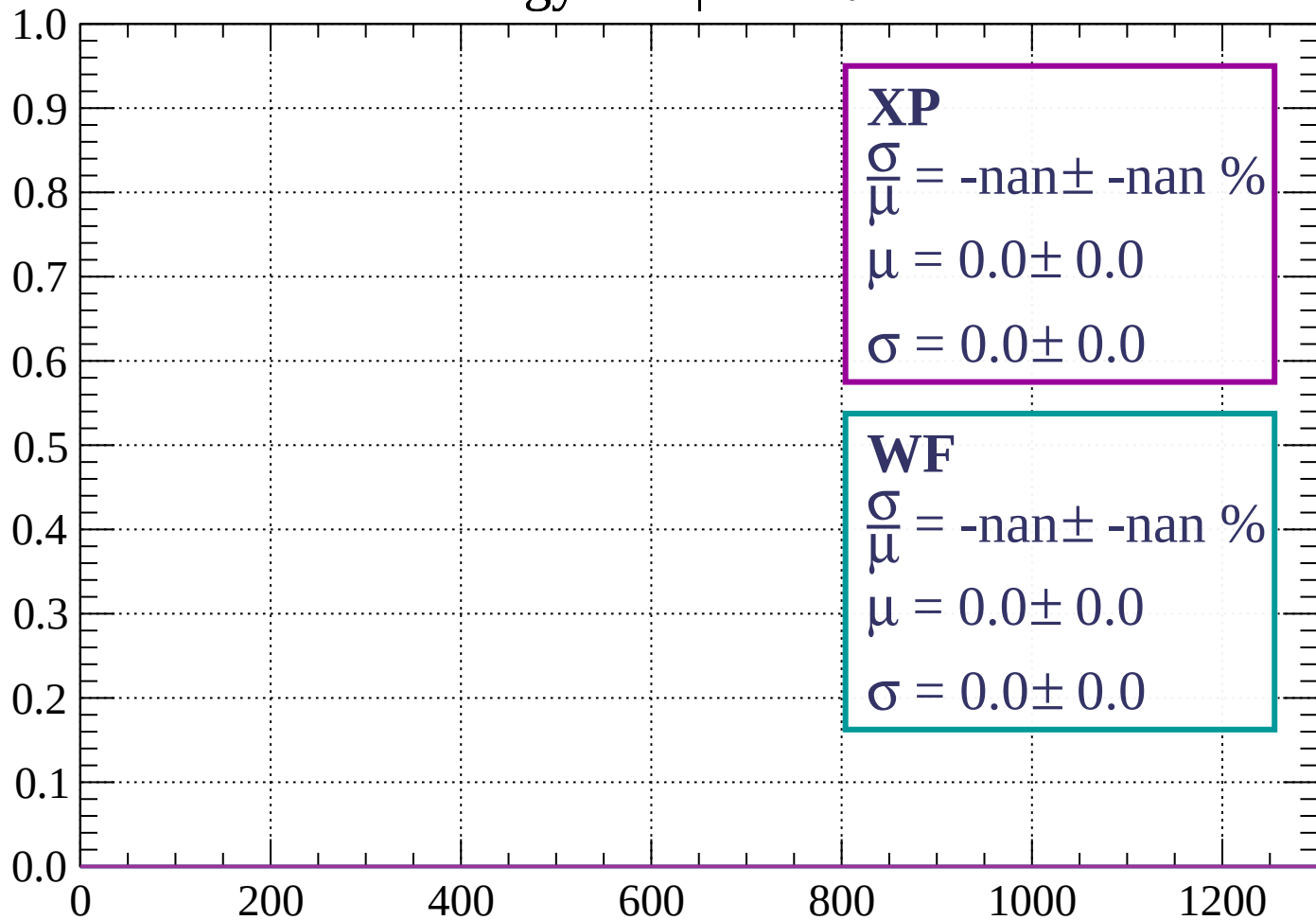
Count



dE/dx (ADC counts/cm)

Energy loss | $63 < \theta < 66$

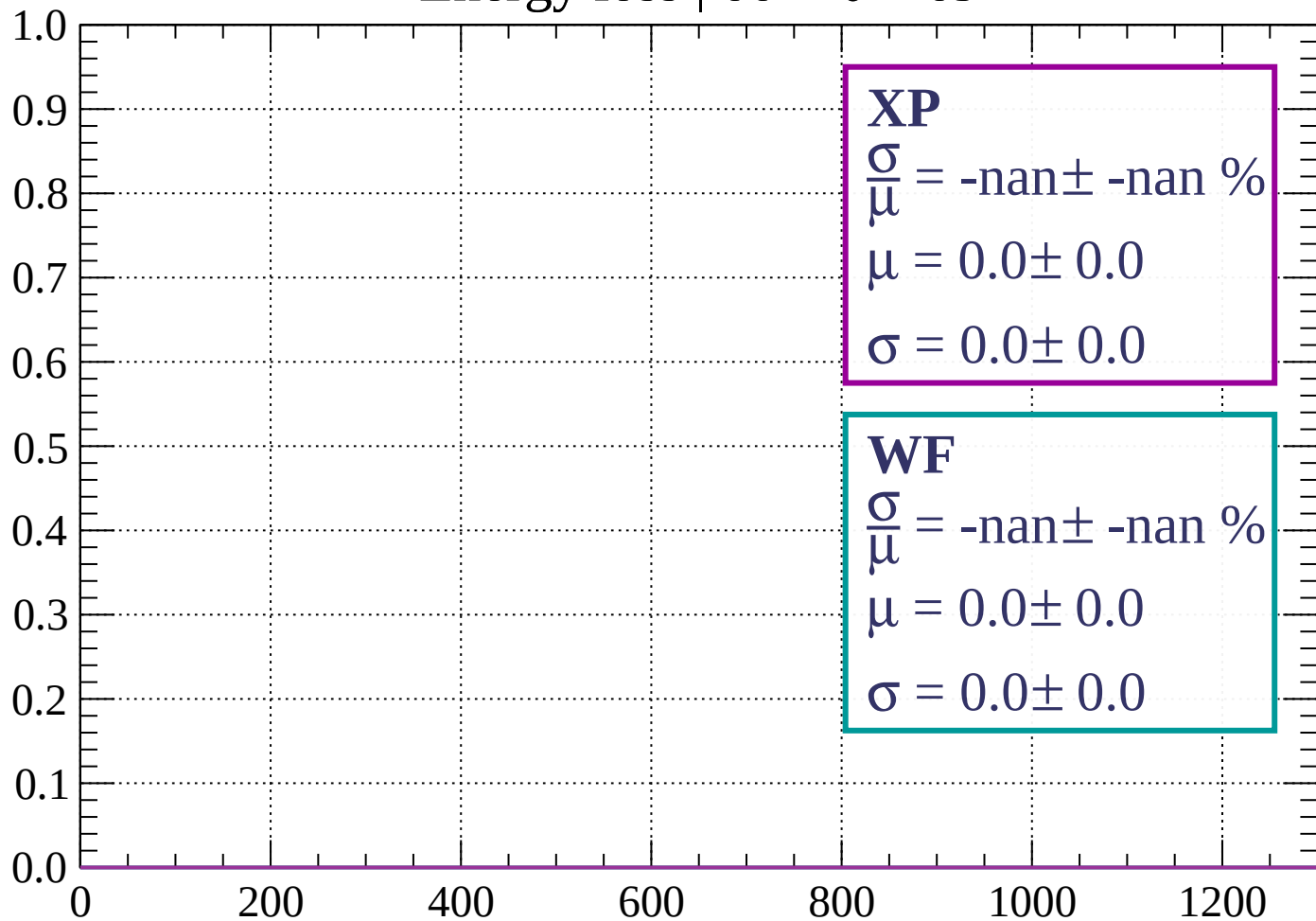
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 69$

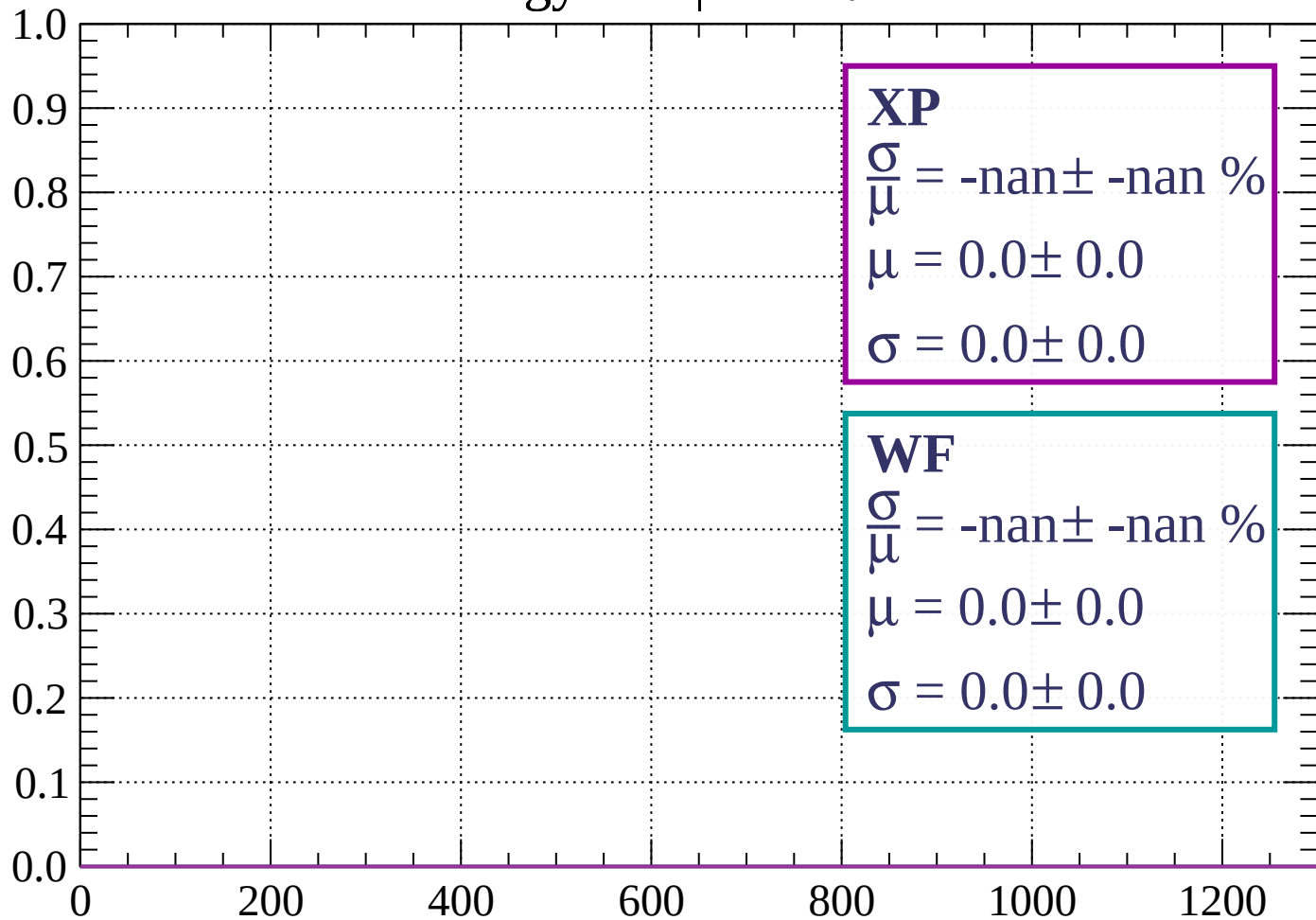
Count



dE/dx (ADC counts/cm)

Energy loss | $69 < \theta < 72$

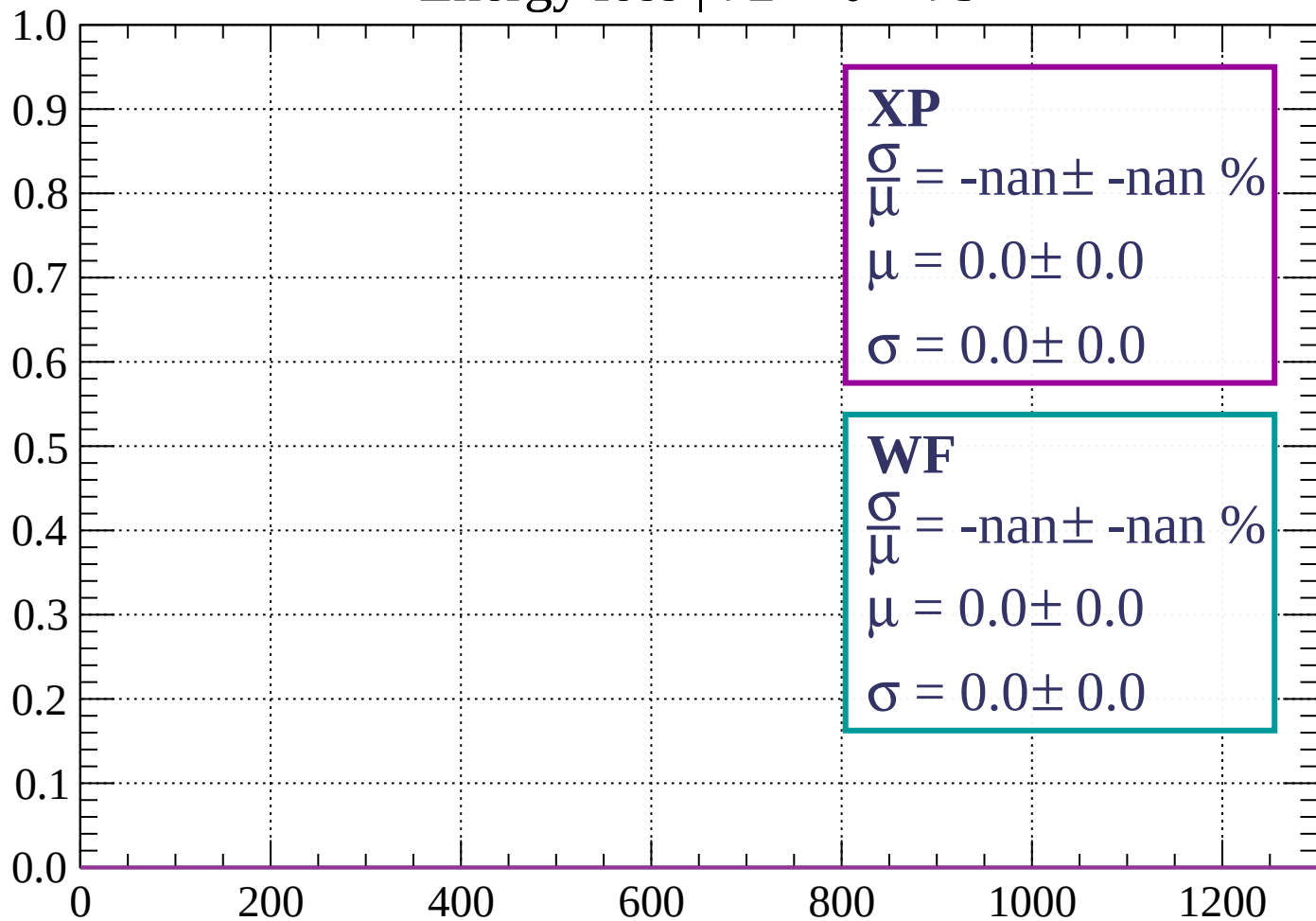
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 75$

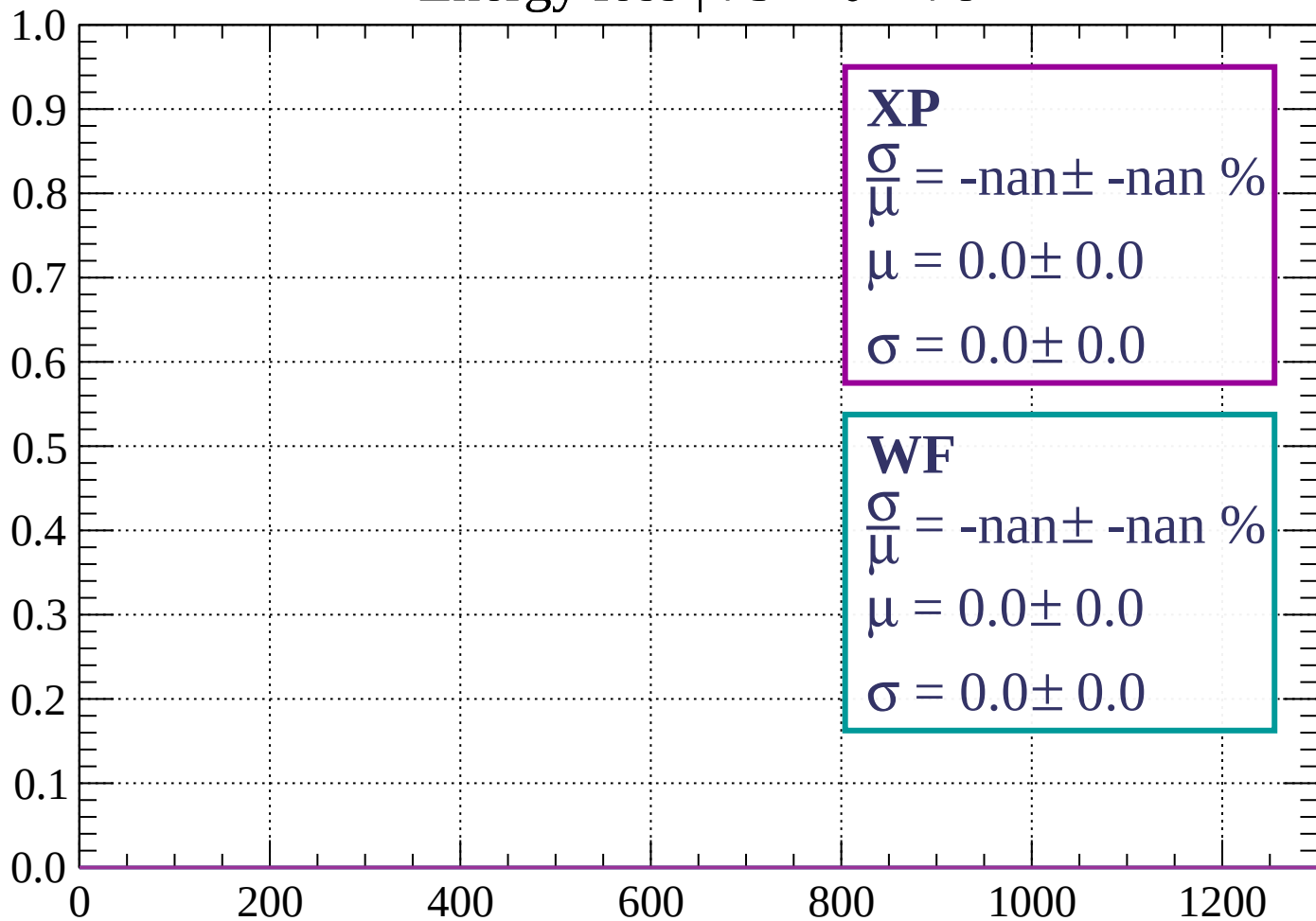
Count



dE/dx (ADC counts/cm)

Energy loss | $75 < \theta < 78$

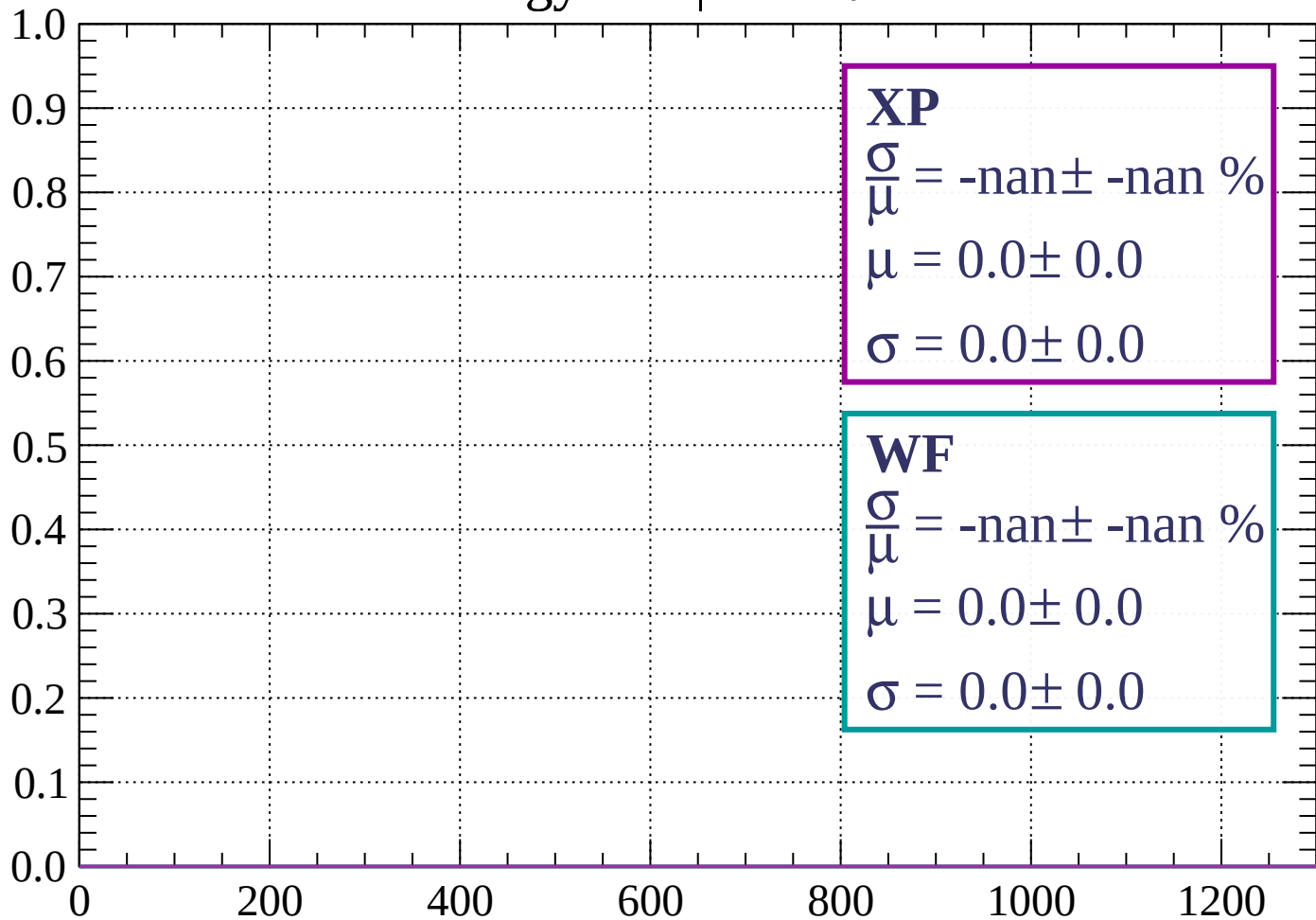
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 81$

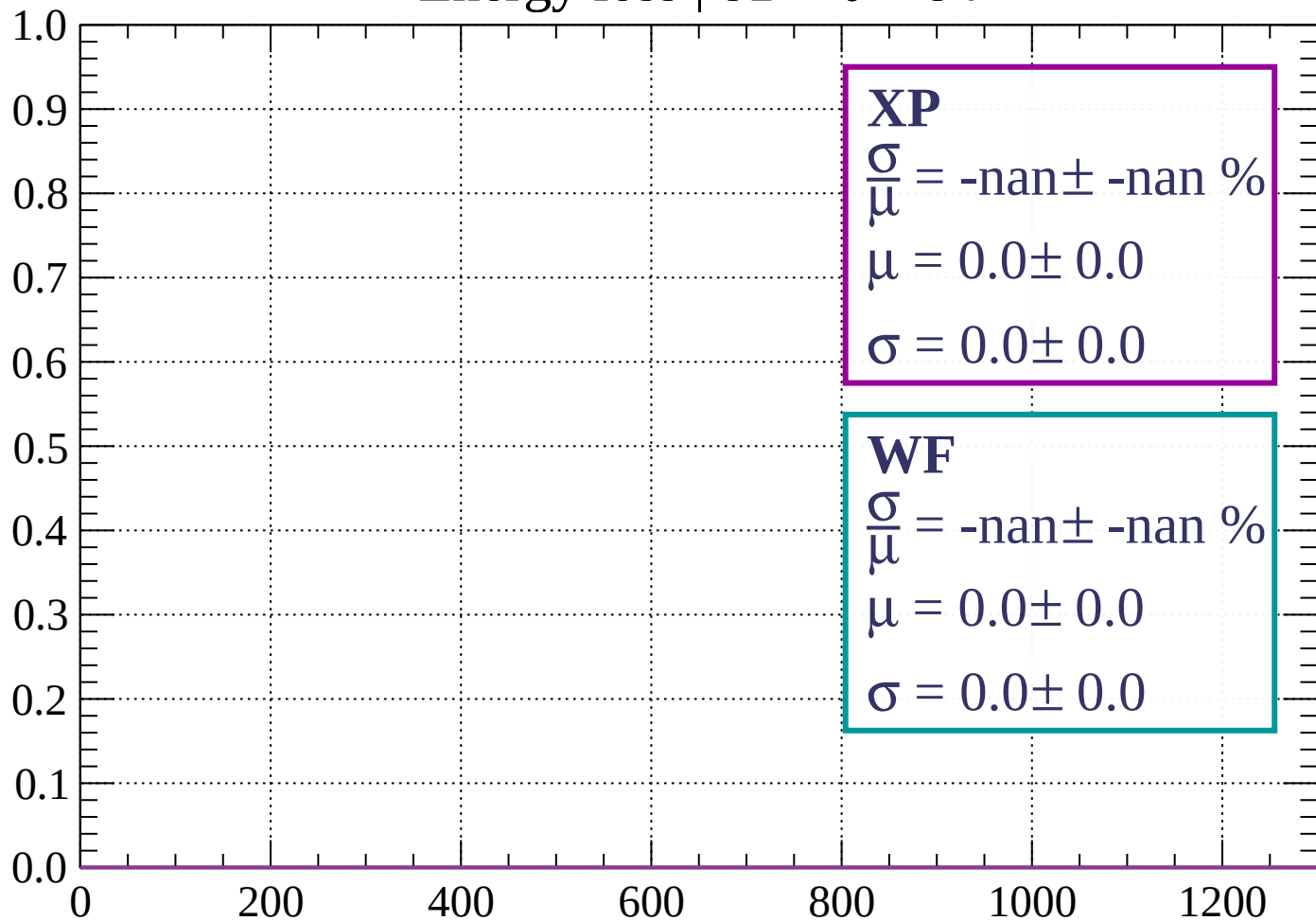
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \theta < 84$

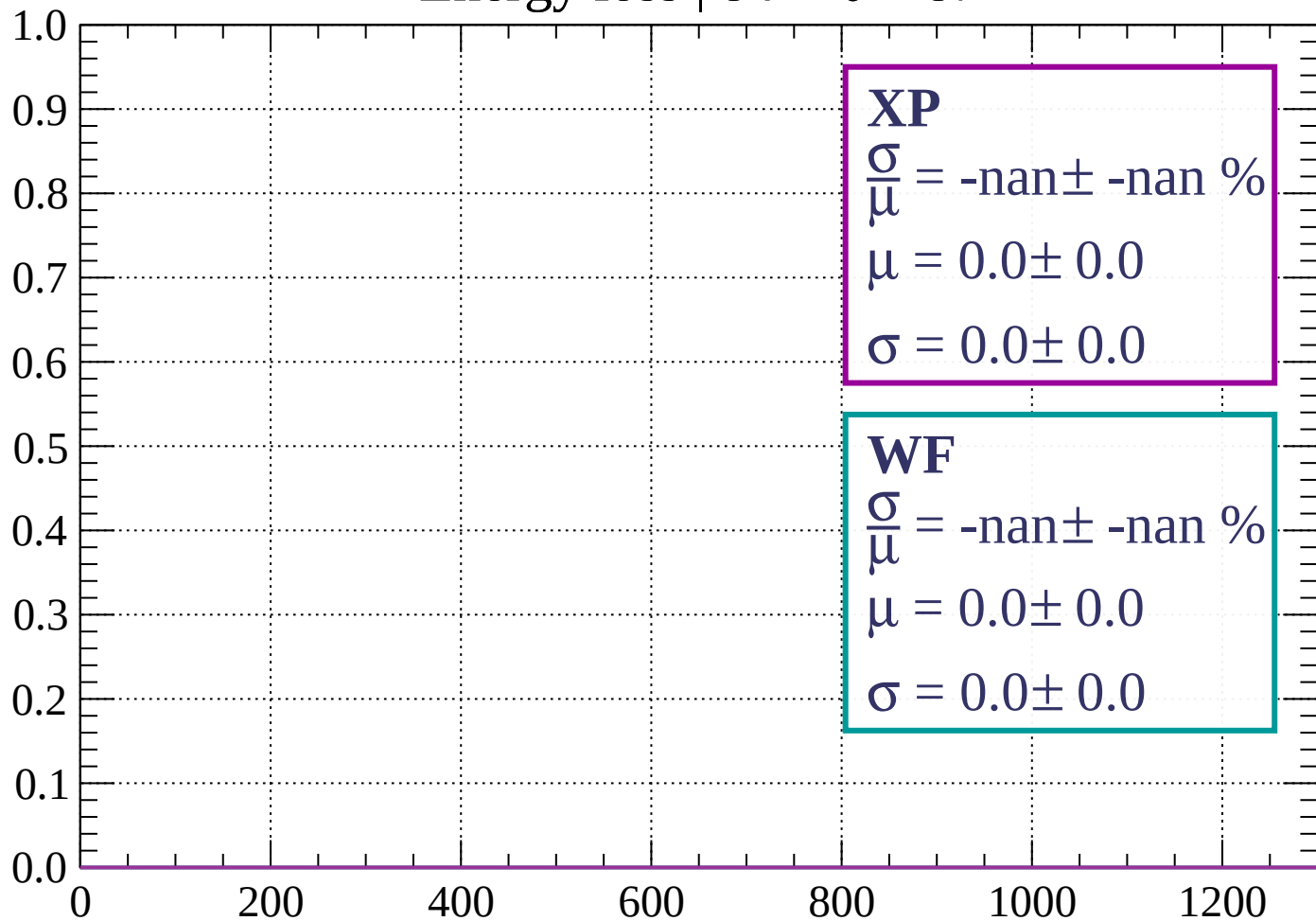
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

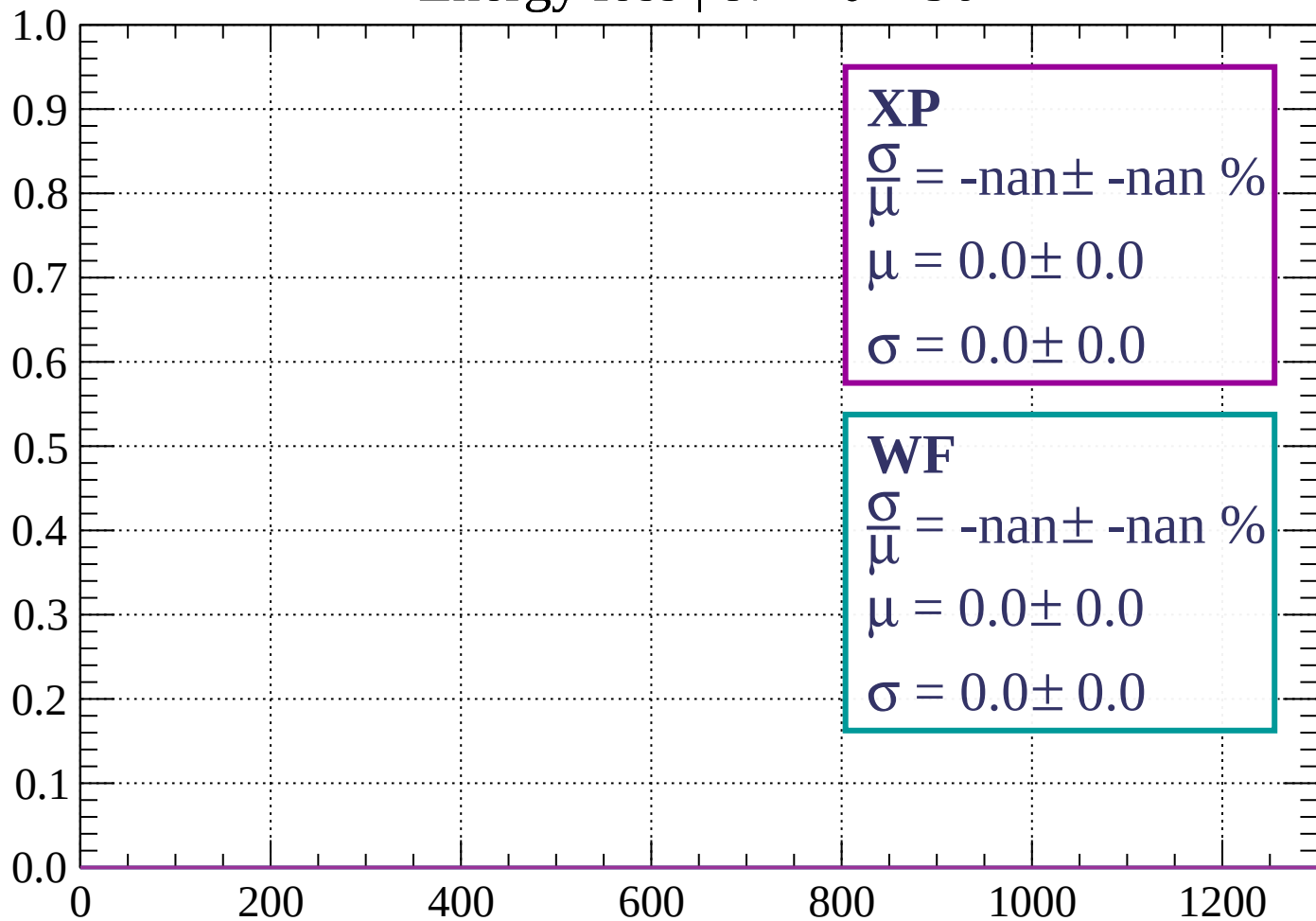
Count



dE/dx (ADC counts/cm)

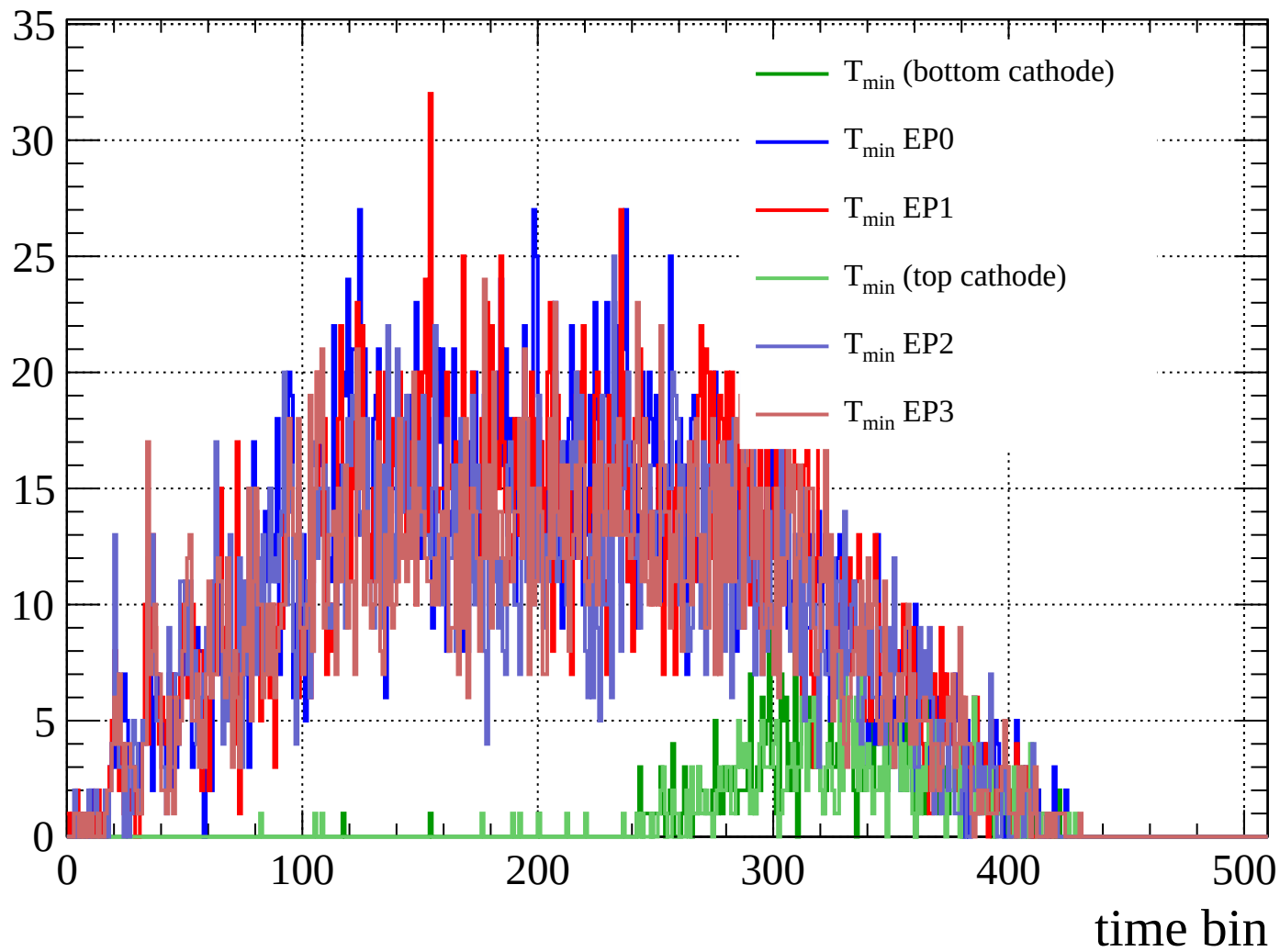
Energy loss | $87 < \theta < 90$

Count



dE/dx (ADC counts/cm)

Count



Count

